

Petrophysics Handbook

User manual for petrolib — the main code for Petrophysics Jan/Feb
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504 functions and 6 classes across 39 modules.

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accuracy

petrolib.ml_stats

```
accuracy(y_true, y_pred)
```

Purpose

Fraction of matching labels.

Input parameters

Parameter	Type	Default
y_true	ArrayLike	required
y_pred	ArrayLike	required

Output

float

Sources

- Article 8: A Comparative Study of Three Supervised Machine-Learning Algorithms for Classifying Carbonate Vuggy Facies in the Kansas Arbuckle Formation. Deng, Xu, Jobe, Xu (2019). DOI: 10.30632/PJV60N6-2019a8. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

acoustic_impedance

petrolib.acoustic_geomech

```
acoustic_impedance(rho, v, *, out='rayl')
```

Purpose

Acoustic impedance $Z = \rho \cdot v$.

out="rayl" returns SI Rayl (kg/m²/s); "mrayl" returns MRayl (Z/1e6).

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
v	ArrayLike	required
out (kw-only)	str	'rayl'

Output

numpy.ndarray of float64

Sources

- Article 1: A Case Study about Formation Evaluation and Rock Physics Modeling of the Bazhenov Shale. Pavel Kulyapin and Tatiana F. Sokolova (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 211-218. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: New 4.75-in. Ultrasonic LWD Technology Provides High-Resolution Caliper and Imaging in Oil-Based and Water-Based Muds. Li, Lee, Coates, Jin, Wong (2019). DOI: 10.30632/PJV60N6-2019a2. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 7: Revealing Hidden Information - High- Resolution Logging-While-Drilling Slowness Measurements and Imaging Using Advanced Dual Ultrasonic Technology. Blyth, Sakiyama, Hori, Yamamoto, Nakajima, Fahim Ud Din, Haecker, Kittridge (2021). DOI: 10.30632/PJV62N1-2021a6. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).
- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Ultrasonic Microscopy Imaging of Carbonate Reservoir Pore Structure. Based on: Chen et al., "New Methodology for Ultrasonic Microscopy Imaging of Carbonate Reservoirs' Pore Structure", Petrophysics, Vol. 66, No. 2, April 2025, pp. 267-282.

acoustic_impedance

petrolib.integrity_drilling

```
acoustic_impedance(rho, v, *, rho_unit='kg/m3')
```

Purpose

Acoustic impedance $Z = \rho \cdot v$ in MRayl.

`rho_unit='kg/m3'` takes SI density; `'g/cc'` takes g/cm^3 (the CBL articles' convention, $Z = \rho_{\text{gcc}} * v / 1000$).

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
v	ArrayLike	required
rho_unit (kw-only)	str	'kg/m3'

Output

numpy.ndarray of float64

Sources

- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).

activity

petrolib.nuclear

```
activity(n, half_life)
```

Purpose

Radioactive activity $A = \lambda N$ with $\lambda = \ln(2)/T_{\text{half}}$.

Input parameters

Parameter	Type	Default
n	ArrayLike	required
half_life	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: In-Situ Evaluation of Vapor Properties Using Condensed Vapor Gamma. O'Sullivan (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 334-345. DOI: none assigned (this issue predates SPWLA DOI assignment).

add_gaussian_noise

petrolib.data_qc_io.signal

```
add_gaussian_noise(x, sigma_fraction, rng=None)
```

Purpose

Add zero-mean Gaussian noise with $\text{sigma} = \text{fraction} * \text{mean}(|x|)$.

The scale is a single scalar tied to the curve's mean absolute level. `rng=None` uses the deterministic `default_rng(0)` of the source article; pass a seed or Generator to vary.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
sigma_fraction	float	required
rng	np.random.Generator int None	None

Output

numpy.ndarray of float64

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).

advect_disperse_1d

petrolib.flow_transport

```
advect_disperse_1d(c_in, *, length, n_cells, t_total, v, D, k_rxn=0.0, n_steps=None)
```

Purpose

Explicit finite-difference 1-D advection-dispersion-reaction transport.

Solves $dC/dt = -v*dC/dx + D*d^2C/dx^2 - k_{rxn}*C$ on $[0, length]$ with a fixed inflow $C(0)=c_{in}$ (upwind advection, central dispersion, explicit first-order reaction), returning (x, C) at t_{total} . The step is CFL/diffusion-stable ($dt = 0.4*\min(dx/v, dx**2/(2*D))$) unless n_{steps} is given, and C is clipped to $[0, c_{in}]$.

Input parameters

Parameter	Type	Default
<code>c_in</code>	float	required
<code>length</code> (kw-only)	float	required
<code>n_cells</code> (kw-only)	int	required
<code>t_total</code> (kw-only)	float	required
<code>v</code> (kw-only)	float	required
<code>D</code> (kw-only)	float	required
<code>k_rxn</code> (kw-only)	float	0.0
<code>n_steps</code> (kw-only)	int None	None

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

affine_rescale

petrolib.ml_stats

```
affine_rescale(x, *, src_lo, src_hi, dst_lo, dst_hi)
```

Purpose

Map the source range [src_lo, src_hi] linearly onto [dst_lo, dst_hi].

The reference-range (histogram) normalization of Shier (2004) used for multi-well GR normalization — unlike minmax the ranges are caller-supplied percentiles, not data extremes.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
src_lo (kw-only)	float	required
src_hi (kw-only)	float	required
dst_lo (kw-only)	float	required
dst_hi (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Article 6 (Technical Note): Normalizing Gamma-Ray Logs Acquired from a Mixture of Vertical and Horizontal Wells in the Haynesville Shale. Xu, Bayer, Wunderle, Bansal (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 638-643. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microresistivity Curve Extraction from Borehole Microimager Data. Roslin (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 140-146. DOI: none assigned (this issue predates SPWLA DOI assignment).

amott_indices

petrolib.relperm_wettability

```
amott_indices(vw_spont, vw_forced, vo_spont, vo_forced)
```

Purpose

Amott-Harvey indices (I_w , I_o , I_{ah}) from displaced volumes.

$I_w = V_{w_spont} / (V_{w_spont} + V_{w_forced})$, I_o likewise, and the Amott-Harvey index $I_{ah} = I_w - I_o$.

Accepts imbibed volumes or the equivalent saturation changes; each ratio is zero-guarded.

Input parameters

Parameter	Type	Default
vw_spont	float	required
vw_forced	float	required
vo_spont	float	required
vo_forced	float	required

Output

tuple[float, float, float]

Sources

- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Impact of Dopants (NaI) on SCAL Experiments. Pairoys et al., 2025, Petrophysics 66(1), 123-133. DOI:10.30632/PJV66N1-2025a9.

anisotropy_coefficient

petrolib.em_dielectric

```
anisotropy_coefficient(rh, rv)
```

Purpose

Resistivity anisotropy coefficient $\lambda = \sqrt{R_v/R_h}$ (canonical).

Input parameters

Parameter	Type	Default
rh	ArrayLike	required
rv	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Inversion-Based Interpretation of Logging-While-Drilling Resistivity and Nuclear Measurements: Field Examples in High-Angle and Horizontal Wells Olabode Ijasan, Carlos Torres-Verdin, William E. Preeg, John Rasmus, Edward J. Stockhausen (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 374-391. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Comparison of the Russian and Western Resistivity Logs in Typical Western Siberian Reservoir Environments - A Numerical Study. Epov, Sukhorukova, Nechaev, Petrov, Rabinovich, Weston, Tyurin, Wang, Abubakar, Claverie (2020). DOI: 10.30632/PJV61N1-2020a1. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 3: Response Characteristics of an Array Induction Tool (HDIL) in Heterogeneous Anisotropic Formations. Liu, Zhang, Zhang, Xu, Kang, Xiao (2020). DOI: 10.30632/PJV61N1-2020a2. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 1: Tutorial - Petrophysics of Thinly Bedded Formations. Aldred (2021). DOI: 10.30632/PJV62N4-2021t1.
- Bower, M., Xie, H., Wang, G. L., Leveque, S., and Dolan, J. (2026). Anisotropy Sensitivity in Ultradeep Azimuthal Resistivity Technologies. Petrophysics, 67(3), 544-559. DOI: 10.30632/PJV67N3-2026a5.

annie_c11

petrolib.acoustic_geomech

```
annie_c11(c33, c44, c66)
```

Purpose

ANNIE approximation $C_{11} = 2*(C_{66} - C_{44}) + C_{33}$.

Input parameters

Parameter	Type	Default
c33	ArrayLike	required
c44	ArrayLike	required
c66	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

annie_c13

petrolib.acoustic_geomech

```
annie_c13(c11, c66)
```

Purpose

ANNIE approximation $C_{13} = C_{11} - 2 \cdot C_{66}$ (assumes $C_{13} = C_{12}$).

Input parameters

Parameter	Type	Default
c11	ArrayLike	required
c66	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

apparent_conductivity_two_zone

petrolib.em_dielectric

```
apparent_conductivity_two_zone(sigma_xo, sigma_t, r_invasion, spacing_m)
```

Purpose

Doll two-zone (invaded + virgin) apparent conductivity.

$\sigma_a = G_{xo} \sigma_{xo} + (1 - G_{xo}) \sigma_t$ with the radial geometric factor G_{xo} of the invasion radius from `doll_radial_geometric_factor`.

Input parameters

Parameter	Type	Default
<code>sigma_xo</code>	ArrayLike	required
<code>sigma_t</code>	ArrayLike	required
<code>r_invasion</code>	ArrayLike	required
<code>spacing_m</code>	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 6: Reminiscences on the Development of the First Commercial Array-Induction Measurement. Peter Elkington (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 618-623. DOI: none assigned (this issue predates SPWLA DOI assignment).

apparent_dip

petrolib.borehole_image

```
apparent_dip(true_dip_deg, section_azimuth_deg)
```

Purpose

Apparent dip at section angle beta to true dip: $\tan(\text{app}) = \tan(\text{true}) \cos(\text{beta})$ (deg).

Input parameters

Parameter	Type	Default
true_dip_deg	float	required
section_azimuth_deg	float	required

Output

float

Sources

- Article 6: Maximizing Net Pay in Penta-Lateral Well With Advanced Proactive Geosteering and 2D Structural Analysis Using Azimuthal Resistivity Measurements. Antonov, Kushnir, Martakov, Pazos, Small, Tropin, Maraj, Itter, Nelson, Rabinovich (2021). DOI: 10.30632/PJV62N3-2021a5. Petrophysics Vol. 62 No. 3 (Jun 2021).

apparent_resistivity_dip

petrolib.em_dielectric

```
apparent_resistivity_dip(rh, rv, dip_deg)
```

Purpose

Apparent resistivity at relative dip theta: $R_h \sqrt{\cos^2 + (R_v/R_h) \sin^2}$.

Input parameters

Parameter	Type	Default
rh	ArrayLike	required
rv	ArrayLike	required
dip_deg	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Response Characteristics of an Array Induction Tool (HDIL) in Heterogeneous Anisotropic Formations. Liu, Zhang, Zhang, Xu, Kang, Xiao (2020). DOI: 10.30632/PJV61N1-2020a2. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 1: Tutorial - Petrophysics of Thinly Bedded Formations. Aldred (2021). DOI: 10.30632/PJV62N4-2021t1.

apparent_water_resistivity

petrolib.saturation_resistivity

```
apparent_water_resistivity(rt, phi, *, a=1.0, m=2.0)
```

Purpose

Apparent water resistivity $R_{wa} = R_t * \phi^m / a$.

Equals R_w in a clean wet zone; the quick-look R_w picker.

Input parameters

Parameter	Type	Default
rt	ArrayLike	required
phi	ArrayLike	required
a (kw-only)	float	1.0
m (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Petrophysics Vol. 56 No. 4 (Aug 2015) (issue-level reference).
- Petrophysics Vol. 65 No. 6 (Dec 2024) (issue-level reference).

apply_depth_shift

petrolib.depth_matching

```
apply_depth_shift(values, depth, shift)
```

Purpose

Apply a continuous (possibly per-depth) shift to values by interpolation.

Resamples values from depth + shift back onto depth with endpoint clamping; shift may be a scalar bulk shift or a per-sample array.

Input parameters

Parameter	Type	Default
values	ArrayLike	required
depth	ArrayLike	required
shift	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Dynamic Depth Alignment of Well Logs: A Continuous Optimization Framework for Enhanced Petrophysical and Rock Physics Interpretation. Westeng et al. (2026), Petrophysics, 67(1), 54-67. DOI: 10.30632/PJV67N1-2026a4.

apply_eec

petrolib.geochem_fluids.mudgas

```
apply_eec(comps, alphas)
```

Purpose

Extraction-efficiency correction: divide each component by its alpha.

Input parameters

Parameter	Type	Default
comps	ArrayLike	required
alphas	ArrayLike	required

Output

numpy.ndarray of float64

apply_integer_shift

petrolib.depth_matching

```
apply_integer_shift(log, lag, *, fill=np.nan)
```

Purpose

Shift log by an integer lag samples without wrapping; vacated ends get fill.

A positive lag moves samples to greater depth (index); the exposed edge is filled with fill.

Input parameters

Parameter	Type	Default
log	ArrayLike	required
lag	int	required
fill (kw-only)	float	np.nan

Output

numpy.ndarray of float64

Sources

- Pan, W., Fu, L., Xu, C., Ashby, M., Lee, H., Lee, J., Meng, F., Chen, S., Ye, Y., Jiang, H., Kim, H., Kong, H., Baek, I., Baek, J., Sun, X., Sun, H., Li, S., Zhao, Z., Ke, Y., ... and Park, J. (2026). Automatic Well-Log Depth Shifting With Data-Driven Approaches: A Contest Summary. *Petrophysics*, 67(2), 437–462. DOI: 10.30632/PJV67N2-2026a12.

archie_conductivity

petrolib.saturation_resistivity

```
archie_conductivity(sw, cw, *, phi, m=2.0, n=2.0)
```

Purpose

Conductivity-domain Archie $C_t = C_w * \phi^{*m} * S_w^{*n}$.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
cw	ArrayLike	required
phi (kw-only)	ArrayLike	required
m (kw-only)	float	2.0
n (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Article 1: Improved Assessment of Hydrocarbon Saturation in Mixed-Wet Rocks With Complex Pore Structure. Garcia, Heidari, Rostami (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 454-469. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Rock Physics Modeling of Gas Hydrate Reservoirs Through Integrated Core and Well-Log Data in NGHP-02 Area, KG Offshore Basin, India. Kumar, Mishra, Chatterjee, Tiwari, Avadhani (2022). DOI: 10.30632/PJV63N2-2022a6. Petrophysics Vol. 63 No. 2 (Apr 2022).

archie_rt

petrolib.saturation_resistivity

```
archie_rt(sw, rw, *, phi, a=1.0, m=2.0, n=2.0)
```

Purpose

Forward Archie resistivity $R_t = a \cdot R_w / (\phi^{**m} * S_w^{**n})$.

Inverse of archie_sw (their round trip is a library property test).

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
rw	ArrayLike	required
phi (kw-only)	ArrayLike	required
a (kw-only)	float	1.0
m (kw-only)	float	2.0
n (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Article 10: Enhanced Assessment of Fluid Saturation in the Wolfcamp Formation of the Permian Basin. Dash, Heidari (2021). DOI: 10.30632/PJV62N6-2021a10. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).

archie_sw

petrolib.saturation_resistivity

```
archie_sw(rt, rw, *, phi, a=1.0, m=2.0, n=2.0, clip=None)
```

Purpose

Archie water saturation $S_w = (a \cdot R_w / (\phi^m \cdot R_t))^{1/n}$.

`clip` is explicit and off by default — hidden `[0, 1]` clipping biases SCAL fits (CONVENTIONS.md rule 3); facades pass their article's historical `clip`.

Input parameters

Parameter	Type	Default
<code>rt</code>	ArrayLike	required
<code>rw</code>	ArrayLike	required
<code>phi</code> (kw-only)	ArrayLike	required
<code>a</code> (kw-only)	float	1.0
<code>m</code> (kw-only)	float	2.0
<code>n</code> (kw-only)	float	2.0
<code>clip</code> (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir
Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Water Saturation Equations for Unconsolidated Reservoirs. Based on: Acosta et al. (2024), "Unveiling the Optimal Water Saturation Equation for Unconsolidated Reservoirs: A Case Study From the Tambaredjo Oil Field, Suriname", Petrophysics, Vol. 65, No. 5, pp. 739-764.

archie_sw_from_conductivity

petrolib.saturation_resistivity

```
archie_sw_from_conductivity(ct, cw, *, phi, m=2.0, n=2.0, clip=None)
```

Purpose

Invert conductivity-domain Archie: $S_w = (C_t / (C_w \cdot \phi^m))^{1/n}$.

The closed-form inverse of `archie_conductivity`.

Input parameters

Parameter	Type	Default
ct	ArrayLike	required
cw	ArrayLike	required
phi (kw-only)	ArrayLike	required
m (kw-only)	float	2.0
n (kw-only)	float	2.0
clip (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 4: A New Resistivity-Based Model for Improved Hydrocarbon Saturation Assessment in Clay-Rich Formations Using Quantitative Geometry of the Clay Network. Garcia, Jagadisan, Rostami, Heidari (2018). DOI: 10.30632/PJV59N3-2018a3. Petrophysics Vol. 59 No. 3 (Jun 2018).

arps_correct

petrolib.geochem_fluids.brine

```
arps_correct(r1, t1, t2, *, unit='F')
```

Purpose

Arps temperature correction $R2 = R1 \cdot (T1+c)/(T2+c)$.

c is 6.77 for unit="F" or 21.5 for unit="C". Applies equally to conductivity with the roles of R inverted by the caller.

Input parameters

Parameter	Type	Default
r1	ArrayLike	required
t1	ArrayLike	required
t2	ArrayLike	required
unit (kw-only)	str	'F'

Output

numpy.ndarray of float64

Sources

- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

assert_matches_original

petrolib.testing

```
assert_matches_original(original, replacement, cases, *, rtol=1e-12, atol=0.0)
```

Purpose

Assert `replacement(*case)` matches `original(*case)` for every case.

Results are compared with `np.testing.assert_allclose` at the given tolerances, recursing through tuples/lists/dicts; strings, bools and `None` must be exactly equal. Returns the number of cases checked (and asserts it is nonzero — an empty case list would vacuously pass).

Input parameters

Parameter	Type	Default
<code>original</code>	<code>Callable[..., Any]</code>	required
<code>replacement</code>	<code>Callable[..., Any]</code>	required
<code>cases</code>	<code>Iterable[Case]</code>	required
<code>rtol</code> (kw-only)	<code>float</code>	<code>1e-12</code>
<code>atol</code> (kw-only)	<code>float</code>	<code>0.0</code>

Output

`int`

attenuation_coefficient

petrolib.acoustic_geomech

```
attenuation_coefficient(a0, ax, x)
```

Purpose

Attenuation coefficient alpha in $A = A0 \cdot \exp(-\alpha \cdot x)$.

$\alpha = \ln(A0/Ax)/x$ (1/length in the units of x).

Input parameters

Parameter	Type	Default
a0	ArrayLike	required
ax	ArrayLike	required
x	ArrayLike	required

Output

numpy.ndarray of float64

attenuation_coefficient

petrolib.integrity_drilling

```
attenuation_coefficient(a0, ax, x_m)
```

Purpose

Exponential attenuation coefficient $\ln(A_0/A_x)/x$ (1/m) from $A = A_0 \cdot \exp(-\alpha \cdot x)$.

Input parameters

Parameter	Type	Default
a0	ArrayLike	required
ax	ArrayLike	required
x_m	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: A Study of the Flexural Attenuation Technique Through Laboratory Measurements and Numerical Simulations. Sirevaag, Johansen, Larsen, Holt (2020). DOI: 10.30632/PJV61N4-2020a1. Petrophysics Vol. 61 No. 4 (Aug 2020).

attenuation_db

petrolib.acoustic_geomech

```
attenuation_db(a1, a2)
```

Purpose

Attenuation in decibels $20 \cdot \log_{10}(A1/A2)$ between two amplitudes.

Input parameters

Parameter	Type	Default
a1	ArrayLike	required
a2	ArrayLike	required

Output

numpy.ndarray of float64

attenuation_db

petrolib.em_dielectric

```
attenuation_db(rho, freq_hz, spacing_m)
```

Purpose

Induction-limit amplitude attenuation over spacing_m: $8.686 \cdot L / \delta$ dB.

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
freq_hz	ArrayLike	required
spacing_m	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).

attenuation_db

petrolib.integrity_drilling

```
attenuation_db(a_near, a_far, spacing_m=None)
```

Purpose

Amplitude attenuation $20 \cdot \log_{10}(A_{\text{near}}/A_{\text{far}})$ in dB (dB/m if spacing_m given).

Input parameters

Parameter	Type	Default
a_near	ArrayLike	required
a_far	ArrayLike	required
spacing_m	float None	None

Output

numpy.ndarray of float64

Sources

- Article 1: A Study of the Flexural Attenuation Technique Through Laboratory Measurements and Numerical Simulations. Sirevaag, Johansen, Larsen, Holt (2020). DOI: 10.30632/PJV61N4-2020a1. Petrophysics Vol. 61 No. 4 (Aug 2020).

attenuation_map

petrolib.nuclear

```
attenuation_map(i, i_ref)
```

Purpose

Optical-density attenuation $-\ln(I/I_{\text{ref}})$ (zero-guarded ratio).

Input parameters

Parameter	Type	Default
i	ArrayLike	required
i_ref	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Simultaneous Neutron and X-Ray Imaging of 3D Structure of Organic Matter and Fracture in Shales. Chiang, LaManna, Hussey, Jacobson, Liu, Zhang, Georgi, Kone, Chen (2018). DOI: 10.30632/PJV59N2-2018a3. Petrophysics Vol. 59 No. 2 (Apr 2018).

attenuation_phase_from_voltages

petrolib.em_dielectric

```
attenuation_phase_from_voltages(v_near, v_far)
```

Purpose

Attenuation (dB) and phase shift (deg) from near/far complex voltages.

ratio = v_near/v_far; atten_db = 20*log10|ratio| and phase_deg = degrees(angle(ratio)).

Input parameters

Parameter	Type	Default
v_near	ArrayLike	required
v_far	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 1: Looking Ahead of the Bit While Drilling: From Vision to Reality. Constable, Antonsen, Stalheim, Olsen, Fjell, Dray, Eikenes, Aarflot, Haldorsen, Digranes, Seydoux, Omeragic, Thiel, Davydychev, Denichou, Salim, Frey, Homan, Tan (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 426-446. DOI: none assigned (this issue predates SPWLA DOI assignment).

average_lethargy_gain

petrolib.nuclear

```
average_lethargy_gain(a_mass)
```

Purpose

Mean logarithmic energy decrement $\xi = 1 + \alpha/(1-\alpha)*\ln(\alpha)$.

The hydrogen limit $A=1$ ($\alpha=0$) returns $\xi=1$.

Input parameters

Parameter	Type	Default
a_mass	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Physical Basis for a Cased-Well Quantitative Gas-Saturation Analysis Method. F. Inanc, W.A. Gilchrist, R. Ansari, D. Chace (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 598-617. DOI: none assigned (this issue predates SPWLA DOI assignment).

balance_ratio

petrolib.geochem_fluids.mudgas

```
balance_ratio(c1, c2, c3, c4, c5)
```

Purpose

Haworth balance $B_h = (C1+...+C5)/(C3+C4+C5)$ (inf where the denominator is 0).

Input parameters

Parameter	Type	Default
c1	ArrayLike	required
c2	ArrayLike	required
c3	ArrayLike	required
c4	ArrayLike	required
c5	ArrayLike	required

Output

numpy.ndarray of float64

bed_sinusoid

petrolib.borehole_image

```
bed_sinusoid(azimuth_deg, z0, radius, dip_deg, dip_azimuth_deg)
```

Purpose

Depth trace of a planar bed on the unrolled borehole wall.

$z(\phi) = z_0 - \text{radius} \cdot \tan(\text{dip}) \cdot \cos(\phi - \text{dip_azimuth})$ -- a sinusoid of amplitude $\text{radius} \cdot \tan(\text{dip})$ whose deepest point is at the dip azimuth. All angles in degrees.

Input parameters

Parameter	Type	Default
azimuth_deg	ArrayLike	required
z0	float	required
radius	float	required
dip_deg	float	required
dip_azimuth_deg	float	required

Output

numpy.ndarray of float64

Sources

- Article 4: Deep-Learning-Based Automated Sedimentary Geometry Characterization From Borehole Images. Lefranc, Bayraktar, Kristensen, Driss, Le Nir, Marza, Kherroubi (2021). DOI: 10.30632/PJV62N6-2021a4. Petrophysics Vol. 62 No. 6 (Dec 2021).

beer_lambert

petrolib.nuclear

```
beer_lambert(i0, mu, x)
```

Purpose

Beer-Lambert transmitted intensity $I = I_0 \cdot \exp(-\mu \cdot x)$.

Input parameters

Parameter	Type	Default
i0	ArrayLike	required
mu	ArrayLike	required
x	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: An Assessment of Fundamentals of Nuclear-Based Alternatives to Conventional Chemical-Source Bulk-Density Measurement. Ahmed Badruzzaman (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 415-434. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Simultaneous Neutron and X-Ray Imaging of 3D Structure of Organic Matter and Fracture in Shales. Chiang, LaManna, Hussey, Jacobson, Liu, Zhang, Georgi, Kone, Chen (2018). DOI: 10.30632/PJV59N2-2018a3. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 7: In-Situ Saturation Monitoring (ISSM) - Recommendations for Improved Processing. Reed, Cense (2019). DOI: 10.30632/PJV60N2-2019a5. Petrophysics Vol. 60 No. 2 (Apr 2019).

bernard_ratio

petrolib.geochem_fluids.mudgas

```
bernard_ratio(c1, c2, c3)
```

Purpose

Bernard ratio $C1/(C2+C3)$ (biogenic vs thermogenic indicator).

Input parameters

Parameter	Type	Default
c1	ArrayLike	required
c2	ArrayLike	required
c3	ArrayLike	required

Output

numpy.ndarray of float64

bet_fit

petrolib.geochem_fluids.adsorption

```
bet_fit(x_rel, v_ads, *, cross_nm2=0.162)
```

Purpose

Linear BET fit -> (Vm, C, SSA_m2_g).

Fits $x/(V*(1-x)) = \text{slope}*x + \text{intercept}$ (least squares); $V_m = 1/(\text{slope}+\text{intercept})$, $C = \text{slope}/\text{intercept} + 1$, and the specific surface area $SSA = (V_m/22414)*N_A*\text{cross}$ (m2/g) with cross in nm2 ($N_2 = 0.162$).

Input parameters

Parameter	Type	Default
x_rel	ArrayLike	required
v_ads	ArrayLike	required
cross_nm2 (kw-only)	float	0.162

Output

tuple[float, float, float]

Sources

- Article 2: Classification of Adsorption Isotherm Curves for Shale Based on Pore Structure. Tian, Chen, Yan, Deng, He (2020). DOI: 10.30632/PJV61N5-2020a2. Petrophysics Vol. 61 No. 5 (Oct 2020).

bet_isotherm

petrolib.geochem_fluids.adsorption

```
bet_isotherm(x_rel, vm, c)
```

Purpose

BET isotherm $V = V_m * C * x / ((1 - x) * (1 + (C - 1) * x))$ with $x = P/P_0$.

Input parameters

Parameter	Type	Default
x_rel	ArrayLike	required
vm	ArrayLike	required
c	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Classification of Adsorption Isotherm Curves for Shale Based on Pore Structure. Tian, Chen, Yan, Deng, He (2020). DOI: 10.30632/PJV61N5-2020a2. Petrophysics Vol. 61 No. 5 (Oct 2020).

biot_coefficient

petrolib.acoustic_geomech

```
biot_coefficient(k_dry, k_mineral)
```

Purpose

Biot-Willis effective-stress coefficient $\alpha = 1 - K_{\text{dry}}/K_{\text{mineral}}$.

Input parameters

Parameter	Type	Default
k_dry	ArrayLike	required
k_mineral	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 6: Petrophysical Analysis of Siliceous- Ooze Sediments, More Basin, Norwegian Sea. Ahmed Awadalkarim, Morten Kanne Sorensen, Ida Lykke Fabricius (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 333-348. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

block_stack

petrolib.data_qc_io.signal

```
block_stack(x, n, axis=-1)
```

Purpose

Block-mean stacking: average consecutive groups of n along axis.
The output length is size // n; a trailing partial block is dropped.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
n	int	required
axis	int	-1

Output

numpy.ndarray of float64

Sources

- Module 2: Fiber-Optic Sensing for Well Integrity Diagnosis Implements ideas from: Bazaid et al., "Pioneering Well Logging: The Role of Fiber Optics in Modern Monitoring for Well Integrity Diagnosis," Petrophysics, vol. 66, no. 4, pp. 555-565, August 2025.

blocky_log

petrolib.data_qc_io.synth

```
blocky_log(n, n_beds=8, *, level_range=(20.0, 150.0), noise=2.0, rng=None)
```

Purpose

Blocky (square) log of `n_beds` constant levels plus Gaussian noise.

Returns (curve, bounds) where bounds are the `n_beds - 1` interior boundary indices, drawn without replacement away from the ends (margin $\max(1, n//15)$). Levels are uniform on `level_range`. Composed from the corpus generators: random breakpoints per [src2022_08/article5](#), curve+bounds return per [src2024_06/article2](#), noise=2.0 per [src2022_02/article2](#).

Input parameters

Parameter	Type	Default
<code>n</code>	int	required
<code>n_beds</code>	int	8
<code>level_range</code> (kw-only)	tuple[float, float]	(20.0, 150.0)
<code>noise</code> (kw-only)	float	2.0
<code>rng</code> (kw-only)	np.random.Generator int None	None

Output

tuple[ndarray[float64], ndarray[int]]

Sources

- Article 5: Log Facies Analysis and Reservoir Properties of Basement Granitic Rocks in the Pearl River Mouth Basin, South China Sea. Hua, Yang, Xu, Lei, Zhong (2022). DOI: 10.30632/PJV63N4-2022a5. Petrophysics Vol. 63 No. 4 (Aug 2022).
- Strobel, J. (2024). "Petrophysical Analyses for Supporting the Search for a Claystone-Hosted Nuclear Repository." Petrophysics 65(3), 302-316. DOI: 10.30632/PJV65N3-2024a2.
- Article 2: Automated Well-Log Depth Matching - 1D Convolutional Neural Networks vs. Classic Cross Correlation. Torres Caceres, Duffaut, Yazidi, Westad, Johansen (2022). DOI: 10.30632/PJV63N1-2022a2. Petrophysics Vol. 63 No. 1 (Feb 2022).

bond_index

petrolib.integrity_drilling

```
bond_index(measured, free_pipe, well_bonded, *, method='linear', input_kind='amplitude', clip=(0.0, 1.0))
```

Purpose

Cement bond index in [0, 1] from a CBL measurement and its two anchors.

method='linear' interpolates between the free_pipe (BI=0) and well_bonded (BI=1) anchors: for input_kind='amplitude' the amplitude falls with bond, $BI = (free_pipe - A) / (free_pipe - well_bonded)$; for 'attenuation' it rises, $BI = (att - free_pipe) / (well_bonded - free_pipe)$. method='log' (amplitude only) uses the logarithmic CBL convention $BI = \log(free_pipe/A) / \log(free_pipe/well_bonded)$ with 0.01 floors on the amplitudes. The result is clipped to clip.

Input parameters

Parameter	Type	Default
measured	ArrayLike	required
free_pipe	float	required
well_bonded	float	required
method (kw-only)	str	'linear'
input_kind (kw-only)	str	'amplitude'
clip (kw-only)	tuple[float, float]	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331-347.

bond_index_combined

petrolib.integrity_drilling

```
bond_index_combined(bi_a, bi_b, *, weights=(0.6, 0.4), corrector=None)
```

Purpose

Weighted combination of two bond indicators, clipped to [0, 1].

corrector (optional callable) maps the raw combination to a corrected value before clipping -- e.g. an eccentricity-correction model.

Input parameters

Parameter	Type	Default
bi_a	ArrayLike	required
bi_b	ArrayLike	required
weights (kw-only)	tuple[float, float]	(0.6, 0.4)
corrector (kw-only)	Any	None

Output

numpy.ndarray of float64

Sources

- Module 10: Through-Tubing Cement Evaluation (SNHR + EMI + ML) Implements ideas from: Zeghlache et al., "Challenges and Solutions for Advanced Through-Tubing Cement Evaluation," Petrophysics, vol. 66, no. 4, pp. 677–688, August 2025.

bond_number

petrolib.relperm_wettability

```
bond_number(*, drho, k, sigma, g=9.81)
```

Purpose

Bond number $N_b = \text{drho} * g * k / \text{sigma}$ (gravity / capillary).

k is the characteristic area — permeability (m^2) in the Darcy form, or a squared length (pass $k=R^2$) in the pore-radius form.

Input parameters

Parameter	Type	Default
drho (kw-only)	ArrayLike	required
k (kw-only)	ArrayLike	required
sigma (kw-only)	float	required
g (kw-only)	float	9.81

Output

numpy.ndarray of float64

Sources

- Article 4: Impact of Wettability on Residual Oil Saturation and Capillary Desaturation Curves K. J. Humphry, B. M. J. M. Suijkerbuijk, H. A. van der Linde, S. G. J. Pieterse, S. K. Masalmeh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 313-318. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Mud-Filtrate Invasion in Laminated and Spatially Heterogeneous Rocks: High-Resolution In-Situ Visualization and Analysis Using Time-Lapse X-Ray Microcomputed Tomography (Micro-CT). Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N5-2022a4. Petrophysics Vol. 63 No. 5 (Oct 2022).

bowers_pore_pressure

petrolib.acoustic_geomech

```
bowers_pore_pressure(velocity, overburden, *, a=10.0, b=0.7, v0=5000.0, unloading=False,
u=3.0, sigma_max=5000.0)
```

Purpose

Bowers (1995) pore pressure from velocity and overburden (field units).

Loading effective stress $\sigma = ((V - V_0)/A)**(1/B)$; the unloading branch adds the Bowers hysteresis $\sigma = \sigma_{\max} * (\sigma_{\text{load}} / \sigma_{\max}) ** (1/U)$. Pore pressure $P_p = \text{overburden} - \sigma$. V/V_0 in ft/s, stresses in psi (the corpus convention); V_0 is the mudline velocity.

Input parameters

Parameter	Type	Default
velocity	ArrayLike	required
overburden	ArrayLike	required
a (kw-only)	float	10.0
b (kw-only)	float	0.7
v0 (kw-only)	float	5000.0
unloading (kw-only)	bool	False
u (kw-only)	float	3.0
sigma_max (kw-only)	float	5000.0

Output

numpy.ndarray of float64

bowers_pore_pressure

petrolib.integrity_drilling

```
bowers_pore_pressure(velocity, overburden, *, A=10.0, B=0.7, unloading=False, U=3.0,
v0=5000.0, sigma_max=5000.0)
```

Purpose

Bowers pore pressure from sonic velocity (ft/s) and overburden (psi).

Inverts the loading curve $V = V_0 + A \cdot \sigma^B$ for the effective stress (floored at 0.01) and subtracts it from the overburden; `unloading=True` uses the flatter unloading branch with maximum stress `sigma_max` and exponent `U`.

Input parameters

Parameter	Type	Default
velocity	ArrayLike	required
overburden	ArrayLike	required
A (kw-only)	float	10.0
B (kw-only)	float	0.7
unloading (kw-only)	bool	False
U (kw-only)	float	3.0
v0 (kw-only)	float	5000.0
sigma_max (kw-only)	float	5000.0

Output

numpy.ndarray of float64

Sources

- Genetic Analysis of Overpressure While Drilling Based on Isotope Logging. Based on: Hu et al., "Genetic Analysis of Overpressure While Drilling Based on Isotope Logging Technology", Petrophysics, Vol. 66, No. 2, April 2025, pp. 283-293.

boyle_grain_volume

petrolib.porosity_lithology

```
boyle_grain_volume(v_cell, v_expansion, p1, p2)
```

Purpose

Boyle's-law grain volume $V_{\text{grain}} = V_{\text{cell}} - V_{\text{exp}}/(p_1/p_2 - 1)$.

Helium-expansion (double-cell) grain volume.

Input parameters

Parameter	Type	Default
v_cell	ArrayLike	required
v_expansion	ArrayLike	required
p1	ArrayLike	required
p2	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: 3D Printing Mudrocks: Experiments in Validating Clay as a Build Material for 3D Printing Porous Micromodels. Hasiuk, Harding (2021). DOI: 10.30632/PJV62N5-2021a4. Petrophysics Vol. 62 No. 5 (Oct 2021).

boyle_porosity

petrolib.porosity_lithology

```
boyle_porosity(rho_bulk, rho_grain)
```

Purpose

Boyle's-law density porosity $\phi = 1 - \rho_{\text{bulk}}/\rho_{\text{grain}}$.

Input parameters

Parameter	Type	Default
rho_bulk	ArrayLike	required
rho_grain	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

bpp_spectral_density

petrolib.nmr

```
bpp_spectral_density(omega, tau_c)
```

Purpose

Bloembergen-Purcell-Pound spectral density $J = \tau_c / (1 + (\omega \tau_c)^2)$.

Input parameters

Parameter	Type	Default
omega	ArrayLike	required
tau_c	float	required

Output

numpy.ndarray of float64

Sources

- Article 1: NMR Relaxometry in Shale and Implications for Logging. Kausik, Fellah, Rylander, Singer, Lewis, Sinclair (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 339-350. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fellah, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).

bpp_t1_t2

petrolib.nmr

```
bpp_t1_t2(omega0, tau_c, *, dipolar_constant=1.0)
```

Purpose

BPP relaxation times (T1, T2) from the correlation time.

$1/T1 = (3/10) * C * (J(w0) + 4 * J(2w0))$, $1/T2 = (3/20) * C * (3 * J(0) + 5 * J(w0) + 2 * J(2w0))$ — the full dipolar prefactors (C the dipolar constant).

Input parameters

Parameter	Type	Default
omega0	float	required
tau_c	float	required
dipolar_constant (kw-only)	float	1.0

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 1: NMR Relaxometry in Shale and Implications for Logging. Kausik, Fella, Rylander, Singer, Lewis, Sinclair (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 339-350. DOI: none assigned (this issue predates SPWLA DOI assignment).

breakdown_pressure

petrolib.acoustic_geomech

```
breakdown_pressure(sh, sH, pp, *, tensile_strength=0.0)
```

Purpose

Hubbert-Willis breakdown pressure $3 \cdot sh - sH - Pp + T0$.

sh minimum and sH maximum horizontal stress; T0 tensile strength.

Input parameters

Parameter	Type	Default
sh	ArrayLike	required
sH	ArrayLike	required
pp	ArrayLike	required
tensile_strength (kw-only)	ArrayLike	0.0

Output

numpy.ndarray of float64

Sources

- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 9: Research of Near-Wellbore Fracture Morphology, Formation Mechanism, and Propagation Law for Different Perforation Modes During the Perforation Process. Wang, Li, Xu, Jia, Zhang (2021). DOI: 10.30632/PJV62N5-2021a9. Petrophysics Vol. 62 No. 5 (Oct 2021).

brie_fluid_modulus

petrolib.acoustic_geomech

```
brie_fluid_modulus(sw, k_liquid, k_gas, *, e=3.0)
```

Purpose

Brie et al. (1995) fluid-mixing modulus $(K_{liq} - K_{gas}) * S_w^e + K_{gas}$.

e is the Brie exponent (default 3.0); Sw is the liquid saturation.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
k_liquid	ArrayLike	required
k_gas	ArrayLike	required
e (kw-only)	float	3.0

Output

numpy.ndarray of float64

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).

brine_density_bw92

petrolib.geochem_fluids.brine

```
brine_density_bw92(nacl_ppm, temp_c, press_mpa=0.1)
```

Purpose

Simplified Batzle-Wang (1992) brine density (kg/m³).

$\rho_w = 1000 \cdot (1 - (T-4)^2/178000)$ (pure water) plus the NaCl term $S \cdot (300 - 2 \cdot T - 60 \cdot S)$ with S the weight fraction; a small linear pressure term $+0.045 \cdot (P - 0.1)$ (P in MPa).

Input parameters

Parameter	Type	Default
nacl_ppm	ArrayLike	required
temp_c	ArrayLike	required
press_mpa	ArrayLike	0.1

Output

numpy.ndarray of float64

Sources

- Laronga, R., Borchardt, E., Hill, B., Velez, E., Klemin, D., Haddad, S., Haddad, E., Chadwick, C., Mahmoodaghdam, E., and Hamichi, F. (2023). "Integrated Formation Evaluation for Site-Specific Evaluation, Optimization, and Permitting of Carbon Storage Projects." *Petrophysics*, 64(5), 580-620. DOI: 10.30632/PJV64N5-2023a1.

brittleness_rickman

petrolib.acoustic_geomech

```
brittleness_rickman(e, nu, *, e_range=(10.0, 80.0), nu_range=(0.15, 0.4), percent=False)
```

Purpose

Rickman (2008) brittleness index from Young's modulus and Poisson's ratio.

Averages the min-max normalised E (high E is brittle) and nu (low nu is brittle):

$0.5 * ((E - E_{min}) / (E_{max} - E_{min}) + (nu - nu_{max}) / (nu_{min} - nu_{max}))$. e_range/nu_range are the normalisation endpoints (defaults are Rickman's GPa / dimensionless brackets); percent scales to 0-100. E must be given in the same units as e_range (GPa by default).

Input parameters

Parameter	Type	Default
e	ArrayLike	required
nu	ArrayLike	required
e_range (kw-only)	tuple[float, float]	(10.0, 80.0)
nu_range (kw-only)	tuple[float, float]	(0.15, 0.4)
percent (kw-only)	bool	False

Output

numpy.ndarray of float64

Sources

- Article 7: Use of Data Analytics to Optimize Hydraulic Fracture Locations Along Borehole. Gupta, Rai, Devegowda, Sondergeld (2018). DOI: 10.30632/PJV59N6Y2018a6. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

brooks_corey_burdine_kr

petrolib.relperm_wettability

```
brooks_corey_burdine_kr(sw, *, swr, lam, snwr=0.0, clip=(0.0, 1.0))
```

Purpose

Brooks-Corey (Burdine) kr pair from the pore-size-distribution index.

$kr_w = Se^{**((2+3*lam)/lam)}$, $kr_{nw} = (1-Se)^{**2} * (1 - Se^{**((2+lam)/lam)})$ — the fixed-exponent form (no endpoint factor; $kr_{max} == 1$). This is NOT free-exponent Corey.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
swr (kw-only)	float	required
lam (kw-only)	float	required
snwr (kw-only)	float	0.0
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 7: Effect of Injection Pressure on the Imbibition Relative Permeability and Capillary Pressure Curves of Shale Gas Matrix. Al-Ameri, Mazeel (2020). DOI: 10.30632/PJV61N2-2020a7. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 4: Evaluation of Relative Permeability From Resistivity Data for Fractal Porous Media. Shi, Meng, Liu, Zhang, Wang (2020). DOI: 10.30632/PJV61N3-2020a4. Petrophysics Vol. 61 No. 3 (Jun 2020).

brooks_corey_pc

petrolib.capillary_pressure

```
brooks_corey_pc(sw, *, pc_entry, lam, swirr, snr=0.0)
```

Purpose

Brooks-Corey capillary pressure $P_c = P_e * S_{wn}^{(-1/\lambda_m)}$.

$S_{wn} = (S_w - S_{wirr}) / (1 - S_{wirr} - S_{nr})$ — the normalized-saturation window is explicit (snr defaults to 0, the drainage convention; the residual-nonwetting window some articles use must be passed, never assumed).

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
pc_entry (kw-only)	ArrayLike	required
lam (kw-only)	float	required
swirr (kw-only)	float	required
snr (kw-only)	float	0.0

Output

numpy.ndarray of float64

Sources

- Article 7: Effect of Injection Pressure on the Imbibition Relative Permeability and Capillary Pressure Curves of Shale Gas Matrix. Al-Ameri, Mazeel (2020). DOI: 10.30632/PJV61N2-2020a7. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 2: Empirical Relation for Capillary Pressure in Shale. Alipour K., Kasha, Sakhaee-Pour, Sadooni, Al-Kuwari (2022). DOI: 10.30632/PJV63N5-2022a2. Petrophysics Vol. 63 No. 5 (Oct 2022).
- Petrophysics Vol. 64 No. 4 (Aug 2023) (issue-level reference).
- Petrophysics Vol. 66 No. 1 (Feb 2025) (issue-level reference).

brooks_corey_sw

petrolib.capillary_pressure

```
brooks_corey_sw(pc, *, pc_entry, lam, swirr, clip=None)
```

Purpose

Brooks-Corey drainage saturation $S_w = S_{wirr} + (1 - S_{wirr}) * (P_e / P_c)^{2\lambda}$.

Returns 1.0 at and below the entry pressure. λ is the pore-size-distribution index — articles using the reciprocal ($1/N$) convention pass $\lambda = 1.0/N$ (section 9 hazard; the parameters are NOT interchangeable).

Input parameters

Parameter	Type	Default
pc	ArrayLike	required
pc_entry (kw-only)	ArrayLike	required
lam (kw-only)	float	required
swirr (kw-only)	float	required
clip (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 8: Saturation-Height Modeling: Assessing Capillary Pressure Stress Corrections. Hulea (2018). DOI: 10.30632/PJV59N3-2018a7 (inferred - see note). Petrophysics Vol. 59 No. 3 (Jun 2018).
- Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics (issue-level reference).
- Article 8: Perched Water Contacts: Understanding Fundamental Controls. Hulea (2019). DOI: 10.30632/PJV60N3-2019a7. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).

bruggeman_symmetric

petrolib.em_dielectric

```
bruggeman_symmetric(fractions, eps_components, *, iterations=60)
```

Purpose

Symmetric Bruggeman effective-medium permittivity (N phases, spheres).

Solves $\sum_i f_i (\epsilon_i - \epsilon_{\text{eff}}) / (\epsilon_i + 2\epsilon_{\text{eff}}) = 0$ by Newton iteration from the arithmetic-mean seed. Complex-capable. `fractions` and `eps_components` index the phases along `axis=0`.

Input parameters

Parameter	Type	Default
<code>fractions</code>	ArrayLike	required
<code>eps_components</code>	ArrayLike	required
<code>iterations</code> (kw-only)	int	60

Output

`_Num`

bulk_density

petrolib.porosity_lithology

```
bulk_density(phi, rho_ma, rho_fl=1.0, *, v_k=0.0, rho_k=1.3)
```

Purpose

Forward bulk density $\rho_b = (1-\phi-v_k)\rho_{ma} + v_k\rho_k + \phi\rho_{fl}$.

The three-component (matrix + kerogen + fluid) mass balance; $v_k=0$ reduces to the two-component form. Keep units consistent (g/cc or kg/m³).

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
rho_ma	ArrayLike	required
rho_fl	ArrayLike	1.0
v_k (kw-only)	ArrayLike	0.0
rho_k (kw-only)	float	1.3

Output

numpy.ndarray of float64

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 6: Enhancing the Detectability of Deep-Sensing Borehole EM Instruments by Joint Inversion of Multiple Logs Within a Probabilistic Geosteering Workflow. Jahani, Alyaev, Ambia, Fossum, Suter, Torres-Verdín (2023). DOI: 10.30632/PJV64N1-2023a6. Petrophysics Vol. 64 No. 1 (Feb 2023).

bulk_volume_water

petrolib.porosity_lithology

```
bulk_volume_water(phi, sw)
```

Purpose

Bulk volume water BVW = $\phi \cdot S_w$ (constant in a Buckles zone).

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
sw	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1 (Tutorial): Organic Mudstone Petrophysics, Part 2: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N2-2019t1.
- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

bulk_volume_water

petrolib.saturation_resistivity

```
bulk_volume_water(phi, sw)
```

Purpose

Bulk volume water $BW = \phi * Sw$.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
sw	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Petrophysical Challenges in Giant Carbonate Tengiz Field, Republic of Kazakhstan. Skalinski, Se, Playton, Theologou, Narr, Sullivan, Mallan (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 615-647. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Workflow to Estimate Storage Capacity (Part 1). Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N1Y2019t1.
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

buoyancy_pc

petrolib.capillary_pressure

```
buoyancy_pc(height, *, delta_rho, g=G_STD)
```

Purpose

Buoyancy capillary pressure $P_c = \text{delta_rho} * g * h$ (SI: Pa).

delta_rho is the water-hydrocarbon density difference (kg/m³), h the height above the free-water level (m).

Input parameters

Parameter	Type	Default
height	ArrayLike	required
delta_rho (kw-only)	ArrayLike	required
g (kw-only)	float	G_STD

Output

numpy.ndarray of float64

Sources

- Article 8: Perched Water Contacts: Understanding Fundamental Controls. Hulea (2019). DOI: 10.30632/PJV60N3-2019a7. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Petrophysics Vol. 64 No. 5 (Oct 2023) — Energy Transition Special Issue (issue-level reference).

buoyancy_pc_gradient

petrolib.capillary_pressure

```
buoyancy_pc_gradient(height_ft, *, sg_water, sg_hc, gradient_psi_per_ft=0.433)
```

Purpose

Oilfield buoyancy Pc: $P_c = \text{gradient} * (SG_w - SG_{hc}) * h$ (psi, ft, SG).

The 0.433 psi/ft/SG freshwater gradient is the conventional rounded field constant, kept visible as a parameter (CONVENTIONS.md rule 5).

Input parameters

Parameter	Type	Default
height_ft	ArrayLike	required
sg_water (kw-only)	ArrayLike	required
sg_hc (kw-only)	ArrayLike	required
gradient_psi_per_ft (kw-only)	float	0.433

Output

numpy.ndarray of float64

Sources

- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: How Pore-Scale Attributes May Be Used to Derive Robust Drainage and Imbibition Water-Saturation Models in Complex Tight-Gas Reservoirs. Merletti, Gramin, Salunke, Hamman, Spain, Shabro, Armitage, Torres-Verdin, Salter, Dacy (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 447-464. DOI: none assigned (this issue predates SPWLA DOI assignment).

bvi_ffl*petrolib.nmr*

```
bvi_ffl(t2_ms, amplitude, *, cutoff_ms=33.0)
```

Purpose

Bound (BVI) and free (FFI) fluid split at a single T2 cutoff.

$BVI = \sum(A[T2 < cutoff])$, $FFI = \sum(A[T2 \geq cutoff])$ — the two-band case of t2_partition.

Input parameters

Parameter	Type	Default
t2_ms	ArrayLike	required
amplitude	ArrayLike	required
cutoff_ms (kw-only)	float	33.0

Output

tuple[float, float]

Sources

- Article 3: NMR Fluid Substitution - A New Method of Reconstructing T2 Distributions Under Primary Drainage and Imbibition Conditions. Li, Kesserwan, Jin, Ma (2021). DOI: 10.30632/PJV62N4-2021a2. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).

bwv_from_permittivity

petrolib.em_dielectric

```
bwv_from_permittivity(eps_meas, phi, *, eps_w, eps_hc, eps_matrix, alpha=0.5)
```

Purpose

Bulk volume water $\phi \cdot S_w$ from crim, unclipped (salinity-robust).

$$BVW = (\text{eps_meas}^{**a} - (1-\phi) \cdot \text{eps_m}^{**a} - \phi \cdot \text{eps_hc}^{**a}) / (\text{eps_w}^{**a} - \text{eps_hc}^{**a}).$$

Input parameters

Parameter	Type	Default
eps_meas	ArrayLike	required
phi	ArrayLike	required
eps_w (kw-only)	float	required
eps_hc (kw-only)	float	required
eps_matrix (kw-only)	float	required
alpha (kw-only)	float	0.5

Output

numpy.ndarray of float64

Sources

- Article 4: Advanced Log Interpretation in Field Development. Merkel, Lessenger (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 479-491. DOI: none assigned (this issue predates SPWLA DOI assignment).

cable_tension

petrolib.depth_correction

```
cable_tension(depth, total_depth, tool_weight, cable_weight_per_length)
```

Purpose

Tension profile $T(z) = W_{\text{tool}} + w (L - z)$ along the cable.

The tension at depth z supports the tool weight plus the buoyant weight of the cable hanging below it ($L = \text{total_depth}$), so it is greatest at surface and equals tool_weight at the tool.

Input parameters

Parameter	Type	Default
depth	ArrayLike	required
total_depth	float	required
tool_weight	float	required
cable_weight_per_length	float	required

Output

numpy.ndarray of float64

Sources

- Article 5: Wireline Logging Depth Quality Improvement: Methodology Review and Elastic-Stretch Correction. Bolt (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 294-310. DOI: none assigned (this issue predates SPWLA DOI assignment).

capillary_desaturation

petrolib.relperm_wettability

```
capillary_desaturation(n, *, sor_max, sor_min, n_crit, exponent=1.0)
```

Purpose

Capillary desaturation curve $S_{or}(N)$.

$S_{or} = S_{or_min} + (S_{or_max} - S_{or_min}) / (1 + (N/N_{crit})^{exponent})$ — the residual falls from the plateau S_{or_max} toward the floor S_{or_min} as the trapping number rises past N_{crit} . Articles that write the exponent as $1/width$ pass $exponent=1.0/width$.

Input parameters

Parameter	Type	Default
n	ArrayLike	required
sor_max (kw-only)	float	required
sor_min (kw-only)	float	required
n_crit (kw-only)	float	required
$exponent$ (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 3: Pore-Scale Insights on Trapped Oil During Waterflooding of Sandstone Rocks of Varying Wettability States. Berthet, Hebert, Barbouteau, Andriamananjaona, Rivenq (2019). DOI: 10.30632/PJV60N2-2019a1. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 4: Impact of Wettability on Residual Oil Saturation and Capillary Desaturation Curves K. J. Humphry, B. M. J. M. Suijkerbuijk, H. A. van der Linde, S. G. J. Pieterse, S. K. Masalmeh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 313-318. DOI: none assigned (this issue predates SPWLA DOI assignment).

capillary_number

petrolib.relperm_wettability

```
capillary_number(*, mu, v, sigma)
```

Purpose

Capillary number $N_{ca} = \mu v / \sigma$ (viscous / capillary).

Keyword-only on purpose: the ~14 article definitions disagree on positional order (four put velocity first) though all compute the same ratio. For the volumetric-rate form $Q \mu / (A \sigma)$ pass $v=Q/A$.

Input parameters

Parameter	Type	Default
mu (kw-only)	float	required
v (kw-only)	ArrayLike	required
sigma (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969–981. DOI: 10.30632/PJV66N6-2025a4.
- Article 1: Flow Regimes During Immiscible Displacement. Armstrong, McClure, Berrill, Rucker, Schluter, Berg (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 10-18. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 55 No. 4 (Aug 2014) (issue-level reference).

capillary_rise_height

petrolib.capillary_pressure

```
capillary_rise_height(radius, *, sigma, theta_deg=0.0, delta_rho, g=G_STD)
```

Purpose

Equilibrium capillary rise $h = 2 \cdot \sigma \cdot \cos(\theta) / (\Delta \rho \cdot g \cdot r)$.

Signed by physics: a wetting fluid ($\theta < 90^\circ$) rises, a non-wetting fluid (mercury) is depressed ($h < 0$).

Input parameters

Parameter	Type	Default
radius	ArrayLike	required
sigma (kw-only)	ArrayLike	required
theta_deg (kw-only)	ArrayLike	0.0
delta_rho (kw-only)	ArrayLike	required
g (kw-only)	float	G_STD

Output

numpy.ndarray of float64

Sources

- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.

casing_condition

petrolib.integrity_drilling

```
casing_condition(loss_pct, *, bands=(10.0, 25.0, 42.5))
```

Purpose

Casing condition class from metal loss %: good / fair / poor / critical.
Strict < at each band edge, matching the corpus convention.

Input parameters

Parameter	Type	Default
loss_pct	ArrayLike	required
bands (kw-only)	tuple[float, float, float]	(10.0, 25.0, 42.5)

Output

NDArray[Any]

Sources

- Casing and Cement Inspection: Logging Two Casing Sizes Simultaneously. Based on: Hawthorn, Ingebretson, Girneata, Delabroy, Winther, Steinsiek, and Leslie (2024), Petrophysics 65(6), pp. 913-918. DOI: 10.30632/PJV65N6-2024a5.

casing_resonance_frequency

petrolib.integrity_drilling

```
casing_resonance_frequency(thickness_m, *, v=V_STEEL, n=1, correction=1.0)
```

Purpose

Casing plate thickness-resonance frequency $f = \text{corr} * n * v / (2 d)$ (Hz).

correction absorbs mode corrections (e.g. 0.95 for the S1 negative-group-velocity minimum).

Input parameters

Parameter	Type	Default
thickness_m	ArrayLike	required
v (kw-only)	float	V_STEEL
n (kw-only)	int	1
correction (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 4: Ultrasonic Logging of Creeping Shale. Diez, Johansen, Larsen (2022). DOI: 10.30632/PJV63N1-2022a4. Petrophysics Vol. 63 No. 1 (Feb 2022).

casing_thickness_from_resonance

petrolib.integrity_drilling

```
casing_thickness_from_resonance(freq_hz, *, v=V_STEEL, n=1, correction=1.0)
```

Purpose

Casing wall thickness (m) from the thickness-resonance frequency (Hz).

Input parameters

Parameter	Type	Default
freq_hz	ArrayLike	required
v (kw-only)	float	V_STEEL
n (kw-only)	int	1
correction (kw-only)	float	1.0

Output

numpy.ndarray of float64

cec_from_qv

petrolib.saturation_resistivity

```
cec_from_qv(qv, *, rho_grain, phi)
```

Purpose

Inverse of qv_from_cec.

Input parameters

Parameter	Type	Default
qv	ArrayLike	required
rho_grain (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).

cement_quality_score

petrolib.integrity_drilling

```
cement_quality_score(z_mrayl, *, cement_type='Portland', thresholds=None)
```

Purpose

Continuous cement-quality score in [0, 1] from annulus impedance (MRayl). Above the good threshold the score is 1; between fair and good it ramps 0.5..1; below fair it ramps toward 0. Thresholds default per cement_type (CEMENT_QUALITY_THRESHOLDS).

Input parameters

Parameter	Type	Default
z_mrayl	ArrayLike	required
cement_type (kw-only)	str	'Portland'
thresholds (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Well Integrity Measurements Throughout the CCS Project Life Cycle. Based on: Valstar, Nettleton, Borchardt, Costeno, Landry, and Laronga (2024), Petrophysics 65(6), pp. 896-912. DOI: 10.30632/PJV65N6-2024a4.

cementation_exponent_at_point

petrolib.saturation_resistivity

```
cementation_exponent_at_point(phi, ff, *, a=1.0)
```

Purpose

Single-point inversion $m = \log(F/a) / \log(1/\phi)$.

The single-sample alternative to `fit_cementation_exponent`.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
ff	ArrayLike	required
a (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Petrophysics Vol. 62 No. 5 (Oct 2021) (issue-level reference).
- Petrophysics Vol. 61 No. 5 (Oct 2020) (issue-level reference).

centrifuge_pc

petrolib.capillary_pressure

```
centrifuge_pc(omega, *, delta_rho, r1, r2)
```

Purpose

Hassler-Brunner inlet-face Pc: $0.5 \cdot \text{delta_rho} \cdot \text{omega}^2 \cdot (r2^2 - r1^2)$.

omega in rad/s (rpm-input articles convert in their facades: $\text{omega} = \text{rpm} \cdot 2 \cdot \pi / 60$); radii in m from the rotation axis.

Input parameters

Parameter	Type	Default
omega	ArrayLike	required
delta_rho (kw-only)	ArrayLike	required
r1 (kw-only)	ArrayLike	required
r2 (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: An Analytical Model for Analysis of Capillary Pressure Measurements by Centrifuge. Andersen, Skjaeveland, Standnes (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 366-375. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 64 No. 3 (June 2023) — Best Papers of the 2022 SCA International Symposium (issue-level reference).
- Petrophysics Vol. 66 No. 6 (Dec 2025) — Best Papers of the 2024 SCA International Symposium (issue-level reference).
- Petrophysics Vol. 61 No. 3 (Jun 2020) (issue-level reference).
- Petrophysics Vol. 61 No. 4 (Aug 2020) (issue-level reference).

cfl_number

petrolib.inversion_numerics.pde

```
cfl_number(alpha, dt, dx)
```

Purpose

Diffusion CFL number $\alpha \cdot dt / dx^2$ (explicit stability needs ≤ 0.5).

Input parameters

Parameter	Type	Default
alpha	float	required
dt	float	required
dx	float	required

Output

float

cgr_api

petrolib.nuclear

```
cgr_api(k_pct, th_ppm, *, coeff=(16.0, 4.0))
```

Purpose

Computed (uranium-free) gamma ray API CGR = $ck \cdot K + cth \cdot Th$.

Input parameters

Parameter	Type	Default
k_pct	ArrayLike	required
th_ppm	ArrayLike	required
coeff (kw-only)	tuple[float, float]	(16.0, 4.0)

Output

numpy.ndarray of float64

Sources

- Article 4: Spectral Gamma-Ray Measurement While Drilling. Xu, Huiszoon, Wang, Adolph, Yi, Cavin, Laughlin, Tollefsen, Jacobsen, Boyce (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 377-389. DOI: none assigned (this issue predates SPWLA DOI assignment).

Chain (class)

petrolib.inversion_numerics.stochastic

Chain()

Purpose

MCMC output: posterior samples and the acceptance fraction.

Input parameters

None.

Output

Chain instance

character_ratio

petrolib.geochem_fluids.mudgas

```
character_ratio(c3, c4, c5)
```

Purpose

Haworth character $Ch = (C4+C5)/C3$ (0 where $C3$ is 0).

Input parameters

Parameter	Type	Default
c3	ArrayLike	required
c4	ArrayLike	required
c5	ArrayLike	required

Output

numpy.ndarray of float64

class_fractions

petrolib.borehole_image

```
class_fractions(image, thresholds)
```

Purpose

Volume fractions between ascending thresholds.

For thresholds = [t0, t1, ...] returns [mean(img<t0), mean(t0<=img<t1), ..., mean(img>=t_last)] -- e.g. pore / clay / grain from two thresholds (dark = pore convention).

Input parameters

Parameter	Type	Default
image	ArrayLike	required
thresholds	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Uncertainty Quantification in Image Segmentation for Image-Based Rock Physics in a Shaly Sandstone. Howard, Lin, Zhang (2019). DOI: 10.30632/PJV60N2-2019a2. Petrophysics Vol. 60 No. 2 (Apr 2019).

classify_annulus

petrolib.integrity_drilling

```
classify_annulus(z_mrayl, *, gas_max=0.5, liquid_max=2.6, cement_min=3.0)
```

Purpose

Classify the annulus fill from its acoustic impedance (MRayl).

'gas' below gas_max, 'liquid' below liquid_max, 'cement' at or above cement_min, else 'transition'. Pass cement_min=liquid_max to remove the transition band.

Input parameters

Parameter	Type	Default
z_mrayl	float	required
gas_max (kw-only)	float	0.5
liquid_max (kw-only)	float	2.6
cement_min (kw-only)	float	3.0

Output

str

Sources

- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).

classify_cement_from_cbl

petrolib.integrity_drilling

```
classify_cement_from_cbl(relative_amp, *, good_max=0.15, medium_max=0.3)
```

Purpose

CBL relative-amplitude cement classes 'Good'/'Medium'/'Poor'.

relative_amp <= good_max is Good, <= medium_max Medium, else Poor.

Input parameters

Parameter	Type	Default
relative_amp	ArrayLike	required
good_max (kw-only)	float	0.15
medium_max (kw-only)	float	0.3

Output

NDArray[Any]

classify_fluid_gor

petrolib.geochem_fluids.mudgas

```
classify_fluid_gor(gor_sm3, *, thresholds=(180, 360, 640, 5000, 15000))
```

Purpose

Fluid type from GOR (Sm³/Sm³) against ascending thresholds.

Labels: black oil / volatile oil / gas condensate / wet gas / dry gas.

Input parameters

Parameter	Type	Default
gor_sm3	float	required
thresholds (kw-only)	tuple[float, ...]	(180, 360, 640, 5000, 15000)

Output

str

classify_fluid_haworth

petrolib.integrity_drilling

```
classify_fluid_haworth(wh, bh, ch=None, *, n_classes=4)
```

Purpose

Classify reservoir fluid from Haworth ratios (wh in percent).

n_classes=4 uses the classic bands (dry gas < 0.5 %, gas / gas-condensate < 17.5 % split on bh > wh, oil < 40 %, else residual oil). n_classes=8 uses the extended 2026 scheme (requires ch).

Input parameters

Parameter	Type	Default
wh	float	required
bh	float	required
ch	float None	None
n_classes (kw-only)	int	4

Output

str

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Luo, P., Li, W., Lu, P., and Qubaisi, K. (2026). An Improved Mud Gas Ratio Method for Enhanced Fluid Identification While Drilling. Petrophysics, 67(3), 582-593. DOI: 10.30632/PJV67N3-2026a8.

classify_fluid_wetness

petrolib.geochem_fluids.mudgas

```
classify_fluid_wetness(wh_pct, bh)
```

Purpose

Haworth wetness/balance fluid typing (wh in PERCENT).

Input parameters

Parameter	Type	Default
wh_pct	float	required
bh	float	required

Output

str

classify_hfu

petrolib.flow_transport

```
classify_hfu(fzi_values, *, n_units=4)
```

Purpose

Classify samples into hydraulic flow units by log-spaced FZI thresholds.
Returns (labels, thresholds) — integer labels in [0, n_units-1] and the n_units+1 FZI threshold values.

Input parameters

Parameter	Type	Default
fzi_values	ArrayLike	required
n_units (kw-only)	int	4

Output

tuple[NDArray[np.intp], ndarray[float64]]

Sources

- Permeability Anisotropy in Carbonates via Digital Rock Petrophysics. Based on: Silva Junior et al. (2024), "Permeability Anisotropy in Brazilian Presalt Carbonates at Core Scale Using Digital Rock Petrophysics", Petrophysics, Vol. 65, No. 5, pp. 711-738.

classify_t1t2

petrolib.nmr

```
classify_t1t2(t1t2, *, cutoff=2.0)
```

Purpose

Boolean hydrocarbon flag T1/T2 $\geq$ cutoff (True = hydrocarbon/bound).

The cutoff varies across the corpus (2.0 sandstone/shale, 4.0 kerogen); it is explicit.

Input parameters

Parameter	Type	Default
t1t2	ArrayLike	required
cutoff (kw-only)	float	2.0

Output

NDArray[np.bool_]

Sources

- Article 2: Estimating Saturations in Organic Shales Using 2D NMR. Nicot, Vorapalawut, Rousseau, Madariaga, Hamon, Korb (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 19-29. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Fluid Typing and Pore Size in Organic Shale Using 2D NMR in Saturated Kerogen Isolates. Singer, Chen, Hirasaki (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 604-619. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Integrated Reservoir Characterization Using Unsupervised Learning on Nuclear Magnetic Resonance (NMR) T1-T2 Logs. Jiang, Bonnie, Correa, Krueger, Kelly, Wasson (2022). DOI: 10.30632/PJV63N3-2022a1. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

classify_wettability_angle

petrolib.relperm_wettability

```
classify_wettability_angle(theta_deg, *, cuts=(75.0, 105.0))
```

Purpose

Classify wettability from the contact angle: water-wet / intermediate / oil-wet.

Default cuts 75 deg / 105 deg (src2014_02/article3); pass cuts to shift the boundaries. Articles using different labels (e.g. wetting/non-wetting) keep their own classifier.

Input parameters

Parameter	Type	Default
theta_deg	float	required
cuts (kw-only)	tuple[float, float]	(75.0, 105.0)

Output

str

Sources

- Article 3: An Evaluation of Spontaneous Imbibition of Water into Oil-Wet Carbonate Reservoir Cores Using Nanofluid. Abbas Roustaei (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 31-37. DOI: none assigned (this issue predates SPWLA DOI assignment).

classify_wettability_index

petrolib.relperm_wettability

```
classify_wettability_index(i, *, scheme='3class')
```

Purpose

Classify wettability from an Amott/USBM-style index in [-1, 1].

scheme="3class": water-wet / mixed-wet / oil-wet at +/-0.3. scheme="5class": adds weakly water-/oil-wet bands at +/-0.1.

Input parameters

Parameter	Type	Default
i	float	required
scheme (kw-only)	str	'3class'

Output

str

Sources

- Aljishi, M.K., Chitrala, Y., Dang, S.T., and Rai, C. (2026). Wettability-Based Pore Partitioning and Its Effects on Oil Recovery and Formation Damage in Unconventional Reservoirs. *Petrophysics*, 67(2), 263-279. DOI: 10.30632/PJV67N2-2026a2.
- Impact of Dopants (NaI) on SCAL Experiments. Pairoys et al., 2025, *Petrophysics* 66(1), 123-133. DOI:10.30632/PJV66N1-2025a9.

cleanup_powerlaw

petrolib.geochem_fluids.contamination

```
cleanup_powerlaw(v, eta0, v_star, *, exponent=5.0 / 12.0)
```

Purpose

Power-law cleanup $\eta(V) = \eta_0 \cdot (1 + V/V_{\text{star}})^{-\text{exponent}}$.

The default exponent 5/12 is the wireline-cleanup value (2/3 for radial).

Input parameters

Parameter	Type	Default
v	ArrayLike	required
eta0	ArrayLike	required
v_star	ArrayLike	required
exponent (kw-only)	float	5.0 / 12.0

Output

numpy.ndarray of float64

Sources

- Article 6: Proxy-Enabled Stochastic Interpretation of Downhole Fluid Sampling Under Immiscible Flow Conditions. Kristensen, Chugunov, Cig, Jackson (2018). DOI: 10.30632/PJV59N5-2018a5. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 5: Innovative Formation Tester Sampling Procedures for Carbon Dioxide and Other Reactive Components. Piazza, Vieira, Sacorague, Jones, Dai, Pearl, Aguiar (2021). DOI: 10.30632/PJV62N1-2021a4. Petrophysics Vol. 62 No. 1 (Feb 2021).

closure_residual

petrolib.data_qc_io.clean

```
closure_residual(*fracs, target=1.0)
```

Purpose

Closure residual $\text{sum}(\text{fracs}) - \text{target}$ (e.g. $S_w + S_o + S_g - 1$).

Zero means the fractions close exactly.

Input parameters

Parameter	Type	Default
*fracs	—	required
target (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 4: Maintaining and Reconstructing In-Situ Saturations: A Comparison Between Whole Core, Sidewall Core, and Pressurized Sidewall Core in the Permian Basin. Blount, McMullen, Durand, Driskill (2019). DOI: 10.30632/PJV60N1Y2019a3. Petrophysics Vol. 60 No. 1 (Feb 2019).

co2_solubility_brine

petrolib.geochem_fluids.solubility

```
co2_solubility_brine(p_mpa, t_k, m_nacl=0.0, *, m_ch4=0.0)
```

Purpose

CO₂ solubility in brine (mol/kg): $P / (H * \gamma) * 1 / (1 + 0.6 * m_{CH_4})$.

H from henry_constant_co2, gamma from setschenow_factor; m_ch4 accounts for methane competition.

Input parameters

Parameter	Type	Default
p_mpa	ArrayLike	required
t_k	ArrayLike	required
m_nacl	ArrayLike	0.0
m_ch4 (kw-only)	ArrayLike	0.0

Output

numpy.ndarray of float64

Sources

- Wang and Ehlig-Economides (2023), Petrophysics 64(6): 970-977. CO₂ solubility in saline water for CO₂ trapping, accounting for pressure, temperature, salinity, and an often-neglected factor (e.g., dissolved CH₄ or the activity-coefficient correction). Implements a Duan-Sun-style solubility model in simplified form.

co_forward_3phase

petrolib.nuclear

```
co_forward_3phase(phi, so, sg, sw, *, c_oil=0.85, c_gas=0.75, c_mat=0.1, o_w=1.0,
o_mat=0.55)
```

Purpose

Inelastic C/O forward model $COR = C / (0+1e-9)$ from a 3-phase saturation.

$C = \phi * (S_o * c_{oil} + S_g * c_{gas}) + (1-\phi) * c_{mat}$; $O = \phi * S_w * o_w + (1-\phi) * o_{mat}$.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
so	ArrayLike	required
sg	ArrayLike	required
sw	ArrayLike	required
c_oil (kw-only)	float	0.85
c_gas (kw-only)	float	0.75
c_mat (kw-only)	float	0.1
o_w (kw-only)	float	1.0
o_mat (kw-only)	float	0.55

Output

numpy.ndarray of float64

Sources

- McGlynn et al. (2023), Petrophysics 64(6): 900-918. New pulsed-neutron spectroscopy instrument with LaBr3 detectors providing simultaneous C/O ratio, capture sigma, and ratio-based gas measurements for two- and three-phase saturation analysis.

co_ratio

petrolib.nuclear

```
co_ratio(c_yield, o_yield)
```

Purpose

Carbon-oxygen ratio COR = Y_C/Y_O .

Input parameters

Parameter	Type	Default
c_yield	ArrayLike	required
o_yield	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Case Studies Demonstrating the Impact of Cement Quality on Carbon/Oxygen and Elemental Analysis From Casedhole Pulsed-Neutron Logging. Wang, Sullivan, Seth, Barnes, Wilson, Lazorek (2020). DOI: 10.30632/PJV61N3-2020a2. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

cole_cole

petrolib.em_dielectric

```
cole_cole(freq_hz, *, eps_inf, eps_s, tau, alpha=0.0)
```

Purpose

Cole-Cole relaxation with distribution exponent alpha (0 -> Debye).

$\text{eps_inf} + (\text{eps_s} - \text{eps_inf}) / (1 + (j * \omega * \tau)^{(1-\alpha)})$.

Input parameters

Parameter	Type	Default
freq_hz	ArrayLike	required
eps_inf (kw-only)	float	required
eps_s (kw-only)	float	required
tau (kw-only)	float	required
alpha (kw-only)	float	0.0

Output

numpy.ndarray of complex128

Sources

- Article 5: Dielectric Permittivity: A Petrophysical Parameter for Shales. Matthew Josh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 319-332. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Al-Qouzi, W., Hassan, A., Attia, H., El- Hussein, A., and Mahmoud, M. (2026). Experimental Study of Sensitivity of Multifrequency (MF) Dielectric Measurements to Hydraulic Fractures in Sandstone and Carbonate Formations. Petrophysics, 67(3), 482-508. DOI: 10.30632/PJV67N3-2026a2 (orig. SPE-227827-MS).

cole_cole_resistivity

petrolib.em_dielectric

```
cole_cole_resistivity(freq_hz, *, rho0, chargeability, tau, c)
```

Purpose

Pelton (1978) Cole-Cole complex resistivity (induced-polarization form).

$\rho(\omega) = \rho_0 * (1 - m * (1 - 1 / (1 + (j * \omega * \tau)^c)))$ with DC resistivity ρ_0 , chargeability m , time constant τ and frequency exponent c .

Input parameters

Parameter	Type	Default
freq_hz	ArrayLike	required
rho0 (kw-only)	float	required
chargeability (kw-only)	float	required
tau (kw-only)	float	required
c (kw-only)	float	required

Output

numpy.ndarray of complex128

Sources

- Article 5: Borehole Measurements of the Complex-Resistivity Dispersion Spectrum: A New Logging Method to Identify Low-Resistivity Reservoirs. Jiang, Ke, Kang, Sun, Yin (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 281-288. DOI: none assigned (this issue predates SPWLA DOI assignment).

collision_parameter

petrolib.nuclear

```
collision_parameter(a_mass)
```

Purpose

Neutron collision parameter $\alpha = ((A-1)/(A+1))^2$.

Input parameters

Parameter	Type	Default
a_mass	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Physical Basis for a Cased-Well Quantitative Gas-Saturation Analysis Method. F. Inanc, W.A. Gilchrist, R. Ansari, D. Chace (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 598-617. DOI: none assigned (this issue predates SPWLA DOI assignment).

combine_clay_indicators

petrolib.porosity_lithology

```
combine_clay_indicators(*vcl, how='mean')
```

Purpose

Combine several clay-volume indicators into one estimate.

how="mean" averages (e.g. ND and uranium-free CGR indicators), "min" takes the most optimistic, "max" the hybrid GR/NMR upper bound.

Input parameters

Parameter	Type	Default
*vcl	—	required
how (kw-only)	str	'mean'

Output

numpy.ndarray of float64

Sources

- Article 2: Integrated Petrofacies Characterization and Interpretation of Depositional Environment of the Bakken Shale in the Williston Basin, North America. Bhattacharya, Carr (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 96-111. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", Petrophysics, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.

combine_relaxation_times

petrolib.nmr

```
combine_relaxation_times(*times_ms)
```

Purpose

Combine relaxation times as parallel rates $1/T = \sum(1/T_i)$.

Variadic over the bulk/surface/diffusion contributions (or any set of component times).

Input parameters

Parameter	Type	Default
*times_ms	—	required

Output

numpy.ndarray of float64

Sources

- Article 1: NMR Relaxometry in Shale and Implications for Logging. Kausik, Fellah, Rylander, Singer, Lewis, Sinclair (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 339-350. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Effects of Bitumen Extraction on the 2D NMR Response of Saturated Kerogen Isolates. Chen, Singer, Kuang, Vargas, Hirasaki (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 470-484. DOI: none assigned (this issue predates SPWLA DOI assignment).

compensated_neutron_porosity

petrolib.nuclear

```
compensated_neutron_porosity(near, far, *, a=0.0, b=-30.0, c=45.0)
```

Purpose

Compensated neutron porosity $\phi_i = a + b \ln R + c \ln R^2$ (p.u.), $R = \text{near}/\text{far}$.

Input parameters

Parameter	Type	Default
near	ArrayLike	required
far	ArrayLike	required
a (kw-only)	float	0.0
b (kw-only)	float	-30.0
c (kw-only)	float	45.0

Output

numpy.ndarray of float64

complex_permittivity

petrolib.em_dielectric

```
complex_permittivity(eps_real, *, sigma=0.0, freq_hz=None, eps_imag=0.0)
```

Purpose

Complex permittivity $\epsilon^* = \epsilon' - j(\epsilon'' + \sigma/(\omega\epsilon_0))$.

ϵ_{real} is the real part ϵ' ; ϵ_{imag} is any dielectric loss ϵ'' already in permittivity units; σ (S/m) adds the ohmic loss $\sigma/(\omega\epsilon_0)$. freq_hz is required only when σ is nonzero. Returns a complex array in the -j engineering convention.

Input parameters

Parameter	Type	Default
ϵ_{real}	ArrayLike	required
σ (kw-only)	ArrayLike	0.0
freq_hz (kw-only)	ArrayLike None	None
ϵ_{imag} (kw-only)	ArrayLike	0.0

Output

numpy.ndarray of complex128

Sources

- Article 7: Bakken Petroleum System Characterization Using Dielectric-Dispersion Logs. Han, Misra (2018). DOI: 10.30632/PJV59N2-2018a6. Petrophysics Vol. 59 No. 2 (Apr 2018).
- AlZoukani, A. M., Al-Hamad, M., and Abdallah, W. (2026). Unlocking the Effect of Pore Types on Dielectric Permittivity in Carbonates. Petrophysics, 67(3), 470-481. DOI: 10.30632/PJV67N3-2026a1 (Best Petrophysics Papers from MEOS GEO 2025; orig. SPE-227798-MS).

complex_wavenumber

petrolib.em_dielectric

```
complex_wavenumber(freq_hz, sigma, eps_r, *, mu_r=1.0)
```

Purpose

Exact lossy-medium wavenumber $(k_r, k_i) = (\text{beta}, \text{alpha})$.

Phase constant beta and attenuation constant alpha of $k = \text{beta} - j*\text{alpha}$ from the loss tangent $p = \text{sigma}/(\omega*\epsilon_0*\epsilon_r)$: $k_r = \text{base}*\sqrt{(\sqrt{1+p^2}+1)/2}$ and $k_i = \text{base}*\sqrt{(\sqrt{1+p^2}-1)/2}$ with $\text{base} = \omega*\sqrt{\mu*\epsilon_0*\epsilon_r}$.

Input parameters

Parameter	Type	Default
freq_hz	ArrayLike	required
sigma	ArrayLike	required
eps_r	ArrayLike	required
mu_r (kw-only)	float	1.0

Output

tuple[ndarray[float64], ndarray[float64]]

condition_number

petrolib.inversion_numerics.linear

```
condition_number(a)
```

Purpose

2-norm condition number $s_{\max}/s_{\min}$ from the singular values.

Input parameters

Parameter	Type	Default
a	ArrayLike	required

Output

float

Sources

- Article 3: Forward Mineral Modeling Using Regularized Least-Squares Regression With Singular Value Decomposition: Case Study From Qusaiba Shale. Xu, McCormick, Herron, Cheshire, Al-Salim, Almarzouq (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 242-269. DOI: none assigned (this issue predates SPWLA DOI assignment).

confusion_matrix

petrolib.ml_stats

```
confusion_matrix(y_true, y_pred, *, labels=None)
```

Purpose

Confusion matrix $C[i, j]$ = count of true class i predicted as j .
labels fixes the class order (default: sorted union of both label sets). For binary 0/1 labels this reproduces the repository's $[[TN, FP], [FN, TP]]$ convention.

Input parameters

Parameter	Type	Default
y_true	ArrayLike	required
y_pred	ArrayLike	required
labels (kw-only)	ArrayLike None	None

Output

NDArray[np.int64]

Sources

- Article 8: A Comparative Study of Three Supervised Machine-Learning Algorithms for Classifying Carbonate Vuggy Facies in the Kansas Arbuckle Formation. Deng, Xu, Jobe, Xu (2019). DOI: 10.30632/PJV60N6-2019a8. Petrophysics Vol. 60 No. 6 (Dec 2019).
- High-Resolution Lithofacies Prediction Using Ensemble Classifiers. Based on: Satti et al. (2024), "Enhancing Reservoir Characterization With High- Resolution Lithofacies Prediction Using Advanced Feature Engineering and Ensemble Classifiers", Petrophysics, Vol. 65, No. 5, pp. 813-834.

contamination_fraction

petrolib.geochem_fluids.contamination

```
contamination_fraction(p, p_v, p_f)
```

Purpose

Contamination fraction $\eta = (P - P_v)/(P_f - P_v)$ (inverse of the mixing rule).

Input parameters

Parameter	Type	Default
p	ArrayLike	required
p_v	ArrayLike	required
p_f	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: A Breakthrough in Accurate Downhole Fluid Sample Contamination Prediction in Real Time. Zuo, Gisolf, Dumont, Dubost, Pfeiffer, Wang, Mishra, Chen, Mullins, Biagi, Gemelli (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 251-265. DOI: none assigned (this issue predates SPWLA DOI assignment).

convert

petrolib.units

```
convert(values, from_unit, to_unit)
```

Purpose

Convert values between two units of the same quantity family.

```
>>> convert(7.0, "bar", "psi") array(101.52641996) >>> convert([100.0, 212.0], "degF", "degC")  
array([ 37.77777778, 100. ])
```

Raises ValueError for an unknown unit or a cross-family conversion (e.g. psi -> metres) — the caller's units are wrong and no answer is better than a plausible one.

Input parameters

Parameter	Type	Default
values	ArrayLike	required
from_unit	str	required
to_unit	str	required

Output

NDArray[np.float64]

convolution_matrix

petrolib.inversion_numerics.linear

```
convolution_matrix(kernel, n)
```

Purpose

Row-normalized (n, n) convolution (blurring) matrix for a centered kernel.

Input parameters

Parameter	Type	Default
kernel	ArrayLike	required
n	int	required

Output

numpy.ndarray of float64

corey_kr

petrolib.relperm_wettability

```
corey_kr(sw, *, swr, sor=0.0, krw_max=1.0, kro_max=1.0, nw=2.0, no=2.0, clip=(0.0, 1.0))
```

Purpose

Two-phase Corey pair (krw, kro) on a shared Se.

The single canonical replacement for the corey_kr / corey_relperm / krw+kro pair variants.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
swr (kw-only)	float	required
sor (kw-only)	float	0.0
krw_max (kw-only)	float	1.0
kro_max (kw-only)	float	1.0
nw (kw-only)	float	2.0
no (kw-only)	float	2.0
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 5: Comparison of Four Numerical Simulators for SCAL Experiments. Lenormand, Lorentzen, Maas, Ruth (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 48-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Using Digital Rock Technology to Quality Control and Reduce Uncertainty in Relative Permeability Measurements. Schembre-McCabe, Kamath (2018). DOI: 10.30632/petro_059_1_a4. Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 4: Mud-Filtrate Invasion in Laminated and Spatially Heterogeneous Rocks: High-Resolution In-Situ Visualization and Analysis Using Time-Lapse X-Ray Microcomputed Tomography (Micro-CT). Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N5-2022a4. Petrophysics Vol. 63 No. 5 (Oct 2022).
- Petrophysics Vol. 65 No. 2 (Apr 2024) (issue-level reference).
- Petrophysics Vol. 66 No. 1 (Feb 2025) (issue-level reference).

corey_krg

petrolib.relpem_wettability

```
corey_krg(sg, *, sgc=0.0, swc=0.0, sorg=0.0, krg_max=1.0, ng=2.0, clip=(0.0, 1.0))
```

Purpose

Corey gas relative permeability $k_{rg} = k_{rg_max} * S_g^{**ng}$.

$S_g^* = (S_g - S_{gc}) / (1 - S_{wc} - S_{gc} - S_{org})$ — the 4-endpoint gas denominator. Sw-framework gas curves $((1 - S_w - S_{gc}) / (1 - S_{wr} - S_{gc}))$ map by passing $sg=1-sw$ and $swc=sw_r$ (bit-exact).

Input parameters

Parameter	Type	Default
sg	ArrayLike	required
sgc (kw-only)	float	0.0
swc (kw-only)	float	0.0
sorg (kw-only)	float	0.0
krg_max (kw-only)	float	1.0
ng (kw-only)	float	2.0
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 11: Loading Effects on Gas Relative Permeability of a Low- Permeability Sandstone. Agostini, Egermann, Jeannin, Portier, Skoczylas, Wang (2019). DOI: 10.30632/PJV60N2-2019a9. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 6: Estimation of Gas-Condensate Relative Permeability Using a Lattice Boltzmann Modeling Approach. Schembre-McCabe, Kamath, Fager, Crouse (2020). DOI: 10.30632/PJV61N2-2020a6. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 1: Drainage Three-Phase Flow Relative Permeability on Oil-Wet Carbonate Reservoir Rock Types: Experiments, Interpretation and Comparison with Standard Correlations. P. Egermann, K. Mejdoub, J.-M. Lombard, O. Vizika, Z. Kalam (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 287-293. DOI: none assigned (this issue predates SPWLA DOI assignment).

corey_kro

petrolib.relperm_wettability

```
corey_kro(sw, *, swr, sor=0.0, kro_max=1.0, no=2.0, clip=(0.0, 1.0))
```

Purpose

Corey oil relative permeability $kro = kro_max * (1 - Se)^{no}$.
Uses the (1-Se) form on the water-normalized Se; articles that normalize oil independently as (1-Sw-Sor)/(1-Swir-Sor) agree to ~1 ULP.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
swr (kw-only)	float	required
sor (kw-only)	float	0.0
kro_max (kw-only)	float	1.0
no (kw-only)	float	2.0
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 1: Drainage Three-Phase Flow Relative Permeability on Oil-Wet Carbonate Reservoir Rock Types: Experiments, Interpretation and Comparison with Standard Correlations. P. Egermann, K. Mejdoub, J.-M. Lombard, O. Vizika, Z. Kalam (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 287-293. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 57 No. 1 (Feb 2016) (issue-level reference).
- Article 5: Comparison of Four Numerical Simulators for SCAL Experiments. Lenormand, Lorentzen, Maas, Ruth (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 48-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

corey_krw

petrolib.relperm_wettability

```
corey_krw(sw, *, swr, sor=0.0, krw_max=1.0, nw=2.0, clip=(0.0, 1.0))
```

Purpose

Corey water relative permeability $krw = krw_max * Se^{nw}$.

$Se = (Sw - Swr) / (1 - Swr - Sor)$.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
swr (kw-only)	float	required
sor (kw-only)	float	0.0
krw_max (kw-only)	float	1.0
nw (kw-only)	float	2.0
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 1: Drainage Three-Phase Flow Relative Permeability on Oil-Wet Carbonate Reservoir Rock Types: Experiments, Interpretation and Comparison with Standard Correlations. P. Egermann, K. Mejdoub, J.-M. Lombard, O. Vizika, Z. Kalam (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 287-293. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO₂-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 58 No. 1 (Feb 2017) (issue-level reference).
- Article 9: Downhole Estimation of Relative Permeability With Integration of Formation-Tester Measurements and Advanced Well Logs. Hadibeik, Azari, Kalawina, Ramakrishna, Eyuboglu, Khan, Al-Rushaid, Al-Rashidi, Ahmad (2018). DOI: 10.30632/PJV59N2-2018a8 (inferred; body beyond extraction). Petrophysics Vol. 59 No. 2 (Apr 2018).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

corrected_depth

petrolib.depth_correction

```
corrected_depth(measured, *, stretch=0.0, thermal=0.0, convention='tally')
```

Purpose

Apply stretch and thermal corrections to a measured depth.

convention='tally' adds the corrections (surface tally reads short); convention='payout' subtracts them (paid-out length overstates depth).

Input parameters

Parameter	Type	Default
measured	ArrayLike	required
stretch (kw-only)	ArrayLike	0.0
thermal (kw-only)	ArrayLike	0.0
convention (kw-only)	str	'tally'

Output

numpy.ndarray of float64

Sources

- Article 5: Wireline Logging Depth Quality Improvement: Methodology Review and Elastic-Stretch Correction. Bolt (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 294-310. DOI: none assigned (this issue predates SPWLA DOI assignment).

corrosion_front_depth

petrolib.integrity_drilling

```
corrosion_front_depth(t, K=2.5)
```

Purpose

Diffusion-limited corrosion front $x = K \cdot \sqrt{t}$ (units follow K).

Input parameters

Parameter	Type	Default
t	ArrayLike	required
K	float	2.5

Output

numpy.ndarray of float64

Sources

- Article 5: Corrosion Behavior and Mechanism Analysis of Oilwell Cement Under CO₂ and H₂S Conditions. Zhou, Zeng, Sun, Zhou, Lei, Wan, Luo, Wu, Zhang, Xiao (2022). DOI: 10.30632/PJV63N5-2022a5. Petrophysics Vol. 63 No. 5 (Oct 2022).

counting_precision

petrolib.nuclear

```
counting_precision(counts)
```

Purpose

Relative counting precision $1/\sqrt{N}$ (Poisson).

Input parameters

Parameter	Type	Default
counts	ArrayLike	required

Output

numpy.ndarray of float64

counting_sigma

petrolib.nuclear

```
counting_sigma(value, counts)
```

Purpose

Absolute counting uncertainty value/sqrt(N) (Poisson).

Input parameters

Parameter	Type	Default
value	ArrayLike	required
counts	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: An Assessment of Fundamentals of Nuclear-Based Alternatives to Conventional Chemical-Source Bulk-Density Measurement. Ahmed Badruzzaman (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 415-434. DOI: none assigned (this issue predates SPWLA DOI assignment).

COW*petrolib.depth_matching*

```
cow(ref, target, *, n_segments=10, slack=8)
```

Purpose

Correlation-optimised warping -> (aligned, warp).

Greedily adjusts `n_segments` interior segment boundaries of target within +/-slack to maximise the piecewise segment correlation with ref, then piecewise-linearly remaps target onto the reference grid.

Input parameters

Parameter	Type	Default
ref	ArrayLike	required
target	ArrayLike	required
n_segments (kw-only)	int	10
slack (kw-only)	int	8

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 9: Automated Well-Log Pattern Alignment and Depth- Matching Techniques: An Empirical Review and Recommendations. Ezenkwu, Guntoro, Starkey, Vaziri, Addario (2023). DOI: 10.30632/PJV64N1-2023a9. Petrophysics Vol. 64 No. 1 (Feb 2023).

cpmg_kernel

petrolib.nmr

```
cpmg_kernel(t_ms, t2_grid_ms)
```

Purpose

CPMG forward kernel $K[i,j] = \exp(-t_i/T2_j)$.

Input parameters

Parameter	Type	Default
t_ms	ArrayLike	required
t2_grid_ms	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.
- Article 3: Discrete Inversion Method for Nuclear Magnetic Resonance Data Processing and Its Applications to Fluid Typing and Quantification. Gao et al. (2026), Petrophysics, 67(1), 38-53. DOI: 10.30632/PJV67N1-2026a3.
- Article 5: New Method to Estimate Porosity More Accurately from NMR Data with Short Relaxation Times. Venkataramanan, Gruber, LaVigne, Habashy, Iglesias, Cohorn, Anand, Rampurawala, Jain, Heaton, Akkurt, Rylander, Lewis (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 147-157. DOI: none assigned (this issue predates SPWLA DOI assignment).

crim

petrolib.em_dielectric

```
crim(phi, sw=1.0, *, eps_w=78.0, eps_hc=2.2, eps_matrix=5.0, alpha=0.5)
```

Purpose

CRIM 3-component rock forward model (water / hydrocarbon / matrix).

$\text{eps_mix} = (\text{phi} * \text{Sw} * \text{eps_w}^a + \text{phi} * (1 - \text{Sw}) * \text{eps_hc}^a + (1 - \text{phi}) * \text{eps_m}^a)^{1/a}$ with $a = \text{alpha}$ (0.5 = classic CRIM square-root mixing).

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
sw	ArrayLike	1.0
eps_w (kw-only)	float	78.0
eps_hc (kw-only)	float	2.2
eps_matrix (kw-only)	float	5.0
alpha (kw-only)	float	0.5

Output

numpy.ndarray of float64

Sources

- Article 6: Application of an Oil-Movability Quicklook Technique Using Dielectric Measurements at Four Depths of Investigation. S.T. Grayson and J.L. Hemingway (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 461-469. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Pore-Scale Evaluation of Dielectric Measurements in Formations with Complex Pore and Grain Structures. Huangye Chen and Zoya Heidari (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 587-597. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Advanced Log Interpretation in Field Development. Merkel, Lessenger (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 479-491. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Improving Dielectric Interpretation by Calibrating Matrix Permittivity and Solving Dielectric Mixing Laws With a New Graphical Method. Wang, Wang, Toumelin, Brown, Crousse (2018). DOI: 10.30632/PJV59N2-2018a5. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 7: Bakken Petroleum System Characterization Using Dielectric-Dispersion Logs. Han, Misra (2018). DOI: 10.30632/PJV59N2-2018a6. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Deducing Electrical Permittivity of Formations From LWD Resistivity Measurements. Stalheim (2019). DOI: 10.30632/PJV60N6-2019a3. Petrophysics Vol. 60 No. 6 (Dec 2019).

ct_porosity

petrolib.porosity_lithology

```
ct_porosity(mu, mu_grain, mu_fluid, *, clip=None)
```

Purpose

Porosity from CT attenuation $\phi = (\mu_{\text{grain}} - \mu) / (\mu_{\text{grain}} - \mu_{\text{fluid}})$.

Input parameters

Parameter	Type	Default
mu	ArrayLike	required
mu_grain	ArrayLike	required
mu_fluid	ArrayLike	required
clip (kw-only)	_Clip	None

Output

numpy.ndarray of float64

Sources

- Article 9: Upscaling of Digital Rock Porosities by Correlation With Whole- Core CT-Scan Histograms. Hertel, Rydzy, Anger, Berg, Appel, de Jong (2018). DOI: 10.30632/PJV59N5-2018a8. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 6: Experimental Method for Time-Lapse Micro-CT Imaging of Mud- Filtrate Invasion and Mudcake Deposition. Schroeder, Torres-Verdin (2019). DOI: 10.30632/PJV60N5-2019a6. Petrophysics Vol. 60 No. 5 (Oct 2019).

ct_saturation

petrolib.porosity_lithology

```
ct_saturation(mu, mu_dry, mu_sat, *, clip=(0.0, 1.0))
```

Purpose

Saturation from CT attenuation $S = (\mu - \mu_{\text{dry}}) / (\mu_{\text{sat}} - \mu_{\text{dry}})$. Clips to [0,1] by default.

Input parameters

Parameter	Type	Default
mu	ArrayLike	required
mu_dry	ArrayLike	required
mu_sat	ArrayLike	required
clip (kw-only)	_Clip	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 6: Experimental Method for Time-Lapse Micro-CT Imaging of Mud- Filtrate Invasion and Mudcake Deposition. Schroeder, Torres-Verdin (2019). DOI: 10.30632/PJV60N5-2019a6. Petrophysics Vol. 60 No. 5 (Oct 2019).

cubic_law_conductivity

petrolib.integrity_drilling

```
cubic_law_conductivity(aperture_m, *, rho=1000.0, mu=0.001, g=9.81)
```

Purpose

Parallel-plate crack hydraulic conductivity $K = \rho \cdot g \cdot w^3 / (12 \cdot \mu)$ (m/s).

Input parameters

Parameter	Type	Default
aperture_m	ArrayLike	required
rho (kw-only)	float	1000.0
mu (kw-only)	float	0.001
g (kw-only)	float	9.81

Output

numpy.ndarray of float64

Sources

- Zhang, X., Zhang, X., Zhang, T., Li, X., Mei, C., Li, B., Jiang, M., Liu, K., and Bai, Y. (2026). New Equipment and Method for Evaluating Anti-Water- Invasion Ability of Cement Slurry. Petrophysics, 67(2), 421–435. DOI: 10.30632/PJV67N2-2026a11.

darcy_gas_permeability

petrolib.flow_transport

```
darcy_gas_permeability(q, *, mu, length, area, p_up, p_down, p_ref)
```

Purpose

Compressible (gas) Darcy permeability, the pressure-squared form $k = 2*q*\mu*L*P_{ref} / (A*(P_{up}^2 - P_{down}^2))$.

The isothermal integration of Darcy's law for an ideal gas; q is the volumetric rate measured at p_ref (e.g. outlet/atmospheric). SI gives k in m2.

Input parameters

Parameter	Type	Default
q	ArrayLike	required
mu (kw-only)	float	required
length (kw-only)	float	required
area (kw-only)	float	required
p_up (kw-only)	float	required
p_down (kw-only)	float	required
p_ref (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).

darcy_permeability

petrolib.flow_transport

```
darcy_permeability(q, *, mu, length, area, dp)
```

Purpose

Darcy permeability of a core plug $k = q \cdot \mu \cdot L / (A \cdot dP)$.

Incompressible (liquid) steady-state Darcy's law. Any consistent unit set works; SI gives k in m^2 .

Input parameters

Parameter	Type	Default
q	ArrayLike	required
mu (kw-only)	float	required
length (kw-only)	float	required
area (kw-only)	float	required
dp (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).

darcy_pressure_drop

petrolib.flow_transport

```
darcy_pressure_drop(q, *, mu, k, area, length, kr=1.0)
```

Purpose

Pressure drop for Darcy flow $dP = q \cdot \mu \cdot L / (k \cdot kr \cdot A)$.

kr is the relative permeability of the flowing phase (1 for single-phase).

Input parameters

Parameter	Type	Default
q	ArrayLike	required
mu (kw-only)	float	required
k (kw-only)	float	required
area (kw-only)	float	required
length (kw-only)	float	required
kr (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 5: Comparison of Four Numerical Simulators for SCAL Experiments. Lenormand, Lorentzen, Maas, Ruth (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 48-56. DOI: none assigned (this issue predates SPWLA DOI assignment).

darcy_rate

petrolib.flow_transport

darcy_rate(k, *, area, dp, mu, length)

Purpose

Darcy volumetric flow rate $q = k \cdot A \cdot dp / (\mu \cdot L)$ (the inverse of darcy_permeability).

Input parameters

Parameter	Type	Default
k	ArrayLike	required
area (kw-only)	float	required
dp (kw-only)	float	required
mu (kw-only)	float	required
length (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.

dean_stark

petrolib.geochem_fluids.core_geochem

```
dean_stark(v_water, v_hc, *, v_bulk=None, v_pore=None)
```

Purpose

Dean-Stark porosity and saturations -> (ϕ , S_w , S_{hc}).

Pore volume is v_{pore} if given, else $v_{water} + v_{hc}$. ϕ is $pore/v_{bulk}$ when v_{bulk} is given (else NaN). $S_w = v_{water}/pore$, $S_{hc} = v_{hc}/pore$.

Input parameters

Parameter	Type	Default
v_{water}	ArrayLike	required
v_{hc}	ArrayLike	required
v_{bulk} (kw-only)	ArrayLike None	None
v_{pore} (kw-only)	ArrayLike None	None

Output

tuple[ndarray[float64], ndarray[float64], ndarray[float64]]

Sources

- Article 3: Petrophysical Characterization of Bitumen-Saturated Karsted Carbonates: Case Study of the Multibillion Barrel Upper Devonian Grosmont Formation, Northern Alberta, Canada. MacNeil (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 592-614. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Maintaining and Reconstructing In-Situ Saturations: A Comparison Between Whole Core, Sidewall Core, and Pressurized Sidewall Core in the Permian Basin. Blount, McMullen, Durand, Driskill (2019). DOI: 10.30632/PJV60N1Y2019a3. Petrophysics Vol. 60 No. 1 (Feb 2019).

debye

petrolib.em_dielectric

```
debye(freq_hz, *, eps_inf, eps_s, tau)
```

Purpose

Single-pole Debye relaxation $\text{eps_inf} + (\text{eps_s} - \text{eps_inf}) / (1 + j * \omega * \tau)$.

Input parameters

Parameter	Type	Default
freq_hz	ArrayLike	required
eps_inf (kw-only)	float	required
eps_s (kw-only)	float	required
tau (kw-only)	float	required

Output

numpy.ndarray of complex128

Sources

- Article 5: Dielectric Permittivity: A Petrophysical Parameter for Shales. Matthew Josh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 319-332. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Funk, J., Myers, M., Hathon, L. "NMR-Mapped Distributions of Dielectric Dispersion" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 421-437 DOI: 10.30632/PJV64N3-2023a7.

decay_constant

petrolib.nuclear

```
decay_constant(half_life)
```

Purpose

Radioactive decay constant $\lambda = \ln(2)/T_{\text{half}}$.

Input parameters

Parameter	Type	Default
half_life	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: In-Situ Evaluation of Vapor Properties Using Condensed Vapor Gamma. O'Sullivan (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 334-345. DOI: none assigned (this issue predates SPWLA DOI assignment).

deconvolve

petrolib.inversion_numerics.linear

```
deconvolve(d, g, rank=None)
```

Purpose

Deconvolve $d = G \cdot x$ for x via (rank-limited) pseudo-inverse.

Input parameters

Parameter	Type	Default
d	ArrayLike	required
g	ArrayLike	required
rank	int None	None

Output

numpy.ndarray of float64

Sources

- Article 3: Accelerated Whole-Core Analysis Optimization With Wellsite Tomography Instrumentation and Bayesian Inversion. Mendoza, Roininen, Girolami, Heikkinen, Haario (2019). DOI: 10.30632/PJV60N3-2019a2. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 1: Inversion of High-Resolution High- Quality Sonic Compressional and Shear Logs for Unconventional Reservoirs. Lei, Zeroug, Bose, Prioul, Donald (2019). DOI: 10.30632/PJV60N6-2019a1. Petrophysics Vol. 60 No. 6 (Dec 2019).

delta_sw_timelapse

petrolib.nuclear

```
delta_sw_timelapse(sigma_base, sigma_mon, phi, *, sigma_w_base, sigma_w_mon, sigma_hc, sw_base)
```

Purpose

Monitor-survey water saturation from a time-lapse Sigma pair.

$Sw_{mon} = [(Sigma_{mon} - Sigma_{base})/\phi + Sw_{base}*(Sigma_{w_base} - Sigma_{hc})] / (Sigma_{w_mon} - Sigma_{hc})$ (e.g. injected brine changing Sigma_w).

Input parameters

Parameter	Type	Default
sigma_base	ArrayLike	required
sigma_mon	ArrayLike	required
phi	ArrayLike	required
sigma_w_base (kw-only)	float	required
sigma_w_mon (kw-only)	float	required
sigma_hc (kw-only)	float	required
sw_base (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

density_from_count

petrolib.nuclear

```
density_from_count(count, *, n0=1000000.0, mu_mass=0.06, spacing_cm=30.0)
```

Purpose

Density from a gamma count $\rho = \ln(N_0/N)/(\mu_{\text{mass}} \cdot \text{spacing})$ (inverse).

Input parameters

Parameter	Type	Default
count	ArrayLike	required
n0 (kw-only)	float	1000000.0
mu_mass (kw-only)	float	0.06
spacing_cm (kw-only)	float	30.0

Output

numpy.ndarray of float64

Sources

- Article 6: A Method of Determining Formation Density Based on Fast-Neutron Gamma Coupled Field Theory. Zhang, Zhang, Liu, Wu, Wu, Jia, Ti, Li (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 411-425. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.

density_from_gradient

petrolib.geochem_fluids.gradients

```
density_from_gradient(dpdz, *, p_unit='Pa')
```

Purpose

Fluid density (kg/m³) from a pressure gradient (no clipping).

p_unit: "Pa" (Pa/m), "psi/ft", or "bar/m". rho = grad_SI/g.

Input parameters

Parameter	Type	Default
dpdz	ArrayLike	required
p_unit (kw-only)	str	'Pa'

Output

numpy.ndarray of float64

density_porosity

petrolib.porosity_lithology

```
density_porosity(rho_b, rho_ma=2.65, rho_fl=1.0, *, clip=None)
```

Purpose

Density porosity $\phi_D = (\rho_{ma} - \rho_b) / (\rho_{ma} - \rho_{fl})$.

ρ_{ma} defaults to sandstone 2.65; carbonate 2.71, dolomite 2.85, and brine fluids 1.03/1.025 are passed explicitly by the caller (the corpus uses all of them). Clipping is opt-in.

Input parameters

Parameter	Type	Default
ρ_b	ArrayLike	required
ρ_{ma}	ArrayLike	2.65
ρ_{fl}	ArrayLike	1.0
clip (kw-only)	_Clip	None

Output

numpy.ndarray of float64

Sources

- Article 1: Improvement in Heavy-Oil Reservoir Evaluation Using Nuclear Magnetic Resonance: Long Lake and Kinosis SAGD Projects, Alberta, Canada. Cheng, Kotov, Pyke, Hanif (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 239-250. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: A Case Study on Integrated Petrophysical Characterization of a Carbonate Reservoir Pore System in the Offshore Red River Basin of Vietnam. Giao, Chung (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 289-301. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Petrophysical Characterization of Bitumen-Saturated Karsted Carbonates: Case Study of the Multibillion Barrel Upper Devonian Grosmont Formation, Northern Alberta, Canada. MacNeil (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 592-614. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: On Error Calculation and Use of First-Order Error Propagation as Integral Part of Petrophysical Calculation. Stalheim (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 465-478. DOI: none assigned (this issue predates SPWLA DOI assignment).

depolarization_spheroid

petrolib.em_dielectric

```
depolarization_spheroid(aspect)
```

Purpose

Osborn depolarization factors (L_x , L_y , L_z) for a spheroid.

aspect = c/a is the polar/equatorial semi-axis ratio: 1 -> sphere (1/3, 1/3, 1/3); <1 -> oblate (disk); >1 -> prolate (needle). L_z is along the symmetry (polar) axis and $L_x = L_y = (1-L_z)/2$. The three factors sum to 1.

Input parameters

Parameter	Type	Default
aspect	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64], ndarray[float64]]

Sources

- Article 5: Coherent Interpretation of Wideband Electromagnetic Measurements in the Millihertz to Gigahertz Frequency Range. Seleznev, Hou, Freed, Habashy, Feng, Fellah, Xu, Nadeev (2018). DOI: 10.30632/PJV59N3-2018a4. Petrophysics Vol. 59 No. 3 (Jun 2018).

despike

petrolib.data_qc_io.clean

```
despike(x, *, threshold=3.0, side='both', replace='interp')
```

Purpose

Remove spikes flagged by the robust MAD criterion.

Spikes are samples further than $\text{threshold} * \text{MAD} * 1.4826$ from the median (see `outlier_mask`). `replace='interp'` fills them by linear interpolation from the surviving samples, `'median'` with the global median (the ferromagnetic-contaminant convention), `'nan'` with NaN.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
threshold (kw-only)	float	3.0
side (kw-only)	str	'both'
replace (kw-only)	str	'interp'

Output

numpy.ndarray of float64

Sources

- Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings. Based on: Banks, J.Y., Tugwell, A.G., and Potter, D.K. (2024), "Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings: A Case Study Constraining Uncertainty in the Culzean Triassic Reservoir," *Petrophysics*, 65(4), pp. 604-623. DOI: 10.30632/PJV65N4-2024a13.

detect_bed_boundaries

petrolib.data_qc_io.filt

```
detect_bed_boundaries(curve, window=11, threshold=None)
```

Purpose

Bed-boundary indices from a derivative-times-local-variability metric.

$\text{metric} = |\text{gradient}(\text{curve})| * \text{sqrt}(\text{moving_var})$ (variance over the centered window, edges left 0); boundaries are the local maxima of the metric above threshold (default adaptive mean + 2*std).

Input parameters

Parameter	Type	Default
curve	ArrayLike	required
window	int	11
threshold	float None	None

Output

NDArray[np.int_]

Sources

- Merletti, G., Rabinovich, M., Al Hajri, S., Dawson, W., Farmer, R., Ambia, J., Torres-Verdín, C. (2023). "New Iterative Resistivity Modeling Workflow Reduces Uncertainty in the Assessment of Water Saturation in Deeply Invaded Reservoirs", *Petrophysics*, Vol. 64, No. 4, pp. 555-567. DOI: 10.30632/PJV64N4-2023a5.
- Thin-Bed NMR Response Characterisation in Horizontal Wells. Based on: Ramadan et al. (2024), "Characterizing Thin-Bed Responses in Horizontal Wells Using LWD NMR Tools: Insights From a Water Tank Experiment", *Petrophysics*, Vol. 65, No. 5, pp. 765-771.
- Strobel, J. (2024). "Petrophysical Analyses for Supporting the Search for a Claystone-Hosted Nuclear Repository." *Petrophysics* 65(3), 302-316. DOI: 10.30632/PJV65N3-2024a2.

diameter_from_molar_volume

petrolib.geochem_fluids.asphaltene

```
diameter_from_molar_volume(va_m3mol)
```

Purpose

Sphere diameter (m) from molar volume = $(6 \cdot (V_a / N_A) / \pi)^{1/3}$.

Input parameters

Parameter	Type	Default
va_m3mol	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).

difference_operator

petrolib.inversion_numerics.linear

```
difference_operator(n, order=1)
```

Purpose

Finite-difference smoothness operator L of shape $(n\text{-order}, n)$.

$\text{order}=1$ is the first difference, $\text{order}=2$ the second difference, and $\text{order}=0$ the identity.

Input parameters

Parameter	Type	Default
n	int	required
order	int	1

Output

numpy.ndarray of float64

diffusion_length

petrolib.flow_transport

```
diffusion_length(D, t, *, geometry_factor=1.0)
```

Purpose

Characteristic diffusion length $L = \sqrt{f \cdot D \cdot t}$.

geometry_factor f is 1 in most copies and 2 in the $\sqrt{2 \cdot D \cdot t}$ displacement form.

Input parameters

Parameter	Type	Default
D	ArrayLike	required
t	ArrayLike	required
geometry_factor (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 2: CO₂ EOR by Diffusive Mixing in Fractured Reservoirs. Eide, Ersland, Brattekas, Haugen, Graue, Ferno (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 23-31. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: The Dynamics of Reservoir Fluids and their Substantial Systematic Variations Oliver C. Mullins, Julian Y. Zuo, Kang Wang, Paul S. Hammond, Ilaria De Santo, Hadrien Dumont, Vinay K. Mishra, Li Chen, Andrew E. Pomerantz, Chengli Dong, Hani Elshahawi, Douglas J. Seifert (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 96-112. DOI: none assigned (this issue predates SPWLA DOI assignment).

diffusion_relaxation_rate

petrolib.nmr

```
diffusion_relaxation_rate(D, *, G, TE, gamma=GAMMA_H)
```

Purpose

Diffusion relaxation rate $1/T2_D = (\gamma * G * TE)^2 * D / 12$.

Restricted/unrestricted diffusion in a field gradient (the CPMG diffusion term). gamma is the proton gyromagnetic ratio in rad/(s*T); pass G, TE, D in a consistent (SI) unit set.

Input parameters

Parameter	Type	Default
D	ArrayLike	required
G (kw-only)	ArrayLike	required
TE (kw-only)	ArrayLike	required
gamma (kw-only)	float	GAMMA_H

Output

numpy.ndarray of float64

Sources

- Article 3: Nuclear Magnetic Resonance and Pore Coupling in Clay-Coated Sandstones With Anomalous Porosity Preservation, Agua Grande Formation, Reconcavo Basin, Brazil. Jacomo, Trindade, de Oliveira, Leite, Montrazi, Andreeta, Bonagamba (2018). DOI: 10.30632/PJV59N2-2018a2. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 1: Pore-Structure Characterization of a Complex Carbonate Reservoir in South Iraq Using Advanced Interpretation of NMR Logs. Saidian, Jain, Milad (2021). DOI: 10.30632/PJV62N2-2021a1. Petrophysics Vol. 62 No. 2 (Apr 2021).

diffusion_step_1d

petrolib.inversion_numerics.pde

```
diffusion_step_1d(u, alpha, dt, dx, source=None, bc='neumann')
```

Purpose

One explicit finite-difference step of the 1D diffusion equation.
 $u_{\text{new}} = u + \alpha \cdot dt / dx^2 * \text{laplacian}(u) + dt \cdot \text{source}$. `bc='neumann'` uses no-flux ends;
`bc='dirichlet'` holds the endpoint values fixed.

Input parameters

Parameter	Type	Default
u	ArrayLike	required
alpha	float	required
dt	float	required
dx	float	required
source	ArrayLike None	None
bc	str	'neumann'

Output

numpy.ndarray of float64

diffusion_time

petrolib.flow_transport

```
diffusion_time(L, D, *, geometry_factor=1.0)
```

Purpose

Diffusion / equilibration time $t = L^2 / (f \cdot D)$ (inverse of diffusion_length). geometry_factor=2 gives $L^2 / (2 \cdot D)$; a sealed-column mixing time uses geometry_factor= π^2 .

Input parameters

Parameter	Type	Default
L	ArrayLike	required
D	ArrayLike	required
geometry_factor (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 3: Gas Diffusion into Oil, Reservoir Baffling and Tar Mats Analyzed by Downhole Fluid Analysis, Pressure Transients, Core Extracts and DSTs. Achourov, Pfeiffer, Kollien, Betancourt, Zuo, di Primio, Mullins (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 346-357. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Differing Equilibration Times of GOR, Asphaltenes and Biomarkers as Determined by Charge History and Reservoir Fluid Geodynamics. Wang, Kauerauf, Zuo, Chen, Dong, Elshahawi, Mullins (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 440-456. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: The Dynamics of Reservoir Fluids and their Substantial Systematic Variations Oliver C. Mullins, Julian Y. Zuo, Kang Wang, Paul S. Hammond, Ilaria De Santo, Hadrien Dumont, Vinay K. Mishra, Li Chen, Andrew E. Pomerantz, Chengli Dong, Hani Elshahawi, Douglas J. Seifert (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 96-112. DOI: none assigned (this issue predates SPWLA DOI assignment).

dip_from_amplitude

petrolib.borehole_image

```
dip_from_amplitude(amplitude, radius, *, sample_spacing=1.0)
```

Purpose

True dip from a sinusoid amplitude: $\arctan(\text{amplitude} \times \text{sample_spacing} / \text{radius})$ (deg).

Input parameters

Parameter	Type	Default
amplitude	float	required
radius	float	required
sample_spacing (kw-only)	float	1.0

Output

float

disk_image

petrolib.data_qc_io.synth

```
disk_image(size, n_features, *, rng=None)
```

Purpose

Grey image with darker random disks; returns (image, mask).
Background $0.6 + 0.05 \cdot N(0,1)$, n_features disks of radius 2-7 px at uniform centers, disk pixels $0.15 + 0.05 \cdot N(0,1)$, clipped to [0, 1] — a synthetic SEM/pore image.

Input parameters

Parameter	Type	Default
size	int	required
n_features	int	required
rng (kw-only)	np.random.Generator int None	None

Output

tuple[ndarray[float64], NDArray[np.bool_]]

Sources

- Chen et al., "Pore Identification and Quantitative Characterization of Shale SEM Images using a Deep-Learning Semantic Segmentation Pore-Net Model", Petrophysics, Vol. 65, No. 2 (April 2024), pp. 233-245.

displacement_efficiency

petrolib.relpem_wettability

```
displacement_efficiency(soi, sor)
```

Purpose

Microscopic displacement efficiency $E_d = (S_{oi} - S_{or})/S_{oi}$.

The same $(S_{oi}-S_{or})/S_{oi}$ form is also written `recovery_factor` in several articles.

Input parameters

Parameter	Type	Default
<code>soi</code>	ArrayLike	required
<code>sor</code>	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 6: Novel Coupling Smart Water-CO₂ Flooding for Sandstone Reservoirs. Al-Saedi, Flori (2019). DOI: 10.30632/PJV60N4-2019a6. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 2: Enhanced Learning of Fundamental Petrophysical Concepts Through Image Processing and 3D Printing. Alyafei, Al Musleh, Bautista, Idris, Seers (2021). DOI: 10.30632/PJV62N5-2021a2. Petrophysics Vol. 62 No. 5 (Oct 2021).

distributed_stretch

petrolib.depth_correction

```
distributed_stretch(length, weight_per_length, ea, *, end_load=0.0, buoyancy=1.0)
```

Purpose

Hanging-string stretch $dL = (F L + 0.5 b w L^2) / (E A)$.

Integrates the tension profile of a string of buoyant weight-per-length w carrying an end_load F at the bottom: the linear term is the end load, the quadratic term the string's own weight (scaled by the buoyancy factor). ea is the lumped $E \cdot A$ stiffness.

Input parameters

Parameter	Type	Default
length	ArrayLike	required
weight_per_length	ArrayLike	required
ea	ArrayLike	required
end_load (kw-only)	ArrayLike	0.0
buoyancy (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 5: Wireline Logging Depth Quality Improvement: Methodology Review and Elastic-Stretch Correction. Bolt (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 294-310. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Correction of Driller's Depth: Field Example Using Driller's Way-Point Depth Correction Methodology. Bolt (2019). DOI: 10.30632/PJV60N1Y2019a7. Petrophysics Vol. 60 No. 1 (Feb 2019).

dogleg_angle

petrolib.wellbore_geometry

```
dogleg_angle(inc1, azi1, inc2, azi2)
```

Purpose

Dogleg (total curvature) angle between two survey stations, in radians.

$\cos(DL) = \cos(i2 - i1) - \sin(i1) \sin(i2) (1 - \cos(a2 - a1))$ with the inclinations i and azimuths a supplied in degrees. This is the numerically well-behaved rearrangement of the spherical law of cosines and is clipped to $[-1, 1]$ before the arccos.

Input parameters

Parameter	Type	Default
inc1	float	required
azi1	float	required
inc2	float	required
azi2	float	required

Output

float

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

doll_radial_geometric_factor

petrolib.em.dielectric

```
doll_radial_geometric_factor(r, spacing_m)
```

Purpose

Doll radial geometric factor $G = r^2 / (r^2 + (L/2)^2)$ (integrated to radius r).

Input parameters

Parameter	Type	Default
<code>r</code>	ArrayLike	required
<code>spacing_m</code>	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 6: Reminiscences on the Development of the First Commercial Array-Induction Measurement. Peter Elkington (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 618-623. DOI: none assigned (this issue predates SPWLA DOI assignment).

drilling_window_margin

petrolib.integrity_drilling

```
drilling_window_margin(pore, frac)
```

Purpose

Drilling-window width $\text{frac} - \text{pore}$ (same units as the inputs).

Input parameters

Parameter	Type	Default
pore	ArrayLike	required
frac	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).

dtw*petrolib.depth_matching*

```
dtw(ref, target, *, band=None, cost='sqdiff', root=False)
```

Purpose

Banded dynamic time warping of target onto ref.

Accumulates $D[i,j] = \text{local}(i,j) + \min(D[i-1,j-1], D[i-1,j], D[i,j-1])$ on a padded matrix, where `local` is the squared (`cost='sqdiff'`) or absolute (`cost='absdiff'`) difference. `band` is the Sakoe-Chiba half-width (`None` = full). Returns the total distance (sqrt of it if `root=True`), the backtracked 0-based path and the accumulated matrix.

Input parameters

Parameter	Type	Default
<code>ref</code>	ArrayLike	required
<code>target</code>	ArrayLike	required
<code>band</code> (kw-only)	int None	None
<code>cost</code> (kw-only)	str	'sqdiff'
<code>root</code> (kw-only)	bool	False

Output

DtwResult

Sources

- Article 10: Machine-Learning-Based Automatic Well-Log Depth Matching. Zimmermann, Liang, Zeroug (2018). DOI: 10.30632/PJV59N6Y2018a9. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 1: A Machine-Learning-Based Approach to Assistive Well-Log Correlation. Brazell, Bayeh, Ashby, Burton (2019). DOI: 10.30632/PJV60N4-2019a1. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 3: A Machine-Learning Framework for Automating Well-Log Depth Matching. Le, Liang, Zimmermann, Zeroug, Heliot (2019). DOI: 10.30632/PJV60N5-2019a3. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 3: Automated Log Data Analytics Workflow - The Value of Data Access and Management to Reduced Turnaround Time for Log Analysis. Torres Caceres, Duffaut, Westad, Stovas, Johansen, Jenssen (2022). DOI: 10.30632/PJV63N1-2022a3. Petrophysics Vol. 63 No. 1 (Feb 2022).
- Article 9: Automated Well-Log Pattern Alignment and Depth- Matching Techniques: An Empirical Review and Recommendations. Ezenkwu, Guntoro, Starkey, Vaziri, Addario (2023). DOI: 10.30632/PJV64N1-2023a9. Petrophysics Vol. 64 No. 1 (Feb 2023).

DtwResult (class)

petrolib.depth_matching

```
DtwResult()
```

Purpose

Result of dtw / rddtw.

Input parameters

None.

Output

DtwResult instance

dual_detector_density

petrolib.nuclear

```
dual_detector_density(near, far, *, a, b)
```

Purpose

Dual-detector density $\rho = a + b \cdot \ln(\text{near}/\text{far})$ (count-ratio calibration).

Input parameters

Parameter	Type	Default
near	ArrayLike	required
far	ArrayLike	required
a (kw-only)	float	required
b (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Article 3: An Assessment of Fundamentals of Nuclear-Based Alternatives to Conventional Chemical-Source Bulk-Density Measurement. Ahmed Badruzzaman (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 415-434. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: A Method of Determining Formation Density Based on Fast-Neutron Gamma Coupled Field Theory. Zhang, Zhang, Liu, Wu, Wu, Jia, Ti, Li (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 411-425. DOI: none assigned (this issue predates SPWLA DOI assignment).

dual_water_conductivity

petrolib.saturation_resistivity

```
dual_water_conductivity(sw, *, cw, cb, swb, phi, m=2.0, n=2.0)
```

Purpose

Dual-water total conductivity, bound-water form (Clavier et al.).

$C_t = S_w^n * \phi^m * C_w + S_w^{n-1} * S_{wb} * \phi^m * (C_b - C_w)$ with bound-water saturation s_{wb} and conductivity c_b . This is NOT the Qv-based dual-water form (`dual_water_conductivity_qv`) — the parameters do not map one-to-one.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
cw (kw-only)	ArrayLike	required
cb (kw-only)	ArrayLike	required
swb (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required
m (kw-only)	float	2.0
n (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).

dual_water_conductivity_qv

petrolib.saturation_resistivity

```
dual_water_conductivity_qv(sw, *, cw, qv, alpha_vqh, beta, phi, m0=2.0, n=2.0)
```

Purpose

Dual-water total conductivity, Qv-based form.

$C_t = \phi^{m_0} * (S_w^n * (1 - \alpha_{vqh} * Q_v) * C_w + S_w^{(n-1)} * \beta * Q_v)$ where α_{vqh} is the product of the salinity-expansion factor and the clay-water volume factor v_{QH} (ml/meq), and β the counterion equivalent conductance.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
cw (kw-only)	ArrayLike	required
qv (kw-only)	ArrayLike	required
alpha_vqh (kw-only)	float	required
beta (kw-only)	float	required
phi (kw-only)	ArrayLike	required
m0 (kw-only)	float	2.0
n (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).

dual_water_sw

petrolib.saturation_resistivity

```
dual_water_sw(ct, *, cw, cb, swb, phi, m=2.0, n=2.0, lo=1e-06, hi=1.0, n_iter=100,
tol=1e-12, clip=None)
```

Purpose

Invert the bound-water dual_water_conductivity for Sw.

Same bisection contract as waxman_smits_sw.

Input parameters

Parameter	Type	Default
ct	ArrayLike	required
cw (kw-only)	ArrayLike	required
cb (kw-only)	ArrayLike	required
swb (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required
m (kw-only)	float	2.0
n (kw-only)	float	2.0
lo (kw-only)	float	1e-06
hi (kw-only)	float	1.0
n_iter (kw-only)	int	100
tol (kw-only)	float	1e-12
clip (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).

dual_water_sw_qv

petrolib.saturation_resistivity

```
dual_water_sw_qv(ct, *, cw, qv, alpha_vqh, beta, phi, m0=2.0, n=2.0, lo=1e-06, hi=1.0,
n_iter=100, tol=1e-12, clip=None)
```

Purpose

Invert the Qv-form dual_water_conductivity_qv for Sw.

Same bisection contract as waxman_smits_sw — the bracket and iteration controls are explicit so facades reproduce their article's solver exactly (src2014_02/article1 uses lo=1e-4, tol=1e-10).

Input parameters

Parameter	Type	Default
ct	ArrayLike	required
cw (kw-only)	ArrayLike	required
qv (kw-only)	ArrayLike	required
alpha_vqh (kw-only)	float	required
beta (kw-only)	float	required
phi (kw-only)	ArrayLike	required
m0 (kw-only)	float	2.0
n (kw-only)	float	2.0
lo (kw-only)	float	1e-06
hi (kw-only)	float	1.0
n_iter (kw-only)	int	100
tol (kw-only)	float	1e-12
clip (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).

early_time_uptake

petrolib.flow_transport

```
early_time_uptake(D, t, half_length)
```

Purpose

Early-time fractional uptake $M_t/M_{inf} = (2/L)*\sqrt{D*t/\pi}$ for a slab of half-length L (valid to ~ 0.6).

Input parameters

Parameter	Type	Default
D	float	required
t	ArrayLike	required
half_length	float	required

Output

numpy.ndarray of float64

Sources

- Article 2: CO2 EOR by Diffusive Mixing in Fractured Reservoirs. Eide, Ersland, Brattekas, Haugen, Graue, Ferno (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 23-31. DOI: none assigned (this issue predates SPWLA DOI assignment).

eaton_pore_pressure

petrolib.acoustic_geomech

```
eaton_pore_pressure(overburden, hydrostatic, observed, normal, *, exponent=3.0,
log_type='sonic')
```

Purpose

Eaton pore pressure $P_p = S_{ov} - (S_{ov} - P_{hydro}) * ratio^{exponent}$.

`log_type="sonic"` -> `ratio = normal/observed` (slowness or velocity); `"resistivity"` -> `ratio = observed/normal`. The ratio-direction flip between the two log types is a documented corpus hazard; the default exponent 3.0 is the sonic value (resistivity typically uses ~1.2).

Input parameters

Parameter	Type	Default
overburden	ArrayLike	required
hydrostatic	ArrayLike	required
observed	ArrayLike	required
normal	ArrayLike	required
exponent (kw-only)	float	3.0
log_type (kw-only)	str	'sonic'

Output

numpy.ndarray of float64

eaton_pore_pressure

petrolib.integrity_drilling

```
eaton_pore_pressure(overburden, hydrostatic, observed, normal, *, exponent=3.0,
log_type='sonic', clip_ratio=None)
```

Purpose

Eaton pore pressure $P_p = OB - (OB - P_n) * ratio^{exponent}$.
log_type='sonic' uses ratio = normal/observed (slowness; slow, undercompacted rock gives ratio < 1 and overpressure); 'resistivity' uses ratio = observed/normal. Works in pressure or gradient space (the output follows the input units). clip_ratio optionally bounds the ratio (e.g. (0.01, 100.0)).

Input parameters

Parameter	Type	Default
overburden	ArrayLike	required
hydrostatic	ArrayLike	required
observed	ArrayLike	required
normal	ArrayLike	required
exponent (kw-only)	float	3.0
log_type (kw-only)	str	'sonic'
clip_ratio (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Genetic Analysis of Overpressure While Drilling Based on Isotope Logging. Based on: Hu et al., "Genetic Analysis of Overpressure While Drilling Based on Isotope Logging Technology", Petrophysics, Vol. 66, No. 2, April 2025, pp. 283-293.

effective_conductivity_2d

petrolib.inversion_numerics.pde

```
effective_conductivity_2d(sigma_map, n_iter=5000, tol=1e-07)
```

Purpose

Effective vertical conductivity of a 2D conductivity map.

Solves the steady Laplace problem $\text{div}(\sigma \text{ grad } V)=0$ with harmonic-mean face conductivities by Jacobi relaxation, top/bottom Dirichlet electrodes ($V=1$ / $V=0$) and no-flux sides, then returns the effective conductivity from the top-layer current.

Input parameters

Parameter	Type	Default
sigma_map	ArrayLike	required
n_iter	int	5000
tol	float	1e-07

Output

float

effective_porosity

petrolib.porosity_lithology

```
effective_porosity(phi_t, vsh, phi_sh, *, clip=(0.0, None))
```

Purpose

Shale-corrected effective porosity $\phi_e = \phi_t - V_{sh} \phi_{sh}$.

ϕ_{sh} (the shale total porosity) is explicit; the corpus uses 0.10, 0.27, 0.30. Clips to non-negative by default; pass `clip=None` for the structural-shale-porosity form.

Input parameters

Parameter	Type	Default
<code>phi_t</code>	ArrayLike	required
<code>vsh</code>	ArrayLike	required
<code>phi_sh</code>	ArrayLike	required
<code>clip</code> (kw-only)	<code>_Clip</code>	(0.0, None)

Output

numpy.ndarray of float64

Sources

- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Part 3: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N3-2019t1.
- Article 6: Untangle Shale and Gas Effects to Estimate Porosity and Net/Gross Ratio Using a Boomerang Workflow - A Case Study in Shoreface Reservoirs in Brunei. Xu, Sharif (2020). DOI: 10.30632/PJV61N1-2020a5. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 2: The Problem With Silt in Low-Resistivity Low-Contrast (LRLC) Pay Reservoirs. Belevich, Bal (2018). DOI: 10.30632/PJV59N2-2018a1. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

effective_stress

petrolib.acoustic_geomech

```
effective_stress(sigma_total, pp, *, biot=1.0)
```

Purpose

Terzaghi/Biot effective stress $\sigma_{\text{eff}} = \sigma_{\text{total}} - \text{biot} * P_p$.

Input parameters

Parameter	Type	Default
<code>sigma_total</code>	ArrayLike	required
<code>pp</code>	ArrayLike	required
<code>biot</code> (kw-only)	ArrayLike	1.0

Output

numpy.ndarray of float64

elastic_stretch

petrolib.depth_correction

```
elastic_stretch(force, length, area, *, E=E_STEEL)
```

Purpose

Point-load Hookean stretch $dL = F L / (E A)$.

force is the axial load (N), length the suspended length (m), area the metal cross-section (m^2), E Young's modulus (Pa).

Input parameters

Parameter	Type	Default
force	ArrayLike	required
length	ArrayLike	required
area	ArrayLike	required
E (kw-only)	float	E_STEEL

Output

numpy.ndarray of float64

Sources

- Article 2: Driller's Depth Quality Improvement: Way- Point Methodology. Bolt (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 564-575. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Depth: A Love and Hate Story. Theys (2019). DOI: 10.30632/PJV60N1Y2019a5. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 8: Correction of Driller's Depth: Field Example Using Driller's Way-Point Depth Correction Methodology. Bolt (2019). DOI: 10.30632/PJV60N1Y2019a7. Petrophysics Vol. 60 No. 1 (Feb 2019).

electron_density_index

petrolib.nuclear

```
electron_density_index(z, a, rho_m)
```

Purpose

Electron density $\rho_e = (2 \cdot Z/A) \cdot \rho_m$ of a single element/compound.

Input parameters

Parameter	Type	Default
z	ArrayLike	required
a	ArrayLike	required
rho_m	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Recharacterization and Validation of Through-the-Bit-Logging Tool Measurements. Slocombe, Bammi, Hunka, Reischman, Schmid (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 58-71. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

electron_density_mixture

petrolib.nuclear

```
electron_density_mixture(z, a, mass_frac, rho_m)
```

Purpose

Mixture electron density $\rho_e = 2 \cdot \sum (w_i \cdot Z_i / A_i) \cdot \rho_m$.

Input parameters

Parameter	Type	Default
z	ArrayLike	required
a	ArrayLike	required
mass_frac	ArrayLike	required
rho_m	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Recharacterization and Validation of Through-the-Bit-Logging Tool Measurements. Slocombe, Bammi, Hunka, Reischman, Schmid (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 58-71. DOI: none assigned (this issue predates SPWLA DOI assignment).

electron_density_to_bulk

petrolib.porosity_lithology

```
electron_density_to_bulk(rho_e, *, a=1.0704, b=-0.1883)
```

Purpose

Electron- to bulk-density calibration $\rho_b = a \cdot \rho_e + b$.

The (1.0704, -0.1883) constants are the standard tool calibration; the (0.1823, 1.07) inverse-form constants are passed by the caller.

Input parameters

Parameter	Type	Default
<code>rho_e</code>	ArrayLike	required
<code>a</code> (kw-only)	float	1.0704
<code>b</code> (kw-only)	float	-0.1883

Output

numpy.ndarray of float64

Sources

- Article 2: Matrix-Adjusted Shale Porosity Measured in Horizontal Wells. Craddock, Mosse, Bernhardt, Ortiz, Gonzalez Tomassini, Pirie, Saldungaray, Pomerantz (2018). DOI: 10.30632/PJV59N3-2018a1. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 5: Recharacterization and Validation of Through-the-Bit-Logging Tool Measurements. Slocombe, Bammi, Hunka, Reischman, Schmid (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 58-71. DOI: none assigned (this issue predates SPWLA DOI assignment).

endpoint_mobility_ratio

petrolib.relpem_wettability

```
endpoint_mobility_ratio(krw_max, kro_max, *, mu_w, mu_o)
```

Purpose

Endpoint mobility ratio $M = (krw_max/mu_w) / (kro_max/mu_o)$.

The waterflood favorability index ($M < 1$ is favorable / piston-like).

Input parameters

Parameter	Type	Default
krw_max	float	required
kro_max	float	required
mu_w (kw-only)	float	required
mu_o (kw-only)	float	required

Output

float

Sources

- Article 9: Downhole Estimation of Relative Permeability With Integration of Formation-Tester Measurements and Advanced Well Logs. Hadibeik, Azari, Kalawina, Ramakrishna, Eyuboglu, Khan, Al-Rushaid, Al-Rashidi, Ahmad (2018). DOI: 10.30632/PJV59N2-2018a8 (inferred; body beyond extraction). Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 6: Novel Coupling Smart Water-CO2 Flooding for Sandstone Reservoirs. Al-Saedi, Flori (2019). DOI: 10.30632/PJV60N4-2019a6. Petrophysics Vol. 60 No. 4 (Aug 2019).

enkf_update

petrolib.inversion_numerics.stochastic

```
enkf_update(ens, obs, obs_cov, obs_op, seed=0)
```

Purpose

Stochastic (perturbed-observation) ensemble-Kalman update of ens.

ens is (n_ens, n_state); obs_op is the observation operator (a callable state->obs or a matrix H).

Returns the updated ensemble.

Input parameters

Parameter	Type	Default
ens	ArrayLike	required
obs	ArrayLike	required
obs_cov	ArrayLike	required
obs_op	Callable[[_Float], Any] ArrayLike	required
seed	int	0

Output

numpy.ndarray of float64

erfc_profile

petrolib.flow_transport

```
erfc_profile(c0, x, *, D, t)
```

Purpose

Semi-infinite diffusion concentration profile $C(x,t) = C_0 \text{erfc}(x/(2\sqrt{D \cdot t}))$ (fixed C_0 at $x=0$, far-field 0).

Uses `scipy.special.erfc` (imported lazily). For the two-plateau step front $C_{\text{far}} + (C_0 - C_{\text{far}}) * 0.5 * \text{erfc}(\dots)$ see the article facade.

Input parameters

Parameter	Type	Default
c0	float	required
x	ArrayLike	required
D (kw-only)	float	required
t (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Article 3: Gas Diffusion into Oil, Reservoir Baffling and Tar Mats Analyzed by Downhole Fluid Analysis, Pressure Transients, Core Extracts and DSTs. Achourov, Pfeiffer, Kollien, Betancourt, Zuo, di Primio, Mullins (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 346-357. DOI: none assigned (this issue predates SPWLA DOI assignment).

fd_gradient

petrolib.inversion_numerics.nonlinear

```
fd_gradient(f, x)
```

Purpose

Finite-difference gradient of a scalar objective f.

Input parameters

Parameter	Type	Default
f	Callable[[_Float], float]	required
x	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: On Error Calculation and Use of First- Order Error Propagation as Integral Part of Petrophysical Calculation. Stalheim (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 465-478. DOI: none assigned (this issue predates SPWLA DOI assignment).

fd_jacobian

petrolib.inversion_numerics.nonlinear

```
fd_jacobian(forward, m, eps=1e-06, scheme='central', relative=True)
```

Purpose

Finite-difference Jacobian $d \text{ forward} / d m$ ($n_{\text{data}} \times n_{\text{param}}$).

Input parameters

Parameter	Type	Default
forward	_Forward	required
m	ArrayLike	required
eps	float	1e-06
scheme	str	'central'
relative	bool	True

Output

numpy.ndarray of float64

feasible_set_sampling

petrolib.inversion_numerics.nonlinear

```
feasible_set_sampling(forward, data, m_center, bounds, noise_level, n_samples=2000,
step_frac=0.15, seed=1)
```

Purpose

Equivalence / feasible-set sampling around a solution for uncertainty bounds.

Random-walks around `m_center` within bounds, keeping models whose RMS data residual is within $2 \times \text{noise_level}$, and returns per-parameter P5/P50/P95 and the accepted models.

Input parameters

Parameter	Type	Default
<code>forward</code>	<code>_Forward</code>	required
<code>data</code>	<code>ArrayLike</code>	required
<code>m_center</code>	<code>ArrayLike</code>	required
<code>bounds</code>	<code>list[tuple[float, float]]</code>	required
<code>noise_level</code>	<code>float</code>	required
<code>n_samples</code>	<code>int</code>	2000
<code>step_frac</code>	<code>float</code>	0.15
<code>seed</code>	<code>int</code>	1

Output

`dict[str, ndarray[float64]]`

fhz_invert_molar_volume

petrolib.geochem_fluids.asphaltene

```
fhz_invert_molar_volume(od1, z1, od2, z2, delta_rho, temp_k)
```

Purpose

Recover the asphaltene molar volume (m³/mol) from two OD/depth points.

Inverts the gravity-only FHZ ratio: $V_a = R \cdot T \cdot \ln(OD2/OD1) / (g \cdot \Delta \rho \cdot (z2 - z1))$.

Input parameters

Parameter	Type	Default
od1	ArrayLike	required
z1	ArrayLike	required
od2	ArrayLike	required
z2	ArrayLike	required
delta_rho	ArrayLike	required
temp_k	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).

fhz_profile

petrolib.geochem_fluids.asphaltene

```
fhz_profile(depth_m, od_ref, depth_ref, va_m3mol, delta_rho, temp_k)
```

Purpose

Optical-density profile $OD(z) = OD_ref * fhz_ratio(z - z_ref, \dots)$ (gravity-only).

Input parameters

Parameter	Type	Default
depth_m	ArrayLike	required
od_ref	ArrayLike	required
depth_ref	ArrayLike	required
va_m3mol	ArrayLike	required
delta_rho	ArrayLike	required
temp_k	ArrayLike	required

Output

numpy.ndarray of float64

fhz_ratio

petrolib.geochem_fluids.asphaltene

```
fhz_ratio(dz_m, va_m3mol, delta_rho, temp_k, *, entropy=0.0, dsol2_pa=0.0)
```

Purpose

Flory-Huggins-Zuo concentration ratio between two depths (gravity + optional terms).

$OD(z+dz)/OD(z) = \exp[Va \cdot g \cdot \Delta \rho \cdot dz / (R \cdot T) + \text{entropy} - Va \cdot dsol2 / (R \cdot T)]$ with dz positive downward, $\Delta \rho = \rho_{\text{asphaltene}} - \rho_{\text{oil}}$ (kg/m³), Va in m³/mol, dsol2_pa the solubility-squared difference (Pa). The entropy and solubility terms default off (gravity-only).

Input parameters

Parameter	Type	Default
dz_m	ArrayLike	required
va_m3mol	ArrayLike	required
delta_rho	ArrayLike	required
temp_k	ArrayLike	required
entropy (kw-only)	ArrayLike	0.0
dsol2_pa (kw-only)	ArrayLike	0.0

Output

numpy.ndarray of float64

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Scanning Electron Micrographs of Tar-Mat Intervals Formed by Asphaltene Phase Transition. Pfeiffer, Di Primio, Achourov, Mullins (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 141-152. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Enigmatic Reservoir Properties Deciphered Using Petroleum System Modeling and Reservoir Fluid Geodynamics. Pierpont, Birkeland, Cely, Yang, Chen, Achourov, Betancourt, Canas, Forsythe, Pomerantz, Yang, Datir, Mullins (2023). DOI: 10.30632/PJV64N1-2023a1. Petrophysics Vol. 64 No. 1 (Feb 2023).

fick_flux

petrolib.flow_transport

```
fick_flux(D, dc, dx)
```

Purpose

Fick's first law steady flux $J = -D*dc/dx$.

Input parameters

Parameter	Type	Default
D	float	required
dc	ArrayLike	required
dx	float	required

Output

numpy.ndarray of float64

Sources

- Article 3: Wettability Effects on Osmosis as an Oil-Mobilization Mechanism During Low-Salinity Waterflooding. Fredriksen, Rognmo, Sandengen, Ferno (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 28-35. DOI: none assigned (this issue predates SPWLA DOI assignment).

fit_cementation_exponent

petrolib.saturation_resistivity

```
fit_cementation_exponent(phi, ff)
```

Purpose

Fit (m, a) from a log-log formation-factor vs porosity regression.

$\log_{10}(F) = \log_{10}(a) - m \cdot \log_{10}(\phi)$; returns (m, a) — in that order, matching the src2014_02/article2 convention.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
ff	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).

fit_corey

petrolib.relpem_wettability

```
fit_corey(sw, krw_obs, krnw_obs, *, swr, sor=0.0)
```

Purpose

Least-squares Corey endpoints and exponents from observed kr.

Fits $krw = krw_max * Se^{**nw}$ and $krnw = krnw_max * (1-Se)^{**nnw}$ on $Se = (Sw-Swr)/(1-Swr-Sor)$ and returns {swr, sor, krw_max, nw, krnw_max, nnw}. Requires scipy (imported lazily).

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
krw_obs	ArrayLike	required
krnw_obs	ArrayLike	required
swr (kw-only)	float	required
sor (kw-only)	float	0.0

Output

dict[str, float]

Sources

- CO2/Brine Relative Permeability: Reconciling SS and USS Methods. Mascle et al., 2025, Petrophysics 66(1), 26-43. DOI:10.30632/PJV66N1-2025a2.

fit_cosine

petrolib.inversion_numerics.fitting

```
fit_cosine(az, y)
```

Purpose

Azimuthal cosine-harmonic fit $y = \text{mean} + \text{amp} \cdot \cos(\text{az} - \text{phase})$.

Least squares on $[1, \cos(\text{az}), \sin(\text{az})]$ (az in degrees). Returns (mean, amplitude, phase_deg).

Input parameters

Parameter	Type	Default
az	ArrayLike	required
y	ArrayLike	required

Output

tuple[float, float, float]

fit_exponential_approach

petrolib.inversion_numerics.fitting

```
fit_exponential_approach(t, y, three_point=False)
```

Purpose

Fit an exponential approach $y = \text{asymptote} + (y_0 - \text{asymptote}) \cdot \exp(-t/\tau)$.

Returns (asymptote, tau). three_point=True uses the closed-form estimate from the first / middle / last (equally spaced) samples; otherwise curve_fit is used.

Input parameters

Parameter	Type	Default
t	ArrayLike	required
y	ArrayLike	required
three_point	bool	False

Output

tuple[float, float]

fit_klinkenberg

petrolib.flow_transport

```
fit_klinkenberg(p_mean, k_app)
```

Purpose

Fit (k_{inf} , b) from the Klinkenberg plot of k_{app} vs $1/P$.

$k_{\text{app}} = k_{\text{inf}} + (k_{\text{inf}}*b)*(1/P)$ — the intercept is k_{inf} and the slope is $k_{\text{inf}}*b$, so $b = \text{slope}/\text{intercept}$.

Input parameters

Parameter	Type	Default
p_mean	ArrayLike	required
k_app	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Reconsidering Klinkenberg's Permeability Data. Ruth, Arabjamaloei (2019). DOI: 10.30632/PJV60N3-2019a5. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 3: Low-Permeability Measurement on Crushed Rock: Insights. Profice, Lenormand (2020). DOI: 10.30632/PJV61N2-2020a3. Petrophysics Vol. 61 No. 2 (Apr 2020).

fit_line

petrolib.inversion_numerics.fitting

```
fit_line(x, y, *, xform=None, yform=None)
```

Purpose

Degree-1 least-squares fit $y = \text{slope} \cdot x + \text{intercept}$ with R^2 .

`xform` / `yform` optionally transform the axes first (e.g. 'log10' for a log-log or semilog fit); slope/intercept/ R^2 are reported in that transformed space.

Input parameters

Parameter	Type	Default
<code>x</code>	ArrayLike	required
<code>y</code>	ArrayLike	required
<code>xform</code> (kw-only)	str None	None
<code>yform</code> (kw-only)	str None	None

Output

LineFit

Sources

- Article 3: An Evaluation of Spontaneous Imbibition of Water into Oil-Wet Carbonate Reservoir Cores Using Nanofluid. Abbas Roustaei (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 31-37. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Desorbed Canister Gas Sampling and Gas Isotopic Analysis Procedures and Practices: A Case Study of Two Coalbed Methane Wells from the Lower Saxony Basin, Germany. Russell W. Spears, Sascha Alles, Alexey Makhonin (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 38-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Multiphase Flow in Porous Rock Imaged Under Dynamic Flow Conditions with Fast X-Ray Computed Microtomography S. Berg, R. Armstrong, H. Ott, A. Georgiadis, S. A. Klapp, A. Schwing, R. Neiteler, N. Brussee, A. Makurat, L. Leu, F. Enzmann, J.-O. Schwarz, M. Wolf, F. Khan, M. Kersten, S. Irvine, M. Stampanoni (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 304-312. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Onset of Oil Mobilization and Nonwetting-Phase Cluster-Size Distribution. Berg, Armstrong, Georgiadis, Ott, Schwing, Neiteler, Brussee, Makurat, Rucker, Leu, Wolf, Khan, Enzmann, Kersten (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 15-22. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: CO2 EOR by Diffusive Mixing in Fractured Reservoirs. Eide, Ersland, Brattekas, Haugen, Graue, Ferno (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 23-31. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Automatically Quantifying Wireline and LWD Pressure-Test Quality. Proett, Musharfi, Gill, Ma, Meridji, Eyuboglu (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 101-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microresistivity Curve Extraction from Borehole Microimager Data. Roslin (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 140-146. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A Breakthrough in Accurate Downhole Fluid Sample Contamination Prediction in Real Time. Zuo, Gisolf, Dumont, Dubost, Pfeiffer, Wang, Mishra, Chen, Mullins, Biagi, Gemelli (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 251-265. DOI: none assigned (this issue predates SPWLA DOI assignment).

fit_line

petrolib.ml_stats

```
fit_line(x, y)
```

Purpose

Degree-1 least-squares fit; returns (slope, intercept) — in that order, always.

The repository's most-duplicated numerical wrapper (~45 files fit a line in some transformed space, disagreeing on return order). Transform-space fits (log-log Archie exponents, semilog Klinkenberg) are built on this in their own domains.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
y	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 1: Automatically Quantifying Wireline and LWD Pressure-Test Quality. Proett, Musharfi, Gill, Ma, Meridji, Eyuboglu (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 101-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: The Impact of Depth and Pressure Measurement Errors on the Estimation of Pressure Gradients. Bowers, Schnacke, Hermance (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 376-396. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Ultrasonic Logging of Creeping Shale. Diez, Johansen, Larsen (2022). DOI: 10.30632/PJV63N1-2022a4. Petrophysics Vol. 63 No. 1 (Feb 2022).

fit_plane

petrolib.borehole_image

```
fit_plane(points_enz)
```

Purpose

SVD plane fit through (E, N, TVD) points -> (dip_deg, dip_azimuth_deg).

The plane normal is the smallest-singular-value direction of the centred points; $\text{dip} = \arccos(|n_z|)$ and $\text{dip_azimuth} = \text{atan2}(n_E, n_N) \bmod 360$ (clockwise from North).

Input parameters

Parameter	Type	Default
points_enz	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 6: Maximizing Net Pay in Penta-Lateral Well With Advanced Proactive Geosteering and 2D Structural Analysis Using Azimuthal Resistivity Measurements. Antonov, Kushnir, Martakov, Pazos, Small, Tropin, Maraj, Itter, Nelson, Rabinovich (2021). DOI: 10.30632/PJV62N3-2021a5. Petrophysics Vol. 62 No. 3 (Jun 2021).

fit_poroperm

petrolib.flow_transport

```
fit_poroperm(phi, k_md)
```

Purpose

Fit (a, b) for poroperm_powerlaw by least squares of $\log_{10}(k)$ on $\log_{10}(\phi)$.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
k_md	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).

fit_powerlaw

petrolib.ml_stats

```
fit_powerlaw(x, y)
```

Purpose

Fit $y = \text{coefficient} * x^{\text{exponent}}$; returns (coefficient, exponent).

A log10-log10 fit_line; inputs must be positive (zeros or negatives propagate as NaN). Note: articles disagree on returning the exponent or its negation — facades apply their historical sign.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
y	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 3: Multiphase Flow in Porous Rock Imaged Under Dynamic Flow Conditions with Fast X-Ray Computed Microtomography S. Berg, R. Armstrong, H. Ott, A. Georgiadis, S. A. Klapp, A. Schwing, R. Neiteler, N. Brussee, A. Makurat, L. Leu, F. Enzmann, J.-O. Schwarz, M. Wolf, F. Khan, M. Kersten, S. Irvine, M. Stampanoni (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 304-312. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: Onset of Oil Mobilization and Nonwetting-Phase Cluster-Size Distribution. Berg, Armstrong, Georgiadis, Ott, Schwing, Neiteler, Brussee, Makurat, Rucker, Leu, Wolf, Khan, Enzmann, Kersten (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 15-22. DOI: none assigned (this issue predates SPWLA DOI assignment).

fit_powerlaw_decay

petrolib.inversion_numerics.fitting

```
fit_powerlaw_decay(x, y, exponent=None)
```

Purpose

Fit $y = a * x^{(-b)}$ -> (a, b).

With exponent given (b known) the amplitude a is recovered by a linear least-squares projection; otherwise both are fit by `curve_fit`.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
y	ArrayLike	required
exponent	float None	None

Output

tuple[float, float]

fit_pressure_gradient

petrolib.geochem_fluids.gradients

```
fit_pressure_gradient(depth_m, p)
```

Purpose

Least-squares pressure-depth gradient -> (dP/dz, P0) (slope, intercept).

Input parameters

Parameter	Type	Default
depth_m	ArrayLike	required
p	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 6: Formation Evaluation With NMR, Resistivity, and Pressure Data - A Case Study of a Carbonate Oil Field Offshore West Africa. Li, Drinkwater, Whittlesey, Condon (2021). DOI: 10.30632/PJV62N1-2021a5. Petrophysics Vol. 62 No. 1 (Feb 2021).

fit_saturation_exponent

petrolib.saturation_resistivity

```
fit_saturation_exponent(sw, ri)
```

Purpose

Fit n from a log-log resistivity-index vs saturation regression.

$\log_{10}(\text{RI}) = -n \cdot \log_{10}(\text{Sw})$.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
ri	ArrayLike	required

Output

float

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir
Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Drainage Capillary Pressure and Resistivity Index from Short-Wait Porous-Plate Experiments.
Dernaika, Wilson, Skjaeveland, Ebeltoft (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 369-376. DOI: none assigned (this issue predates SPWLA DOI assignment).

fit_sinusoid

petrolib.borehole_image

```
fit_sinusoid(azimuth_deg, z, *, mask=None)
```

Purpose

Least-squares sinusoid fit on [1, cos, sin] -> (z0, amplitude, phase_deg).

Fits $z = z_0 + c_1 \cos(\phi) + c_2 \sin(\phi)$ (ϕ from azimuth_deg), returning the offset z_0 , amplitude = $\text{hypot}(c_1, c_2)$ and phase_deg = $\text{atan2}(c_2, c_1)$. An optional boolean mask selects valid samples.

Input parameters

Parameter	Type	Default
azimuth_deg	ArrayLike	required
z	ArrayLike	required
mask (kw-only)	ArrayLike None	None

Output

tuple[float, float, float]

fit_stress_permeability

petrolib.flow_transport

```
fit_stress_permeability(ncs, k)
```

Purpose

Fit (k_0 , γ) from measured k vs net confining stress, a line in (NCS, $\ln k$): $k_0 = \exp(\text{intercept})$, $\gamma = -\text{slope}$.

Input parameters

Parameter	Type	Default
ncs	ArrayLike	required
k	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).

fit_t1

petrolib.nmr

```
fit_t1(t_ms, signal, *, model='saturation')
```

Purpose

Fit (M_0 , T_1) from a T_1 recovery curve (scipy curve_fit, lazy).

model="saturation" fits $M_0*(1-\exp(-t/T_1))$; "inversion" fits $M_0*(1-2*\exp(-t/T_1))$.

Input parameters

Parameter	Type	Default
t_ms	ArrayLike	required
signal	ArrayLike	required
model (kw-only)	str	'saturation'

Output

tuple[float, float]

Sources

- Article 5: Application of a Fast NMR T_1 Relaxation Time Measurement to Sedimentary Rock Cores. Mitchell, Valori (2019). DOI: 10.30632/PJV60N3-2019a4. Petrophysics Vol. 60 No. 3 (Jun 2019).

flow_regime

petrolib.flow_transport

```
flow_regime(kn, *, cuts=(0.001, 0.1, 10.0))
```

Purpose

Gas flow regime from the Knudsen number: "continuum" ($Kn < cuts[0]$), "slip" ($< cuts[1]$), "transition" ($< cuts[2]$), else "free-molecular".

Input parameters

Parameter	Type	Default
kn	float	required
cuts (kw-only)	tuple[float, float, float]	(0.001, 0.1, 10.0)

Output

str

Sources

- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 10: Review of Micro/Nanofluidic Insights on Fluid Transport Controls in Tight Rocks. Mehmani, Kelly, Torres-Verdin (2019). DOI: 10.30632/PJV60N6-2019a10. Petrophysics Vol. 60 No. 6 (Dec 2019).

fluid_contact

petrolib.geochem_fluids.gradients

```
fluid_contact(depth_a, p_a, depth_b, p_b)
```

Purpose

Contact depth (m) where two fitted pressure gradients intersect.

(depth_a, p_a) and (depth_b, p_b) are the point sets for the two fluid columns; each is fit to a line and the intersection depth returned.

Input parameters

Parameter	Type	Default
depth_a	ArrayLike	required
p_a	ArrayLike	required
depth_b	ArrayLike	required
p_b	ArrayLike	required

Output

float

Sources

- Article 6: Formation Evaluation With NMR, Resistivity, and Pressure Data - A Case Study of a Carbonate Oil Field Offshore West Africa. Li, Drinkwater, Whittlesey, Condon (2021). DOI: 10.30632/PJV62N1-2021a5. Petrophysics Vol. 62 No. 1 (Feb 2021).

fluid_density

petrolib.porosity_lithology

```
fluid_density(saturations, rhos)
```

Purpose

Pore-fluid density $\rho_{fl} = \sum(S_i \cdot \rho_i)$ over the last axis.

Input parameters

Parameter	Type	Default
saturations	ArrayLike	required
rhos	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Enhancing the Detectability of Deep-Sensing Borehole EM Instruments by Joint Inversion of Multiple Logs Within a Probabilistic Geosteering Workflow. Jahani, Alyaev, Ambia, Fossum, Suter, Torres-Verdín (2023). DOI: 10.30632/PJV64N1-2023a6. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 4: A Fast and Transparent Bayesian Log Interpretation Method. Spalburg (2022). DOI: 10.30632/PJV63N4-2022a4. Petrophysics Vol. 63 No. 4 (Aug 2022).

fluid_summation_porosity

petrolib.porosity_lithology

```
fluid_summation_porosity(bv_oil, bv_water, bv_gas=0.0)
```

Purpose

Fluid-summation porosity $\phi = BV_{oil} + BV_{water} + BV_{gas}$.

Input parameters

Parameter	Type	Default
bv_oil	ArrayLike	required
bv_water	ArrayLike	required
bv_gas	ArrayLike	0.0

Output

numpy.ndarray of float64

Sources

- Article 4: Crushed-Rock Analysis Workflow Based on Advanced Fluid Characterization for Improved Interpretation of Core Data. Nikitin, Durand, McMullen, Blount, Driskill, Hows (2019). DOI: 10.30632/PJV60N6-2019a4. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

formation_factor

petrolib.saturation_resistivity

```
formation_factor(phi, *, a=1.0, m=2.0)
```

Purpose

Archie formation resistivity factor $F = a / \phi^{**m}$.

NOT the Waxman-Smiths $F^* = \phi^{**m} - m^*$ — m^* is a different calibration (module docstring).

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
a (kw-only)	float	1.0
m (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).

fractional_flow

petrolib.relperm_wettability

```
fractional_flow(kr_w, kr_nw, *, mu_w, mu_nw)
```

Purpose

Buckley-Leverett fractional flow of the wetting phase (zero-safe).

$f_w = (k_{r_w}/\mu_w) / (k_{r_w}/\mu_w + k_{r_{nw}}/\mu_{nw})$; returns 0 where the total mobility is 0 (no gravity/capillary term — none of the article fw kernels include one). Takes kr *values*; use fractional_flow_curve from Sw.

Input parameters

Parameter	Type	Default
kr_w	ArrayLike	required
kr_nw	ArrayLike	required
mu_w (kw-only)	float	required
mu_nw (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969-981. DOI: 10.30632/PJV66N6-2025a4.
- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Workflow to Estimate Storage Capacity (Part 1). Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N1Y2019t1.

fractional_flow_curve

petrolib.relperm_wettability

```
fractional_flow_curve(*, swr, sor=0.0, krw_max=1.0, kro_max=1.0, nw=2.0, no=2.0, mu_w,
mu_nw, n=2000)
```

Purpose

Corey fractional-flow curve (sw , fw) over the mobile window.

Convenience wrapper: builds $sw = \text{linspace}(swr, 1-sor, n)$, evaluates the Corey pair, and returns the wetting-phase fractional flow.

Input parameters

Parameter	Type	Default
swr (kw-only)	float	required
sor (kw-only)	float	0.0
krw_max (kw-only)	float	1.0
kro_max (kw-only)	float	1.0
nw (kw-only)	float	2.0
no (kw-only)	float	2.0
mu_w (kw-only)	float	required
mu_nw (kw-only)	float	required
n (kw-only)	int	2000

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).

free_gas

petrolib.geochem_fluids.adsorption

```
free_gas(phi, sw, bg)
```

Purpose

Free-gas volume per bulk volume = $\phi \cdot (1 - S_w) / B_g$.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
sw	ArrayLike	required
bg	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 10: New Perspectives on the Effects of Gas Adsorption on Storage and Production of Natural Gas From Shale Formations. Tinni, Sondergeld, Rai (2018). DOI: 10.30632/petro_059_1_a9 (contents-only - see note). Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 2: More Accurate Quantification of Free and Adsorbed Gas in Shale Reservoirs. Ansari, Merletti, Gramin, Armitage (2019). DOI: 10.30632/PJV60N5-2019a2. Petrophysics Vol. 60 No. 5 (Oct 2019).

fzi

petrolib.flow_transport

```
fzi(k_md, phi, *, c=0.0314)
```

Purpose

Amaefule Flow Zone Indicator FZI = RQI/phi_z (um).

Input parameters

Parameter	Type	Default
k_md	ArrayLike	required
phi	ArrayLike	required
c (kw-only)	float	0.0314

Output

numpy.ndarray of float64

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 5: Integrated Petrophysical Rock Classification in the McElroy Field, West Texas, USA. Saneifar, Skalinski, Theologou, Kenter, Cuffey, Salazar-Tio (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 493-510. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Permeability Anisotropy in Carbonates via Digital Rock Petrophysics. Based on: Silva Junior et al. (2024), "Permeability Anisotropy in Brazilian Presalt Carbonates at Core Scale Using Digital Rock Petrophysics", Petrophysics, Vol. 65, No. 5, pp. 711-738.

gamma_count

petrolib.nuclear

```
gamma_count(rho, *, n0=1000000.0, mu_mass=0.06, spacing_cm=30.0)
```

Purpose

Gamma count rate vs density $N = N_0 \cdot \exp(-\mu_{\text{mass}} \cdot \rho \cdot \text{spacing})$.

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
n0 (kw-only)	float	1000000.0
mu_mass (kw-only)	float	0.06
spacing_cm (kw-only)	float	30.0

Output

numpy.ndarray of float64

Sources

- Article 6: A Method of Determining Formation Density Based on Fast-Neutron Gamma Coupled Field Theory. Zhang, Zhang, Liu, Wu, Wu, Jia, Ti, Li (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 411-425. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.

gamma_ray_index

petrolib.porosity_lithology

```
gamma_ray_index(gr, gr_clean, gr_shale, *, clip=(0.0, 1.0))
```

Purpose

Gamma-ray shale index $IGR = (GR - GR_{clean}) / (GR_{shale} - GR_{clean})$.

The clean and shale endpoints carry any GR unit (API/GAPI); the index is dimensionless and clipped to [0,1] by default.

Input parameters

Parameter	Type	Default
gr	ArrayLike	required
gr_clean	ArrayLike	required
gr_shale	ArrayLike	required
clip (kw-only)	_Clip	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 1: Improvement in Heavy-Oil Reservoir Evaluation Using Nuclear Magnetic Resonance: Long Lake and Kinosis SAGD Projects, Alberta, Canada. Cheng, Kotov, Pyke, Hanif (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 239-250. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: On Error Calculation and Use of First-Order Error Propagation as Integral Part of Petrophysical Calculation. Stalheim (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 465-478. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.

gas_density

petrolib.geochem_fluids.pvt

```
gas_density(p_pa, t_k, *, m_kg_mol=MW_METHANE, z=1.0)
```

Purpose

Real-gas density $\rho = P \cdot M / (z \cdot R \cdot T)$ (kg/m³).

Input parameters

Parameter	Type	Default
p_pa	ArrayLike	required
t_k	ArrayLike	required
m_kg_mol (kw-only)	ArrayLike	MW_METHANE
z (kw-only)	ArrayLike	1.0

Output

numpy.ndarray of float64

Sources

- Article 3: The Compressibility Factor (Z) of Shale Gas at the Core Scale. Tran, Sakhaee-Pour (2019). DOI: 10.30632/PJV60N4-2019a3. Petrophysics Vol. 60 No. 4 (Aug 2019).

gas_gravity

petrolib.geochem_fluids.pvt

gas_gravity(mw_kg_mol)

Purpose

Gas specific gravity MW_{gas}/MW_{air} (air = 0.028964 kg/mol).

Input parameters

Parameter	Type	Default
mw_kg_mol	ArrayLike	required

Output

numpy.ndarray of float64

gas_in_place

petrolib.geochem_fluids.adsorption

```
gas_in_place(area_m2, h_m, phi, sg, bg)
```

Purpose

Volumetric free gas-in-place = $A \cdot h \cdot \phi \cdot S_g / B_g$.

Input parameters

Parameter	Type	Default
area_m2	ArrayLike	required
h_m	ArrayLike	required
phi	ArrayLike	required
sg	ArrayLike	required
bg	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 7 (Technical Note): Connecting the Dots - The Relevance of Proper Depth Control in the Discovery of the Groningen Field. Fokkema, Visser (2019). DOI: 10.30632/PJV60N1Y2019a6. Petrophysics Vol. 60 No. 1 (Feb 2019).

gassmann_kdry

petrolib.acoustic_geomech

```
gassmann_kdry(*, k_sat, k_mineral, k_fluid, phi)
```

Purpose

Gassmann dry-frame bulk modulus from the saturated modulus (inverse).

$K_{dry} = (K_{sat} * (\phi * K_{min} / K_{fl} + 1 - \phi) - K_{min}) / (\phi * K_{min} / K_{fl} + K_{sat} / K_{min} - 1 - \phi)$. Keyword-only.

Input parameters

Parameter	Type	Default
k_sat (kw-only)	ArrayLike	required
k_mineral (kw-only)	ArrayLike	required
k_fluid (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

gassmann_ksat

petrolib.acoustic_geomech

```
gassmann_ksat(*, k_dry, k_mineral, k_fluid, phi)
```

Purpose

Gassmann saturated bulk modulus from the dry-frame modulus.

$$K_{\text{sat}} = K_{\text{dry}} + (1 - K_{\text{dry}}/K_{\text{min}})^2 / (\phi/K_{\text{fl}} + (1 - \phi)/K_{\text{min}} - K_{\text{dry}}/K_{\text{min}}^2).$$

Keyword-only to defeat the k_dry/k_sat argument collision.

Input parameters

Parameter	Type	Default
k_dry (kw-only)	ArrayLike	required
k_mineral (kw-only)	ArrayLike	required
k_fluid (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).
- Article 6: Rock Physics Modeling of Gas Hydrate Reservoirs Through Integrated Core and Well-Log Data in NGHP-02 Area, KG Offshore Basin, India. Kumar, Mishra, Chatterjee, Tiwari, Avadhani (2022). DOI: 10.30632/PJV63N2-2022a6. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

gaussian_loglik

petrolib.inversion_numerics.stochastic

```
gaussian_loglik(obs, pred, sigma, weights=None, log_space=False)
```

Purpose

Gaussian log-likelihood $-0.5 \cdot \sum(w \cdot ((obs - pred) / sigma)^2)$.

Input parameters

Parameter	Type	Default
obs	ArrayLike	required
pred	ArrayLike	required
sigma	ArrayLike	required
weights	ArrayLike None	None
log_space	bool	False

Output

float

gaussian_mixture_spectrum

petrolib.data_qc_io.synth

```
gaussian_mixture_spectrum(axis, centres, amps, widths, *, noise=0.0, log_axis=True,
                           rng=None)
```

Purpose

Sum-of-Gaussians spectrum (synthetic NMR T2 / pore-size distributions).

Each component contributes $\text{amp} * \exp(-0.5*((u - c)/\text{width})^2)$ where u is $\log_{10}(\text{axis})$ and c is $\log_{10}(\text{centre})$ when `log_axis=True` (widths in decades), or the raw values otherwise — the standard-deviation form of `src2026_06/a03` (`gaussian_t2_model`) and `src2022_06/article1`. `noise>0` adds Gaussian noise and clips at zero (the `src2022_06/article9` convention).

Input parameters

Parameter	Type	Default
<code>axis</code>	ArrayLike	required
<code>centres</code>	ArrayLike	required
<code>amps</code>	ArrayLike	required
<code>widths</code>	ArrayLike	required
<code>noise</code> (kw-only)	float	0.0
<code>log_axis</code> (kw-only)	bool	True
<code>rng</code> (kw-only)	<code>np.random.Generator</code> <code>int</code> <code>None</code>	None

Output

`numpy.ndarray` of `float64`

Sources

- Manuaba, I. B. G. H., Najrani, H., Cavalleri, C., Moge, A., and Chapura, M. (2026). Effect of Pore-Size Distribution on Fluid Movement in Complex Reservoirs. *Petrophysics*, 67(3), 509-524. DOI: 10.30632/PJV67N3-2026a3 (orig. SPE-227259-MS).
- Article 1: Integrated Reservoir Characterization Using Unsupervised Learning on Nuclear Magnetic Resonance (NMR) T1-T2 Logs. Jiang, Bonnie, Correa, Krueger, Kelly, Wasson (2022). DOI: 10.30632/PJV63N3-2022a1. *Petrophysics* Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.
- Article 9: NMR-Based Wettability Index for Unconventional Rocks. Dick, Veselinovic, Bonnie, Kelly (2022). DOI: 10.30632/PJV63N3-2022a9. *Petrophysics* Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

gibbs_excess

petrolib.geochem_fluids.adsorption

```
gibbs_excess(gc, rho_free, rho_ads)
```

Purpose

Gibbs excess correction $G_{\text{excess}} = G_c \cdot (1 - \rho_{\text{free}} / \rho_{\text{adsorbed}})$.

Input parameters

Parameter	Type	Default
gc	ArrayLike	required
rho_free	ArrayLike	required
rho_ads	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: More Accurate Quantification of Free and Adsorbed Gas in Shale Reservoirs. Ansari, Merletti, Gramin, Armitage (2019). DOI: 10.30632/PJV60N5-2019a2. Petrophysics Vol. 60 No. 5 (Oct 2019).

glcm

petrolib.borehole_image

```
glcm(image, *, levels=16, offset=(0, 1), symmetric=True)
```

Purpose

Normalised grey-level co-occurrence matrix.

Quantises image to levels grey levels by min-max scaling, then counts co-occurrences at offset = (drow, dcol). symmetric=True counts both directions; the result is normalised to sum to one.

Input parameters

Parameter	Type	Default
image	ArrayLike	required
levels (kw-only)	int	16
offset (kw-only)	tuple[int, int]	(0, 1)
symmetric (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 7: Integrated Multiphysics Workflow for Automatic Rock Classification and Formation Evaluation Using Multiscale Image Analysis and Conventional Well Logs. Gonzalez, Kanyan, Heidari, Lopez (2020). DOI: 10.30632/PJV61N5-2020a7. Petrophysics Vol. 61 No. 5 (Oct 2020).

glcm_features

petrolib.borehole_image

glcm_features(p)

Purpose

Haralick contrast / energy / correlation of a normalised GLCM p.

Input parameters

Parameter	Type	Default
p	ArrayLike	required

Output

dict[str, float]

Sources

- Article 7: Integrated Multiphysics Workflow for Automatic Rock Classification and Formation Evaluation Using Multiscale Image Analysis and Conventional Well Logs. Gonzalez, Kanyan, Heidari, Lopez (2020). DOI: 10.30632/PJV61N5-2020a7. Petrophysics Vol. 61 No. 5 (Oct 2020).

gr_api*petrolib.nuclear*

```
gr_api(k_pct, u_ppm, th_ppm, *, coeff=(16.0, 8.0, 4.0))
```

Purpose

Total (spectral) gamma ray API SGR = $ck \cdot K + cu \cdot U + cth \cdot Th$.

Default coeff=(16, 8, 4) (K wt%, U ppm, Th ppm); pass e.g. (16.32, 8.09, 3.93) for the alternative calibration.

Input parameters

Parameter	Type	Default
k_pct	ArrayLike	required
u_ppm	ArrayLike	required
th_ppm	ArrayLike	required
coeff (kw-only)	tuple[float, float, float]	(16.0, 8.0, 4.0)

Output

numpy.ndarray of float64

Sources

- Article 4: Spectral Gamma-Ray Measurement While Drilling. Xu, Huiszoon, Wang, Adolph, Yi, Cavin, Laughlin, Tollefsen, Jacobsen, Boyce (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 377-389. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): What is it about Shaly Sands? Shaly Sand Tutorial No. 2 of 3. Thomas (2018). DOI: 10.30632/PJV59N2-2018t1. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

gradient_descent

petrolib.inversion_numerics.optimize

```
gradient_descent(f, x0, lr=None, backtracking=False, max_iter=4000, tol=1e-10)
```

Purpose

Finite-difference gradient descent of a scalar objective f .

`backtracking=True` shrinks the step by an Armijo condition each iteration; otherwise a fixed `lr` (default $1e-2$) is used.

Input parameters

Parameter	Type	Default
<code>f</code>	Callable[[<code>_Float</code>], float]	required
<code>x0</code>	ArrayLike	required
<code>lr</code>	float None	None
<code>backtracking</code>	bool	False
<code>max_iter</code>	int	4000
<code>tol</code>	float	$1e-10$

Output

numpy.ndarray of float64

gravimetric_porosity

petrolib.porosity_lithology

```
gravimetric_porosity(m_dry, m_sat, v_bulk, *, rho_fluid=1.0)
```

Purpose

Mass-balance porosity $\phi = (m_{\text{sat}} - m_{\text{dry}}) / (\rho_{\text{fluid}} \cdot V_{\text{bulk}})$.

ρ_{fluid} and the mass/volume units must be consistent (the corpus mixes g/cc·cm³ and kg/m³·m³).

Input parameters

Parameter	Type	Default
m_dry	ArrayLike	required
m_sat	ArrayLike	required
v_bulk	ArrayLike	required
rho_fluid (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Eichmann, S.L., Bouchard, J., Ow, H., Petkie, D., Poitzsch, M.E. "THz Imaging to Map the Lateral Microporosity Distribution in Carbonate Rocks" *Petrophysics*, Vol. 64, No. 3 (June 2023), pp. 438-447 DOI: 10.30632/PJV64N3-2023a8.
- Zhao, Y. (2026). Effect of Water-Rock Interactions on Mechanical Properties and Acoustic Emission Characteristics of Sandstones. *Petrophysics*, 67(2), 280-293. DOI: 10.30632/PJV67N2-2026a3.

grid_search

petrolib.inversion_numerics.nonlinear

```
grid_search(forward, data, grids, misfit='log_l2')
```

Purpose

Brute-force grid search minimizing the misfit over the Cartesian grids.

Input parameters

Parameter	Type	Default
forward	_Forward	required
data	ArrayLike	required
grids	list[ArrayLike]	required
misfit	str	'log_l2'

Output

tuple[ndarray[float64], float]

hanai_bruggeman

petrolib.em_dielectric

```
hanai_bruggeman(eps_host, eps_incl, f_incl, *, iterations=80)
```

Purpose

Asymmetric (Hanai) Bruggeman effective permittivity by robust Newton solve.

Solves $((\epsilon_{\text{eff}} - \epsilon_{\text{i}}) / (\epsilon_{\text{h}} - \epsilon_{\text{i}})) * (\epsilon_{\text{h}} / \epsilon_{\text{eff}})^{1/3} = 1 - f$ for the effective permittivity ϵ_{eff} (inclusion fraction f). Complex-capable; no absolute-value damping.

Input parameters

Parameter	Type	Default
eps_host	ArrayLike	required
eps_incl	ArrayLike	required
f_incl	ArrayLike	required
iterations (kw-only)	int	80

Output

_Num

havriliak_negami

petrolib.em_dielectric

```
havriliak_negami(freq_hz, *, eps_inf, eps_s, tau, alpha=1.0, beta=1.0)
```

Purpose

Havriliak-Negami relaxation (alpha=beta=1 -> Debye).
$$\text{eps_inf} + (\text{eps_s} - \text{eps_inf}) / (1 + (j * \omega * \tau)^{\alpha})^{\beta}$$

Input parameters

Parameter	Type	Default
freq_hz	ArrayLike	required
eps_inf (kw-only)	float	required
eps_s (kw-only)	float	required
tau (kw-only)	float	required
alpha (kw-only)	float	1.0
beta (kw-only)	float	1.0

Output

numpy.ndarray of complex128

Sources

- Funk, J., Myers, M., Hathon, L. "NMR-Mapped Distributions of Dielectric Dispersion" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 421-437 DOI: 10.30632/PJV64N3-2023a7.

haworth_ratios

petrolib.integrity_drilling

```
haworth_ratios(c1, c2, c3, c4, c5, *, percent=True)
```

Purpose

Haworth wetness / balance / character ratios (Wh, Bh, Ch).
Wh = (C2+...+C5)/total (x100 if percent), Bh = (C1+C2)/(C3+C4+C5), Ch = (C4+C5)/C3. Any consistent gas units. Empty gas returns NaN wetness; zero denominators return +inf.

Input parameters

Parameter	Type	Default
c1	float	required
c2	float	required
c3	float	required
c4	float	required
c5	float	required
percent (kw-only)	bool	True

Output

tuple[float, float, float]

Sources

- Luo, P., Li, W., Lu, P., and Qubaisi, K. (2026). An Improved Mud Gas Ratio Method for Enhanced Fluid Identification While Drilling. Petrophysics, 67(3), 582-593. DOI: 10.30632/PJV67N3-2026a8.

height_above_fwl

petrolib.capillary_pressure

```
height_above_fwl(pc, *, delta_rho, g=G_STD)
```

Purpose

Height above the free-water level from Pc: inverse of buoyancy_pc.

Input parameters

Parameter	Type	Default
pc	ArrayLike	required
delta_rho (kw-only)	ArrayLike	required
g (kw-only)	float	G_STD

Output

numpy.ndarray of float64

Sources

- Article 8: Saturation-Height Modeling: Assessing Capillary Pressure Stress Corrections. Hulea (2018). DOI: 10.30632/PJV59N3-2018a7 (inferred - see note). Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 2 - The Path Out of the Jungle. Thomas (2018). DOI: 10.30632/PJV59N5-2018t1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 3 - The Jungle Gives Us Many Things. Murphy (2018). DOI: 10.30632/PJV59N6Y2018t1. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 8: Perched Water Contacts: Understanding Fundamental Controls. Hulea (2019). DOI: 10.30632/PJV60N3-2019a7. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Laronga, R., Borchardt, E., Hill, B., Velez, E., Klemin, D., Haddad, S., Haddad, E., Chadwick, C., Mahmoodaghdam, E., and Hamichi, F. (2023). "Integrated Formation Evaluation for Site-Specific Evaluation, Optimization, and Permitting of Carbon Storage Projects." Petrophysics, 64(5), 580-620. DOI: 10.30632/PJV64N5-2023a1.

henry_constant_co2

petrolib.geochem_fluids.solubility

```
henry_constant_co2(t_k)
```

Purpose

CO2 Henry constant (MPa/molality), Duan-Sun style vs temperature (K).

$H = 10 \cdot \exp(-9.4234 + 4.0087e-2 \cdot T - 4.5e-5 \cdot T^2)$.

Input parameters

Parameter	Type	Default
t_k	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Wang and Ehlig-Economides (2023), Petrophysics 64(6): 970-977. CO2 solubility in saline water for CO2 trapping, accounting for pressure, temperature, salinity, and an often-neglected factor (e.g., dissolved CH4 or the activity-coefficient correction). Implements a Duan-Sun-style solubility model in simplified form.

henry_solubility_ln

petrolib.geochem_fluids.solubility

```
henry_solubility_ln(p_mpa, t_k, a, b, dh_j_mol)
```

Purpose

Krichevsky-Kasarnovsky $\ln(x) = a + b \cdot \ln(P) - dH/(R \cdot T)$ (gas-in-oil mole fraction).

Input parameters

Parameter	Type	Default
p_mpa	ArrayLike	required
t_k	ArrayLike	required
a	ArrayLike	required
b	ArrayLike	required
dh_j_mol	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Experimental Investigation on the Effect of Methane Solubility in Oil-Based Mud Under Downhole Conditions. Song, Sukari, Wang, Jiang, Cai, Xu, Huang (2022). DOI: 10.30632/PJV63N2-2022a5. Petrophysics Vol. 63 No. 2 (Apr 2022).

hi_mix

petrolib.nuclear

```
hi_mix(phi, *, hi_fluid=1.0, hi_matrix=0.0)
```

Purpose

Volumetric hydrogen-index log $HI = \phi * HI_{fluid} + (1-\phi) * HI_{matrix}$.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
hi_fluid (kw-only)	float	1.0
hi_matrix (kw-only)	float	0.0

Output

numpy.ndarray of float64

Sources

- Article 6: Petrophysical Analysis of Siliceous- Ooze Sediments, More Basin, Norwegian Sea. Ahmed Awadalkarim, Morten Kanne Sorensen, Ida Lykke Fabricius (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 333-348. DOI: none assigned (this issue predates SPWLA DOI assignment).

hydrocarbon_pore_volume

petrolib.porosity_lithology

```
hydrocarbon_pore_volume(phi, sw)
```

Purpose

Hydrocarbon pore volume fraction HCPV = $\phi \cdot (1 - S_w)$.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
sw	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Characterizing Hydrocarbon Storage in Organic-Rich Mudstones by Integrating Core Measurements, Kinetic Modeling, and Pore-Scale Observations: Application to South Texas Organic-Rich Mudstones. Capsan, Sanchez-Ramirez (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 97-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Lessons Learned in Permian Core Analysis: Comparison Between Retort, GRI, and Routine Methodologies. Blount, Croft, Driskill, Tepper (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 517-527. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

hydrogen_index

petrolib.nmr

```
hydrogen_index(rho, n_protons, mol_weight, *, rho_w=1.0, n_w=2.0, m_w=18.02)
```

Purpose

Hydrogen index vs water HI = $(\rho \cdot n / M) / (\rho_w \cdot n_w / M_w)$.

The water reference is 2 protons / 18.02 g/mol at 1 g/cc, so pure water has HI = 1.

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
n_protons	ArrayLike	required
mol_weight	float	required
rho_w (kw-only)	float	1.0
n_w (kw-only)	float	2.0
m_w (kw-only)	float	18.02

Output

numpy.ndarray of float64

Sources

- Article 2: A Novel Determination of Total Gas-In-Place (TGIP) for Gas Shale From Magnetic Resonance Logs. Kausik, Kleinberg, Rylander, Lewis, Sibbit, Westacott (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 232-241. DOI: none assigned (this issue predates SPWLA DOI assignment).

hydrogen_index_chemical

petrolib.nuclear

```
hydrogen_index_chemical(rho, n_protons, mol_weight, *, rho_ref=1.0, n_ref=2.0,  
mw_ref=18.015)
```

Purpose

Hydrogen index from chemistry, normalised to fresh water.

$HI = (\rho * n_{\text{protons}} / M) / (\rho_{\text{ref}} * n_{\text{ref}} / m_{\text{w_ref}})$.

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
n_protons	ArrayLike	required
mol_weight	ArrayLike	required
rho_ref (kw-only)	float	1.0
n_ref (kw-only)	float	2.0
mw_ref (kw-only)	float	18.015

Output

numpy.ndarray of float64

hydrogen_index_fluid

petrolib.nuclear

```
hydrogen_index_fluid(fluid='water', *, rho_gas=0.2, hi_oil=0.9)
```

Purpose

Hydrogen index of a pore fluid (HI, not $HI \cdot \phi$).

water -> 1.0, gas -> rho_gas, oil -> hi_oil.

Input parameters

Parameter	Type	Default
fluid	str	'water'
rho_gas (kw-only)	float	0.2
hi_oil (kw-only)	float	0.9

Output

float

hydrostatic_pressure

petrolib.acoustic_geomech

```
hydrostatic_pressure(depth, *, rho=1000.0, g=G_STANDARD)
```

Purpose

Hydrostatic pressure $P = \rho \cdot g \cdot \text{depth}$ (SI, Pa).

ρ is the pore-fluid density (default fresh water 1000 kg/m³).

Input parameters

Parameter	Type	Default
depth	ArrayLike	required
rho (kw-only)	ArrayLike	1000.0
g (kw-only)	float	G_STANDARD

Output

numpy.ndarray of float64

hydrostatic_pressure

petrolib.integrity_drilling

```
hydrostatic_pressure(tvd_m, *, rho=1030.0)
```

Purpose

Hydrostatic pressure $\rho \cdot g \cdot \text{TVD}$ (Pa), $g = 9.80665$.

Input parameters

Parameter	Type	Default
tvd_m	ArrayLike	required
rho (kw-only)	float	1030.0

Output

numpy.ndarray of float64

hydrostatic_pressure_bar

petrolib.integrity_drilling

```
hydrostatic_pressure_bar(tvd_m, *, sg=1.0)
```

Purpose

Hydrostatic pressure in bar from TVD in m: $0.0980665 \cdot SG \cdot TVD$.

Input parameters

Parameter	Type	Default
tvd_m	ArrayLike	required
sg (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 8: The Impact of Overbalanced Drilling From Exploration/Appraisal Wells to Field Development Plan. Mohammadlou, Reppert, Del Negro, Jones (2021). DOI: 10.30632/PJV62N6-2021a8. Petrophysics Vol. 62 No. 6 (Dec 2021).

hydrostatic_pressure_psi

petrolib.integrity_drilling

```
hydrostatic_pressure_psi(tvd_ft, *, sg=1.0)
```

Purpose

Hydrostatic pressure in psi from TVD in ft: $0.433 * SG * TVD$.

Input parameters

Parameter	Type	Default
tvd_ft	ArrayLike	required
sg (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 1: Deepwater Exploration and Production in the Gulf of Mexico - Challenges and Opportunities. Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 81-87. DOI: none assigned (this issue predates SPWLA DOI assignment).

imag_permittivity_from_sigma

petrolib.em_dielectric

```
imag_permittivity_from_sigma(sigma, freq_hz)
```

Purpose

Loss permittivity from conductivity: $\epsilon'' = \sigma / (2\pi f \epsilon_0)$.

Input parameters

Parameter	Type	Default
sigma	ArrayLike	required
freq_hz	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- AlZoukani, A. M., Al-Hamad, M., and Abdallah, W. (2026). Unlocking the Effect of Pore Types on Dielectric Permittivity in Carbonates. *Petrophysics*, 67(3), 470-481. DOI: 10.30632/PJV67N3-2026a1 (Best Petrophysics Papers from MEOS GEO 2025; orig. SPE-227798-MS).

impedivity

petrolib.em_dielectric

```
impedivity(eps_r, sigma, freq_hz)
```

Purpose

Complex impedivity $1/(j*\omega*\epsilon_0*\epsilon_r + \sigma)$ (inverse admittivity).

Input parameters

Parameter	Type	Default
eps_r	ArrayLike	required
sigma	ArrayLike	required
freq_hz	ArrayLike	required

Output

numpy.ndarray of complex128

Sources

- Implements a simplified version of the high-definition OBM borehole-imager forward / inverse model from: Chen, Y.-H., Zhang, T., Bloemenkamp, R., Liang, L. (2023). "Fracture Imaging and Response Characterization of the High-Definition Oil-Based Mud Borehole Imagers Through Modeling and Inversion", Petrophysics, Vol. 64, No. 4, pp. 544-554. DOI: 10.30632/PJV64N4-2023a4.

impute_gaps

petrolib.data_qc_io.clean

```
impute_gaps(x, index=None)
```

Purpose

Fill NaN runs by linear interpolation over the valid samples.

`index` gives the sample coordinates (default `arange`); leading and trailing NaNs are held flat at the nearest valid value (`np.interp` endpoint behavior).

Input parameters

Parameter	Type	Default
<code>x</code>	ArrayLike	required
<code>index</code>	ArrayLike None	None

Output

`numpy.ndarray` of float64

Sources

- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

indonesia_sw

petrolib.saturation_resistivity

```
indonesia_sw(rt, rw, *, phi, vcl, rcl, a=1.0, m=2.0, n=2.0, clip=None)
```

Purpose

Indonesia (Poupon-Leveaux) water saturation, closed form.
Solves $1/\sqrt{Rt} = (Vcl^{**}(1-Vcl/2))/\sqrt{Rcl} + \sqrt{\phi^{**}m/(a^{**}Rw)}) * Sw^{**}(n/2)$ for Sw.

Input parameters

Parameter	Type	Default
rt	ArrayLike	required
rw	ArrayLike	required
phi (kw-only)	ArrayLike	required
vcl (kw-only)	ArrayLike	required
rcl (kw-only)	ArrayLike	required
a (kw-only)	float	1.0
m (kw-only)	float	2.0
n (kw-only)	float	2.0
clip (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

induction_number

petrolib.em_dielectric

```
induction_number(spacing_m, rho, freq_hz)
```

Purpose

Induction number L/δ (near-field vs far-field discriminator).

Input parameters

Parameter	Type	Default
spacing_m	ArrayLike	required
rho	ArrayLike	required
freq_hz	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Bower, M., Xie, H., Wang, G. L., Leveque, S., and Dolan, J. (2026). Anisotropy Sensitivity in Ultradeep Azimuthal Resistivity Technologies. *Petrophysics*, 67(3), 544-559. DOI: 10.30632/PJV67N3-2026a5.

interval_thickness

petrolib.porosity_lithology

```
interval_thickness(depth, flag)
```

Purpose

Summed thickness $\sum(|dz|)$ over samples where `flag` is true.

`dz` is the absolute midpoint gradient of depth (works for irregular sampling).

Input parameters

Parameter	Type	Default
depth	ArrayLike	required
flag	ArrayLike	required

Output

float

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).

InvResult (class)

petrolib.inversion_numerics.nonlinear

`InvResult()`

Purpose

Result of a deterministic nonlinear inversion.

Input parameters

None.

Output

InvResult instance

ista_l1

petrolib.inversion_numerics.linear

```
ista_l1(a, b, eta=0.1, nonneg=True, max_iter=500)
```

Purpose

Sparse L1 inversion by ISTA (iterative soft-thresholding).

Minimizes $0.5 * ||Ax - b||^2 + \text{eta} * ||x||_1$ with an optional non-negativity projection.

Input parameters

Parameter	Type	Default
a	ArrayLike	required
b	ArrayLike	required
eta	float	0.1
nonneg	bool	True
max_iter	int	500

Output

numpy.ndarray of float64

k_from_fzi

petrolib.flow_transport

```
k_from_fzi(phi, fzi_val, *, c=0.0314)
```

Purpose

Permeability (mD) from FZI and porosity — the inverse of fzi: $k = fzi^{**2} \cdot phi^{**3} / (c^{**2} \cdot (1 - phi)^{**2})$.

Consistent with rqi/fzi for a shared c.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
fzi_val	ArrayLike	required
c (kw-only)	float	0.0314

Output

numpy.ndarray of float64

Sources

- Article 4: Petrophysical Challenges in Giant Carbonate Tengiz Field, Republic of Kazakhstan. Skalinski, Se, Playton, Theologou, Narr, Sullivan, Mallan (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 615-647. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

kerogen_mass_fraction

petrolib.porosity_lithology

```
kerogen_mass_fraction(toc, *, k=1.2)
```

Purpose

Organic-matter mass fraction from TOC $OM = k \cdot TOC$ (carbon fraction $1/k$).

k is 1.2 (carbon 0.83) or 1.25 depending on source; explicit.

Input parameters

Parameter	Type	Default
toc	ArrayLike	required
k (kw-only)	float	1.2

Output

numpy.ndarray of float64

Sources

- Article 2: Matrix-Adjusted Shale Porosity Measured in Horizontal Wells. Craddock, Mosse, Bernhardt, Ortiz, Gonzalez Tomassini, Pirie, Saldungaray, Pomerantz (2018). DOI: 10.30632/PJV59N3-2018a1. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

kerogen_volume_from_toc

petrolib.porosity_lithology

```
kerogen_volume_from_toc(toc, rho_ref, *, rho_k=1.3, carbon_frac=0.8)
```

Purpose

Kerogen volume fraction $V_k = (TOC/carbon_frac) * rho_ref / rho_k$.

rho_ref is the bulk density (most copies) or the matrix density (TMALI form) per the source model;
TOC is a fraction (divide wt% by 100 in the facade).

Input parameters

Parameter	Type	Default
toc	ArrayLike	required
rho_ref	ArrayLike	required
rho_k (kw-only)	float	1.3
carbon_frac (kw-only)	float	0.8

Output

numpy.ndarray of float64

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 1 (Tutorial): Organic-Mudstone Petrophysics: Part 3: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N3-2019t1.
- Article 1 (Tutorial): Organic Mudstone Petrophysics, Part 2: Workflow to Estimate Storage Capacity. Newsham, Comisky, Chemali (2019). DOI: 10.30632/PJV60N2-2019t1.
- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

kirsch_hoop_stress

petrolib.acoustic_geomech

```
kirsch_hoop_stress(sH, sh, pw, theta_deg)
```

Purpose

Kirsch tangential (hoop) stress at the borehole wall.

$\sigma_{\theta} = (sH + sh) - 2 \cdot (sH - sh) \cdot \cos(2 \cdot \theta) - Pw$ with θ measured from the sH azimuth (degrees).

Input parameters

Parameter	Type	Default
sH	ArrayLike	required
sh	ArrayLike	required
pw	ArrayLike	required
theta_deg	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 9: Research of Near-Wellbore Fracture Morphology, Formation Mechanism, and Propagation Law for Different Perforation Modes During the Perforation Process. Wang, Li, Xu, Jia, Zhang (2021). DOI: 10.30632/PJV62N5-2021a9. Petrophysics Vol. 62 No. 5 (Oct 2021).

klinkenberg_apparent

petrolib.flow_transport

```
klinkenberg_apparent(k_inf, *, b, p_mean, c2=0.0)
```

Purpose

Apparent gas permeability $k_{app} = k_{inf} * (1 + b/P + c2/P**2)$.

The first-order Klinkenberg (1941) slip correction; c2 adds the second-order term (default 0). b and P share units.

Input parameters

Parameter	Type	Default
k_inf	ArrayLike	required
b (kw-only)	float	required
p_mean (kw-only)	ArrayLike	required
c2 (kw-only)	float	0.0

Output

numpy.ndarray of float64

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microstructural Investigation of Stress-Dependent Permeability in Tight-Oil Rocks. King, Sansone, Kortunov, Xu, Callen, Chhatre, Sahoo, Buono (2018). DOI: 10.30632/petro_059_1_a3. Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 6: Reconsidering Klinkenberg's Permeability Data. Ruth, Arabjamaloei (2019). DOI: 10.30632/PJV60N3-2019a5. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 3: Low-Permeability Measurement on Crushed Rock: Insights. Profice, Lenormand (2020). DOI: 10.30632/PJV61N2-2020a3. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 3: Reliable Measurement of Saturation-Dependent Relative Permeability in Tight Gas Sand Formations. Gonzalez, Tandon, Heidari, Gramin, Merle (2020). DOI: 10.30632/PJV61N3-2020a3. Petrophysics Vol. 61 No. 3 (Jun 2020).

klinkenberg_corrected

petrolib.flow_transport

```
klinkenberg_corrected(k_app, *, b, p_mean)
```

Purpose

Liquid-equivalent permeability $k_{inf} = k_{app}/(1 + b/P)$ — the explicit inverse of klinkenberg_apparent (no c2 term).

Distinct from the forward slip law: this **corrects** a measured apparent gas permeability back to the intrinsic value.

Input parameters

Parameter	Type	Default
k_app	ArrayLike	required
b (kw-only)	float	required
p_mean (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 8: The Impact of Overbalanced Drilling From Exploration/Appraisal Wells to Field Development Plan. Mohammadlou, Reppert, Del Negro, Jones (2021). DOI: 10.30632/PJV62N6-2021a8. Petrophysics Vol. 62 No. 6 (Dec 2021).

kmeans

petrolib.ml_stats

```
kmeans(x, k, *, weights=None, max_iter=100)
```

Purpose

Deterministic k-means; returns (labels, centroids).

Uses the repository's farthest-point initialization (first centre = point farthest from the data mean, then greedy max-min), Lloyd iterations, convergence on unchanged labels, and optional per-sample weights in the centroid update. Deterministic by construction — no RNG — so results are exactly reproducible; note cluster NUMBERING is an artifact of the init, not a physical ordering. The quantile init of src2015_10 stays article-local.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
k	int	required
weights (kw-only)	ArrayLike None	None
max_iter (kw-only)	int	100

Output

tuple[NDArray[np.int64], ndarray[float64]]

Sources

- Petrophysics Vol. 56 No. 5 (Oct 2015) (issue-level reference).
- Article 5: An Unsupervised Learning Algorithm to Compute Fluid Volumes From NMR T1-T2 Logs in Unconventional Reservoirs. Venkataramanan, Evirgen, Allen, Mutina, Cai, Johnson, Green, Jiang (2018). DOI: 10.30632/PJV59N5-2018a4. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 3: Automatic Detection of Anomalous Density Measurements due to Wellbore Cave-in. Sen, Ong, Kainkaryam, Sharma (2020). DOI: 10.30632/PJV61N5-2020a3. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Article 7: Integrated Multiphysics Workflow for Automatic Rock Classification and Formation Evaluation Using Multiscale Image Analysis and Conventional Well Logs. Gonzalez, Kanyan, Heidari, Lopez (2020). DOI: 10.30632/PJV61N5-2020a7. Petrophysics Vol. 61 No. 5 (Oct 2020).

knudsen_number

petrolib.flow_transport

```
knudsen_number(mfp, pore_size)
```

Purpose

Knudsen number $Kn = \lambda/L_{\text{pore}}$.

`pore_size` is the characteristic length the article uses — pore radius in some copies, diameter in others (a factor-2 choice the caller owns).

Input parameters

Parameter	Type	Default
<code>mfp</code>	ArrayLike	required
<code>pore_size</code>	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.

kozeny_carman

petrolib.flow_transport

```
kozeny_carman(phi, *, specific_surface, tau=1.0, c=5.0, grain_term=True)
```

Purpose

Kozeny-Carman permeability from porosity and specific surface area.

$k = \phi^3 / (c \tau^2 S^2 (1-\phi)^2)$ with `grain_term=True` (the classic grain-volume form), or $k = \phi^3 / (c \tau^2 S^2)$ with `grain_term=False`. S is the specific surface per unit volume (1/m); k is returned in m². The Kozeny constant c (~2-5) and tortuosity τ are explicit.

Input parameters

Parameter	Type	Default
<code>phi</code>	ArrayLike	required
<code>specific_surface</code> (kw-only)	ArrayLike	required
<code>tau</code> (kw-only)	float	1.0
<code>c</code> (kw-only)	float	5.0
<code>grain_term</code> (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 4: Finite-Volume Computations of Shale Tortuosity and Permeability From 3D Pore Networks Extracted From Scanning Electron Tomographic Images. Almasoodi, Reza (2019). DOI: 10.30632/PJV60N3-2019a3. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 2: Permeability Modeling in Clay-Rich Carbonate Reservoir. Storebo, Meireles, Fabricius (2022). DOI: 10.30632/PJV63N2-2022a2. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Article 7: Using Digital Rock Physics to Evaluate Novel Percussion Core Quality. Lakshtanov, Zapata, Saucier, Cook, Eve, Lancaster, Lane, Gettemy, Sincok, Liu, Geetan, Draper, Gill (2022). DOI: 10.30632/PJV63N6-2022a7. Petrophysics Vol. 63 No. 6 (Dec 2022) — Best Papers of the 2022 SPWLA Annual Symposium special issue.

kozeny_carman_ratio

petrolib.flow_transport

```
kozeny_carman_ratio(k0, phi, phi0, *, grain_term=True)
```

Purpose

Kozeny-Carman porosity-permeability sensitivity (ratio/update form).

$k = k0 * (\phi / \phi0)^3 * ((1 - \phi0) / (1 - \phi))^2$ with `grain_term=True`, or the simplified $k = k0 * (\phi / \phi0)^3$ with `grain_term=False`.

Input parameters

Parameter	Type	Default
k0	ArrayLike	required
phi	ArrayLike	required
phi0	float	required
grain_term (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 4: In-Situ Visualization and Characterization of Filter-Cake Deposition Using Time-Lapse Micro-CT Imaging. Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N2-2022a4. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Shafiq, M. U., Ben Mahmud, H., Khan, M., Gishkori, S. N., Wang, L., and Jamil, M. (2023). "Effect of Chelating Agents on Tight Sandstone Formation Mineralogy During Sandstone Acidizing." Petrophysics, 64(5), 796-817. DOI: 10.30632/PJV64N5-2023a11.
- Article 9: Chemically Induced Formation Damage in Shale. Wick, Taneja, Gupta, Sondergeld, Rai (2020). DOI: 10.30632/PJV61N2-2020a9. Petrophysics Vol. 61 No. 2 (Apr 2020).

lame_from_velocity

petrolib.acoustic_geomech

```
lame_from_velocity(vp, vs, rho)
```

Purpose

Lame parameters from velocities and density -> (λ , μ).

$\mu = \rho * V_s^2$; $\lambda = \rho * V_p^2 - 2 * \mu$ (Pa with SI inputs).

Input parameters

Parameter	Type	Default
vp	ArrayLike	required
vs	ArrayLike	required
rho	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 4: Nonlinear Acoustics Applications for Near-Wellbore Formation Evaluation. Skelt, TenCate, Guyer, Johnson, Larmat, Le Bas, Nihei, Vu (2021). DOI: 10.30632/PJV62N2-2021a4. Petrophysics Vol. 62 No. 2 (Apr 2021).

land_c

petrolib.relpem_wettability

```
land_c(s_i_max, s_r_max)
```

Purpose

Land trapping coefficient $C = 1/Sr_{max} - 1/Si_{max}$.

The single-endpoint form $1/Sgt_{max} - 1 (Si_{max} = 1)$ is `land_c(1.0, Sgt_max)`.

Input parameters

Parameter	Type	Default
<code>s_i_max</code>	float	required
<code>s_r_max</code>	float	required

Output

float

Sources

- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO₂-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Gao, Y., Sorop, T., Brussee, N., van der Linde, H., Coorn, A., Appel, M., Berg, S. "Advanced Digital-SCAL Measurements of Gas Trapped in Sandstone" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 368-383 DOI: 10.30632/PJV64N3-2023a4.
- SCAL Model for CCS: LET Correlations for Relative Permeability and Capillary Pressure. Ebeltoft et al., 2025, Petrophysics 66(1), 10-25. DOI:10.30632/PJV66N1-2025a1.
- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).

land_trapped

petrolib.relpem_wettability

```
land_trapped(s_i, *, C=None, s_r_max=None)
```

Purpose

Land trapped saturation $S_r = S_i / (1 + C \cdot S_i)$.

Pass the coefficient C directly, or s_r_max to use the $C = 1/S_r_max - 1$ ($S_{i_max} = 1$) reduced form.

Input parameters

Parameter	Type	Default
s_i	ArrayLike	required
C (kw-only)	float None	None
s_r_max (kw-only)	float None	None

Output

numpy.ndarray of float64

Sources

- Article 2: How Pore-Scale Attributes May Be Used to Derive Robust Drainage and Imbibition Water-Saturation Models in Complex Tight-Gas Reservoirs. Merletti, Gramin, Salunke, Hamman, Spain, Shabro, Armitage, Torres-Verdin, Salter, Dacy (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 447-464. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 66 No. 1 (Feb 2025) (issue-level reference).

langmuir

petrolib.geochem_fluids.adsorption

```
langmuir(p, v_l, p_l, *, rho_b=None)
```

Purpose

Langmuir isotherm $V = V_L * P / (P + P_L)$ (half capacity at $P = P_L$).

With ρ_b given, returns the sorbed gas *content* $\rho_b * V_L * P / (P + P_L)$.

Input parameters

Parameter	Type	Default
p	ArrayLike	required
v_l	ArrayLike	required
p_l	ArrayLike	required
rho_b (kw-only)	ArrayLike None	None

Output

numpy.ndarray of float64

Sources

- Article 10: New Perspectives on the Effects of Gas Adsorption on Storage and Production of Natural Gas From Shale Formations. Tinni, Sondergeld, Rai (2018). DOI: 10.30632/petro_059_1_a9 (contents-only - see note). Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 2: Composition of the Shales in Niutitang Formation at Huijunba Syncline and its Influence on Microscopic Pore Structure and Gas Adsorption. Fu, Xu, Tian, Qin, Yang (2019). DOI: 10.30632/PJV60N3-2019a1. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 2: More Accurate Quantification of Free and Adsorbed Gas in Shale Reservoirs. Ansari, Merletti, Gramin, Armitage (2019). DOI: 10.30632/PJV60N5-2019a2. Petrophysics Vol. 60 No. 5 (Oct 2019).
- CO₂ Uptake Capacity Measurement in Source-Rock Shales for GCS. Petrophysics, 66(6), 982-994. DOI: 10.30632/PJV66N6-2025a5.

larmor_frequency

petrolib.nmr

```
larmor_frequency(b0_T, *, gamma=GAMMA_H)
```

Purpose

Larmor frequency $f = \gamma \cdot B_0 / (2 \cdot \pi)$ (Hz). γ in rad/(s*T).

Input parameters

Parameter	Type	Default
b0_T	ArrayLike	required
gamma (kw-only)	float	GAMMA_H

Output

numpy.ndarray of float64

Sources

- Article 3: Magnetic Resonance Core-Plug Analysis with the Three-Magnet Array Unilateral Magnet Juan C. Garcia-Naranjo, Pan Guo, Florin Marica, Guangzhi Liao, Bruce J. Balcom (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 229-239. DOI: none assigned (this issue predates SPWLA DOI assignment).

leak_rate_gas

petrolib.integrity_drilling

```
leak_rate_gas(aperture_m, r_casing_m, p_in_pa, p_out_pa, length_m, mu_pa_s)
```

Purpose

Gas microannulus leak rate (m^3/s at outlet), isothermal compressible flow.

$Q = (\pi / (16 \mu L P_2)) * \Omega * (P_1^2 - P_2^2)$; gravity neglected.

Input parameters

Parameter	Type	Default
aperture_m	float	required
r_casing_m	float	required
p_in_pa	float	required
p_out_pa	float	required
length_m	float	required
mu_pa_s	float	required

Output

float

Sources

- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331-347.

leak_rate_liquid

petrolib.integrity_drilling

```
leak_rate_liquid(aperture_m, r_casing_m, dp_pa, length_m, mu_pa_s, *, rho=1000.0,
inclination_deg=90.0)
```

Purpose

Liquid microannulus leak rate (m^3/s), gravity-corrected Hagen-Poiseuille.

$Q = (\pi / (8 \mu L)) * \Omega * (dP - \rho g L \cos(\theta))$ with θ the well inclination from horizontal (90 deg = vertical). Returns 0 when the applied pressure cannot overcome gravity (or the gap is closed).

Input parameters

Parameter	Type	Default
aperture_m	float	required
r_casing_m	float	required
dp_pa	float	required
length_m	float	required
mu_pa_s	float	required
rho (kw-only)	float	1000.0
inclination_deg (kw-only)	float	90.0

Output

float

Sources

- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331-347.

let_kr

petrolib.relperm_wettability

```
let_kr(sw, *, swr, L, E, T, kr_max=1.0, phase='wetting', clip=(0.0, 1.0))
```

Purpose

Lomeland-Ebeltoft-Thomas (LET) relative permeability.

phase="wetting": $kr = kr_max * Se^{**L} / (Se^{**L} + E*(1-Se)^{**T})$. phase="nonwetting": $kr = kr_max * (1-Se)^{**L} / ((1-Se)^{**L} + E*Se^{**T})$. $Se = (Sw - Swr)/(1 - Swr)$ (LET articles use the drainage denominator); the denominator is zero-guarded. Call once per phase.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
swr (kw-only)	float	required
L (kw-only)	float	required
E (kw-only)	float	required
T (kw-only)	float	required
kr_max (kw-only)	float	1.0
phase (kw-only)	str	'wetting'
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- SCAL Model for CCS: LET Correlations for Relative Permeability and Capillary Pressure. Ebeltoft et al., 2025, Petrophysics 66(1), 10-25. DOI:10.30632/PJV66N1-2025a1.
- Enhanced Gas Recovery (EGR) by CO2 Injection. Jones et al., 2025, Petrophysics 66(1), 54-66. DOI:10.30632/PJV66N1-2025a4.

levenberg_marquardt

petrolib.inversion_numerics.nonlinear

```
levenberg_marquardt(forward, data, m0, *, bounds=None, log_params=False, lam0=0.001,
lam_up=5.0, lam_dn=0.5, damping='diag', max_iter=50, tol=1e-08)
```

Purpose

Levenberg-Marquardt least-squares inversion of $\text{forward}(m) \sim \text{data}$.

`bounds=(lo, hi)` box-constrains the (physical) model; `log_params=True` inverts for $\log(m)$; `damping` is 'diag' (Marquardt) or 'identity'. Returns an `InvResult`.

Input parameters

Parameter	Type	Default
<code>forward</code>	<code>_Forward</code>	required
<code>data</code>	<code>ArrayLike</code>	required
<code>m0</code>	<code>ArrayLike</code>	required
<code>bounds</code> (kw-only)	<code>tuple[ArrayLike, ArrayLike] None</code>	None
<code>log_params</code> (kw-only)	<code>bool</code>	False
<code>lam0</code> (kw-only)	<code>float</code>	0.001
<code>lam_up</code> (kw-only)	<code>float</code>	5.0
<code>lam_dn</code> (kw-only)	<code>float</code>	0.5
<code>damping</code> (kw-only)	<code>str</code>	'diag'
<code>max_iter</code> (kw-only)	<code>int</code>	50
<code>tol</code> (kw-only)	<code>float</code>	1e-08

Output

`InvResult`

leverett_j

petrolib.capillary_pressure

```
leverett_j(pc, *, sigma, theta_deg=0.0, k, phi, absolute=True)
```

Purpose

Leverett J-function $J = Pc / (\sigma \cos(\theta)) * \sqrt{k/\phi}$.
Use consistent units (Pc and sigma as pressure and force/length, k as an area, e.g. m2); J is dimensionless. NOT src2024_12's empirical leverett_j_function(Sw) correlation, and NOT src2017_02's pc_scaling (which drops sigma*cos) — those stay article-local.

Input parameters

Parameter	Type	Default
pc	ArrayLike	required
sigma (kw-only)	ArrayLike	required
theta_deg (kw-only)	ArrayLike	0.0
k (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required
absolute (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Petrophysics Vol. 65 No. 6 (Dec 2024) (issue-level reference).
- Petrophysics Vol. 58 No. 1 (Feb 2017) (issue-level reference).
- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 63 No. 5 (Oct 2022) (issue-level reference).
- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO2-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).

LineFit (class)

petrolib.inversion_numerics.fitting

LineFit()

Purpose

Result of fit_line (in the transformed space).

Input parameters

None.

Output

LineFit instance

lm_enrml

petrolib.inversion_numerics.stochastic

```
lm_enrml(prior_mean, prior_cov, obs, obs_cov, forward, n_ens=80, n_iter=12, lam0=1.0,
         lam_up=4.0, lam_dn=0.4, seed=0)
```

Purpose

Levenberg-Marquardt ensemble randomized maximum likelihood (LM-EnRML).

Samples a Gaussian prior ensemble, forms the empirical sensitivity by SVD, and applies damped Kalman-type updates with an adaptive λ schedule. Returns the final ensemble (n_{ens} , n_{param}).

Input parameters

Parameter	Type	Default
prior_mean	ArrayLike	required
prior_cov	ArrayLike	required
obs	ArrayLike	required
obs_cov	ArrayLike	required
forward	Callable[[_Float], Any]	required
n_ens	int	80
n_iter	int	12
lam0	float	1.0
lam_up	float	4.0
lam_dn	float	0.4
seed	int	0

Output

numpy.ndarray of float64

local_shifts

petrolib.depth_matching

```
local_shifts(ref, target, *, window=60, step=30, max_lag=20)
```

Purpose

Windowed non-wrapping lag profile of target relative to ref.

Over sliding windows (length window, stride step) picks the lag that maximises the mean-removed full cross-correlation, clipped to [-max_lag, max_lag].

Input parameters

Parameter	Type	Default
ref	ArrayLike	required
target	ArrayLike	required
window (kw-only)	int	60
step (kw-only)	int	30
max_lag (kw-only)	int	20

Output

numpy.ndarray of float64

Sources

- Article 10: Machine-Learning-Based Automatic Well-Log Depth Matching. Zimmermann, Liang, Zeroug (2018). DOI: 10.30632/PJV59N6Y2018a9. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

log_response

petrolib.porosity_lithology

```
log_response(volumes, endpoints)
```

Purpose

Linear tool-response mixing $M = \sum(V_j * R_j) = \text{endpoints} @ \text{volumes}$.

Used for matrix GR, PEF, Sigma, and forward multi-mineral log synthesis.

Input parameters

Parameter	Type	Default
volumes	ArrayLike	required
endpoints	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Using Advanced Logging Measurements to Develop a Robust Petrophysical Model for the Bakken Petroleum System. Simpson, Hohman, Pirie, Horkowitz (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 457-478. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Petrophysical Interpretation of LWD, Neutron-Induced Gamma-Ray Spectroscopy Measurements: An Inversion-Based Approach. Ajayi, Torres-Verdin, Preeg (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 358-378. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", Petrophysics, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.

log_suite

petrolib.data_qc_io.synth

```
log_suite(n=600, n_facies=4, *, rng=None)
```

Purpose

Five-log synthetic well (GR, RD, RHOB, DT, NPHI) with facies blocks.

Returns (logs, boundaries, facies_labels): a dict of the LOG_SUITE_CURVES, the interior breakpoint indices, and the per-sample facies index. Each block draws each curve from $N(\mu, 0.05*|\mu| + 1e-3)$ with μ from LOG_SUITE_FACIES_PROPS; breakpoints are drawn from `arange(40, n-40)` without replacement.

Input parameters

Parameter	Type	Default
n	int	600
n_facies	int	4
rng (kw-only)	np.random.Generator int None	None

Output

tuple[dict[str, ndarray[float64]], ndarray[int], ndarray[int]]

Sources

- Article 5: Log Facies Analysis and Reservoir Properties of Basement Granitic Rocks in the Pearl River Mouth Basin, South China Sea. Hua, Yang, Xu, Lei, Zhong (2022). DOI: 10.30632/PJV63N4-2022a5. Petrophysics Vol. 63 No. 4 (Aug 2022).

loss_tangent

petrolib.em_dielectric

```
loss_tangent(eps_star)
```

Purpose

Loss tangent $\tan(\delta) = -\text{Im}(\epsilon^*)/\text{Re}(\epsilon^*)$ (-j convention).

Input parameters

Parameter	Type	Default
eps_star	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- AlZoukani, A. M., Al-Hamad, M., and Abdallah, W. (2026). Unlocking the Effect of Pore Types on Dielectric Permittivity in Carbonates. *Petrophysics*, 67(3), 470-481. DOI: 10.30632/PJV67N3-2026a1 (Best Petrophysics Papers from MEOS GEO 2025; orig. SPE-227798-MS).

lucas_washburn_length

petrolib.capillary_pressure

```
lucas_washburn_length(t, *, sigma, radius, theta_deg=0.0, mu)
```

Purpose

Lucas-Washburn imbibition front $L(t) = \sqrt{\sigma r \cos(\theta) t / (2\mu)}$.

Signed cos: an oil-wet system ($\cos < 0$) yields NaN rather than a fabricated zero — src2020_04's clip-to-zero behavior lives in its facade.

Input parameters

Parameter	Type	Default
t	ArrayLike	required
sigma (kw-only)	ArrayLike	required
radius (kw-only)	ArrayLike	required
theta_deg (kw-only)	ArrayLike	0.0
mu (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Petrophysics Vol. 61 No. 2 (Apr 2020) (issue-level reference).
- Article 8: Spontaneous Gas-Water Imbibition in Mixed-Wet Pores. Wang, He, Xiao, Wang, Ma (2020). DOI: 10.30632/PJV61N2-2020a8. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Wettability Effect on Adsorption and Capillary Condensation in Nanopores. Petrophysics, 66(6), 1061-1071. DOI: 10.30632/PJV66N6-2025a10.

lucia_permeability

petrolib.flow_transport

```
lucia_permeability(phi_g, rfn)
```

Purpose

Lucia (2007) rock-fabric-number permeability transform.

$\log_{10}(k) = (9.7982 - 12.0838 \cdot \log_{10}(RFN)) + (8.6711 - 8.2965 \cdot \log_{10}(RFN)) \cdot \log_{10}(\phi_g)$ — ϕ_g interparticle porosity (fraction), RFN the rock-fabric number; k in mD.

Input parameters

Parameter	Type	Default
phi_g	ArrayLike	required
rfn	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

lucia_rfn_from_swi

petrolib.flow_transport

```
lucia_rfn_from_swi(phi_g, swi)
```

Purpose

Lucia rock-fabric number from interparticle porosity and irreducible water saturation: $\log(RFN) = (3.1107 + 1.8834 \cdot \log(\phi_g) + \log(Swi)) / (3.0634 + 1.4045 \cdot \log(\phi_g))$.

Input parameters

Parameter	Type	Default
phi_g	ArrayLike	required
swi	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

macroscopic_sigma

petrolib.nuclear

```
macroscopic_sigma(number_densities, micro_barns, *, units='cu')
```

Purpose

Macroscopic capture Sigma from number densities and micro cross-sections.

Sigma = $\sum_i N_i * \sigma_i$ with sigma in barns (1 barn = $1e-24$ cm²); units='cu' returns c.u. (cm⁻¹ * 1e3), units='cm' returns cm⁻¹.

Input parameters

Parameter	Type	Default
number_densities	ArrayLike	required
micro_barns	ArrayLike	required
units (kw-only)	str	'cu'

Output

numpy.ndarray of float64

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

mae

petrolib.ml_stats

```
mae(y_true, y_pred)
```

Purpose

Mean absolute error.

Input parameters

Parameter	Type	Default
y_true	ArrayLike	required
y_pred	ArrayLike	required

Output

float

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 4: Borehole Resistivity Measurement Modeling Using Machine-Learning Techniques. Xu, Sun, Xie, Zhong, Mirto, Feng, Hong (2018). DOI: 10.30632/PJV59N6Y2018a3. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

mala

petrolib.inversion_numerics.stochastic

```
mala(log_post, x0, step, n_samples, seed=0)
```

Purpose

Metropolis-adjusted Langevin (MALA) sampler with a finite-difference drift.

Input parameters

Parameter	Type	Default
log_post	_LogPost	required
x0	ArrayLike	required
step	float	required
n_samples	int	required
seed	int	0

Output

Chain

mannie2_c66

petrolib.acoustic_geomech

```
mannie2_c66(c11, c33, c44, *, k=0.93)
```

Purpose

Modified-ANNIE-2 shear closure $C_{66} = C_{44} * (1 + k * (C_{11} - C_{33}) / C_{33})$.

k is the gamma/epsilon anisotropy ratio (corpus default 0.93).

Input parameters

Parameter	Type	Default
c11	ArrayLike	required
c33	ArrayLike	required
c44	ArrayLike	required
k (kw-only)	float	0.93

Output

numpy.ndarray of float64

mannie_c11

petrolib.acoustic_geomech

```
mannie_c11(c33, c44, c66, *, kp=1.0)
```

Purpose

Modified-ANNIE C11 = $kp \cdot 2 \cdot (C66 - C44) + C33$ (kp=1 recovers ANNIE).

Input parameters

Parameter	Type	Default
c33	ArrayLike	required
c44	ArrayLike	required
c66	ArrayLike	required
kp (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

mannie_c13

petrolib.acoustic_geomech

```
mannie_c13(c11, c66, *, k=1.0)
```

Purpose

Modified-ANNIE C13 = $k \cdot (C11 - 2 \cdot C66)$ ($k=1$ recovers ANNIE).

Input parameters

Parameter	Type	Default
c11	ArrayLike	required
c66	ArrayLike	required
k (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 1: Shale Fracturing Characterization and Optimization by Using Anisotropic Acoustic Interpretation, 3D Fracture Modeling, and Supervised Machine Learning. Gu, Gokaraju, Chen, Quirein (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 573-587. DOI: none assigned (this issue predates SPWLA DOI assignment).

map_estimate

petrolib.inversion_numerics.linear

```
map_estimate(g, d, noise_var, prior_strength, ell=None, m_prior=None)
```

Purpose

Bayesian MAP estimate and posterior covariance for a linear-Gaussian model.

$A = G^T G / \text{noise_var} + \text{prior_strength} * L^T L$; the MAP model is $A^{-1} (G^T d / \text{noise_var} + \text{prior_strength} * L^T L \text{ m_prior})$ with posterior covariance A^{-1} . Returns (x, posterior_cov).

Input parameters

Parameter	Type	Default
g	ArrayLike	required
d	ArrayLike	required
noise_var	float	required
prior_strength	float	required
ell	ArrayLike None	None
m_prior	ArrayLike None	None

Output

tuple[ndarray[float64], ndarray[float64]]

mape

petrolib.ml_stats

```
mape(y_true, y_pred, *, as_percent=True)
```

Purpose

Mean absolute percentage error (the articles' AAPE).

Division by `y_true` is deliberately unguarded — a zero observation yields `inf/NaN` rather than a silently biased score (CONVENTIONS.md rule 6).

Input parameters

Parameter	Type	Default
<code>y_true</code>	ArrayLike	required
<code>y_pred</code>	ArrayLike	required
<code>as_percent</code> (kw-only)	bool	True

Output

float

Sources

- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).

match_moments

petrolib.data_qc_io.scale

```
match_moments(x, target_mean, target_std)
```

Purpose

Affine transform of x to the target mean and (population) std.

$$\text{out} = (x - \text{mean}(x)) / \text{std}(x) * \text{target_std} + \text{target_mean};$$
 a constant input divides by zero rather than guessing a scale.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
target_mean	float	required
target_std	float	required

Output

numpy.ndarray of float64

Sources

- Article 6 (Technical Note): Normalizing Gamma-Ray Logs Acquired from a Mixture of Vertical and Horizontal Wells in the Haynesville Shale. Xu, Bayer, Wunderle, Bansal (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 638-643. DOI: none assigned (this issue predates SPWLA DOI assignment).

matrix_density_from_masses

petrolib.porosity_lithology

```
matrix_density_from_masses(w, rho, *, w_kerogen=0.0, rho_kerogen=1.43)
```

Purpose

Grain density by mass-weighted (harmonic) mixing.

$1/\rho_{ma} = \sum(w_i/\rho_i) + w_{kerogen}/\rho_{kerogen}$ — the reciprocal-mass average with an optional organic term.

Input parameters

Parameter	Type	Default
w	ArrayLike	required
rho	ArrayLike	required
w_kerogen (kw-only)	float	0.0
rho_kerogen (kw-only)	float	1.43

Output

numpy.ndarray of float64

Sources

- Article 2: Matrix-Adjusted Shale Porosity Measured in Horizontal Wells. Craddock, Mosse, Bernhardt, Ortiz, Gonzalez Tomassini, Pirie, Saldungaray, Pomerantz (2018). DOI: 10.30632/PJV59N3-2018a1. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 3: A New Approach to Measuring Organic Density. Dang, Sondergeld, Rai (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 112-120. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

matrix_density_from_volumes

petrolib.porosity_lithology

```
matrix_density_from_volumes(v, rho)
```

Purpose

Grain density by volume-weighted (arithmetic) mixing $\sum(v_i \cdot \rho_i)$.

v volume fractions and rho grain densities along the last axis.

Input parameters

Parameter	Type	Default
v	ArrayLike	required
rho	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Experimental Quantification of the Impact of Thermal Maturity on Kerogen Density. Jagadisan, Yang, Heidari (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 603-612. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Enhanced Mineral Quantification and Uncertainty Analysis From Downhole Spectroscopy Logs Using Variational Autoencoders. Craddock, Srivastava, Datir, Rose, Zhou, Mosse, Venkataramanan (2021). DOI: 10.30632/PJV62N6-2021a2. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

maxwell_garnett

petrolib.em_dielectric

```
maxwell_garnett(eps_host, eps_incl, f_incl, *, depol=1.0 / 3.0)
```

Purpose

Maxwell-Garnett effective permittivity with depolarization depol.
$$\text{eps_eff} = \text{eps_h} * (1 + f * \text{ds} / (\text{eps_h} + N * (1 - f) * \text{ds}))$$
 where $\text{ds} = \text{eps_i} - \text{eps_h}$ and $N = \text{depol} \cdot (1/3)$ for spheres). Complex-safe -- also valid for conductivity mixing.

Input parameters

Parameter	Type	Default
eps_host	ArrayLike	required
eps_incl	ArrayLike	required
f_incl	ArrayLike	required
depol (kw-only)	float	1.0 / 3.0

Output

_Num

Sources

- Article 4: A New Resistivity-Based Model for Improved Hydrocarbon Saturation Assessment in Clay-Rich Formations Using Quantitative Geometry of the Clay Network. Garcia, Jagadisan, Rostami, Heidari (2018). DOI: 10.30632/PJV59N3-2018a3. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Albenayyan, N., Hassan, A., El-Husseiny, A., and Mahmoud, M. (2026). A Method to Characterize the Emulsions in Oil-Water Production Wells Using Multifrequency Dielectric Technique. Petrophysics, 67(3), 594-618. DOI: 10.30632/PJV67N3-2026a9 (cover paper).

md_to_tvd

petrolib.wellbore_geometry

```
md_to_tvd(md, inc_deg, *, method='min_curvature')
```

Purpose

Cumulative true vertical depth from measured depth and inclination.

method='min_curvature' uses the balanced minimum-curvature vertical increments (azimuth-independent, so any constant azimuth is used internally); method='tangential' reproduces the legacy single- inclination form $TVD_k = \sum_{j \leq k} dMD_j \cos(inc_j)$ (each segment takes the inclination at its lower station). Returns a cumulative TVD array the same length as md, starting at $md[0] * \cos(inc[0])$ for tangential and at 0 increment for the first sample.

Input parameters

Parameter	Type	Default
md	ArrayLike	required
inc_deg	ArrayLike	required
method (kw-only)	str	'min_curvature'

Output

numpy.ndarray of float64

mean_free_path

petrolib.flow_transport

```
mean_free_path(*, pressure, temperature, mu=None, molar_mass=None, d_collision=None)
```

Purpose

Gas mean free path, in one of two physically distinct forms.

Viscosity form (pass `mu` and `molar_mass` in kg/mol): $\lambda = (\mu/P) \cdot \sqrt{\pi \cdot R \cdot T / (2 \cdot M)}$.

Kinetic-theory form (pass `d_collision`, the molecular collision diameter): $\lambda = k_B \cdot T / (\sqrt{2} \cdot \pi \cdot d^2 \cdot P)$. SI throughout.

Input parameters

Parameter	Type	Default
pressure (kw-only)	ArrayLike	required
temperature (kw-only)	float	required
mu (kw-only)	float None	None
molar_mass (kw-only)	float None	None
d_collision (kw-only)	float None	None

Output

numpy.ndarray of float64

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Incorporating Flow Regimes Into Crushed-Rock Analysis to Better Understand Matrix Permeability and Pore Structure in Shales. Royer, Hobbs, Bonar (2018). DOI: 10.30632/PJV59V4-2018a7. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 10: Review of Micro/Nanofluidic Insights on Fluid Transport Controls in Tight Rocks. Mehmani, Kelly, Torres-Verdin (2019). DOI: 10.30632/PJV60N6-2019a10. Petrophysics Vol. 60 No. 6 (Dec 2019).

median_filter2d

petrolib.data_qc_io.filt

```
median_filter2d(img, size=3)
```

Purpose

Pure-numpy 2-D median filter with edge-replicated padding.
Builds the size*size stack of shifted views of the padded image and takes the median across it.

Input parameters

Parameter	Type	Default
img	ArrayLike	required
size	int	3

Output

numpy.ndarray of float64

Sources

- Aerens, P., Torres-Verdin, C., Espinoza, N. "Experimental Time-Lapse Visualization of Mud-Filtrate Invasion and Mudcake Deposition Using X-Ray Radiography" Petrophysics, Vol. 64, No. 3 (June 2023), pp. 448-461 DOI: 10.30632/PJV64N3-2023a9.

metal_loss_pct

petrolib.integrity_drilling

```
metal_loss_pct(measured, nominal)
```

Purpose

Casing metal loss $(1 - \text{measured}/\text{nominal}) \times 100$, clipped to $[0, 100]$ %.

Input parameters

Parameter	Type	Default
measured	ArrayLike	required
nominal	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Module 3: Seven-Pipe Electromagnetic Corrosion Evaluation Implements ideas from: Fouda et al., "First-Ever Seven-Pipe Corrosion Evaluation for Comprehensive Assessment of Pipe Integrity in Complex Well Completions," Petrophysics, vol. 66, no. 4, pp. 566-577, August 2025.

metropolis

petrolib.inversion_numerics.stochastic

```
metropolis(log_post, x0, step, n_samples=4000, *, log_space=False, burn_in=0, seed=0)
```

Purpose

Random-walk Metropolis sampler of log_post.

log_space=True proposes multiplicative moves $x \cdot \exp(\text{step} \cdot N(0, 1))$; otherwise additive $x + \text{step} \cdot N(0, 1)$. Returns a Chain.

Input parameters

Parameter	Type	Default
log_post	_LogPost	required
x0	ArrayLike	required
step	ArrayLike	required
n_samples	int	4000
log_space (kw-only)	bool	False
burn_in (kw-only)	int	0
seed (kw-only)	int	0

Output

Chain

micp_apex

petrolib.flow_transport

micp_apex(shg, pc)

Purpose

Pittman (1992) MICP apex: the maximum Shg/Pc ratio and its index.

Returns (apex_ratio, index) — the apex feeds swanson_permeability.

Input parameters

Parameter	Type	Default
shg	ArrayLike	required
pc	ArrayLike	required

Output

tuple[ndarray[float64], int]

Sources

- Article 3: Stress Sensitivity of Mercury-Injection Measurements. Guise, Grattoni, Allshorn, Fisher, Schiffer (2018). DOI: 10.30632/petro_059_1_a2. Petrophysics Vol. 59 No. 1 (Feb 2018).

microannulus_omega

petrolib.integrity_drilling

microannulus_omega(r_casing_m, aperture_m)

Purpose

Annular-gap fourth moment 0mega (m^4) of a microannulus.

For the gap between R1 = r_casing_m and R2 = R1 + aperture: 0mega = R2^4 - R1^4 - (R2^2 - R1^2)^2 / ln(R2/R1); zero for a closed (non-positive) aperture.

Input parameters

Parameter	Type	Default
r_casing_m	float	required
aperture_m	float	required

Output

float

Sources

- Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification Through Microannuli. Based on: Machicote et al., "The Road Through Microannuli: Advanced Ultrasonic Log Analysis and Mechanistic Modeling for Leak Rate Quantification", Petrophysics, Vol. 66, No. 2, April 2025, pp. 331-347.

millington_quirk

petrolib.flow_transport

```
millington_quirk(D0, phi, sw=1.0)
```

Purpose

Millington-Quirk effective diffusivity $D_{eff} = (\phi * S_w)^{(10/3)} / \phi^{*2} * D_0$.
Equivalent to $D_0 * \phi^{(4/3)} * S_w^{(10/3)}$; the $(\phi * S_w)^{(10/3)} / \phi^{*2}$ structure is kept to reproduce the article float path.

Input parameters

Parameter	Type	Default
D0	float	required
phi	ArrayLike	required
sw	ArrayLike	1.0

Output

numpy.ndarray of float64

Sources

- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

min_horizontal_stress

petrolib.acoustic_geomech

```
min_horizontal_stress(sv, pp, nu, *, biot=1.0, e=0.0, eps_h=0.0, eps_H=0.0, tectonic=0.0)
```

Purpose

Minimum horizontal stress -- poroelastic uniaxial-strain superset.

$sh = \nu / (1 - \nu) * (Sv - biot * Pp) + biot * Pp + E / (1 - \nu^2) * (eps_h + \nu * eps_H) + tectonic$. With $e=eps=0$ this reduces to the uniaxial-strain-plus-constant-tectonic form; with $tectonic=0$ and the strain terms it is the Thiercelin-Plumb poroelastic form (the corpus writes both under this name and as `closure_stress`). e is Young's modulus in the same pressure unit as the stresses.

Input parameters

Parameter	Type	Default
sv	ArrayLike	required
pp	ArrayLike	required
nu	ArrayLike	required
biot (kw-only)	ArrayLike	1.0
e (kw-only)	ArrayLike	0.0
eps_h (kw-only)	ArrayLike	0.0
eps_H (kw-only)	ArrayLike	0.0
tectonic (kw-only)	ArrayLike	0.0

Output

numpy.ndarray of float64

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). *Petrophysics* Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). *Petrophysics* Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Use of Data Analytics to Optimize Hydraulic Fracture Locations Along Borehole. Gupta, Rai, Devegowda, Sondergeld (2018). DOI: 10.30632/PJV59N6Y2018a6. *Petrophysics* Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

minimum_curvature_step

petrolib.wellbore_geometry

```
minimum_curvature_step(md1, incl, azi1, md2, inc2, azi2)
```

Purpose

One minimum-curvature step, returning (dTVD, dNorth, dEast).

$\text{half} = (\text{dMD}/2) * \text{RF}$ scales the balanced-tangential increments $\text{dN} = \text{half} (\sin i1 \cos a1 + \sin i2 \cos a2)$, $\text{dE} = \text{half} (\sin i1 \sin a1 + \sin i2 \sin a2)$, $\text{dTVD} = \text{half} (\cos i1 + \cos i2)$ with RF the ratio_factor of the dogleg_angle. Angles in degrees, lengths in the units of md.

Input parameters

Parameter	Type	Default
md1	float	required
incl	float	required
azi1	float	required
md2	float	required
inc2	float	required
azi2	float	required

Output

tuple[float, float, float]

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

minmax

petrolib.ml_stats

```
minmax(x, *, axis=None, lo=0.0, hi=1.0, eps=0.0)
```

Purpose

Rescale to the range [lo, hi] (min-max normalization).

eps is the additive guard on the range denominator; default 0.0 yields NaN for a constant input.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
axis (kw-only)	int None	None
lo (kw-only)	float	0.0
hi (kw-only)	float	1.0
eps (kw-only)	float	0.0

Output

numpy.ndarray of float64

Sources

- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 9: Application of Artificial Neural Network to Predict Formation Bulk Density While Drilling. Gowida, Elkatatny, Abdulraheem (2019). DOI: 10.30632/PJV60N5-2019a9. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 5: Estimation of Reservoir Porosity From Drilling Parameters Using Artificial Neural Networks. Al-Abduljabbar, Al-Azani, Elkatatny (2020). DOI: 10.30632/PJV61N3-2020a5. Petrophysics Vol. 61 No. 3 (Jun 2020).

misfit

petrolib.inversion_numerics.costs

```
misfit(sim, obs, *, weights=None, kind='l2', log_space=False)
```

Purpose

Data misfit between a simulation and observations.

kind: * 'l2' -- weighted sum of squares $\sum(w*(sim-obs)^2)$ * 'rms' -- $\sqrt{\text{mean}(w*(sim-obs)^2)}$
 * 'rel_data'-- $\sum(w*((sim-obs)/obs)^2)$ (per-datum relative) * 'rel_norm'-- $||sim-obs|| / ||obs||$ (relative 2-norm) * 'chi2' -- $\sum(w*(sim-obs)^2)$ with $w = 1/\sigma^2$

log_space=True compares log10 of the values first.

Input parameters

Parameter	Type	Default
sim	ArrayLike	required
obs	ArrayLike	required
weights (kw-only)	ArrayLike None	None
kind (kw-only)	str	'l2'
log_space (kw-only)	bool	False

Output

float

Sources

- Article 1: Inversion-Based Interpretation of Logging-While-Drilling Resistivity and Nuclear Measurements: Field Examples in High-Angle and Horizontal Wells Olabode Ijasan, Carlos Torres-Verdin, William E. Preeg, John Rasmus, Edward J. Stockhausen (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 374-391. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).

mitra_short_time

petrolib.nmr

```
mitra_short_time(d0, t, s_over_v, *, normalized=True)
```

Purpose

Mitra short-time restricted diffusion $D(t)/D0 = 1 - (4/(9*\sqrt{\pi}))*(S/V)*\sqrt{D0*t}$.

normalized=True returns the ratio $D(t)/D0$; False returns the apparent diffusivity $D(t)$.

Input parameters

Parameter	Type	Default
d0	float	required
t	ArrayLike	required
s_over_v	float	required
normalized (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 3: Coupling Multiphase Hydrodynamic and NMR Pore-Scale Modeling for Advanced Characterization of Saturated Rocks. Evseev, Dinariev, Hurlimann, Safonov (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 32-44. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).

mix_linear

petrolib.geochem_fluids.contamination

```
mix_linear(p_v, p_f, eta)
```

Purpose

Linear mixing $P = (1 - \text{eta}) * P_{\text{virgin}} + \text{eta} * P_{\text{filtrate}}$.

Input parameters

Parameter	Type	Default
p_v	ArrayLike	required
p_f	ArrayLike	required
eta	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: A Breakthrough in Accurate Downhole Fluid Sample Contamination Prediction in Real Time. Zuo, Gisolf, Dumont, Dubost, Pfeiffer, Wang, Mishra, Chen, Mullins, Biagi, Gemelli (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 251-265. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Advances in Quantification of Miscible Contamination in Hydrocarbon and Water Samples From Downhole to Surface Laboratories. Zuo, Gisolf, Pfeiffer, Achourov, Chen, Mullins, Edmundson, Partouche (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 397-410. DOI: none assigned (this issue predates SPWLA DOI assignment).

mix_power_law

petrolib.em_dielectric

```
mix_power_law(fractions, eps_components, *, alpha=0.5)
```

Purpose

Lichtenecker-Rother power-law mixing over the leading (component) axis.

$\text{eps_mix} = (\sum_i f_i * \text{eps}_i^{\alpha})^{1/\alpha}$. $\alpha=0.5$ is the Complex Refractive Index Method (CRIM / time-average of sqrt). Complex-safe: `eps_components` may carry loss. `fractions` and `eps_components` broadcast with the component index along `axis=0`.

Input parameters

Parameter	Type	Default
<code>fractions</code>	ArrayLike	required
<code>eps_components</code>	ArrayLike	required
<code>alpha</code> (kw-only)	float	0.5

Output

`_Num`

Sources

- Article 4: Pore-Scale Evaluation of Dielectric Measurements in Formations with Complex Pore and Grain Structures. Huangye Chen and Zoya Heidari (2014). *Petrophysics* Vol. 55, No. 6 (December 2014), pp. 587-597. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Improving Dielectric Interpretation by Calibrating Matrix Permittivity and Solving Dielectric Mixing Laws With a New Graphical Method. Wang, Wang, Toumelin, Brown, Crousse (2018). DOI: 10.30632/PJV59N2-2018a5. *Petrophysics* Vol. 59 No. 2 (Apr 2018).
- Article 4: Identifying Fracture-Filling Material in Oil-Based Mud With Dielectric Borehole Imaging. Schlicht, Zhang, Luling, Graham, Cournot, Sadownyk (2021). DOI: 10.30632/PJV62N1-2021a3. *Petrophysics* Vol. 62 No. 1 (Feb 2021).

mixture_mw

petrolib.geochem_fluids.pvt

```
mixture_mw(y, mw, *, axis=-1)
```

Purpose

Mixture molecular weight $\sum(y_i * MW_i)$ over axis.

Input parameters

Parameter	Type	Default
y	ArrayLike	required
mw	ArrayLike	required
axis (kw-only)	int	-1

Output

numpy.ndarray of float64

Sources

- Article 2: A Novel Determination of Total Gas-In-Place (TGIP) for Gas Shale From Magnetic Resonance Logs. Kausik, Kleinberg, Rylander, Lewis, Sibbit, Westacott (2017). Petrophysics Vol. 58, No. 3 (June 2017), pp. 232-241. DOI: none assigned (this issue predates SPWLA DOI assignment).

moderating_power

petrolib.nuclear

```
moderating_power(a_mass, sigma_s)
```

Purpose

Moderating power $\xi \cdot \Sigma_s$ (slowing-down power).

Input parameters

Parameter	Type	Default
a_mass	ArrayLike	required
sigma_s	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Physical Basis for a Cased-Well Quantitative Gas-Saturation Analysis Method. F. Inanc, W.A. Gilchrist, R. Ansari, D. Chace (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 598-617. DOI: none assigned (this issue predates SPWLA DOI assignment).

moduli_from_velocity

petrolib.acoustic_geomech

`moduli_from_velocity(vp, vs, rho)`

Purpose

Bulk and shear modulus from P/S velocities and density -> (K, G).

$G = \rho v_s^2$; $K = \rho v_p^2 - 4/3 G$. Returns (K, G) in Pa (with SI inputs). The corpus writes the pair both (K, G) and (G, K) -- the order here is fixed as (K, G).

Input parameters

Parameter	Type	Default
vp	ArrayLike	required
vs	ArrayLike	required
rho	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 3: Characterization of Pore Structure Variation and Permeability Heterogeneity in Carbonate Rocks Using MICP and Sonic Logs: Puguang Gas Field, China. Huang, Dou, Sun (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 576-591. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

molar_volume_from_diameter

petrolib.geochem_fluids.asphaltene

```
molar_volume_from_diameter(d_m)
```

Purpose

Molar volume (m3/mol) of a sphere of diameter $d_m = (\pi/6) \cdot d^3 \cdot N_A$.

Input parameters

Parameter	Type	Default
d_m	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Fault Block Migrations Inferred from Asphaltene Gradients. Chengli Dong, Melton P. Hows, Peter M.W. Cornelisse, Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 113-123. DOI: none assigned (this issue predates SPWLA DOI assignment).

moving_stat

petrolib.data_qc_io.filt

```
moving_stat(x, window, stat='mean', *, center=True)
```

Purpose

Moving-window statistic with shrinking edge windows.

center=True uses the symmetric window $x[i - \text{window} // 2 : i + \text{window} // 2 + 1]$ (clamped at the edges); center=False uses the causal/trailing window $x[i - \text{window} + 1 : i + 1]$. stat is 'mean', 'std', 'var' (all ddof=0), 'mad' (mean absolute deviation about the window mean), or 'cv' (std/mean, 0 where the mean is 0).

Input parameters

Parameter	Type	Default
x	ArrayLike	required
window	int	required
stat	str	'mean'
center (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 3: Automatic Detection of Anomalous Density Measurements due to Wellbore Cave-in. Sen, Ong, Kainkaryam, Sharma (2020). DOI: 10.30632/PJV61N5-2020a3. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Automatic Permeability Estimation: ML Methods vs Conventional Models. Based on: Raheem et al. (2024), "Best Practices in Automatic Permeability Estimation: Machine- Learning Methods vs. Conventional Petrophysical Models", Petrophysics, Vol. 65, No. 5, pp. 789-812.
- Liu, Z., Zhang, X., Fan, Q., Zhou, L., Zhang, Y., Zhao, Z., and Zhang, Z. (2026). Intelligent Sensors and Algorithms for Diagnosing Downhole Operating Conditions of Wireline Logging Instruments. Petrophysics, 67(2), 295–317. DOI: 10.30632/PJV67N2-2026a4.

mu_from_intensity

petrolib.nuclear

```
mu_from_intensity(i0, i, x)
```

Purpose

Linear attenuation coefficient $\mu = \ln(I_0/I)/x$ from an intensity pair.

Input parameters

Parameter	Type	Default
i0	ArrayLike	required
i	ArrayLike	required
x	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 7: In-Situ Saturation Monitoring (ISSM) - Recommendations for Improved Processing. Reed, Cense (2019). DOI: 10.30632/PJV60N2-2019a5. Petrophysics Vol. 60 No. 2 (Apr 2019).

mudcake_thickness

petrolib.integrity_drilling

```
mudcake_thickness(t_s, *, k_mc_m2=0.0, dp_pa=0.0, mu_pa_s=0.001, solids_ratio=0.6,
model='dewan', rate_const=0.0, v=0.5)
```

Purpose

Mudcake thickness growth (m) in one of the corpus conventions.

model='dewan' is the closed-form filtration solution $h = \sqrt{2 k_{mc} dp_{pa} t / (\mu_{pa} S)}$ with $S = \text{solids_ratio}$ the cake solids term. model='sqrt_k' is the empirical $h = \text{rate_const} * \sqrt{t}$. model='chin_ode' integrates the Chin thickness ODE with a pressure-decaying cake permeability $K(t) = k_{mc} * ((t + 1e-3)/1e-3)^{-v}$ (10-micron seed cake, solids_ratio is the mud solid fraction fs).

Input parameters

Parameter	Type	Default
t_s	ArrayLike	required
k_mc_m2 (kw-only)	float	0.0
dp_pa (kw-only)	float	0.0
mu_pa_s (kw-only)	float	0.001
solids_ratio (kw-only)	float	0.6
model (kw-only)	str	'dewan'
rate_const (kw-only)	float	0.0
v (kw-only)	float	0.5

Output

numpy.ndarray of float64

Sources

- Article 6: Experimental Method for Time-Lapse Micro-CT Imaging of Mud-Filtrate Invasion and Mudcake Deposition. Schroeder, Torres-Verdin (2019). DOI: 10.30632/PJV60N5-2019a6. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 4: In-Situ Visualization and Characterization of Filter-Cake Deposition Using Time-Lapse Micro-CT Imaging. Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N2-2022a4. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Implements the rock/fluid/mudcake equations and the radial 1-D mud-filtrate invasion + Archie-resistivity workflow described in: Merletti, G., Al Hajri, S., Rabinovich, M., Farmer, R., Bennis, M., Torres-Verdin, C. (2023). "Assessment of True Formation Resistivity and Water Saturation in Deeply Invaded Tight-Gas Sandstones Based on the Combined Numerical Simulation of Mud-Filtrate Invasion and Resistivity Logs", Petrophysics, Vol. 64, No. 4, pp. 502-517. DOI: 10.30632/PJV64N4-2023a2.

multiexp_decay

petrolib.nmr

```
multiexp_decay(t_ms, amplitudes, t2_ms, *, noise=0.0, rng=None)
```

Purpose

Multi-exponential CPMG decay $M(t) = \sum_i A_i \exp(-t/T2_i)$.

Scalars give a mono-exponential decay. $\text{noise} > 0$ adds zero-mean Gaussian noise and requires an explicit `rng` for reproducibility.

Input parameters

Parameter	Type	Default
<code>t_ms</code>	ArrayLike	required
<code>amplitudes</code>	ArrayLike	required
<code>t2_ms</code>	ArrayLike	required
<code>noise</code> (kw-only)	float	0.0
<code>rng</code> (kw-only)	<code>np.random.Generator</code> None	None

Output

`numpy.ndarray` of float64

Sources

- Article 3: Nuclear Magnetic Resonance and Pore Coupling in Clay-Coated Sandstones With Anomalous Porosity Preservation, Agua Grande Formation, Reconcavo Basin, Brazil. Jacomo, Trindade, de Oliveira, Leite, Montrazi, Andreeta, Bonagamba (2018). DOI: 10.30632/PJV59N2-2018a2. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 1: Subsurface Fluid Characterization Using Downhole and Core NMR T1-T2 Maps Combined with Pore-Scale Imaging Techniques. Lessenger, Merkel, Medina, Ramakrishna, Chen, Balliet, Xie, Bhattad, Carnerup, Knackstedt (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 313-333. DOI: none assigned (this issue predates SPWLA DOI assignment).

multimineral_solve

petrolib.porosity_lithology

```
multimineral_solve(measured, endpoints, *, sigma=None, closure=True,
closure_weight=1000.0, nonneg=False, method='lstsq')
```

Purpose

Solve tool responses endpoints @ v = measured for mineral volumes v .

endpoints is the (n_logs, n_minerals) response matrix. With closure a unity-sum row weighted by closure_weight (1e3, or 100 in some copies) is appended so $\sum(v)=1$. sigma (per-log) applies inverse-variance weighting. method: "lstsq" (least squares), "nnls" (non-negative, lazy scipy), or "simplex" (projected-gradient onto $v \geq 0$, $\sum(v)=1$). nonneg clips a lstsq result to ≥ 0 .

Input parameters

Parameter	Type	Default
measured	ArrayLike	required
endpoints	ArrayLike	required
sigma (kw-only)	ArrayLike None	None
closure (kw-only)	bool	True
closure_weight (kw-only)	float	1000.0
nonneg (kw-only)	bool	False
method (kw-only)	str	'lstsq'

Output

numpy.ndarray of float64

Sources

- Article 2: Integrated Petrofacies Characterization and Interpretation of Depositional Environment of the Bakken Shale in the Williston Basin, North America. Bhattacharya, Carr (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 96-111. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: A Case Study about Formation Evaluation and Rock Physics Modeling of the Bazhenov Shale. Pavel Kulyapin and Tatiana F. Sokolova (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 211-218. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Integration of Nuclear Spectroscopy Technology and Core Data Results for Through-Casing TOC Measurement and Saturation Analysis - A Case Study in Najmah- Sargelu Reservoir, South Kuwait. Bouchou, Abughneej, Ghioca, Alarcon, Mendez (2020). DOI: 10.30632/PJV61N6-2020a5. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Article 2: Enhanced Mineral Quantification and Uncertainty Analysis From Downhole Spectroscopy Logs Using Variational Autoencoders. Craddock, Srivastava, Datir, Rose, Zhou, Mosse, Venkataramanan (2021). DOI: 10.30632/PJV62N6-2021a2. Petrophysics Vol. 62 No. 6 (Dec 2021).

multistart

petrolib.inversion_numerics.nonlinear

```
multistart(solver, bounds, n_starts=50, seed=0, aggregate='misfit_weighted')
```

Purpose

Run solver from many random starts and aggregate the models.

aggregate='best' returns the lowest-misfit model; 'misfit_weighted' returns a softmax(-misfit) weighted average.

Input parameters

Parameter	Type	Default
solver	Callable[[_Float], InvResult tuple[ArrayLike, float]]	required
bounds	list[tuple[float, float]]	required
n_starts	int	50
seed	int	0
aggregate	str	'misfit_weighted'

Output

numpy.ndarray of float64

nacl_meq_per_liter

petrolib.geochem_fluids.brine

```
nacl_meq_per_liter(nacl_ppm)
```

Purpose

NaCl concentration in meq/L = ppm/58.44 (monovalent, brine density ~1).

Input parameters

Parameter	Type	Default
nacl_ppm	ArrayLike	required

Output

numpy.ndarray of float64

nearest_yen_mullins

petrolib.geochem_fluids.asphaltene

```
nearest_yen_mullins(d_m, *, rtol=0.25)
```

Purpose

Nearest Yen-Mullins class for a particle diameter (m) -> (name, ref_d, agrees).

agrees is True when the diameter is within rtol (relative) of the class.

Input parameters

Parameter	Type	Default
d_m	float	required
rtol (kw-only)	float	0.25

Output

tuple[str, float, bool]

net_confining_stress

petrolib.flow_transport

```
net_confining_stress(total_stress, *, pore_pressure, biot=1.0)
```

Purpose

Terzaghi/Biot net confining stress $\sigma_{\text{eff}} = \sigma_{\text{total}} - \text{biot} \cdot P_p$.

Input parameters

Parameter	Type	Default
total_stress	ArrayLike	required
pore_pressure (kw-only)	ArrayLike	required
biot (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 11: Loading Effects on Gas Relative Permeability of a Low- Permeability Sandstone. Agostini, Egermann, Jeannin, Portier, Skoczylas, Wang (2019). DOI: 10.30632/PJV60N2-2019a9. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 4: Microstructural Investigation of Stress-Dependent Permeability in Tight-Oil Rocks. King, Sansone, Kortunov, Xu, Callen, Chhatre, Sahoo, Buono (2018). DOI: 10.30632/petro_059_1_a3. Petrophysics Vol. 59 No. 1 (Feb 2018).

net_to_gross

petrolib.porosity_lithology

```
net_to_gross(depth, net_flag, gross_flag=None)
```

Purpose

Net-to-gross ratio $\text{net_thickness} / \text{gross_thickness}$.

gross is the whole logged interval unless gross_flag (e.g. a reservoir flag) restricts it to a subset.

Input parameters

Parameter	Type	Default
depth	ArrayLike	required
net_flag	ArrayLike	required
gross_flag	ArrayLike None	None

Output

float

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 6: Untangle Shale and Gas Effects to Estimate Porosity and Net/Gross Ratio Using a Boomerang Workflow - A Case Study in Shoreface Reservoirs in Brunei. Xu, Sharif (2020). DOI: 10.30632/PJV61N1-2020a5. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).

neutron_density_porosity

petrolib.porosity_lithology

```
neutron_density_porosity(phi_n, phi_d, *, method='rms')
```

Purpose

Combine neutron and density porosity.

method="rms" gives the gas-corrected root-mean-square $\sqrt{(\phi_n^2 + \phi_d^2)/2}$; "mean" the arithmetic average $(\phi_n + \phi_d)/2$ (shale/clay corrections stay in the caller).

Input parameters

Parameter	Type	Default
phi_n	ArrayLike	required
phi_d	ArrayLike	required
method (kw-only)	str	'rms'

Output

numpy.ndarray of float64

Sources

- Article 6: Untangle Shale and Gas Effects to Estimate Porosity and Net/Gross Ratio Using a Boomerang Workflow - A Case Study in Shoreface Reservoirs in Brunei. Xu, Sharif (2020). DOI: 10.30632/PJV61N1-2020a5. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Article 4: First Hexa-Combo Logging-While-Drilling Run in Kuwait: A Case Study. Saleh, Al-Khudari, Al-Azmi, Al-Otaibi, Patnaik, Joshi, Abdulkarim, Aki, Fahri, Sanyal, Sainuddin (2023). DOI: 10.30632/PJV64N1-2023a4. Petrophysics Vol. 64 No. 1 (Feb 2023).

nmr_saturation

petrolib.nmr

```
nmr_saturation(v_fluid, v_total)
```

Purpose

NMR fluid saturation $S = V_{\text{fluid}}/V_{\text{total}}$.

Input parameters

Parameter	Type	Default
v_fluid	ArrayLike	required
v_total	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Estimating Saturations in Organic Shales Using 2D NMR. Nicot, Vorapalawut, Rousseau, Madariaga, Hamon, Korb (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 19-29. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Bulk Saturation Measurement Using ^{13}C and ^1H Magnetic Resonance. Ansaribaranghar et al., 2025, Petrophysics 66(1), 155-168. DOI:10.30632/PJV66N1-2025a11.

nmr_wettability_index

petrolib.relperm_wettability

```
nmr_wettability_index(w_signal, o_signal)
```

Purpose

NMR wettability index $I_w = (w - o)/(w + o)$.

Works on any matched water/oil pair — signal amplitudes, surface relaxation rates, or T2 peak areas.

Input parameters

Parameter	Type	Default
w_signal	ArrayLike	required
o_signal	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Water-Wet or Oil-Wet: is it Really That Simple in Shales?. Gupta, Jernigen, Curtis, Rai, Sondergeld (2018). DOI: 10.30632/PJV59N3-2018a2. Petrophysics Vol. 59 No. 3 (Jun 2018).
- Article 4: Practical Approach to Derive Wettability Index by NMR in Core Analysis Experiments. Looyestijn (2019). DOI: 10.30632/PJV60N4-2019a4. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 9: NMR-Based Wettability Index for Unconventional Rocks. Dick, Veselinovic, Bonnie, Kelly (2022). DOI: 10.30632/PJV63N3-2022a9. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

nnls_solve

petrolib.inversion_numerics.linear

```
nnls_solve(a, b, reg=0.0)
```

Purpose

Non-negative least squares (scipy), optionally ridge-augmented by reg.

Input parameters

Parameter	Type	Default
a	ArrayLike	required
b	ArrayLike	required
reg	float	0.0

Output

numpy.ndarray of float64

normalize_composition

petrolib.geochem_fluids.mudgas

```
normalize_composition(comps, *, axis=-1)
```

Purpose

Sum-to-one (zero-safe) closure of a composition along axis.

Input parameters

Parameter	Type	Default
comps	ArrayLike	required
axis (kw-only)	int	-1

Output

numpy.ndarray of float64

normalize_gas

petrolib.integrity_drilling

```
normalize_gas(total_gas, rop, flow, bit_diameter, *, mud_weight=None, units='metric',
reference=(30.0, 500.0, 10.0))
```

Purpose

Normalize a total-gas reading for drilling parameters.

units='metric' returns the drilled-rock gas index $G \cdot Q / (ROP \cdot A_{bit})$ (ROP m/hr, flow L/min, bit inches; ROP floored at $1e-6$). units='field' divides the reading by the ratio-to-reference factors $(ROP/ROP_0) \cdot (Q/Q_0) \cdot (MW/MW_0)^2$ (ft/hr, gpm, ppg; bit_diameter unused) to bring it to reference conditions.

Input parameters

Parameter	Type	Default
total_gas	ArrayLike	required
rop	ArrayLike	required
flow	ArrayLike	required
bit_diameter	ArrayLike	required
mud_weight (kw-only)	ArrayLike None	None
units (kw-only)	str	'metric'
reference (kw-only)	tuple[float, float, float]	(30.0, 500.0, 10.0)

Output

numpy.ndarray of float64

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- What Causes Mud-Logging Mud Gas Response to Vary and Two Techniques to Quantify. Based on: Donovan, W.S. (2024), "What Causes Mud-Logging Mud Gas Response to Vary and Two Techniques to Quantify Mud Gas," Petrophysics, 65(4), pp. 565-584. DOI: 10.30632/PJV65N4-2024a10.

normalize_to_reference

petrolib.data_qc_io.scale

```
normalize_to_reference(x, ref_lo, ref_hi, *, in_lo=None, in_hi=None, pct=(5.0, 95.0))
```

Purpose

Two-point (Shier 2004) normalization onto reference-well endpoints.

$$\text{out} = \text{ref_lo} + (\text{ref_hi} - \text{ref_lo}) * (\text{x} - \text{in_lo}) / (\text{in_hi} - \text{in_lo}).$$

in_lo/in_hi are the input-well endpoints; when omitted they are taken as the *pct* percentiles of *x* (the Shier 5th/95th convention), or the min/max when *pct=None* (the image-histogram convention). No clipping is applied. The affine map itself is `petrolib.ml_stats.affine_rescale` (bit-identical arrangement).

Input parameters

Parameter	Type	Default
<i>x</i>	ArrayLike	required
<i>ref_lo</i>	float	required
<i>ref_hi</i>	float	required
<i>in_lo</i> (kw-only)	float None	None
<i>in_hi</i> (kw-only)	float None	None
<i>pct</i> (kw-only)	tuple[float, float] None	(5.0, 95.0)

Output

numpy.ndarray of float64

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 6 (Technical Note): Normalizing Gamma-Ray Logs Acquired from a Mixture of Vertical and Horizontal Wells in the Haynesville Shale. Xu, Bayer, Wunderle, Bansal (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 638-643. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microresistivity Curve Extraction from Borehole Microimager Data. Roslin (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 140-146. DOI: none assigned (this issue predates SPWLA DOI assignment).

normalized_saturation

petrolib.relperm_wettability

```
normalized_saturation(s, sr, snr=0.0, *, clip=(0.0, 1.0))
```

Purpose

Normalized (effective) saturation $S_e = (S - S_r)/(1 - S_r - S_{nr})$.

$snr=0$ gives the $(1-S_r)$ drainage convention; the residual of the other phase (S_{or} , S_{nwr} , S_{org}) goes in snr . $clip=(0.0, 1.0)$ bounds S_e to the mobile window (the common Corey behavior); $clip=None$ disables it (some articles leave S_e unclipped).

Input parameters

Parameter	Type	Default
s	ArrayLike	required
sr	float	required
snr	float	0.0
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 9: Downhole Estimation of Relative Permeability With Integration of Formation-Tester Measurements and Advanced Well Logs. Hadibeik, Azari, Kalawina, Ramakrishna, Eyuboglu, Khan, Al-Rushaid, Al-Rashidi, Ahmad (2018). DOI: 10.30632/PJV59N2-2018a8 (inferred; body beyond extraction). Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Permeability Interpretation from Wireline Formation Testing Measurements with Consideration of Effective Thickness. Yang, Yang (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 251-269. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Effect of Injection Pressure on the Imbibition Relative Permeability and Capillary Pressure Curves of Shale Gas Matrix. Al-Ameri, Mazeel (2020). DOI: 10.30632/PJV61N2-2020a7. Petrophysics Vol. 61 No. 2 (Apr 2020).

number_density

petrolib.nuclear

```
number_density(rho_g_cc, wfrac, atomic_mass)
```

Purpose

Atomic number density $N = \rho \cdot N_A \cdot w / A$ (1/cm³) for a mass fraction w .

Input parameters

Parameter	Type	Default
rho_g_cc	ArrayLike	required
wfrac	ArrayLike	required
atomic_mass	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

occam

petrolib.inversion_numerics.nonlinear

```
occam(forward, data, m0, noise_level, *, reg_order=2, lam0=100.0, max_iter=50)
```

Purpose

Occam smoothest-model iteration toward a target data misfit.

Minimizes the reg_order roughness $||L m||^2$ subject to fitting data to within the noise_level (chi-square target = number of data).

Input parameters

Parameter	Type	Default
forward	_Forward	required
data	ArrayLike	required
m0	ArrayLike	required
noise_level	float	required
reg_order (kw-only)	int	2
lam0 (kw-only)	float	100.0
max_iter (kw-only)	int	50

Output

numpy.ndarray of float64

ols

petrolib.ml_stats

```
ols(X, y, *, intercept=True)
```

Purpose

Multivariate ordinary least squares; returns (coef, intercept).

The intercept is returned separately — never at beta[0] or coef[-1] — because the repository holds both conventions and a silent mix-up is a wrong prediction (module docstring).

Input parameters

Parameter	Type	Default
X	ArrayLike	required
y	ArrayLike	required
intercept (kw-only)	bool	True

Output

tuple[ndarray[float64], float]

Sources

- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 4: Borehole Resistivity Measurement Modeling Using Machine-Learning Techniques. Xu, Sun, Xie, Zhong, Mirto, Feng, Hong (2018). DOI: 10.30632/PJV59N6Y2018a3. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.

osi*petrolib.geochem_fluids.core_geochem*`osi(sl_mg_g, toc_wt_pct)`**Purpose**

Oil saturation index $OSI = 100 \cdot S1 / TOC$ (mg HC / g TOC; Jarvie producibility).

Input parameters

Parameter	Type	Default
sl_mg_g	ArrayLike	required
toc_wt_pct	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: The Reservoir Producibility Index: a Metric to Assess Reservoir Quality in Tight-Oil Plays from Logs. Reeder, Craddock, Rylander, Pirie, Lewis, Kausik, Kleinberg, Yang, Pomerantz (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 83-95. DOI: none assigned (this issue predates SPWLA DOI assignment).

otsu_threshold

petrolib.borehole_image

```
otsu_threshold(image, *, bins=256)
```

Purpose

Otsu threshold on the image's native data range, returning a bin-centre value.

Maximises the between-class variance $\sigma_b^2(t) = (\mu_T \omega - \mu)^2 / [\omega(1-\omega)]$ over the histogram of *image* (bins spanning $[\min, \max]$) and returns the value at the optimal bin centre.

Input parameters

Parameter	Type	Default
<i>image</i>	ArrayLike	required
<i>bins</i> (kw-only)	int	256

Output

float

Sources

- Article 4: Uncertainty Quantification in Image Segmentation for Image-Based Rock Physics in a Shaly Sandstone. Howard, Lin, Zhang (2019). DOI: 10.30632/PJV60N2-2019a2. Petrophysics Vol. 60 No. 2 (Apr 2019).

outlier_mask

petrolib.data_qc_io.clean

```
outlier_mask(x, method='zscore', *, threshold=3.0, k=1.5, side='both')
```

Purpose

Boolean outlier mask by z-score, IQR whiskers, or robust MAD.

'zscore': $|z| > \text{threshold}$ with population std (guarded $>1\text{e-}12$). 'iqr': outside ($Q1 - k \cdot \text{IQR}$, $Q3 + k \cdot \text{IQR}$). 'mad': further than $\text{threshold} * \text{MAD} * 1.4826$ from the median. side restricts to 'upper'/'lower' exceedances (the ferromagnetic-contaminant screen is upper-only).

Input parameters

Parameter	Type	Default
x	ArrayLike	required
method	str	'zscore'
threshold (kw-only)	float	3.0
k (kw-only)	float	1.5
side (kw-only)	str	'both'

Output

numpy.ndarray of bool

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings. Based on: Banks, J.Y., Tugwell, A.G., and Potter, D.K. (2024), "Applying Magnetic Susceptibility to Estimate Permeability From Drill Cuttings: A Case Study Constraining Uncertainty in the Culzean Triassic Reservoir," Petrophysics, 65(4), pp. 604-623. DOI: 10.30632/PJV65N4-2024a13.

overburden_pressure

petrolib.integrity_drilling

```
overburden_pressure(tvd_m, rho_bulk, *, water_depth_m=0.0, rho_sw=1030.0)
```

Purpose

Overburden pressure (Pa): seawater column plus constant-density sediment.

$P = g * (rho_sw * water_depth + rho_bulk * tvd)$ with tvd below mudline and $g = 9.80665$.

Input parameters

Parameter	Type	Default
tvd_m	ArrayLike	required
rho_bulk	ArrayLike	required
water_depth_m (kw-only)	float	0.0
rho_sw (kw-only)	float	1030.0

Output

numpy.ndarray of float64

overburden_pressure_psi

petrolib.integrity_drilling

```
overburden_pressure_psi(water_depth_ft, sediment_depth_ft, *, sw_sg=1.025,  
sediment_sg=2.3)
```

Purpose

Deepwater overburden in psi: $0.433 * (SG_{sw} * wd + SG_{sed} * sd)$.

Input parameters

Parameter	Type	Default
water_depth_ft	ArrayLike	required
sediment_depth_ft	ArrayLike	required
sw_sg (kw-only)	float	1.025
sediment_sg (kw-only)	float	2.3

Output

numpy.ndarray of float64

Sources

- Article 1: Deepwater Exploration and Production in the Gulf of Mexico - Challenges and Opportunities. Hani Elshahawi (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 81-87. DOI: none assigned (this issue predates SPWLA DOI assignment).

overburden_stress

petrolib.acoustic_geomech

```
overburden_stress(depth, rho_bulk, *, g=G_STANDARD)
```

Purpose

Overburden (vertical) stress $\sigma_v = \rho_{bulk} \cdot g \cdot depth$ (SI, Pa).
 ρ_{bulk} is the (average or per-sample) overburden bulk density and g defaults to standard gravity; the cumulative depth-column integration with a seawater layer stays in the caller (a field-unit corpus form).

Input parameters

Parameter	Type	Default
depth	ArrayLike	required
rho_bulk	ArrayLike	required
g (kw-only)	float	G_STANDARD

Output

numpy.ndarray of float64

Sources

- Desroches, J., Peyret, E., Gisolf, A., Wilcox, A., Di Giovanni, M., Schram de Jong, A., Sepehri, S., Garrard, R., and Giger, S. (2023). "Stress Measurement Campaign in Scientific Deep Boreholes: Focus on Tools and Methods." *Petrophysics*, 64(5), 621-639. DOI: 10.30632/PJV64N5-2023a2.
- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: *Petrophysics*, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

oxide_closure

petrolib.geochem_fluids.core_geochem

```
oxide_closure(oxides, *, axis=-1)
```

Purpose

Normalize an oxide (or elemental) suite to sum to one -> (closed, factor).

factor = 1/sum(oxides); closed = factor*oxides.

Input parameters

Parameter	Type	Default
oxides	ArrayLike	required
axis (kw-only)	int	-1

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

partition_t1t2_map

petrolib.nmr

```
partition_t1t2_map(t1t2, amplitudes, *, cutoff=2.0)
```

Purpose

Split a T1-T2 amplitude map into (v_hc, v_water) at a T1/T2 cutoff.

v_hc sums the amplitude where T1/T2 >= cutoff; v_water is the rest.

Input parameters

Parameter	Type	Default
t1t2	ArrayLike	required
amplitudes	ArrayLike	required
cutoff (kw-only)	float	2.0

Output

tuple[float, float]

Sources

- Article 2: Estimating Saturations in Organic Shales Using 2D NMR. Nicot, Vorapalawut, Rousseau, Madariaga, Hamon, Korb (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 19-29. DOI: none assigned (this issue predates SPWLA DOI assignment).

path_depth_shifts

petrolib.depth_matching

```
path_depth_shifts(path, depth_ref, depth_target)
```

Purpose

Per-depth correction from a DTW path: (depth_ref[i], depth_target[j]-depth_ref[i]).

Input parameters

Parameter	Type	Default
path	list[tuple[int, int]]	required
depth_ref	ArrayLike	required
depth_target	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Pan, W., Fu, L., Xu, C., Ashby, M., Lee, H., Lee, J., Meng, F., Chen, S., Ye, Y., Jiang, H., Kim, H., Kong, H., Baek, I., Baek, J., Sun, X., Sun, H., Li, S., Zhao, Z., Ke, Y., ... and Park, J. (2026). Automatic Well-Log Depth Shifting With Data-Driven Approaches: A Contest Summary. Petrophysics, 67(2), 437–462. DOI: 10.30632/PJV67N2-2026a12.

pay_flag

petrolib.porosity_lithology

```
pay_flag(phi, vsh=None, sw=None, *, phi_cut=0.08, vsh_cut=0.4, sw_cut=0.6)
```

Purpose

Boolean net-pay flag from porosity, shale, and saturation cutoffs.

$\text{phi} \geq \text{phi_cut}$ AND ($\text{vsh} \leq \text{vsh_cut}$ if vsh given) AND ($\text{sw} \leq \text{sw_cut}$ if sw given). Cutoffs vary by study (sw 0.5-0.65, phi 0.05-0.15) and are explicit.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
vsh	ArrayLike None	None
sw	ArrayLike None	None
phi_cut (kw-only)	float	0.08
vsh_cut (kw-only)	float	0.4
sw_cut (kw-only)	float	0.6

Output

NDArray[np.bool_]

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

pc_convert_system

petrolib.capillary_pressure

```
pc_convert_system(pc, *, sigma_from, theta_from_deg, sigma_to, theta_to_deg,
absolute=True)
```

Purpose

Convert P_c between fluid systems by the $\sigma \cos(\theta)$ ratio.

The lab-to-reservoir conversion (air-mercury -> gas-brine etc.); `absolute=True` (the `src2018_10/src2018_02` convention) keeps mercury conversions positive; `src2016_10` uses signed $\cos$.

Input parameters

Parameter	Type	Default
<code>pc</code>	ArrayLike	required
<code>sigma_from</code> (kw-only)	ArrayLike	required
<code>theta_from_deg</code> (kw-only)	ArrayLike	required
<code>sigma_to</code> (kw-only)	ArrayLike	required
<code>theta_to_deg</code> (kw-only)	ArrayLike	required
<code>absolute</code> (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- `src2018_10/src2018_02`
- Petrophysics Vol. 57 No. 5 (Oct 2016) (issue-level reference).
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 2 - The Path Out of the Jungle. Thomas (2018). DOI: 10.30632/PJV59N5-2018t1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 2: How Pore-Scale Attributes May Be Used to Derive Robust Drainage and Imbibition Water-Saturation Models in Complex Tight-Gas Reservoirs. Merletti, Gramin, Salunke, Hamman, Spain, Shabro, Armitage, Torres-Verdin, Salter, Dacy (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 447-464. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 59 No. 1 (Feb 2018) (issue-level reference).

pc_from_leverett_j

petrolib.capillary_pressure

```
pc_from_leverett_j(j, *, sigma, theta_deg=0.0, k, phi, absolute=True)
```

Purpose

Invert leverett_j for capillary pressure.

Input parameters

Parameter	Type	Default
j	ArrayLike	required
sigma (kw-only)	ArrayLike	required
theta_deg (kw-only)	ArrayLike	0.0
k (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required
absolute (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 1: The Impact of Reservoir Conditions and Rock Heterogeneity on CO₂-Brine Multiphase Flow in Permeable Sandstone. Krevor, Reynolds, Al-Menhali, Niu (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 12-18. DOI: none assigned (this issue predates SPWLA DOI assignment).

pca*petrolib.ml_stats*`pca(x, *, n_components=None)`**Purpose**

Principal component analysis via SVD of the centered data.

Returns (scores, components, mean, explained) where components rows are the principal axes and explained is the fraction of variance per retained component. Article variants return subsets of this tuple — facades slice it.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
n_components (kw-only)	int None	None

Output

tuple[ndarray[float64], ndarray[float64], ndarray[float64], ndarray[float64]]

Sources

- Article 2: Predicting Carbonate Rock Properties Using NMR Data and Generalized Interpolation-Based Techniques. Kwak, Hursan, Shao, Chen, Balliet, Eid, Guergueb (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 351-368. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Enhanced Mineral Quantification and Uncertainty Analysis From Downhole Spectroscopy Logs Using Variational Autoencoders. Craddock, Srivastava, Datir, Rose, Zhou, Mosse, Venkataramanan (2021). DOI: 10.30632/PJV62N6-2021a2. Petrophysics Vol. 62 No. 6 (Dec 2021).

pearson_r

petrolib.ml_stats

```
pearson_r(x, y, *, eps=0.0)
```

Purpose

Pearson correlation coefficient R (NOT R^2).

eps is the additive denominator guard; the default 0.0 returns NaN when either input is constant. Articles with a **conditional** fallback (return 0.0 if $\text{denom} \leq 1e-12$) keep that branch in their facade.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
y	ArrayLike	required
eps (kw-only)	float	0.0

Output

float

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).

permeability_average

petrolib.flow_transport

```
permeability_average(k, *, method='geometric', weights=None)
```

Purpose

Average permeability by method = "arithmetic" (parallel flow), "harmonic" (series flow), or "geometric" (default).

Optional weights give thickness/volume-weighted means.

Input parameters

Parameter	Type	Default
k	ArrayLike	required
method (kw-only)	str	'geometric'
weights (kw-only)	ArrayLike None	None

Output

float

Sources

- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Estimation of Permeability in the McMurray Formation Using High- Resolution Data Sources. Manchuk, Garner, Deutsch (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 125-139. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Permeability Anisotropy in Carbonates via Digital Rock Petrophysics. Based on: Silva Junior et al. (2024), "Permeability Anisotropy in Brazilian Presalt Carbonates at Core Scale Using Digital Rock Petrophysics", Petrophysics, Vol. 65, No. 5, pp. 711-738.

phase_mobility

petrolib.relperm_wettability

phase_mobility(kr, mu)

Purpose

Phase mobility $\lambda = k_r / \mu$.

Input parameters

Parameter	Type	Default
kr	ArrayLike	required
mu	float	required

Output

numpy.ndarray of float64

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969-981. DOI: 10.30632/PJV66N6-2025a4.

phase_saturation

petrolib.borehole_image

phase_saturation(phase, pore)

Purpose

Saturation of a phase within the pore space: |phase & pore| / |pore|.

Input parameters

Parameter	Type	Default
phase	ArrayLike	required
pore	ArrayLike	required

Output

float

Sources

- Article 2: Enhanced Learning of Fundamental Petrophysical Concepts Through Image Processing and 3D Printing. Alyafei, Al Musleh, Bautista, Idris, Seers (2021). DOI: 10.30632/PJV62N5-2021a2. Petrophysics Vol. 62 No. 5 (Oct 2021).

phase_shift_deg

petrolib.em_dielectric

```
phase_shift_deg(rho, freq_hz, spacing_m)
```

Purpose

Induction-limit phase shift over spacing_m: degrees(L/delta).

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
freq_hz	ArrayLike	required
spacing_m	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).

phi_from_hi

petrolib.nuclear

```
phi_from_hi(hi_log, *, hi_matrix, hi_fluid=1.0)
```

Purpose

Porosity from a hydrogen-index log $\phi = (HI - HI_{matrix}) / (HI_{fluid} - HI_{matrix})$.

Input parameters

Parameter	Type	Default
hi_log	ArrayLike	required
hi_matrix (kw-only)	float	required
hi_fluid (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 6: Petrophysical Analysis of Siliceous- Ooze Sediments, More Basin, Norwegian Sea. Ahmed Awadalkarim, Morten Kanne Sorensen, Ida Lykke Fabricius (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 333-348. DOI: none assigned (this issue predates SPWLA DOI assignment).

phi_from_ls

petrolib.nuclear

```
phi_from_ls(ls, e_mev, *, ls0=20.0, e_ref=4.5)
```

Purpose

Porosity from slowing-down length (inverse of slowing_down_length_empirical).

Input parameters

Parameter	Type	Default
ls	ArrayLike	required
e_mev	ArrayLike	required
ls0 (kw-only)	float	20.0
e_ref (kw-only)	float	4.5

Output

numpy.ndarray of float64

phi_hi_correction

petrolib.nuclear

```
phi_hi_correction(phi_apparent, hi)
```

Purpose

Hydrogen-index porosity correction $\phi_i = \phi_{\text{apparent}}/\text{HI}$ (neutron & NMR).

Input parameters

Parameter	Type	Default
phi_apparent	ArrayLike	required
hi	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fellah, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).

phi_n_from_lm

petrolib.nuclear

```
phi_n_from_lm(lm_star, *, lithology='limestone')
```

Purpose

Neutron porosity from migration length $\phi_i = (Lm_{ma} - Lm)/(Lm_{ma} - Lm_{fl})$.

Input parameters

Parameter	Type	Default
lm_star	ArrayLike	required
lithology (kw-only)	str	'limestone'

Output

numpy.ndarray of float64

phi_z

petrolib.flow_transport

`phi_z(phi)`

Purpose

Normalized porosity index $\text{phi_z} = \text{phi} / (1 - \text{phi})$.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 5: Integrated Petrophysical Rock Classification in the McElroy Field, West Texas, USA. Saneifar, Skalinski, Theologou, Kenter, Cuffey, Salazar-Tio (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 493-510. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Presenting a Multifaceted Approach to Unconventional Rock Typing and Technical Validation - Case Study in the Permian Basin and Impacts on Reservoir Characterization Workflows. Perry, Hayes (2019). DOI: 10.30632/PJV60N5-2019a8. Petrophysics Vol. 60 No. 5 (Oct 2019).

pixler_ratios

petrolib.geochem_fluids.mudgas

```
pixler_ratios(c1, c2, c3, c4, c5)
```

Purpose

Pixler ratios {C1/C2, C1/C3, C1/C4, C1/C5} (inf where a denominator is 0).

Input parameters

Parameter	Type	Default
c1	ArrayLike	required
c2	ArrayLike	required
c3	ArrayLike	required
c4	ArrayLike	required
c5	ArrayLike	required

Output

dict[str, ndarray[float64]]

pixler_ratios

petrolib.integrity_drilling

```
pixler_ratios(c1, c2, c3, c4, c5)
```

Purpose

Pixler component ratios $C1/C2 \dots C1/C5$ plus the Bernard ratio $C1/(C2+C3)$.

Zero denominators return +inf.

Input parameters

Parameter	Type	Default
c1	float	required
c2	float	required
c3	float	required
c4	float	required
c5	float	required

Output

dict[str, float]

Sources

- Article 1 (Tutorial): Maximizing Value From Mudlogs - Integrated Approach to Determine Net Pay. Malik, Hanson, Clinch (2021). DOI: 10.30632/PJV62N1-2021t1. Petrophysics Vol. 62 No. 1 (Feb 2021).

pnc_decay

petrolib.nuclear

```
pnc_decay(t_us, n0, sigma_cu, *, background=0.0)
```

Purpose

Pulsed-neutron capture decay $N(t) = N0 \cdot \exp(-\text{Sigma} \cdot v \cdot t) + \text{background}$.
Sigma (c.u.) is converted to $\text{cm}^{-1} \cdot 10^{-3}$, v is the thermal-neutron velocity, and t (us) to seconds $\cdot 10^{-6}$.

Input parameters

Parameter	Type	Default
t_us	ArrayLike	required
n0	float	required
sigma_cu	float	required
background (kw-only)	float	0.0

Output

numpy.ndarray of float64

Sources

- Badruzzaman, A. (2024). "Nuclear Logging in Geological Probing for a Low-Carbon Energy Future - A New Frontier?" Petrophysics 65(3), 274-301. DOI: 10.30632/PJV65N3-2024a1.

poisson_from_kg

petrolib.acoustic_geomech

poisson_from_kg(k, g)

Purpose

Poisson's ratio from bulk and shear modulus $\nu = (3K - 2G)/(2*(3K + G))$.

Input parameters

Parameter	Type	Default
k	ArrayLike	required
g	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 5: Analysis of Influencing Factors of Poisson's Ratio in Deep Shale Gas Reservoir Based on Digital Core Simulation. Liu, Wang, Lai, Wang, Zhang, Zhang, Ou (2023). DOI: 10.30632/PJV64N1-2023a5. Petrophysics Vol. 64 No. 1 (Feb 2023).

poisson_from_velocity

petrolib.acoustic_geomech

`poisson_from_velocity(vp, vs)`

Purpose

Poisson's ratio from P/S velocities $(V_p^{**2} - 2*V_s^{**2}) / (2*(V_p^{**2} - V_s^{**2}))$.

Input parameters

Parameter	Type	Default
vp	ArrayLike	required
vs	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: A Rigorous Data-Driven Approach to Predict Poisson's Ratio of Carbonate Rocks Using a Functional Network. Tariq, Abdulraheem, Mahmoud, Ahmed (2018). DOI: 10.30632/PJV59N6Y2018a2. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 6: Prediction of Sonic Wave Transit Times From Drilling Parameters While Horizontal Drilling in Carbonate Rocks Using Neural Networks. Gowida, Elkatatny (2020). DOI: 10.30632/PJV61N5-2020a6. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

pore_radius_from_t2

petrolib.nmr

```
pore_radius_from_t2(t2_ms, *, rho, shape_factor=3.0, t2_bulk=np.inf)
```

Purpose

Pore radius from a relaxation time $r = \text{shape_factor} / (S/V)$.

With $t2_bulk=\text{inf}$ this is $r = \text{shape_factor} * \rho * T2$. shape_factor is 3 for a sphere radius, 2 for a cylinder radius, 6 for a sphere diameter (explicit; the corpus mixes all three).

Input parameters

Parameter	Type	Default
t2_ms	ArrayLike	required
rho (kw-only)	float	required
shape_factor (kw-only)	float	3.0
t2_bulk (kw-only)	ArrayLike	np.inf

Output

numpy.ndarray of float64

Sources

- Article 3: A Laboratory Study of the Link Between NMR Relaxation Data and Pore Size In Carbonate Skeletal Grains and Micrite. El-Husseiny, Knight (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 116-125. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: NMR Fluid Substitution - A New Method of Reconstructing T2 Distributions Under Primary Drainage and Imbibition Conditions. Li, Kesserwan, Jin, Ma (2021). DOI: 10.30632/PJV62N4-2021a2. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).

pore_velocity

petrolib.flow_transport

```
pore_velocity(u_darcy, phi, sw=1.0)
```

Purpose

Interstitial (pore) velocity $v = u_darcy / (\phi * S_w)$.

Input parameters

Parameter	Type	Default
u_darcy	ArrayLike	required
phi	ArrayLike	required
sw	ArrayLike	1.0

Output

numpy.ndarray of float64

Sources

- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

poroperm_powerlaw

petrolib.flow_transport

```
poroperm_powerlaw(phi, *, a, b)
```

Purpose

Log-log poro-perm power law $k = 10^{(a + b \cdot \log_{10}(\phi))}$.

The $a + b \cdot \log_{10}(\phi)$ (log-log) trend; k in mD, ϕ a fraction. For the semilog $10^{(a \cdot \phi + b)}$ trend see the article facades directly.

Input parameters

Parameter	Type	Default
<code>phi</code>	ArrayLike	required
<code>a</code> (kw-only)	float	required
<code>b</code> (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Article 2: Defining Net-Pay Cutoffs in Carbonates Using Advanced Petrophysical Methods. Skalinski, Mallan, Edwards, Sun, Toumelin, Kelly, Wushur, Sullivan (2019). DOI: 10.30632/PJV60N1Y2019a1. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 1: Heterogeneous Carbonate Reservoirs: Ensuring Consistency of Subsurface Models by Maximizing the use of Saturation-Height Models and Dynamic Data. Hulea, Frese, Ramaswami (2016). Petrophysics Vol. 57, No. 3 (June 2016), pp. 223-232. DOI: none assigned (this issue predates SPWLA DOI assignment).

porosity_from_mask

petrolib.borehole_image

```
porosity_from_mask(pore)
```

Purpose

Porosity as the fraction of voxels flagged pore: `pore.sum()/pore.size`.

Input parameters

Parameter	Type	Default
pore	ArrayLike	required

Output

float

Sources

- Article 4: Uncertainty Quantification in Image Segmentation for Image-Based Rock Physics in a Shaly Sandstone. Howard, Lin, Zhang (2019). DOI: 10.30632/PJV60N2-2019a2. Petrophysics Vol. 60 No. 2 (Apr 2019).

porosity_from_volumes

petrolib.porosity_lithology

```
porosity_from_volumes(v_bulk, v_grain)
```

Purpose

Core porosity $\phi = (V_{\text{bulk}} - V_{\text{grain}})/V_{\text{bulk}}$.

Bulk volume first, grain volume second (the corpus writes both orders under `helium_porosity/porosity`; here the order is fixed).

Input parameters

Parameter	Type	Default
<code>v_bulk</code>	ArrayLike	required
<code>v_grain</code>	ArrayLike	required

Output

`numpy.ndarray` of `float64`

Sources

- Article 6: Impact of Different Cleaning Methods on Petrophysical Measurements. Gupta, Rai, Tinni, Sondergeld (2017). *Petrophysics* Vol. 58, No. 6 (December 2017), pp. 613-622. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Lessons Learned in Permian Core Analysis: Comparison Between Retort, GRI, and Routine Methodologies. Blount, Croft, Driskill, Tepper (2017). *Petrophysics* Vol. 58, No. 5 (October 2017), pp. 517-527. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Crushed-Rock Analysis Workflow Based on Advanced Fluid Characterization for Improved Interpretation of Core Data. Nikitin, Durand, McMullen, Blount, Driskill, Hows (2019). DOI: 10.30632/PJV60N6-2019a4. *Petrophysics* Vol. 60 No. 6 (Dec 2019).
- Article 4: 3D Printing Mudrocks: Experiments in Validating Clay as a Build Material for 3D Printing Porous Micromodels. Hasiuk, Harding (2021). DOI: 10.30632/PJV62N5-2021a4. *Petrophysics* Vol. 62 No. 5 (Oct 2021).

porosity_from_voxel_count

petrolib.porosity_lithology

```
porosity_from_voxel_count(pore_voxels, total_voxels)
```

Purpose

Digital-rock porosity $\phi_i = \text{pore_voxels} / \text{total_voxels}$.

The caller counts pore voxels under its own segmentation convention (some label pore==0, some pore==True).

Input parameters

Parameter	Type	Default
pore_voxels	ArrayLike	required
total_voxels	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: 3D Printing Berea Sandstone: Testing a New Tool for Petrophysical Analysis of Reservoirs. Ishutov, Hasiuk (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 592-602. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Multiphase Flow in Porous Rock Imaged Under Dynamic Flow Conditions with Fast X-Ray Computed Microtomography S. Berg, R. Armstrong, H. Ott, A. Georgiadis, S. A. Klapp, A. Schwing, R. Neiteler, N. Brussee, A. Makurat, L. Leu, F. Enzmann, J.-O. Schwarz, M. Wolf, F. Khan, M. Kersten, S. Irvine, M. Stampanoni (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 304-312. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Application of Digital Core Technology in Wellbore Stability Research. Zhou, Ye, Zhu, Cheng, Song, Wang, Cai (2022). DOI: 10.30632/PJV63N2-2022a7. Petrophysics Vol. 63 No. 2 (Apr 2022).
- Article 7: Using Digital Rock Physics to Evaluate Novel Percussion Core Quality. Lakshtanov, Zapata, Saucier, Cook, Eve, Lancaster, Lane, Gettemy, Sincock, Liu, Geetan, Draper, Gill (2022). DOI: 10.30632/PJV63N6-2022a7. Petrophysics Vol. 63 No. 6 (Dec 2022) — Best Papers of the 2022 SPWLA Annual Symposium special issue.

porosity_hi_correction

petrolib.nmr

```
porosity_hi_correction(phi_apparent, hi)
```

Purpose

HI-corrected porosity $\phi_{\text{true}} = \phi_{\text{apparent}}/\text{HI}$.
The apparent NMR porosity under-reads when the fluid HI < 1 (gas, light oil).

Input parameters

Parameter	Type	Default
phi_apparent	ArrayLike	required
hi	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fella, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Laboratory Measurement of Total Porosity and Fluid Saturations for Unconventional Tight Rocks. Based on: Cheng et al., "Laboratory Measurement of Total Porosity and Fluid Saturations for Unconventional Tight Rocks: Methodologies, Challenges, and Comparison", Petrophysics, Vol. 66, No. 2, April 2025, pp. 250-266.

precision_recall_f1

petrolib.ml_stats

```
precision_recall_f1(y_true, y_pred, *, positive=1, zero_division=0.0)
```

Purpose

Binary precision, recall and F1 for the positive class.

zero_division is returned when a denominator is empty (the repository convention is 0.0).

Input parameters

Parameter	Type	Default
y_true	ArrayLike	required
y_pred	ArrayLike	required
positive (kw-only)	object	1
zero_division (kw-only)	float	0.0

Output

tuple[float, float, float]

Sources

- Article 8: A Comparative Study of Three Supervised Machine-Learning Algorithms for Classifying Carbonate Vuggy Facies in the Kansas Arbuckle Formation. Deng, Xu, Jobe, Xu (2019). DOI: 10.30632/PJV60N6-2019a8. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).

predict_linear

petrolib.ml_stats

```
predict_linear(X, coef, intercept=0.0)
```

Purpose

Evaluate the linear model from ols.

Input parameters

Parameter	Type	Default
X	ArrayLike	required
coef	ArrayLike	required
intercept	float	0.0

Output

numpy.ndarray of float64

Sources

- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).

pressure_from_gas_density

petrolib.geochem_fluids.pvt

```
pressure_from_gas_density(rho, t_k, *, m_kg_mol=MW_METHANE, z=1.0)
```

Purpose

Pressure (Pa) from gas density -- inverse of gas_density.

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
t_k	ArrayLike	required
m_kg_mol (kw-only)	ArrayLike	MW_METHANE
z (kw-only)	ArrayLike	1.0

Output

numpy.ndarray of float64

project_simplex

petrolib.inversion_numerics.linear

```
project_simplex(v, total=1.0)
```

Purpose

Euclidean projection of v onto the simplex {x>=0, sum(x)=total}.

Input parameters

Parameter	Type	Default
v	ArrayLike	required
total	float	1.0

Output

numpy.ndarray of float64

pseudo_reduced

petrolib.geochem_fluids.pvt

pseudo_reduced(p, t, ppc, tpc)

Purpose

Pseudo-reduced pressure and temperature (P/Ppc, T/Tpc).

Input parameters

Parameter	Type	Default
p	ArrayLike	required
t	ArrayLike	required
ppc	ArrayLike	required
tpc	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 3: The Compressibility Factor (Z) of Shale Gas at the Core Scale. Tran, Sakhaee-Pour (2019). DOI: 10.30632/PJV60N4-2019a3. Petrophysics Vol. 60 No. 4 (Aug 2019).

ps0

petrolib.inversion_numerics.optimize

```
ps0(objective, bounds, n_particles=30, n_iter=200, omega=0.7, inertia_decay=None, c1=1.5, c2=1.5, seed=0)
```

Purpose

Particle-swarm optimization over box bounds.

Returns (x_best, f_best, history) where history is the best objective per iteration.

inertia_decay optionally multiplies *omega* each step.

Input parameters

Parameter	Type	Default
objective	Callable[[_Float], float]	required
bounds	list[tuple[float, float]]	required
n_particles	int	30
n_iter	int	200
omega	float	0.7
inertia_decay	float None	None
c1	float	1.5
c2	float	1.5
seed	int	0

Output

tuple[ndarray[float64], float, ndarray[float64]]

pwave_modulus

petrolib.acoustic_geomech

pwave_modulus(k, g)

Purpose

P-wave (constrained) modulus $M = K + 4/3 \cdot G$.

Input parameters

Parameter	Type	Default
k	ArrayLike	required
g	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Log Interpretation for Petrophysical and Elastic Properties of Fine-Grained Sedimentary Rocks Authors: Ermis Proestakis and Ida Lykke Fabricius Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 705-727. DOI: 10.30632/PJV66N5-2025a1.

qv_from_cec

petrolib.saturation_resistivity

```
qv_from_cec(cec, *, rho_grain, phi)
```

Purpose

Cation exchange capacity per pore volume, grain-based.

$Q_v = CEC * \rho_{\text{grain}} * (1 - \phi) / \phi$ with CEC in meq/g, grain density in g/cm³, Q_v in meq/ml.

Input parameters

Parameter	Type	Default
cec	ArrayLike	required
rho_grain (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Solving Complex Dual-Water Equation using Dielectric-NMR- Spectroscopy and Conventional Logs. Willy Tan, Ryan Lafferty, Thomas J. Neville (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 14-23. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 61 No. 4 (Aug 2020) (issue-level reference).

qv_juhasz

petrolib.saturation_resistivity

```
qv_juhasz(v_clay_dry, *, rho_clay, cec_clay, phit)
```

Purpose

Juhasz clay-based Qv normalization.

$Qv = V_{cl_dry} * \rho_{clay} * CEC_{clay} / \phi_{it}$ — physically distinct from the grain-based qv_{from_cec} (dry-clay properties, total porosity).

Input parameters

Parameter	Type	Default
v_clay_dry	ArrayLike	required
rho_clay (kw-only)	ArrayLike	required
cec_clay (kw-only)	ArrayLike	required
phit (kw-only)	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).

r2_score

petrolib.ml_stats

```
r2_score(y_true, y_pred, *, eps=0.0)
```

Purpose

Coefficient of determination R^2 (NOT the correlation coefficient R).

eps is the additive denominator guard several articles use ($ss_{tot} + 1e-12$); the default 0.0 returns a non-finite value (-inf, or NaN when the fit is also perfect) for constant y_true.

Input parameters

Parameter	Type	Default
y_true	ArrayLike	required
y_pred	ArrayLike	required
eps (kw-only)	float	0.0

Output

float

Sources

- Article 5: Synthetic Sonic Log Generation With Machine Learning - A Contest Summary From Five Methods. Yu, Xu, Misra, Li, Ashby, et al. (2021). DOI: 10.30632/PJV62N4-2021a4. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 7: New Robust Model to Estimate Formation Tops in Real Time Using Artificial Neural Networks (ANN). Elkatatny, Al-Abduljabbar, Mahmoud (2019). DOI: 10.30632/PJV60N6-2019a7. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 1: A Guide to Nanoindentation. Sondergeld, Rai (2022). DOI: 10.30632/PJV63N5-2022a1. Petrophysics Vol. 63 No. 5 (Oct 2022).

rachford_rice

petrolib.geochem_fluids.pvt

```
rachford_rice(z, k, *, tol=1e-12, max_iter=200)
```

Purpose

Rachford-Rice vapor fraction beta solving $\sum(z*(k-1)/(1+beta*(k-1))) = 0$.
Bisection on beta in [0, 1]; returns the root (clamped to the bracket).

Input parameters

Parameter	Type	Default
z	ArrayLike	required
k	ArrayLike	required
tol (kw-only)	float	1e-12
max_iter (kw-only)	int	200

Output

float

radioactive_decay

petrolib.nuclear

```
radioactive_decay(n0, t, half_life)
```

Purpose

Radioactive decay $N = N_0 \cdot \exp(-\lambda \cdot t)$ with $\lambda = \ln(2)/T_{\text{half}}$.

Input parameters

Parameter	Type	Default
n0	ArrayLike	required
t	ArrayLike	required
half_life	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: In-Situ Evaluation of Vapor Properties Using Condensed Vapor Gamma. O'Sullivan (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 334-345. DOI: none assigned (this issue predates SPWLA DOI assignment).

ratio_factor

petrolib.wellbore_geometry

```
ratio_factor(dogleg_rad)
```

Purpose

Minimum-curvature ratio (radius) factor $RF = (2/DL) \tan(DL/2)$.

dogleg_rad is the dogleg angle in radians; for a near-straight step ($DL < 1e-9$) the factor is exactly 1.0 (the balanced-tangential limit).

Input parameters

Parameter	Type	Default
dogleg_rad	float	required

Output

float

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

rddtw

petrolib.depth_matching

```
rddtw(ref, target, *, tau=4.0, lam=0.5, band=None)
```

Purpose

Regularized derivative DTW (RDDTW) with an excessive-warping penalty.

The local cost combines shape and Keogh-Pazzani derivative (trend) absolute differences $|s - t| + \tau * |s' - t'|$ plus an off-diagonal penalty $\lambda * |i/n - j/m|$ that discourages excessive warping.

Input parameters

Parameter	Type	Default
ref	ArrayLike	required
target	ArrayLike	required
tau (kw-only)	float	4.0
lam (kw-only)	float	0.5
band (kw-only)	int None	None

Output

DtwResult

reflection_coefficient

petrolib.acoustic_geomech

```
reflection_coefficient(z1, z2)
```

Purpose

Normal-incidence reflection coefficient $(Z2 - Z1)/(Z2 + Z1)$.

z1 is the incident-side impedance, z2 the transmitted side.

Input parameters

Parameter	Type	Default
z1	ArrayLike	required
z2	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Permeability Estimation Using Ultrasonic Borehole Image Logs in Dual-Porosity Carbonate Reservoirs. Menezes de Jesus, Martins Compan, Surmas (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 620-637. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Novel Composite Cement for Improved Well Integrity Evaluation. Elshahawi, Huang, Pollock, Veedu (2019). DOI: 10.30632/PJV60N1Y2019a4. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 2: New 4.75-in. Ultrasonic LWD Technology Provides High-Resolution Caliper and Imaging in Oil-Based and Water-Based Muds. Li, Lee, Coates, Jin, Wong (2019). DOI: 10.30632/PJV60N6-2019a2. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 7: Revealing Hidden Information - High- Resolution Logging-While-Drilling Slowness Measurements and Imaging Using Advanced Dual Ultrasonic Technology. Blyth, Sakiyama, Hori, Yamamoto, Nakajima, Fahim Ud Din, Haecker, Kittridge (2021). DOI: 10.30632/PJV62N1-2021a6. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 4: Ultrasonic Logging of Creeping Shale. Diez, Johansen, Larsen (2022). DOI: 10.30632/PJV63N1-2022a4. Petrophysics Vol. 63 No. 1 (Feb 2022).
- Article 1: Full-Waveform Inversion of Fiber-Optic VSP Data From Deviated Wells. Podgornova, Bettinelli, Liang, Le Calvez, Leaney, Perez, Soliman (2022). DOI: 10.30632/PJV63N6-2022a1. Petrophysics Vol. 63 No. 6 (Dec 2022) — Best Papers of the 2022 SPWLA Annual Symposium special issue.
- Ultrasonic Microscopy Imaging of Carbonate Reservoir Pore Structure. Based on: Chen et al., "New Methodology for Ultrasonic Microscopy Imaging of Carbonate Reservoirs' Pore Structure", Petrophysics, Vol. 66, No. 2, April 2025, pp. 267-282.

reflection_coefficient

petrolib.integrity_drilling

```
reflection_coefficient(z1, z2)
```

Purpose

Normal-incidence reflection coefficient $R = (Z2 - Z1)/(Z2 + Z1)$ (z1 incident side).

Input parameters

Parameter	Type	Default
z1	ArrayLike	required
z2	ArrayLike	required

Output

numpy.ndarray of float64

reg_lambda_brd

petrolib.inversion_numerics.costs

```
reg_lambda_brd(a, b, chi2_target, bracket=(1e-06, 1000.0), *, max_iter=50)
```

Purpose

Discrepancy-principle regularization weight by geometric bisection.
Finds λ such that the Tikhonov data misfit $\|A x_{\lambda} - b\|^2$ equals χ^2_{target} (e.g. $n \cdot \sigma^2$), bisecting on $\log(\lambda)$ within bracket.

Input parameters

Parameter	Type	Default
a	ArrayLike	required
b	ArrayLike	required
chi2_target	float	required
bracket	tuple[float, float]	(1e-06, 1000.0)
max_iter (kw-only)	int	50

Output

float

reg_lambda_multiplicative

petrolib.inversion_numerics.costs

```
reg_lambda_multiplicative(misfit_value, alpha, beta, lam_max=np.inf)
```

Purpose

Habashy-Abubakar multiplicative cooling $\lambda = \min(\alpha \cdot \text{misfit}^\beta, \lambda_{\text{max}})$.

Input parameters

Parameter	Type	Default
misfit_value	float	required
alpha	float	required
beta	float	required
lam_max	float	np.inf

Output

float

Sources

- Article 2: Inversion-Based Workflow for Quantitative Interpretation of the New-Generation Oil-Based-Mud Resistivity Imager Yong-Hua Chen, Dzevat Omeragic, Tarek Habashy, Richard Bloemenkamp, Tianhua Zhang, Phillip Cheung, Robert Laronga (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 554-571. DOI: none assigned (this issue predates SPWLA DOI assignment).

relative_discrepancy

petrolib.data_qc_io.clean

```
relative_discrepancy(a, b)
```

Purpose

Relative discrepancy $2|a-b| / (a+b)$ (difference over the mean).

Both corpus arrangements ($2|a-b|/(a+b)$ and $|a-b|/(0.5(a+b))$) round identically in IEEE arithmetic.

Input parameters

Parameter	Type	Default
a	ArrayLike	required
b	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Low-Permeability Measurements: Insights. Profice, Hamon, Nicot (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 30-40. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Lessons Learned in Permian Core Analysis: Comparison Between Retort, GRI, and Routine Methodologies. Blount, Croft, Driskill, Tepper (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 517-527. DOI: none assigned (this issue predates SPWLA DOI assignment).

relaxation_rate

petrolib.nmr

```
relaxation_rate(*, t2_bulk=np.inf, rho=0.0, s_over_v=0.0, D=0.0, G=0.0, TE=0.0,
gamma=GAMMA_H)
```

Purpose

Brownstein-Tarr relaxation rate (bulk + surface + diffusion).

$1/T2 = 1/T2_bulk + \rho*(S/V) + (\gamma*G*TE)**2*D/12$. The default $t2_bulk=inf$ and zero surface/diffusion terms let a caller build any subset (surface-only, bulk+surface, or the full three-mechanism rate). Use a consistent unit set.

Input parameters

Parameter	Type	Default
t2_bulk (kw-only)	ArrayLike	np.inf
rho (kw-only)	ArrayLike	0.0
s_over_v (kw-only)	ArrayLike	0.0
D (kw-only)	ArrayLike	0.0
G (kw-only)	ArrayLike	0.0
TE (kw-only)	ArrayLike	0.0
gamma (kw-only)	float	GAMMA_H

Output

numpy.ndarray of float64

Sources

- Article 6: Influence of Magnetic Susceptibility Contrast on NMR Studies - Experimental Analysis From Siliciclastic Reservoirs. Sarkar, Chatterjee, Lal, Kumar, Deo (2019). DOI: 10.30632/PJV60N6-2019a6. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozlowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 2: A New Apparatus for Coupled Low-Field NMR and Ultrasonic Measurements in Rocks at Reservoir Conditions. Connolly, Sarout, Dautriat, May, Johns (2020). DOI: 10.30632/PJV61N2-2020a2. Petrophysics Vol. 61 No. 2 (Apr 2020).
- Article 3: Magnetic Resonance Core-Plug Analysis with the Three-Magnet Array Unilateral Magnet Juan C. Garcia-Naranjo, Pan Guo, Florin Marica, Guangzhi Liao, Bruce J. Balcom (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 229-239. DOI: none assigned (this issue predates SPWLA DOI assignment).

remaining_life_years

petrolib.integrity_drilling

```
remaining_life_years(thickness_mm, min_acceptable_mm, rate_mm_per_yr)
```

Purpose

Remaining casing life $(t - t_{min}) / \text{rate}$ (years), floored at 0.
A non-positive corrosion rate returns the 999.0 sentinel (no wear).

Input parameters

Parameter	Type	Default
thickness_mm	float	required
min_acceptable_mm	float	required
rate_mm_per_yr	float	required

Output

float

renormalize

petrolib.data_qc.io.clean

```
renormalize(fracs, axis=-1)
```

Purpose

Rescale fractions to sum to one along axis (compositional closure).
A zero sum yields NaN/inf rather than a guessed composition.

Input parameters

Parameter	Type	Default
fracs	ArrayLike	required
axis	int	-1

Output

numpy.ndarray of float64

Sources

- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", Petrophysics, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.
- Article 2: Application and Quality Control of Core Data for the Development and Validation of Elemental Spectroscopy Log Interpretation Susan Herron, Michael Herron, Iain Pirie, Pablo Saldungaray, Paul Craddock, Alyssa Charsky, Marina Polyakov, Frank Shray, Ting Li (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 392-414. DOI: none assigned (this issue predates SPWLA DOI assignment).

reopening_pressure

petrolib.acoustic_geomech

reopening_pressure(sh, sH, pp)

Purpose

Fracture reopening pressure $3 \cdot sh - sH - Pp$ (breakdown with $T0=0$).

Input parameters

Parameter	Type	Default
sh	ArrayLike	required
sH	ArrayLike	required
pp	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).

resistivity_from_phase

petrolib.em_dielectric

```
resistivity_from_phase(phase_deg, freq_hz, spacing_m)
```

Purpose

Induction-limit inverse of phase_shift_deg -> apparent rho.

$\rho = L^2 \omega \mu_0 / (2 \text{phase_rad}^2)$ with $\text{phase_rad} = \text{radians}(\text{phase_deg})$.

Input parameters

Parameter	Type	Default
phase_deg	ArrayLike	required
freq_hz	ArrayLike	required
spacing_m	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).

resistivity_index

petrolib.saturation_resistivity

resistivity_index(rt, ro)

Purpose

Resistivity index by definition, $RI = Rt/Ro$.

Input parameters

Parameter	Type	Default
rt	ArrayLike	required
ro	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2 (Tutorial): Introduction to Resistivity Principles for Formation Evaluation: A Tutorial Primer. Kennedy, Garcia (2019). DOI: 10.30632/PJV60N2-2019t2. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Petrophysics Vol. 66 No. 6 (Dec 2025) — Best Papers of the 2024 SCA International Symposium (issue-level reference).

resistivity_index_from_sw

petrolib.saturation_resistivity

```
resistivity_index_from_sw(sw, *, n=2.0, b=1.0)
```

Purpose

Resistivity-index power law $RI = b * Sw^{*-n}$ ($b=1$ is Archie).

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
n (kw-only)	float	2.0
b (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir
Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Drainage Capillary Pressure and Resistivity Index from Short-Wait Porous-Plate Experiments.
Dernaika, Wilson, Skjaeveland, Ebeltoft (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 369-376. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 61 No. 4 (Aug 2020) (issue-level reference).

reuss

petrolib.acoustic_geomech

```
reuss(f, m, *, axis=-1)
```

Purpose

Reuss (harmonic) bound $1/\sum(f_i/m_i)$ over the mixing axis.

Input parameters

Parameter	Type	Default
f	ArrayLike	required
m	ArrayLike	required
axis (kw-only)	int	-1

Output

numpy.ndarray of float64

Sources

- Article 5: Thermal Conductivity Estimation From Elastic-Wave Velocity - Application of a Petrographic-Coded Model. Nina Gegenhuber, Jurgen Schon (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 51-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).

rhob_from_rhoe

petrolib.nuclear

```
rhob_from_rhoe(rho_e)
```

Purpose

Bulk density from electron density $\rho_b = 1.0704 \cdot \rho_e - 0.1883$.

Input parameters

Parameter	Type	Default
rho_e	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

rhoe_from_rhob

petrolib.nuclear

```
rhoe_from_rhob(rho_b)
```

Purpose

Electron density from bulk density $\rho_e = (\rho_b + 0.1883)/1.0704$ (inverse).

Input parameters

Parameter	Type	Default
rho_b	ArrayLike	required

Output

numpy.ndarray of float64

rmse*petrolib.ml_stats*

```
rmse(y_true, y_pred)
```

Purpose

Root-mean-square error.

Input parameters

Parameter	Type	Default
y_true	ArrayLike	required
y_pred	ArrayLike	required

Output

float

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Article 5: Synthetic Sonic Log Generation With Machine Learning - A Contest Summary From Five Methods. Yu, Xu, Misra, Li, Ashby, et al. (2021). DOI: 10.30632/PJV62N4-2021a4. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 7: New Robust Model to Estimate Formation Tops in Real Time Using Artificial Neural Networks (ANN). Elkatatny, Al-Abduljabbar, Mahmoud (2019). DOI: 10.30632/PJV60N6-2019a7. Petrophysics Vol. 60 No. 6 (Dec 2019).

rqi*petrolib.flow_transport*`rqi(k_md, phi, *, c=0.0314)`**Purpose**

Amaefule Reservoir Quality Index RQI = $c \cdot \sqrt{k_{md}/\phi}$ (um).

$c = 0.0314 = \sqrt{1/1014}$ converts mD to um; copies that write $\sqrt{k/(1014 \cdot \phi)}$ map by passing $c=(1/1014)**0.5$.

Input parameters

Parameter	Type	Default
k_md	ArrayLike	required
phi	ArrayLike	required
c (kw-only)	float	0.0314

Output

numpy.ndarray of float64

Sources

- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 5: Integrated Petrophysical Rock Classification in the McElroy Field, West Texas, USA. Saneifar, Skalinski, Theologou, Kenter, Cuffey, Salazar-Tio (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 493-510. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Presenting a Multifaceted Approach to Unconventional Rock Typing and Technical Validation - Case Study in the Permian Basin and Impacts on Reservoir Characterization Workflows. Perry, Hayes (2019). DOI: 10.30632/PJV60N5-2019a8. Petrophysics Vol. 60 No. 5 (Oct 2019).

rss*petrolib.ml_stats*`rss(y_true, y_pred)`**Purpose**

Residual sum of squares.

Input parameters

Parameter	Type	Default
y_true	ArrayLike	required
y_pred	ArrayLike	required

Output

float

Sources

- Article 1: Automatically Quantifying Wireline and LWD Pressure-Test Quality. Proett, Musharfi, Gill, Ma, Meridji, Eyuboglu (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 101-115. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).

rw75_from_salinity

petrolib.geochem_fluids.brine

```
rw75_from_salinity(nacl_ppm)
```

Purpose

Bateman-Konen water resistivity at 75 degF $R_{75} = 0.0123 + 3647.5/C^{0.955}$.

Input parameters

Parameter	Type	Default
nacl_ppm	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).

rw_from_salinity

petrolib.geochem_fluids.brine

```
rw_from_salinity(nacl_ppm, temp, *, unit='F')
```

Purpose

Water resistivity at temp from NaCl ppm (75 degF reference).

Computes R75 (Bateman-Konen) then Arps-corrects from 75 degF to temp. unit="C" converts temp to Fahrenheit first (75 degF reference kept).

Input parameters

Parameter	Type	Default
nacl_ppm	ArrayLike	required
temp	ArrayLike	required
unit (kw-only)	str	'F'

Output

numpy.ndarray of float64

Sources

- Article 5 (Technical Note): The Bateman- Konen Resistivity-Salinity Transform II. Kennedy (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 379-381. DOI: none assigned (this issue predates SPWLA DOI assignment).

salinity_from_rw

petrolib.geochem_fluids.brine

```
salinity_from_rw(rw, temp, *, unit='F')
```

Purpose

NaCl ppm from water resistivity at temp -- inverse of rw_from_salinity.

Input parameters

Parameter	Type	Default
rw	ArrayLike	required
temp	ArrayLike	required
unit (kw-only)	str	'F'

Output

numpy.ndarray of float64

Sources

- Article 5 (Technical Note): The Bateman- Konen Resistivity-Salinity Transform II. Kennedy (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 379-381. DOI: none assigned (this issue predates SPWLA DOI assignment).

salinity_from_rw75

petrolib.geochem_fluids.brine

```
salinity_from_rw75(rw75_ohmm)
```

Purpose

NaCl ppm from the 75 degF resistivity -- exact inverse of rw75_from_salinity.

Input parameters

Parameter	Type	Default
rw75_ohmm	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4 (Technical Note): The Bateman-Konen Resistivity-Salinity Transform. Kennedy (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 282-283. DOI: none assigned (this issue predates SPWLA DOI assignment).

sdr

petrolib.nmr

```
sdr(phi, t2lm_ms, *, a=4.0, m=4.0, n=2.0, rho_um_s=None)
```

Purpose

Schlumberger-Doll-Research NMR permeability (mD).

$k = a \cdot \phi^m \cdot T2LM^n$; if `rho_um_s` is given, the KSDR form $a \cdot \phi^m \cdot (\rho \cdot T2LM)^n$. The prefactor `a` is calibration-specific (4.0, 4.6, ...) and explicit.

Input parameters

Parameter	Type	Default
<code>phi</code>	ArrayLike	required
<code>t2lm_ms</code>	ArrayLike	required
<code>a</code> (kw-only)	float	4.0
<code>m</code> (kw-only)	float	4.0
<code>n</code> (kw-only)	float	2.0
<code>rho_um_s</code> (kw-only)	float None	None

Output

numpy.ndarray of float64

Sources

- Article 1: Pore-Structure Characterization of a Complex Carbonate Reservoir in South Iraq Using Advanced Interpretation of NMR Logs. Saidian, Jain, Milad (2021). DOI: 10.30632/PJV62N2-2021a1. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.
- Article 3: Using Advanced Logging Measurements to Develop a Robust Petrophysical Model for the Bakken Petroleum System. Simpson, Hohman, Pirie, Horkowitz (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 457-478. DOI: none assigned (this issue predates SPWLA DOI assignment).

sentinels_to_nan

petrolib.data_qc.io.clean

```
sentinels_to_nan(x, sentinels=SENTINELS, *, rtol=1e-05, atol=1e-08)
```

Purpose

Replace LAS/DLIS no-data sentinels (e.g. -999.25) with NaN.

Matching uses `np.isclose` (defaults `rtol=1e-5`, `atol=1e-8`) so float round-trips of the sentinel still match.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
sentinels	Sequence[float]	SENTINELS
rtol (kw-only)	float	1e-05
atol (kw-only)	float	1e-08

Output

`numpy.ndarray` of `float64`

Sources

- Article 1: Data Quality Considerations for Petrophysical Machine-Learning Models. McDonald (2021). DOI: 10.30632/PJV62N6-2021a1. Petrophysics Vol. 62 No. 6 (Dec 2021).

setschenow_factor

petrolib.geochem_fluids.solubility

```
setschenow_factor(m_nacl, t_k, *, ks25=0.11)
```

Purpose

Setschenow salting-out activity factor $\exp(2*ks*m)$.

$ks = ks25 + 1e-4*(T - 298)$; m is the NaCl molality (mol/kg).

Input parameters

Parameter	Type	Default
m_nacl	ArrayLike	required
t_k	ArrayLike	required
ks25 (kw-only)	float	0.11

Output

numpy.ndarray of float64

shear_wave_splitting

petrolib.acoustic_geomech

```
shear_wave_splitting(v_fast, v_slow)
```

Purpose

Shear-wave splitting (anisotropy) $(V_{\text{fast}} - V_{\text{slow}})/V_{\text{fast}}$.

Input parameters

Parameter	Type	Default
v_fast	ArrayLike	required
v_slow	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: A Case Study about Formation Evaluation and Rock Physics Modeling of the Bazhenov Shale. Pavel Kulyapin and Tatiana F. Sokolova (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 211-218. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Geomechanics of Orthorhombic Media. Far, Quirein, Mekic (2016). Petrophysics Vol. 57, No. 6 (December 2016), pp. 588-596. DOI: none assigned (this issue predates SPWLA DOI assignment).

shifted_pair

petrolib.data_qc_io.synth

```
shifted_pair(n, shift, *, noise=0.0, rng=None)
```

Purpose

Reference curve and a copy lagged by shift samples.

Both come from one multi-harmonic base with $\text{other}[i] = \text{ref}[i + \text{shift}]$ (noise-free) — aligning other to ref (e.g. with `petrolib.depth_matching.xcorr_shift`) recovers shift. noise adds independent Gaussian noise to each curve. Pattern: src2024_10/rddtw_depth_matching (integer-shift core/log pair), src2022_02/article2 (rigid-lag windows).

Input parameters

Parameter	Type	Default
n	int	required
shift	int	required
noise (kw-only)	float	0.0
rng (kw-only)	np.random.Generator int None	None

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Core-Log Depth Adaptive Matching Using RDDTW. Based on: Fang et al. (2024), "Core-Log Depth Adaptive Matching Using RDDTW", Petrophysics, Vol. 65, No. 5, pp. 835-851.
- Article 2: Automated Well-Log Depth Matching - 1D Convolutional Neural Networks vs. Classic Cross Correlation. Torres Caceres, Duffaut, Yazidi, Westad, Johansen (2022). DOI: 10.30632/PJV63N1-2022a2. Petrophysics Vol. 63 No. 1 (Feb 2022).

ShiftResult (class)

petrolib.depth_matching

```
ShiftResult()
```

Purpose

Result of xcorr_shift.

Input parameters

None.

Output

ShiftResult instance

shmax_from_reopening

petrolib.acoustic_geomech

```
shmax_from_reopening(pr, sh, pp)
```

Purpose

Maximum horizontal stress from the reopening pressure $sH = 3 \cdot sh - Pr - Pp$.

Input parameters

Parameter	Type	Default
pr	ArrayLike	required
sh	ArrayLike	required
pp	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: How Can Microfracturing Improve Reservoir Management?. Malik, Jones, Boratko (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 492-507. DOI: none assigned (this issue predates SPWLA DOI assignment).

sigma_eps_from_wavenumber

petrolib.em_dielectric

```
sigma_eps_from_wavenumber(k_r, k_i, freq_hz, *, mu_r=1.0)
```

Purpose

Closed-form inverse of complex_wavenumber -> (sigma, eps_r).

From $k^2 = \omega^2 \mu \epsilon$: $\epsilon_r = (k_r^2 - k_i^2) / (\omega^2 \mu \epsilon_0)$ and $\sigma = 2 k_r k_i / (\omega \mu)$.

Input parameters

Parameter	Type	Default
k_r	ArrayLike	required
k_i	ArrayLike	required
freq_hz	ArrayLike	required
mu_r (kw-only)	float	1.0

Output

tuple[ndarray[float64], ndarray[float64]]

sigma_forward

petrolib.nuclear

```
sigma_forward(phi, sw, *, sigma_ma=10.0, sigma_w=55.0, sigma_hc=21.0, vsh=0.0,
sigma_sh=27.0)
```

Purpose

Volumetric capture cross-section Sigma (c.u.) of a rock.

$\text{Sigma} = (1 - \phi - V_{\text{sh}}) * \text{Sigma}_{\text{ma}} + V_{\text{sh}} * \text{Sigma}_{\text{sh}} + \phi * (S_{\text{w}} * \text{Sigma}_{\text{w}} + (1 - S_{\text{w}}) * \text{Sigma}_{\text{hc}})$. With $v_{\text{sh}}=0$ this is the clean two-phase form.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
sw	ArrayLike	required
sigma_ma (kw-only)	float	10.0
sigma_w (kw-only)	float	55.0
sigma_hc (kw-only)	float	21.0
vsh (kw-only)	ArrayLike	0.0
sigma_sh (kw-only)	float	27.0

Output

numpy.ndarray of float64

Sources

- Article 7: Multidetector Pulsed-Neutron Tool Application in a Low-Porosity Reservoir - A Case Study in Mutiara Field, Indonesia. Wijaya, Aulianagara, Guo, Naibaho, Asriwan, Amirudin (2020). DOI: 10.30632/PJV61N6-2020a7. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.
- Laronga, R., Swager, L., and Bustos, U. (2023). "Time-Lapse Pulsed- Neutron Logs for Carbon Capture and Sequestration: Practical Learnings and Key Insights." Petrophysics, 64(5), 680-699. DOI: 10.30632/PJV64N5-2023a5.

sigma_forward_3phase

petrolib.nuclear

```
sigma_forward_3phase(phi, so, sg, sw, *, sigma_oil=22.0, sigma_gas=8.0, sigma_w=80.0,
sigma_ma=10.0)
```

Purpose

Three-phase (oil/gas/water) capture cross-section Sigma (c.u.).

$$\text{Sigma} = (1-\text{phi}) * \text{Sigma_ma} + \text{phi} * (\text{So} * \text{Sigma_oil} + \text{Sg} * \text{Sigma_gas} + \text{Sw} * \text{Sigma_w}).$$

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
so	ArrayLike	required
sg	ArrayLike	required
sw	ArrayLike	required
sigma_oil (kw-only)	float	22.0
sigma_gas (kw-only)	float	8.0
sigma_w (kw-only)	float	80.0
sigma_ma (kw-only)	float	10.0

Output

numpy.ndarray of float64

Sources

- McGlynn et al. (2023), Petrophysics 64(6): 900-918. New pulsed-neutron spectroscopy instrument with LaBr3 detectors providing simultaneous C/O ratio, capture sigma, and ratio-based gas measurements for two- and three-phase saturation analysis.

sigma_from_decay_fit

petrolib.nuclear

```
sigma_from_decay_fit(t_us, counts, *, fit_window=None)
```

Purpose

Capture cross-section from a log-linear fit of a capture-decay curve.

Fits $\ln(\text{counts})$ vs t (seconds); $\text{Sigma} = -\text{slope}/v * 1e3$ (c.u.). `fit_window` optionally selects a (start, stop) sample slice.

Input parameters

Parameter	Type	Default
<code>t_us</code>	ArrayLike	required
<code>counts</code>	ArrayLike	required
<code>fit_window</code> (kw-only)	tuple[int, int] None	None

Output

numpy.ndarray of float64

sigma_from_imag_permittivity

petrolib.em_dielectric

```
sigma_from_imag_permittivity(eps_imag, freq_hz)
```

Purpose

Conductivity from loss permittivity: $\sigma = \epsilon'' * 2 * \pi * f * \epsilon_0$.

Input parameters

Parameter	Type	Default
eps_imag	ArrayLike	required
freq_hz	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Al-Qouzi, W., Hassan, A., Attia, H., El- Hussein, A., and Mahmoud, M. (2026). Experimental Study of Sensitivity of Multifrequency (MF) Dielectric Measurements to Hydraulic Fractures in Sandstone and Carbonate Formations. *Petrophysics*, 67(3), 482-508. DOI: 10.30632/PJV67N3-2026a2 (orig. SPE-227827-MS).

sigma_from_tau

petrolib.nuclear

```
sigma_from_tau(tau_us)
```

Purpose

Capture cross-section from thermal decay time $\Sigma = 4550/\tau$ (c.u.).

Input parameters

Parameter	Type	Default
tau_us	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

sigma_sensitivity

petrolib.nuclear

```
sigma_sensitivity(phi, sigma_w, sigma_hc)
```

Purpose

Saturation sensitivity $d\text{Sigma}/d\text{Sw} = \text{phi} * (\text{Sigma}_w - \text{Sigma}_{hc})$ (c.u.).

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
sigma_w	float	required
sigma_hc	float	required

Output

numpy.ndarray of float64

Sources

- Article 5: 'Log-Soak-Log' Experiment in Tengiz Field: Novel Technology for In-Situ Imbibition Measurements to Support an Improved Oil Recovery Project. Seth, Villegas, Iskakov, Playton, Lindsell, Cordova, Turmanbekova, Wang (2019). DOI: 10.30632/PJV60N5-2019a5. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 7: Multidetector Pulsed-Neutron Tool Application in a Low-Porosity Reservoir - A Case Study in Mutiara Field, Indonesia. Wijaya, Aulianagara, Guo, Naibaho, Asriwan, Amirudin (2020). DOI: 10.30632/PJV61N6-2020a7. Petrophysics Vol. 61 No. 6 (Dec 2020).

sigma_w_from_salinity

petrolib.geochem_fluids.brine

```
sigma_w_from_salinity(nacl_ppm, temp_c=24.0)
```

Purpose

Thermal-neutron capture cross-section of brine (c.u.).
Sigma_w = (22 + 750*w)*(1 - 8e-4*(T - 75)) with w the NaCl weight fraction; 22 c.u. is the fresh-water baseline.

Input parameters

Parameter	Type	Default
nacl_ppm	ArrayLike	required
temp_c	ArrayLike	24.0

Output

numpy.ndarray of float64

Sources

- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.

sigma_w_from_salinity

petrolib.nuclear

```
sigma_w_from_salinity(ppm_nacl, *, temperature_c=75.0, model='fitz2023')
```

Purpose

Brine capture cross-section Σ_w (c.u.) from NaCl salinity (ppm).
model='fitz2023': $(22 + 750 \cdot s) \cdot (1 - 8e-4 \cdot (T-75))$ with $s = \text{ppm}/1e6$. model='linear450k': $22 + (220-22) \cdot \text{ppm}/450000$.

Input parameters

Parameter	Type	Default
ppm_nacl	ArrayLike	required
temperature_c (kw-only)	float	75.0
model (kw-only)	str	'fitz2023'

Output

numpy.ndarray of float64

Sources

- Article 5: 'Log-Soak-Log' Experiment in Tengiz Field: Novel Technology for In-Situ Imbibition Measurements to Support an Improved Oil Recovery Project. Seth, Villegas, Iskakov, Playton, Lindsell, Cordova, Turmanbekova, Wang (2019). DOI: 10.30632/PJV60N5-2019a5. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.

silhouette_score

petrolib.ml_stats

```
silhouette_score(x, labels)
```

Purpose

Mean silhouette coefficient over all samples ($O(N^2)$ pairwise).

Returns 0.0 for fewer than two clusters; a singleton's own-cluster distance counts as 0.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
labels	ArrayLike	required

Output

float

Sources

- Article 11: A New Hierarchical Method for Rock Classification Using Well- Log-Based Rock Fabric Quantification. Purba, Garcia, Heidari (2018). DOI: 10.30632/PJV59N5-2018a10. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 4: Detecting Specific Facies in Well-Log Data Sets Using Knowledge- Driven Hierarchical Clustering. Emelyanova, Peyaud, Dance, Pervukhina (2020). DOI: 10.30632/PJV61N4-2020a4. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 2: Data-Driven Algorithms for Image-Based Rock Classification and Formation Evaluation in Formations With Rapid Spatial Variation in Rock Fabric. Gonzalez, Heidari, and Lopez (2023). DOI: 10.30632/PJV64N2-2023a2. Petrophysics Vol. 64 No. 2 (Apr 2023) — SPWLA AI/ML Special Issue.

simandoux_sw

petrolib.saturation_resistivity

```
simandoux_sw(rt, rw, *, phi, vsh, rsh, a=1.0, m=2.0, clip=None)
```

Purpose

Simandoux (1963) water saturation, closed-form for $n = 2$.

Solves $1/R_t = S_w^{*2} * \phi^{*m}/(a*R_w) + V_{sh}*S_w/R_{sh}$ — the quadratic in S_w . The equation is only quadratic for $n = 2$, which is why this function has no n parameter (some article variants accept an n kwarg they silently ignore; the canonical API refuses to).

Input parameters

Parameter	Type	Default
rt	ArrayLike	required
rw	ArrayLike	required
phi (kw-only)	ArrayLike	required
vsh (kw-only)	ArrayLike	required
rsh (kw-only)	ArrayLike	required
a (kw-only)	float	1.0
m (kw-only)	float	2.0
clip (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Water Saturation Equations for Unconsolidated Reservoirs. Based on: Acosta et al. (2024), "Unveiling the Optimal Water Saturation Equation for Unconsolidated Reservoirs: A Case Study From the Tambaredjo Oil Field, Suriname", Petrophysics, Vol. 65, No. 5, pp. 739-764.
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Petrophysics Vol. 62 No. 6 (Dec 2021) (issue-level reference).

skin_depth

petrolib.em_dielectric

```
skin_depth(rho, freq_hz, *, mu_r=1.0)
```

Purpose

EM skin depth $\delta = \sqrt{2 \cdot \rho / (\omega \cdot \mu)}$ ($\mu = \mu_0 \cdot \mu_r$).

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
freq_hz	ArrayLike	required
mu_r (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 1: Looking Ahead of the Bit While Drilling: From Vision to Reality. Constable, Antonsen, Stalheim, Olsen, Fjell, Dray, Eikenes, Aarflot, Haldorsen, Digranes, Seydoux, Omeragic, Thiel, Davydychev, Denichou, Salim, Frey, Homan, Tan (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 426-446. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Advanced Log Interpretation in Field Development. Merkel, Lessenger (2016). Petrophysics Vol. 57, No. 5 (October 2016), pp. 479-491. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Comparison of the Russian and Western Resistivity Logs in Typical Western Siberian Reservoir Environments - A Numerical Study. Epov, Sukhorukova, Nechaev, Petrov, Rabinovich, Weston, Tyurin, Wang, Abubakar, Claverie (2020). DOI: 10.30632/PJV61N1-2020a1. Petrophysics Vol. 61 No. 1 (Feb 2020).
- Article 8: Mapping Complex Injectite Bodies With Multiwell Electromagnetic 3D Inversion Data. Clegg, Eriksen, Best, Tollefsen, Kowicki, Marchant (2021). DOI: 10.30632/PJV62N1-2021a7. Petrophysics Vol. 62 No. 1 (Feb 2021).
- Article 5: First LWD Co-Located Antenna Sensors for Real-Time Anisotropy and Dip Angle Determination, Yielding Better Look-Ahead Detection. Bittar, Wu, Ma, Pan, Fan, Griffing, Lozinsky (2021). DOI: 10.30632/PJV62N3-2021a4. Petrophysics Vol. 62 No. 3 (Jun 2021).
- UDAR Look-Ahead-While-Drilling Technology Assessment. Based on: Cuadros et al., "Look-Ahead-While-Drilling Technology Assessment for Early Hazards Identification in Presalt Offshore Brazil", Petrophysics, Vol. 66, No. 2, April 2025, pp. 190-211.
- Bower, M., Xie, H., Wang, G. L., Leveque, S., and Dolan, J. (2026). Anisotropy Sensitivity in Ultradeep Azimuthal Resistivity Technologies. Petrophysics, 67(3), 544-559. DOI: 10.30632/PJV67N3-2026a5.

slowing_down_length_empirical

petrolib.nuclear

```
slowing_down_length_empirical(phi, e_mev, *, ls0=20.0, e_ref=4.5)
```

Purpose

Empirical slowing-down length $L_s = L_{s0} \cdot (1 - 0.6 \cdot \phi) \cdot \sqrt{E/E_{ref}}$ (cm).

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
e_mev	ArrayLike	required
ls0 (kw-only)	float	20.0
e_ref (kw-only)	float	4.5

Output

numpy.ndarray of float64

slowness_from_velocity

petrolib.acoustic_geomech

```
slowness_from_velocity(v, *, dt_unit='us/ft')
```

Purpose

Acoustic slowness from velocity (m/s) -- inverse of velocity_from_slowness.

dt_unit="us/ft" -> $0.3048 * 1e6 / v$ (us/ft); "us/m" -> $1e6 / v$.

Input parameters

Parameter	Type	Default
v	ArrayLike	required
dt_unit (kw-only)	str	'us/ft'

Output

numpy.ndarray of float64

slowness_to_velocity

petrolib.units

```
slowness_to_velocity(slowness, unit='us/ft')
```

Purpose

Sonic slowness (us/ft or us/m) to velocity in m/s.

Input parameters

Parameter	Type	Default
slowness	ArrayLike	required
unit	str	'us/ft'

Output

NDArray[np.float64]

smooth

petrolib.data_qc_io.filt

```
smooth(x, window=5, kind='boxcar', *, sigma=None)
```

Purpose

Smooth a 1-D curve with a boxcar, Gaussian, or median filter.

'boxcar': centered moving average, `np.convolve(mode='same')` (zero-padded edges); `window<=1` returns a copy. 'gaussian': normalized Gaussian kernel of standard deviation `sigma` samples (kernel length `max(3, round(6*sigma)|1)`), same convolution. 'median': running median over an odd window with edge-replicated padding.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
window	int	5
kind	str	'boxcar'
sigma (kw-only)	float None	None

Output

`numpy.ndarray` of `float64`

Sources

- Okwoli, E. and Potter, D. K. (2023). "Probe Screening Techniques for Rapid, High-Resolution Core Analysis and Their Potential Usefulness for Energy Transition Applications." *Petrophysics*, 64(5), 640-655. DOI: 10.30632/PJV64N5-2023a3.
- Liu, Z., Zhang, X., Fan, Q., Zhou, L., Zhang, Y., Zhao, Z., and Zhang, Z. (2026). Intelligent Sensors and Algorithms for Diagnosing Downhole Operating Conditions of Wireline Logging Instruments. *Petrophysics*, 67(2), 295-317. DOI: 10.30632/PJV67N2-2026a4.
- Strobel, J. (2024). "Petrophysical Analyses for Supporting the Search for a Claystone-Hosted Nuclear Repository." *Petrophysics* 65(3), 302-316. DOI: 10.30632/PJV65N3-2024a2.

snell_angle

petrolib.acoustic_geomech

```
snell_angle(v1, v2, *, degrees=True)
```

Purpose

Snell critical/refraction angle $\arcsin(V1/V2)$ from two velocities.

With $V1$ the slower medium this is the critical angle. Returned in degrees by default (`degrees=False` for radians).

Input parameters

Parameter	Type	Default
<code>v1</code>	ArrayLike	required
<code>v2</code>	ArrayLike	required
<code>degrees</code> (kw-only)	bool	True

Output

numpy.ndarray of float64

snr_db

petrolib.data_qc_io.signal

```
snr_db(signal, *, noise=None, noise_std=None, guard=0.0)
```

Purpose

Signal-to-noise ratio in dB: $10 \cdot \log_{10}(\text{mean}(s^2)/P_{\text{noise}})$.

Give either a noise record/residual (noise, power mean(n^2)+guard) or a known standard deviation (noise_std, power noise_std**2). Zero noise power returns inf; guard reproduces the additive $+1e-30$ convention of the VMD paper.

Input parameters

Parameter	Type	Default
signal	ArrayLike	required
noise (kw-only)	ArrayLike None	None
noise_std (kw-only)	float None	None
guard (kw-only)	float	0.0

Output

float

Sources

- Module 2: Fiber-Optic Sensing for Well Integrity Diagnosis Implements ideas from: Bazaid et al., "Pioneering Well Logging: The Role of Fiber Optics in Modern Monitoring for Well Integrity Diagnosis," Petrophysics, vol. 66, no. 4, pp. 555–565, August 2025.
- Module 8: Automated Anomaly Detection via Signal Mode Decomposition Implements ideas from: Wang et al., "Automated Anomaly Detection of Multimetallic Tubulars in Well Integrity Logs Using Signal Mode Decomposition and Physics-Informed Decision Making," Petrophysics, vol. 66, no. 4, pp. 647–661, August 2025.

so_from_co

petrolib.nuclear

```
so_from_co(cor, cor_water, cor_oil, *, clip=(0.0, 1.0))
```

Purpose

Oil saturation from C/O endpoint interpolation $S_o = (C_{OR} - C_{OR_w}) / (C_{OR_o} - C_{OR_w})$.

Input parameters

Parameter	Type	Default
cor	ArrayLike	required
cor_water	ArrayLike	required
cor_oil	ArrayLike	required
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 2: Case Studies Demonstrating the Impact of Cement Quality on Carbon/Oxygen and Elemental Analysis From Casedhole Pulsed-Neutron Logging. Wang, Sullivan, Seth, Barnes, Wilson, Lazorek (2020). DOI: 10.30632/PJV61N3-2020a2. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 4: New Generation of Pulsed-Neutron Multidetector Comparison in a Challenging Multistack Clastic Reservoir - A Case Study in a Brown Field, Malaysia. Johare, Mohd Amin, Prasodjo, Afandi, Din (2020). DOI: 10.30632/PJV61N6-2020a4. Petrophysics Vol. 61 No. 6 (Dec 2020).

sobel_gradient

petrolib.borehole_image

```
sobel_gradient(image)
```

Purpose

3x3 Sobel gradient -> (gx, gy, magnitude) with zero borders.

gx is the horizontal (column) derivative, gy the vertical (row) derivative; magnitude = $\sqrt{gx^2 + gy^2}$. Interior pixels only; the one-pixel border stays zero.

Input parameters

Parameter	Type	Default
image	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64], ndarray[float64]]

Sources

- Ultrasonic Microscopy Imaging of Carbonate Reservoir Pore Structure. Based on: Chen et al., "New Methodology for Ultrasonic Microscopy Imaging of Carbonate Reservoirs' Pore Structure", Petrophysics, Vol. 66, No. 2, April 2025, pp. 267-282.

soft_envelope_logprior

petrolib.inversion_numerics.stochastic

```
soft_envelope_logprior(x, lo, hi)
```

Purpose

Soft box log-prior: 0 inside [lo, hi] and a quadratic penalty outside.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
lo	ArrayLike	required
hi	ArrayLike	required

Output

float

sphere_pack_volume

petrolib.data_qc_io.synth

```
sphere_pack_volume(shape, r_range, target_phi, *, rng=None, max_grains=100000)
```

Purpose

Random sphere pack on a voxel grid (0 = pore, 1 = grain).

Spheres with integer voxel radii uniform on `r_range` (inclusive) are stamped at uniform random centers until the pore fraction drops to `target_phi` (or `max_grains` is hit). Grain fill uses the squared-distance broadcast of `src2022_12/article7` (`make_sand_pack`).

Input parameters

Parameter	Type	Default
<code>shape</code>	<code>tuple[int, int, int]</code>	required
<code>r_range</code>	<code>tuple[int, int]</code>	required
<code>target_phi</code>	<code>float</code>	required
<code>rng</code> (kw-only)	<code>np.random.Generator int None</code>	None
<code>max_grains</code> (kw-only)	<code>int</code>	100000

Output

`NDArray[np.uint8]`

Sources

- Article 7: Using Digital Rock Physics to Evaluate Novel Percussion Core Quality. Lakshtanov, Zapata, Saucier, Cook, Eve, Lancaster, Lane, Gettemy, Sincok, Liu, Geetan, Draper, Gill (2022). DOI: 10.30632/PJV63N6-2022a7. Petrophysics Vol. 63 No. 6 (Dec 2022) — Best Papers of the 2022 SPWLA Annual Symposium special issue.

spine_ribs

petrolib.nuclear

```
spine_ribs(rho_ls, rho_ss, *, rib_coeffs=(0.0, 1.0, 0.0))
```

Purpose

Spine-and-ribs compensated density -> (rho_b, drho).

$drho = c0 + c1*d + c2*d**2$ with $d = rho_ls - rho_ss$ (long-short spacing); $rho_b = rho_ls + drho$.

Input parameters

Parameter	Type	Default
rho_ls	ArrayLike	required
rho_ss	ArrayLike	required
rib_coeffs (kw-only)	tuple[float, float, float]	(0.0, 1.0, 0.0)

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 2: A Novel X-Ray Tool for True Sourceless Density Logging. Simon, Tkabladze, Beekman, Atobatele, De Looz, Grover, Hamichi, Jundt, McFarland, Mlcak, Reijonen, Revol, Stewart, Yeboah, Zhang (2018). DOI: 10.30632/PJV59N5-2018a1. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.

stack_gain

petrolib.data_qc_io.signal

```
stack_gain(n)
```

Purpose

Amplitude SNR gain $\sqrt{n}$ from stacking n repeat measurements.

Input parameters

Parameter	Type	Default
n	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozlowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).
- ^{13}C Magnetic Resonance Relaxation-Time Wettability Characterisation. Petrophysics, 66(6), 1073–1089. DOI: 10.30632/PJV66N6-2025a11.

stiffness_from_velocity

petrolib.acoustic_geomech

```
stiffness_from_velocity(rho, v)
```

Purpose

Stiffness coefficient $C = \rho \cdot V^2$ (the P-wave modulus M for V_p).

Input parameters

Parameter	Type	Default
rho	ArrayLike	required
v	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).
- Article 1: Predicting In-Situ Physical Properties for Gas Condensates From Fluid Pressure Gradients. Bryndzia, Kittridge (2022). DOI: 10.30632/PJV63N4-2022a1. Petrophysics Vol. 63 No. 4 (Aug 2022).

stokes_einstein

petrolib.flow_transport

```
stokes_einstein(T_K, mu, radius)
```

Purpose

Stokes-Einstein diffusivity $D = k_B T / (6 \pi \mu r)$ of a sphere of radius r in a fluid of viscosity μ . The diameter form $k_B T / (3 \pi \mu d)$ is identical with $d=2r$.

Input parameters

Parameter	Type	Default
T_K	float	required
mu	float	required
radius	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Wettability Effects on Osmosis as an Oil-Mobilization Mechanism During Low-Salinity Waterflooding. Fredriksen, Rognmo, Sandengen, Ferno (2017). Petrophysics Vol. 58, No. 1 (February 2017), pp. 28-35. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Quantifying the Impact of Petrophysical Properties on Spatial Distribution of Contrasting Nanoparticle Agents in the Near-Wellbore Region. Kai Cheng, Aderonke Aderibigbe, Masoud Alfi, Zoya Heidari, John Killough (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 447-460. DOI: none assigned (this issue predates SPWLA DOI assignment).

stress_permeability

petrolib.flow_transport

```
stress_permeability(k0, *, gamma, ncs, ncs0=0.0)
```

Purpose

Exponential permeability decline $k = k_0 \cdot \exp(-\gamma \cdot (NCS - NCS_0))$.

k_0 is the permeability at the reference net confining stress NCS_0 (0 for the plain $k_0 \cdot \exp(-\gamma \cdot NCS)$ form).

Input parameters

Parameter	Type	Default
k_0	ArrayLike	required
γ (kw-only)	float	required
ncs (kw-only)	ArrayLike	required
ncs_0 (kw-only)	float	0.0

Output

numpy.ndarray of float64

Sources

- Article 2: Steady-State Stress-Dependent Permeability Measurements of Tight Oil-Bearing Rocks. Chhatre, Braun, Sinha, Determan, Passey, Zirkle, Wood, Boros, Berry, Leonardi, Kudva (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 116-124. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Microstructural Investigation of Stress-Dependent Permeability in Tight-Oil Rocks. King, Sansone, Kortunov, Xu, Callen, Chhatre, Sahoo, Bueno (2018). DOI: 10.30632/petro_059_1_a3. Petrophysics Vol. 59 No. 1 (Feb 2018).
- Article 11: Loading Effects on Gas Relative Permeability of a Low- Permeability Sandstone. Agostini, Egermann, Jeannin, Portier, Skoczylas, Wang (2019). DOI: 10.30632/PJV60N2-2019a9. Petrophysics Vol. 60 No. 2 (Apr 2019).
- Article 8: Saturation-Height Modeling: Assessing Capillary Pressure Stress Corrections. Hulea (2018). DOI: 10.30632/PJV59N3-2018a7 (inferred - see note). Petrophysics Vol. 59 No. 3 (Jun 2018).

surface_relaxivity_from_pore

petrolib.nmr

```
surface_relaxivity_from_pore(t2_ms, radius_um, *, shape_factor=3.0)
```

Purpose

Surface relaxivity from a relaxation time and known pore size.

$\rho = \text{radius} / (\text{shape_factor} * T_2)$ — the inverse of `pore_radius_from_t2` in the surface-only limit (`shape_factor=1` gives the raw r/T_2).

Input parameters

Parameter	Type	Default
<code>t2_ms</code>	ArrayLike	required
<code>radius_um</code>	ArrayLike	required
<code>shape_factor</code> (kw-only)	float	3.0

Output

numpy.ndarray of float64

Sources

- Article 5: NMR T1-T2 Logging in Unconventional Reservoirs: Pore-Size. Distribution, Permeability, and Reservoir Quality. Ijasan, Macquaker, Luycx, Alzobaidi, Oyewole, Rudnicki (2022). DOI: 10.30632/PJV63N3-2022a5. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

surface_to_volume

petrolib.nmr

```
surface_to_volume(t2_ms, *, rho, t2_bulk=np.inf)
```

Purpose

Surface-to-volume ratio from a relaxation time, inverting Brownstein-Tarr.

$S/V = (1/T2 - 1/T2_bulk)/rho$ ($t2_bulk=inf$ gives the surface-only $1/(rho*T2)$).

Input parameters

Parameter	Type	Default
t2_ms	ArrayLike	required
rho (kw-only)	float	required
t2_bulk (kw-only)	ArrayLike	np.inf

Output

numpy.ndarray of float64

Sources

- Article 3: Nuclear Magnetic Resonance and Pore Coupling in Clay-Coated Sandstones With Anomalous Porosity Preservation, Agua Grande Formation, Reconcavo Basin, Brazil. Jacomo, Trindade, de Oliveira, Leite, Montrazi, Andreeta, Bonagamba (2018). DOI: 10.30632/PJV59N2-2018a2. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Using Advanced Logging Measurements to Develop a Robust Petrophysical Model for the Bakken Petroleum System. Simpson, Hohman, Pirie, Horkowitz (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 457-478. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Predicting Carbonate Rock Properties Using NMR Data and Generalized Interpolation-Based Techniques. Kwak, Hursan, Shao, Chen, Balliet, Eid, Guergueb (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 351-368. DOI: none assigned (this issue predates SPWLA DOI assignment).

survey_to_path

petrolib.wellbore_geometry

```
survey_to_path(md, inc, azi)
```

Purpose

Cumulative minimum-curvature trajectory, shape (n, 3).

Integrates minimum_curvature_step from the origin (0, 0, 0) over consecutive stations of a survey given as parallel md/inc/azi arrays (degrees). Column order is (TVD, North, East).

Input parameters

Parameter	Type	Default
md	ArrayLike	required
inc	ArrayLike	required
azi	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 9: Wellbore Positioning While Drilling With LWD Measurements. Poedjono, Nwosu, Martin (2019). DOI: 10.30632/PJV60N3-2019a8. Petrophysics Vol. 60 No. 3 (Jun 2019).

svd_solve

petrolib.inversion_numerics.linear

```
svd_solve(a, b, rank=None)
```

Purpose

Least-squares / truncated-SVD solve of $A \cdot x = b$ (rank-limited pseudo-inverse).

Input parameters

Parameter	Type	Default
a	ArrayLike	required
b	ArrayLike	required
rank	int None	None

Output

numpy.ndarray of float64

sw_from_permittivity

petrolib.em_dielectric

```
sw_from_permittivity(eps_meas, phi, *, eps_w, eps_hc, eps_matrix, alpha=0.5, clip=True)
```

Purpose

Water saturation by inverting crim for Sw.

$$Sw = (\epsilon_{\text{meas}}^a - (1-\phi)\epsilon_m^a - \phi\epsilon_{\text{hc}}^a) / (\phi(\epsilon_w^a - \epsilon_{\text{hc}}^a))$$

clip=True bounds the result to [0, 1].

Input parameters

Parameter	Type	Default
eps_meas	ArrayLike	required
phi	ArrayLike	required
eps_w (kw-only)	float	required
eps_hc (kw-only)	float	required
eps_matrix (kw-only)	float	required
alpha (kw-only)	float	0.5
clip (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 3: Experimental Study of the Effects of Wettability and Fluid Saturation on NMR and Dielectric Measurements in Limestone Lalitha Venkataramanan, Martin D. Hurlimann, Jeffrey A. Tarvin, Kamilla Fellah, Diana Acero-Allard, Nikita V. Seleznev (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 572-586. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Pore-Scale Evaluation of Dielectric Measurements in Formations with Complex Pore and Grain Structures. Huangye Chen and Zoya Heidari (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 587-597. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Bakken Petroleum System Characterization Using Dielectric-Dispersion Logs. Han, Misra (2018). DOI: 10.30632/PJV59N2-2018a6. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 3: Deducing Electrical Permittivity of Formations From LWD Resistivity Measurements. Stalheim (2019). DOI: 10.30632/PJV60N6-2019a3. Petrophysics Vol. 60 No. 6 (Dec 2019).

sw_from_resistivity_index

petrolib.saturation_resistivity

```
sw_from_resistivity_index(ri, *, n=2.0, b=1.0)
```

Purpose

Water saturation from the resistivity index, $S_w = (RI/b)^{(-1/n)}$.

Input parameters

Parameter	Type	Default
ri	ArrayLike	required
n (kw-only)	float	2.0
b (kw-only)	float	1.0

Output

numpy.ndarray of float64

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir
Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Drainage Capillary Pressure and Resistivity Index from Short-Wait Porous-Plate Experiments.
Dernaika, Wilson, Skjaeveland, Ebeltoft (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 369-376. DOI: none assigned (this issue predates SPWLA DOI assignment).

sw_from_sigma

petrolib.nuclear

```
sw_from_sigma(sigma_t, phi, *, sigma_ma, sigma_w, sigma_hc, vsh=0.0, sigma_sh=27.0,
clip=(0.0, 1.0))
```

Purpose

Water saturation by inverting sigma_forward for Sw.

$$Sw = (\text{Sigma}_t - (1 - \phi - V_{sh}) * \text{Sigma}_{ma} - V_{sh} * \text{Sigma}_{sh} - \phi * \text{Sigma}_{hc}) / (\phi * (\text{Sigma}_w - \text{Sigma}_{hc}))$$
clip bounds the result (None to skip).

Input parameters

Parameter	Type	Default
sigma_t	ArrayLike	required
phi	ArrayLike	required
sigma_ma (kw-only)	float	required
sigma_w (kw-only)	float	required
sigma_hc (kw-only)	float	required
vsh (kw-only)	ArrayLike	0.0
sigma_sh (kw-only)	float	27.0
clip (kw-only)	tuple[float, float] None	(0.0, 1.0)

Output

numpy.ndarray of float64

Sources

- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 5: 'Log-Soak-Log' Experiment in Tengiz Field: Novel Technology for In-Situ Imbibition Measurements to Support an Improved Oil Recovery Project. Seth, Villegas, Iskakov, Playton, Lindsell, Cordova, Turmanbekova, Wang (2019). DOI: 10.30632/PJV60N5-2019a5. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 2: Case Studies Demonstrating the Impact of Cement Quality on Carbon/Oxygen and Elemental Analysis From Casedhole Pulsed-Neutron Logging. Wang, Sullivan, Seth, Barnes, Wilson, Lazorek (2020). DOI: 10.30632/PJV61N3-2020a2. Petrophysics Vol. 61 No. 3 (Jun 2020).
- Article 4: New Generation of Pulsed-Neutron Multidetector Comparison in a Challenging Multistack Clastic Reservoir - A Case Study in a Brown Field, Malaysia. Johare, Mohd Amin, Prasodjo, Afandi, Din (2020). DOI: 10.30632/PJV61N6-2020a4. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Article 7: Multidetector Pulsed-Neutron Tool Application in a Low-Porosity Reservoir - A Case Study in Mutiara Field, Indonesia. Wijaya, Aulianagara, Guo, Naibaho, Asriwan, Amirudin (2020). DOI: 10.30632/PJV61N6-2020a7. Petrophysics Vol. 61 No. 6 (Dec 2020).

swanson_permeability

petrolib.flow_transport

```
swanson_permeability(sb_pc_apex, *, c=399.0, d=1.691)
```

Purpose

Swanson (1981) permeability from the MICP apex $k = c \cdot (S_b/P_c)_{\text{apex}} \cdot d$.

$(S_b/P_c)_{\text{apex}}$ is the maximum bulk-mercury-saturation / injection-pressure ratio; k in mD. The Wolfcamp copy uses $c=399$, the stress-MICP copy $c=339$ (pass $c=339$).

Input parameters

Parameter	Type	Default
sb_pc_apex	ArrayLike	required
c (kw-only)	float	399.0
d (kw-only)	float	1.691

Output

numpy.ndarray of float64

Sources

- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Stress Sensitivity of Mercury-Injection Measurements. Guise, Grattoni, Allshorn, Fisher, Schiffer (2018). DOI: 10.30632/petro_059_1_a2. Petrophysics Vol. 59 No. 1 (Feb 2018).

t1_inversion_recovery

petrolib.nmr

```
t1_inversion_recovery(t_ms, m0, t1_ms)
```

Purpose

Inversion-recovery magnetization $M(t) = M0 \cdot (1 - 2 \cdot \exp(-t/T1))$.

Input parameters

Parameter	Type	Default
t_ms	ArrayLike	required
m0	float	required
t1_ms	float	required

Output

numpy.ndarray of float64

Sources

- Article 5: Application of a Fast NMR T1 Relaxation Time Measurement to Sedimentary Rock Cores. Mitchell, Valori (2019). DOI: 10.30632/PJV60N3-2019a4. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Althaus, S., Chen, J., Sun, Q., and Broyles, J. D. (2024). "Determine Oil and Water Saturations in Preserved Source Rocks From 2D T1-T2 NMR." Petrophysics 65(3), 388-396. DOI: 10.30632/PJV65N3-2024a6.

t1_saturation_recovery

petrolib.nmr

```
t1_saturation_recovery(t_ms, m0, t1_ms)
```

Purpose

Saturation-recovery magnetization $M(t) = M_0 \cdot (1 - \exp(-t/T_1))$.

Input parameters

Parameter	Type	Default
t_ms	ArrayLike	required
m0	float	required
t1_ms	float	required

Output

numpy.ndarray of float64

Sources

- Article 5: Application of a Fast NMR T1 Relaxation Time Measurement to Sedimentary Rock Cores. Mitchell, Valori (2019). DOI: 10.30632/PJV60N3-2019a4. Petrophysics Vol. 60 No. 3 (Jun 2019).
- Article 2: A New Apparatus for Coupled Low-Field NMR and Ultrasonic Measurements in Rocks at Reservoir Conditions. Connolly, Sarout, Dautriat, May, Johns (2020). DOI: 10.30632/PJV61N2-2020a2. Petrophysics Vol. 61 No. 2 (Apr 2020).

t1_t2_ratio

petrolib.nmr

```
t1_t2_ratio(t1, t2)
```

Purpose

T1/T2 ratio for fluid typing.

Input parameters

Parameter	Type	Default
t1	ArrayLike	required
t2	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Frequency and Temperature Dependence of 2D NMR T1-T2 Maps of Shale. Kausik, Freed, Fella, Feng, Ling, Simpson (2019). DOI: 10.30632/PJV60N1Y2019a2. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1: High- and Low-Field NMR Relaxometry and Diffusometry of the Bakken Petroleum System. Kausik, Fella, Feng, Simpson (2017). Petrophysics Vol. 58, No. 4 (August 2017), pp. 341-351. DOI: none assigned (this issue predates SPWLA DOI assignment).

t1t2_kernel

petrolib.nmr

```
t1t2_kernel(t_echo_ms, t_wait_ms, t1_grid_ms, t2_grid_ms, *, mode='saturation')
```

Purpose

2D T1-T2 Kronecker kernel $kron(K_{T1}, K_{T2})$.
 $K_{T2} = \exp(-t_echo/T2)$; $K_{T1} = 1 - \exp(-tw/T1)$ for mode="saturation" or $1 - 2*\exp(-tw/T1)$ for mode="inversion" (the corpus disagrees on the T1-axis recovery form, so it is explicit).

Input parameters

Parameter	Type	Default
t_echo_ms	ArrayLike	required
t_wait_ms	ArrayLike	required
t1_grid_ms	ArrayLike	required
t2_grid_ms	ArrayLike	required
mode (kw-only)	str	'saturation'

Output

numpy.ndarray of float64

Sources

- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.
- Althaus, S., Chen, J., Sun, Q., and Broyles, J. D. (2024). "Determine Oil and Water Saturations in Preserved Source Rocks From 2D T1-T2 NMR." Petrophysics 65(3), 388-396. DOI: 10.30632/PJV65N3-2024a6.

t2_apparent

petrolib.nmr

```
t2_apparent(*, t2_bulk=np.inf, rho=0.0, s_over_v=0.0, D=0.0, G=0.0, TE=0.0, gamma=GAMMA_H)
```

Purpose

Apparent relaxation time $T2 = 1/\text{relaxation_rate}(\dots)$ — the inverse of `relaxation_rate` with the same keywords.

Input parameters

Parameter	Type	Default
t2_bulk (kw-only)	ArrayLike	np.inf
rho (kw-only)	ArrayLike	0.0
s_over_v (kw-only)	ArrayLike	0.0
D (kw-only)	ArrayLike	0.0
G (kw-only)	ArrayLike	0.0
TE (kw-only)	ArrayLike	0.0
gamma (kw-only)	float	GAMMA_H

Output

numpy.ndarray of float64

Sources

- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).
- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozlowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).

t2_logmean

petrolib.nmr

t2_logmean(t2_ms, amplitude)

Purpose

Amplitude-weighted geometric mean T2LM = $\exp(\text{sum}(A*\ln T2)/\text{sum}(A))$.

Returns nan on zero (or negative) total amplitude.

Input parameters

Parameter	Type	Default
t2_ms	ArrayLike	required
amplitude	ArrayLike	required

Output

float

Sources

- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Predicting Carbonate Rock Properties Using NMR Data and Generalized Interpolation-Based Techniques. Kwak, Hursan, Shao, Chen, Balliet, Eid, Guergueb (2016). Petrophysics Vol. 57, No. 4 (August 2016), pp. 351-368. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Manuaba, I. B. G. H., Najrani, H., Cavalleri, C., Moge, A., and Chapura, M. (2026). Effect of Pore-Size Distribution on Fluid Movement in Complex Reservoirs. Petrophysics, 67(3), 509-524. DOI: 10.30632/PJV67N3-2026a3 (orig. SPE-227259-MS).

t2_partition

petrolib.nmr

```
t2_partition(t2_ms, amplitude, *, cutoffs_ms=(33.0,), fractions=False)
```

Purpose

Partition T2 amplitude into $\text{len}(\text{cutoffs})+1$ bands by T2 cutoffs.

Band i sums amplitude where $\text{cutoffs}[i-1] \leq T2 < \text{cutoffs}[i]$ (the first band is $T2 < \text{cutoffs}[0]$, the last $T2 \geq \text{cutoffs}[-1]$). One cutoff gives BVI/FFI (33 ms sandstone, 100 ms carbonate); two give the CBW/capillary/free split (3/33 ms). $\text{fractions}=\text{True}$ normalizes by the total.

Input parameters

Parameter	Type	Default
t2_ms	ArrayLike	required
amplitude	ArrayLike	required
cutoffs_ms (kw-only)	tuple[float, ...]	(33.0,)
fractions (kw-only)	bool	False

Output

tuple[float, ...]

Sources

- Article 1: Reservoir Characterization Challenges Due to the Multiscale Spatial Heterogeneity in the Presalt Carbonate Sag Formation, North Campos Basin, Brazil. Chitale, Alabi, Gramin, Lepley, Piccoli (2015). Petrophysics Vol. 56, No. 6 (December 2015), pp. 552-576. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: An Integrated Petrophysical Workflow for Fluid Characterization and Contacts Identification Using NMR Continuous and Stationary Measurements in a High-Porosity Sandstone Formation, Offshore Norway. Kozlowski, Chakraborty, Jambunathan, Lowrey, Balliet, Engelman, Anensen, Kotwicki, Johansen (2021). DOI: 10.30632/PJV62N2-2021a5. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 3: Discrete Inversion Method for Nuclear Magnetic Resonance Data Processing and Its Applications to Fluid Typing and Quantification. Gao et al. (2026), Petrophysics, 67(1), 38-53. DOI: 10.30632/PJV67N1-2026a3.

tau_from_sigma

petrolib.nuclear

```
tau_from_sigma(sigma_cu)
```

Purpose

Thermal decay time from capture cross-section $\tau = 4550/\text{Sigma}$ (us).

Input parameters

Parameter	Type	Default
sigma_cu	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: A History of Nuclear Spectroscopy in Well Logging. Pemper (2020). DOI: 10.30632/PJV61N6-2020a1. Petrophysics Vol. 61 No. 6 (Dec 2020).

thermal_elongation

petrolib.depth_correction

```
thermal_elongation(length, dT, *, alpha=ALPHA_STEEL)
```

Purpose

Thermal elongation $dL = \alpha L dT$.

dT is the temperature rise over the calibration temperature; α the linear expansion coefficient (1/K).

Input parameters

Parameter	Type	Default
length	ArrayLike	required
dT	ArrayLike	required
alpha (kw-only)	float	ALPHA_STEEL

Output

numpy.ndarray of float64

Sources

- Article 2: Driller's Depth Quality Improvement: Way- Point Methodology. Bolt (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 564-575. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Depth: A Love and Hate Story. Theys (2019). DOI: 10.30632/PJV60N1Y2019a5. Petrophysics Vol. 60 No. 1 (Feb 2019).
- Article 8: Correction of Driller's Depth: Field Example Using Driller's Way-Point Depth Correction Methodology. Bolt (2019). DOI: 10.30632/PJV60N1Y2019a7. Petrophysics Vol. 60 No. 1 (Feb 2019).

thomas_stieber_phit

petrolib.porosity_lithology

```
thomas_stieber_phit(v_lam, phi_sand, phi_sh, *, v_disp=0.0)
```

Purpose

Thomas-Stieber total porosity vs laminar shale volume.

$\phi_t = \phi_{\text{sand}}(1-V_{\text{lam}}) + \phi_{\text{sh}}V_{\text{lam}} - V_{\text{disp}}$ — the laminated trend (dispersed shale subtracts pore-filling clay).

Input parameters

Parameter	Type	Default
v_lam	ArrayLike	required
phi_sand	ArrayLike	required
phi_sh	ArrayLike	required
v_disp (kw-only)	ArrayLike	0.0

Output

numpy.ndarray of float64

Sources

- Article 6: Applying a Consistent Evaluation Approach to Thin-Bedded Sands in a Gulf of Mexico Deepwater Field. Salunke, Hamman (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 511-520. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

thomas_stieber_sand_porosity

petrolib.porosity_lithology

```
thomas_stieber_sand_porosity(phi_t, v_lam, phi_sh)
```

Purpose

Sand (net) porosity $\phi_{\text{sand}} = (\phi_{\text{t}} - V_{\text{lam}} \cdot \phi_{\text{sh}}) / (1 - V_{\text{lam}})$.

Input parameters

Parameter	Type	Default
ϕ_{t}	ArrayLike	required
v_{lam}	ArrayLike	required
ϕ_{sh}	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: The Problem With Silt in Low-Resistivity Low-Contrast (LRLC) Pay Reservoirs. Belevich, Bal (2018). DOI: 10.30632/PJV59N2-2018a1. Petrophysics Vol. 59 No. 2 (Apr 2018).
- Article 6: Applying a Consistent Evaluation Approach to Thin-Bedded Sands in a Gulf of Mexico Deepwater Field. Salunke, Hamman (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 511-520. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

thomas_stieber_vlam

petrolib.porosity_lithology

```
thomas_stieber_vlam(phi_t, phi_sand, phi_sh, *, clip=None)
```

Purpose

Inverse Thomas-Stieber laminar shale $V_{lam} = (\phi_{sand} - \phi_t) / (\phi_{sand} - \phi_{sh})$.

The net-to-grain fraction is $1 - V_{lam}$.

Input parameters

Parameter	Type	Default
phi_t	ArrayLike	required
phi_sand	ArrayLike	required
phi_sh	ArrayLike	required
clip (kw-only)	_Clip	None

Output

numpy.ndarray of float64

Sources

- Article 6: Applying a Consistent Evaluation Approach to Thin-Bedded Sands in a Gulf of Mexico Deepwater Field. Salunke, Hamman (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 511-520. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Evaluating Petrophysical Properties and Volumetrics Uncertainties of Sand Injectite Reservoirs - Norwegian North Sea. Kotwicki, Baig, Johansen, Leirdal, Aftret, Sandstad, Anthonsen, Gianotten, Hansen, Firinu (2022). DOI: 10.30632/PJV63N1-2022a5. Petrophysics Vol. 63 No. 1 (Feb 2022).

thomeer_shg

petrolib.capillary_pressure

```
thomeer_shg(pc, *, bv_inf, g, pd, log_base=10.0)
```

Purpose

Thomeer hyperbola $Sh_g = Bv_inf * \exp(-G / \log_base(Pc/Pd))$; 0 below Pd.

$\log_base$ is explicit because the articles disagree — src2021_04 and src2016_10 use $\log_{10}$, src2021_08 uses the natural log — and G values differ by $\ln(10) = 2.303$ between the conventions (section 9 hazard).

Input parameters

Parameter	Type	Default
pc	ArrayLike	required
bv_inf (kw-only)	float	required
g (kw-only)	float	required
pd (kw-only)	float	required
log_base (kw-only)	float	10.0

Output

numpy.ndarray of float64

Sources

- Petrophysics Vol. 62 No. 2 (Apr 2021) (issue-level reference).
- Petrophysics Vol. 57 No. 5 (Oct 2016) (issue-level reference).
- Petrophysics Vol. 62 No. 4 (Aug 2021) (issue-level reference).
- Article 3: Free or Bound? Thomeer and NMR Porosity Partitioning in Carbonate Reservoirs, Alta Discovery, Southwestern Barents Sea. Gianotten, Rameil, Foyn, Kollien, Marre, Looyestijn, Zhang, Hebing (2021). DOI: 10.30632/PJV62N2-2021a3. Petrophysics Vol. 62 No. 2 (Apr 2021).
- Article 1: Tutorial - Petrophysics of Thinly Bedded Formations. Aldred (2021). DOI: 10.30632/PJV62N4-2021t1.

thomsen_delta

petrolib.acoustic_geomech

thomsen_delta(c13, c33, c44)

Purpose

Thomsen delta = ((C13 + C44)**2 - (C33 - C44)**2)/(2*C33*(C33 - C44)).

Input parameters

Parameter	Type	Default
c13	ArrayLike	required
c33	ArrayLike	required
c44	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).

thomsen_epsilon

petrolib.acoustic_geomech

```
thomsen_epsilon(c11, c33)
```

Purpose

Thomsen epsilon = $(C11 - C33)/(2*C33)$ (P-wave anisotropy).

Input parameters

Parameter	Type	Default
c11	ArrayLike	required
c33	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

thomsen_gamma

petrolib.acoustic_geomech

```
thomsen_gamma(c66, c44)
```

Purpose

Thomsen gamma = $(C66 - C44) / (2 * C44)$ (S-wave anisotropy).

Input parameters

Parameter	Type	Default
c66	ArrayLike	required
c44	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Untangling Acoustic Anisotropy. Market, Mejia, Mutlu, Shahri, Tudge (2015). Petrophysics Vol. 56, No. 5 (October 2015), pp. 420-439. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Method for Acoustic Anisotropy Interpretation in Shales When the Stoneley-Wave Velocity is Missing. Gu, Quirein, Murphy, Rivera Barraza, Ou (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 140-156. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Machine-Learning-Enabled Automatic Sonic Shear Processing. Liang, Lei (2021). DOI: 10.30632/PJV62N3-2021a3. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

tikhonov_solve

petrolib.inversion_numerics.linear

```
tikhonov_solve(a, b, lam=1.0, *, reg_op=None, x_ref=None, sigma=None, nonneg=False)
```

Purpose

Tikhonov-regularized least squares.

$\text{argmin}_x ||W(Ax - b)||^2 + \text{lam} * ||L(x - x_{\text{ref}})||^2$ with $L = \text{reg_op}$ (identity by default), data weights $W = \text{diag}(1/\text{sigma})$, and lam applied ****once****. $\text{nonneg}=\text{True}$ solves the augmented system with `nnls_solve`.

Input parameters

Parameter	Type	Default
a	ArrayLike	required
b	ArrayLike	required
lam	float	1.0
reg_op (kw-only)	ArrayLike None	None
x_ref (kw-only)	ArrayLike None	None
sigma (kw-only)	ArrayLike None	None
nonneg (kw-only)	bool	False

Output

numpy.ndarray of float64

Sources

- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). *Petrophysics* Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: New Method to Estimate Porosity More Accurately from NMR Data with Short Relaxation Times. Venkataramanan, Gruber, LaVigne, Habashy, Iglesias, Cohorn, Anand, Rampurawala, Jain, Heaton, Akkurt, Rylander, Lewis (2015). *Petrophysics* Vol. 56, No. 2 (April 2015), pp. 147-157. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Petrophysical Interpretation of LWD, Neutron-Induced Gamma-Ray Spectroscopy Measurements: An Inversion-Based Approach. Ajayi, Torres-Verdin, Preeg (2015). *Petrophysics* Vol. 56, No. 4 (August 2015), pp. 358-378. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Forward Mineral Modeling Using Regularized Least-Squares Regression With Singular Value Decomposition: Case Study From Qusaiba Shale. Xu, McCormick, Herron, Cheshire, Al-Salim, Almarzouq (2017). *Petrophysics* Vol. 58, No. 3 (June 2017), pp. 242-269. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: A Physics-Driven Deep-Learning Network for Solving Nonlinear Inverse Problems. Jin, Shen, Wu, Chen, Huang (2020). DOI: 10.30632/PJV61N1-2020a3. *Petrophysics* Vol. 61 No. 1 (Feb 2020).
- Article 8: Mapping Complex Injectite Bodies With Multiwell Electromagnetic 3D Inversion Data. Clegg, Eriksen, Best, Tollefsen, Kowicki, Marchant (2021). DOI: 10.30632/PJV62N1-2021a7. *Petrophysics* Vol. 62 No. 1 (Feb 2021).
- Article 3: Discrete Inversion Method for Nuclear Magnetic Resonance Data Processing and Its Applications to Fluid Typing and Quantification. Gao et al. (2026), *Petrophysics*, 67(1), 38-53. DOI: 10.30632/PJV67N1-2026a3.

timur

petrolib.nmr

```
timur(phi, swirr, *, a=4800.0, b=4.4, c=2.0)
```

Purpose

Timur (1968) NMR permeability $k = a \cdot \phi^b / Swirr^c$ (mD).
The prefactor a is a regional calibration (4800, 8581, 1360, ...) and explicit.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
swirr	ArrayLike	required
a (kw-only)	float	4800.0
b (kw-only)	float	4.4
c (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Cross-Calibrated Permeabilities: A Reliable and Physically Coherent Methodology Implements the methodology from: Sifontes, V., Leal, L.A., Farfan, M., and Mussio Arias, A., 2025, "Cross-Calibrated Permeabilities: A Reliable and Physically Coherent Methodology," Petrophysics, Vol. 66, No. 3, pp. 449–466.
- Yu, H., Pan, B., Guo, Y., Wang, Y., Zhou, Y., and Li, Y. (2026). A Knowledge- Guided Data-Driven Method for Predicting Reservoir Parameters. Petrophysics, 67(2), 351–373. DOI: 10.30632/PJV67N2-2026a7.
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.

timur_coates

petrolib.nmr

```
timur_coates(phi, ffi, bvi, *, C=10.0, m=4.0, n=2.0, form='classic')
```

Purpose

Timur-Coates NMR permeability (mD).

form="classic": $k = (\phi/C)^m (FFI/BVI)^n$ (the quartic-ratio form, $m=4/n=2$).

form="prefactor": $k = C \phi^m (FFI/BVI)^n$. These are algebraically distinct families sharing the name; copies that report in different units apply their scale in the facade.

Input parameters

Parameter	Type	Default
phi	ArrayLike	required
ffi	ArrayLike	required
bvi	ArrayLike	required
C (kw-only)	float	10.0
m (kw-only)	float	4.0
n (kw-only)	float	2.0
form (kw-only)	str	'classic'

Output

numpy.ndarray of float64

Sources

- Article 4: Method for Predicting Permeability of Complex Carbonate Reservoirs Using NMR Logging Measurements Willian Trevizan, Paulo Neto, Bernardo Coutinho, Vinicius F. Machado, Edmilson H. Rios, Songhua Chen, Wei Shao, Pedro Romero (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 240-252. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: NMR Logging Data Processing. Shao, Balliet (2022). DOI: 10.30632/PJV63N3-2022a3. Petrophysics Vol. 63 No. 3 (Jun 2022) — NMR SIG Special Issue.

toc_from_yield

petrolib.nuclear

```
toc_from_yield(y_toc, calib)
```

Purpose

Total organic carbon from a carbon yield $TOC = calib * Y_{TOC}$.

Input parameters

Parameter	Type	Default
y_toc	ArrayLike	required
calib	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Borehole Carbon Corrections Enable Accurate TOC Determination from Nuclear Spectroscopy. Jeffrey Miles and Rob Badry (2014). Petrophysics Vol. 55, No. 3 (June 2014), pp. 219-228. DOI: none assigned (this issue predates SPWLA DOI assignment).

toc_passey_dlogr

petrolib.porosity_lithology

```
toc_passey_dlogr(rt, overlay, rt_base, overlay_base, *, lom=10.0, k_overlay=0.02)
```

Purpose

Passey Delta-log-R TOC TOC = dlogR * 10**(2.297 - 0.1688*L0M).
dlogR = log10(Rt/Rt_base) + k_overlay*(overlay - overlay_base). The overlay curve and its coefficient are explicit: sonic +0.02 us/ft, density -2.50 g/cc, neutron +4.0 v/v, GR +0.01 API.

Input parameters

Parameter	Type	Default
rt	ArrayLike	required
overlay	ArrayLike	required
rt_base	ArrayLike	required
overlay_base	ArrayLike	required
lom (kw-only)	float	10.0
k_overlay (kw-only)	float	0.02

Output

numpy.ndarray of float64

Sources

- Article 2: Total Organic Carbon Characterization Using Neural-Network Analysis of XRF Data. Lawal, Mahmoud, Alade, Abdulraheem (2019). DOI: 10.30632/PJV60N4-2019a2. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning Implements the methodology from: Dong, M., Shang, J., Tian, L., Wu, M., and Nie, X., 2025, "Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning - A Case Study of the Ordos Basin-Yanchang Formation," Petrophysics, Vol. 66, No. 3, pp. 425-448.

toc_schmoker

petrolib.porosity_lithology

```
toc_schmoker(rho_b, *, a=154.497, b=57.261)
```

Purpose

Schmoker density TOC $\text{TOC}[\text{wt}\%] = a/\rho_b - b$.

The (154.497, 57.261) Schmoker-Hester constants are the corpus default; other basins pass their own. The density-difference Schmoker variant $(\rho_{\text{min}} - \rho_b) / (\rho_{\text{min}} - \rho_{\text{org}}) * 100 * f_c$ is a distinct form kept in its facade.

Input parameters

Parameter	Type	Default
ρ_b	ArrayLike	required
a (kw-only)	float	154.497
b (kw-only)	float	57.261

Output

numpy.ndarray of float64

Sources

- Article 2: Integrated Petrofacies Characterization and Interpretation of Depositional Environment of the Bakken Shale in the Williston Basin, North America. Bhattacharya, Carr (2016). Petrophysics Vol. 57, No. 2 (April 2016), pp. 96-111. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Investigating Delaware Basin Bone Spring and Wolfcamp Observations Through Core-Based Quantification: Case Study in the Integrated Workflow, Including Closed Retort Comparisons. Perry, Zumberge, Cheng (2022). DOI: 10.30632/PJV63N1-2022a6. Petrophysics Vol. 63 No. 1 (Feb 2022).

tool_response

petrolib.data_qc_io.filt

```
tool_response(x, dz, fwhm)
```

Purpose

Convolve a fine-scale curve with a Gaussian tool vertical response.

$\sigma = \text{fwhm} / (2 \cdot \sqrt{2 \cdot \ln 2})$; the kernel spans $\sim 6 \sigma$ with an odd length ($\max(3, \text{round}(6 \cdot \sigma / \text{dz}) | 1)$), is unit-normalized, and is applied with `np.convolve(mode='same')`. This is the log-simulation / upscaling operator for a tool of aperture `fwhm` on a grid of spacing `dz` (same length units).

Input parameters

Parameter	Type	Default
x	ArrayLike	required
dz	float	required
fwhm	float	required

Output

numpy.ndarray of float64

Sources

- Bennis, M. and Torres-Verdin, C. (2023). "Numerical Simulation of Well Logs Based on Core Measurements: An Effective Method for Data Quality Control and Improved Petrophysical Interpretation." *Petrophysics*, 64(5), 753-772. DOI: 10.30632/PJV64N5-2023a9.

tortuosity

petrolib.nmr

```
tortuosity(d0, d_inf)
```

Purpose

Diffusive tortuosity $\tau = D0/D_inf$ (free / restricted-plateau diffusivity).

Input parameters

Parameter	Type	Default
d0	ArrayLike	required
d_inf	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Pore Size, Tortuosity, and Permeability From NMR Restricted Diffusion in Organic-Rich Chalks. Wang, Singer, Liu, Chen, Hirasaki, Vinegar (2021). DOI: 10.30632/PJV62N3-2021a1. Petrophysics Vol. 62 No. 3 (Jun 2021).
- Article 5: NMR Evaluation of Light-Hydrocarbon Composition, Pore Size, and Tortuosity in Organic-Rich Chalks. Chen, Singer, Wang, Vinegar, Nguyen, Hirasaki (2019). DOI: 10.30632/PJV60N6-2019a5. Petrophysics Vol. 60 No. 6 (Dec 2019).

total_mobility

petrolib.relperm_wettability

```
total_mobility(kr_w, kr_nw, *, mu_w, mu_nw)
```

Purpose

Total two-phase mobility $\lambda_t = k_{r_w}/\mu_w + k_{r_{nw}}/\mu_{nw}$.

Input parameters

Parameter	Type	Default
kr_w	ArrayLike	required
kr_nw	ArrayLike	required
mu_w (kw-only)	float	required
mu_nw (kw-only)	float	required

Output

numpy.ndarray of float64

Sources

- Analog Two-Phase Relative Permeability for CO₂/Brine Estimation. Petrophysics, 66(6), 969-981. DOI: 10.30632/PJV66N6-2025a4.

total_porosity

petrolib.nmr

```
total_porosity(amplitude)
```

Purpose

Total NMR porosity as the summed T2 amplitude $\phi_i = \text{sum}(A)$.

Input parameters

Parameter	Type	Default
amplitude	ArrayLike	required

Output

float

Sources

- Article 3: NMR Fluid Substitution - A New Method of Reconstructing T2 Distributions Under Primary Drainage and Imbibition Conditions. Li, Kesserwan, Jin, Ma (2021). DOI: 10.30632/PJV62N4-2021a2. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 5: New Method to Estimate Porosity More Accurately from NMR Data with Short Relaxation Times. Venkataramanan, Gruber, LaVigne, Habashy, Iglesias, Cohorn, Anand, Rampurawala, Jain, Heaton, Akkurt, Rylander, Lewis (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 147-157. DOI: none assigned (this issue predates SPWLA DOI assignment).

transmission_energy

petrolib.acoustic_geomech

transmission_energy(z1, z2)

Purpose

Transmitted energy fraction $1 - R^2$ at a normal-incidence interface.

Input parameters

Parameter	Type	Default
z1	ArrayLike	required
z2	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 7: Case Studies on Multistring Isolation Evaluation in P&A Operations. Zhang, Mueller, Bryce, Brockway, Iskander (2021). DOI: 10.30632/PJV62N6-2021a7. Petrophysics Vol. 62 No. 6 (Dec 2021).

transmission_energy

petrolib.integrity_drilling

```
transmission_energy(z1, z2)
```

Purpose

Transmitted energy fraction across an interface, $1 - R^2$.

Input parameters

Parameter	Type	Default
z1	ArrayLike	required
z2	ArrayLike	required

Output

numpy.ndarray of float64

transport_length_mix

petrolib.nuclear

```
transport_length_mix(vol_fracs, lengths)
```

Purpose

Harmonic transport/migration-length mix $L = 1/\sum(v_i/L_i)$.

Input parameters

Parameter	Type	Default
vol_fracs	ArrayLike	required
lengths	ArrayLike	required

Output

numpy.ndarray of float64

trapping_number

petrolib.relperm_wettability

```
trapping_number(nca, nb)
```

Purpose

Total trapping number $N_t = N_{ca} + N_b$ (the additive form used repo-wide).

Input parameters

Parameter	Type	Default
nca	ArrayLike	required
nb	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Impact of Wettability on Residual Oil Saturation and Capillary Desaturation Curves K. J. Humphry, B. M. J. M. Suijkerbuijk, H. A. van der Linde, S. G. J. Pieterse, S. K. Masalmeh (2014). Petrophysics Vol. 55, No. 4 (August 2014), pp. 313-318. DOI: none assigned (this issue predates SPWLA DOI assignment).

uniform_logprior

petrolib.inversion_numerics.stochastic

```
uniform_logprior(x, bounds)
```

Purpose

Uniform (box) log-prior: 0 inside bounds, -inf outside.

Input parameters

Parameter	Type	Default
x	ArrayLike	required
bounds	list[tuple[float, float]]	required

Output

float

unmix

petrolib.inversion_numerics.linear

```
unmix(measured, response_matrix, *, sigma=None, closure=None, nonneg='nnls',
      normalize=False)
```

Purpose

Mineral / spectral unmixing $\text{measured} \sim \text{response_matrix} @ x$.

`closure` in `{None, 'row', 'simplex'}` enforces sum-to-one ('row' appends a heavily weighted unit-sum equation; 'simplex' projects the solution). `nonneg` in `{None, 'clip', 'nnls', 'simplex'}` handles non-negativity. `sigma` applies $1/\text{sigma}$ data weights; `normalize` rescales the result to sum to one.

Input parameters

Parameter	Type	Default
<code>measured</code>	ArrayLike	required
<code>response_matrix</code>	ArrayLike	required
<code>sigma</code> (kw-only)	ArrayLike None	None
<code>closure</code> (kw-only)	str None	None
<code>nonneg</code> (kw-only)	str None	'nnls'
<code>normalize</code> (kw-only)	bool	False

Output

numpy.ndarray of float64

usbm_index

petrolib.relpem_wettability

```
usbm_index(area_drainage, area_imbibition)
```

Purpose

USBM wettability index $W = \log_{10}(A_{\text{drainage}} / A_{\text{imbibition}})$.

Positive is water-wet.

Input parameters

Parameter	Type	Default
area_drainage	float	required
area_imbibition	float	required

Output

float

Sources

- Article 8: Application of an Optimization Method for the Restoration of Core Samples for SCAL Experiments. Sripal, James (2018). DOI: 10.30632/petro_059_1_a7. Petrophysics Vol. 59 No. 1 (Feb 2018).

velocity_from_moduli

petrolib.acoustic_geomech

velocity_from_moduli(k, g, rho)

Purpose

P/S velocities from bulk/shear modulus and density -> (Vp, Vs).

$V_p = \sqrt{(K + 4/3G)/\rho}$; $V_s = \sqrt{G/\rho}$ (m/s with SI inputs).

Input parameters

Parameter	Type	Default
k	ArrayLike	required
g	ArrayLike	required
rho	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: Characterization of Pore Structure Variation and Permeability Heterogeneity in Carbonate Rocks Using MICP and Sonic Logs: Puguang Gas Field, China. Huang, Dou, Sun (2017). Petrophysics Vol. 58, No. 6 (December 2017), pp. 576-591. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: A New Apparatus for Coupled Low-Field NMR and Ultrasonic Measurements in Rocks at Reservoir Conditions. Connolly, Sarout, Dautriat, May, Johns (2020). DOI: 10.30632/PJV61N2-2020a2. Petrophysics Vol. 61 No. 2 (Apr 2020).

velocity_from_slowness

petrolib.acoustic_geomech

```
velocity_from_slowness(dt, *, dt_unit='us/ft')
```

Purpose

Velocity (m/s) from acoustic slowness.

`dt_unit="us/ft" -> 0.3048 * 1e6 / dt; "us/m" -> 1e6 / dt.`

Input parameters

Parameter	Type	Default
dt	ArrayLike	required
dt_unit (kw-only)	str	'us/ft'

Output

numpy.ndarray of float64

velocity_to_slowness

petrolib.units

```
velocity_to_slowness(velocity, unit='us/ft')
```

Purpose

Velocity in m/s to sonic slowness (us/ft or us/m).

Input parameters

Parameter	Type	Default
velocity	ArrayLike	required
unit	str	'us/ft'

Output

NDArray[np.float64]

voigt

petrolib.acoustic_geomech

```
voigt(f, m, *, axis=-1)
```

Purpose

Voigt (arithmetic) bound $\sum(f_i m_i)$ over the mixing axis.

f volume fractions and m component moduli along axis.

Input parameters

Parameter	Type	Default
f	ArrayLike	required
m	ArrayLike	required
axis (kw-only)	int	-1

Output

numpy.ndarray of float64

Sources

- Article 5: Thermal Conductivity Estimation From Elastic-Wave Velocity - Application of a Petrographic-Coded Model. Nina Gegenhuber, Jurgen Schon (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 51-56. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Low-Field NMR Spectrometry of Chalk and Argillaceous Sandstones: Rock-Fluid Affinity Assessed from T1/T2 Ratio. Katika, Saidian, Prasad, Fabricius (2017). Petrophysics Vol. 58, No. 2 (April 2017), pp. 126-140. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).

voigt_reuss_hill

petrolib.acoustic_geomech

```
voigt_reuss_hill(f, m, *, axis=-1)
```

Purpose

Voigt-Reuss-Hill average $0.5 \times (\text{Voigt} + \text{Reuss})$.

Input parameters

Parameter	Type	Default
f	ArrayLike	required
m	ArrayLike	required
axis (kw-only)	int	-1

Output

numpy.ndarray of float64

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.

volume_to_target

petrolib.geochem_fluids.contamination

```
volume_to_target(eta0, v_star, eta_t, *, exponent=5.0 / 12.0)
```

Purpose

Pumped volume to reach contamination eta_t -- inverse of cleanup_powerlaw.

$$V = V_star * ((eta0/eta_t)**(1/exponent) - 1).$$

Input parameters

Parameter	Type	Default
eta0	ArrayLike	required
v_star	ArrayLike	required
eta_t	ArrayLike	required
exponent (kw-only)	float	5.0 / 12.0

Output

numpy.ndarray of float64

Sources

- Article 6: Proxy-Enabled Stochastic Interpretation of Downhole Fluid Sampling Under Immiscible Flow Conditions. Kristensen, Chugunov, Cig, Jackson (2018). DOI: 10.30632/PJV59N5-2018a5. Petrophysics Vol. 59 No. 5 (Oct 2018) - "Best of 2018 SPWLA Symposium" issue.
- Article 5: Innovative Formation Tester Sampling Procedures for Carbon Dioxide and Other Reactive Components. Piazza, Vieira, Sacorague, Jones, Dai, Pearl, Aguiar (2021). DOI: 10.30632/PJV62N1-2021a4. Petrophysics Vol. 62 No. 1 (Feb 2021).

volume_to_weight_fractions

petrolib.porosity_lithology

```
volume_to_weight_fractions(v, rho)
```

Purpose

Convert volume fractions to weight fractions $w_i = v_i \cdot \rho_i / \sum(v \cdot \rho)$.

Input parameters

Parameter	Type	Default
v	ArrayLike	required
rho	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Petrophysical Interpretation of LWD, Neutron-Induced Gamma-Ray Spectroscopy Measurements: An Inversion-Based Approach. Ajayi, Torres-Verdin, Preeg (2015). Petrophysics Vol. 56, No. 4 (August 2015), pp. 358-378. DOI: none assigned (this issue predates SPWLA DOI assignment).

vs_from_vp

petrolib.acoustic_geomech

```
vs_from_vp(vp, *, method='castagna_ls')
```

Purpose

Empirical shear velocity from compressional velocity (Vp in km/s).

method: "castagna_ls" limestone $-0.05050 \cdot V_p^{**2} + 1.10168 \cdot V_p - 1.0305$; "pickett_ls" $V_p/1.9$; "pickett_dol" $V_p/1.8$; "carroll" $0.756090 \cdot V_p^{**0.81846}$; "brocher" the Brocher (2005) 4th-order polynomial. Vs in km/s.

Input parameters

Parameter	Type	Default
vp	ArrayLike	required
method (kw-only)	str	'castagna_ls'

Output

numpy.ndarray of float64

Sources

- Article 3: Real-Time Prediction of Acoustic Velocities While Drilling Vertical Complex Lithology Using AI Technique. Alsaihati, Elkatatny (2021). DOI: 10.30632/PJV62N3-2021a2. Petrophysics Vol. 62 No. 3 (Jun 2021).

vshale_from_gr

petrolib.porosity_lithology

```
vshale_from_gr(gr, gr_clean, gr_shale, *, method='linear', gcur=2.0, clip=None)
```

Purpose

Shale volume from the gamma-ray index by one of several transforms.

method: "linear" ($V_{sh}=IGR$), "larionov_tertiary" $0.083*(2^{(3.7*IGR)}-1)$, "larionov_older" $0.33*(2^{(2*IGR)}-1)$, "clavier" $1.7-\sqrt{3.38-(IGR+0.7)^2}$, "steiber" $0.5*IGR/(1.5-IGR)$, or "stieber_gcur" the normalised $(2^{(gcur*IGR)}-1)/(2^{gcur}-1)$. The IGR is computed and clipped to [0,1] internally, so copies that pass an already-computed IGR call this with gr_clean=0, gr_shale=1. clip applies to the final Vsh (opt-in; the corpus Larionov copies leave it unclipped, Clavier clips [0,1]).

Input parameters

Parameter	Type	Default
gr	ArrayLike	required
gr_clean	ArrayLike	required
gr_shale	ArrayLike	required
method (kw-only)	str	'linear'
gcur (kw-only)	float	2.0
clip (kw-only)	_Clip	None

Output

numpy.ndarray of float64

Sources

- Article 9: An Integrated Petrophysical Characterization of a Siliciclastic Tight Gas Reservoir in Neuquen Basin, Western Argentina. Carrizo, Santiago, Saldungaray (2021). DOI: 10.30632/PJV62N6-2021a9. Petrophysics Vol. 62 No. 6 (Dec 2021).
- Fitz, D.E. (2023). "Evolution of Casedhole Nuclear Surveillance Logging Through Time", Petrophysics, Vol. 64, No. 4 (August 2023), pp. 473-501. DOI: 10.30632/PJV64N4-2023a1.
- Article 6: Enhanced Learning Experience for New Petrophysicists Using Open-Source Carbonate Data and Python Programming Author: Imran M. Fadhil Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 807-838. DOI: 10.30632/PJV66N5-2025a6.
- Article 1: Improvement in Heavy-Oil Reservoir Evaluation Using Nuclear Magnetic Resonance: Long Lake and Kinosis SAGD Projects, Alberta, Canada. Cheng, Kotov, Pyke, Hanif (2015). Petrophysics Vol. 56, No. 3 (June 2015), pp. 239-250. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 3: An Efficient Approach for Predicting Young's Modulus of Sandstones Using Well Logs Without Shear-Wave Traveltime Authors: Mabkhout M. Al-Dousari, Yousef M. Al- Enezi, Ali A. Garrouch Ref: Petrophysics, Vol. 66, No. 5 (October 2025), pp. 741-762. DOI: 10.30632/PJV66N5-2025a3.
- Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning Implements the methodology from: Dong, M., Shang, J., Tian, L., Wu, M., and Nie, X., 2025, "Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning – A Case Study of the Ordos Basin-Yanchang Formation," Petrophysics, Vol. 66, No. 3, pp. 425–448.

vshale_neutron_density

petrolib.porosity_lithology

```
vshale_neutron_density(phi_n, phi_d, phi_n_sh, phi_d_sh, *, clip=None)
```

Purpose

Shale volume from neutron-density separation.

$V_{sh} = (\phi_n - \phi_d) / (\phi_{n_sh} - \phi_{d_sh})$ — the sand separation scaled by the shale-point separation.

Input parameters

Parameter	Type	Default
phi_n	ArrayLike	required
phi_d	ArrayLike	required
phi_n_sh	ArrayLike	required
phi_d_sh	ArrayLike	required
clip (kw-only)	_Clip	None

Output

numpy.ndarray of float64

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Jácomo, M.H., Hartmann, G.A., Rebelo, T.B., Mattos, N.H., Batezelli, A., Leite, E.P. (2023). "Mineralogical Modeling and Petrophysical Properties of the Barra Velha Formation, Santos Basin, Brazil", Petrophysics, Vol. 64, No. 4, pp. 518-543. DOI: 10.30632/PJV64N4-2023a3.

vti_engineering_moduli

petrolib.acoustic_geomech

```
vti_engineering_moduli(c11, c33, c44, c66, c13, *, c12=None)
```

Purpose

VTI engineering moduli from the stiffness constants.

Returns {"Ev", "Eh", "nu_v", "nu_h", "Gvh", "Ghh"} -- vertical/horizontal Young's moduli, Poisson's ratios, and shear moduli. c12 defaults to $c11 - 2*c66$. Standard VTI relations::

$E_v = c33 - 2*c13**2/(c11 + c12)$ $E_h = (c11 - c12)*(c11*c33 - 2*c13**2 + c12*c33)/(c11*c33 - c13**2)$ $\nu_v = c13/(c11 + c12)$ $\nu_h = (c12*c33 - c13**2)/(c11*c33 - c13**2)$ $G_{vh} = c44$, $G_{hh} = c66$

Input parameters

Parameter	Type	Default
c11	ArrayLike	required
c33	ArrayLike	required
c44	ArrayLike	required
c66	ArrayLike	required
c13	ArrayLike	required
c12 (kw-only)	ArrayLike None	None

Output

dict[str, ndarray[float64]]

Sources

- Article 1: Lessons Learned From Casedhole Formation Evaluation Along Unconventional Horizontal Wells. Sullivan, Wang, Bolshakov, Song, Lazorek, Tohidi, Seth (2020). DOI: 10.30632/PJV61N3-2020a1. Petrophysics Vol. 61 No. 3 (Jun 2020).

warp_to_reference

petrolib.depth_matching

```
warp_to_reference(target, path, n_ref, *, reduce='mean')
```

Purpose

Collapse a DTW path into a target curve aligned to the reference grid.
For each reference index the matched target samples are combined (reduce='mean'); reference indices absent from the path are filled from the nearest populated index.

Input parameters

Parameter	Type	Default
target	ArrayLike	required
path	list[tuple[int, int]]	required
n_ref	int	required
reduce (kw-only)	str	'mean'

Output

numpy.ndarray of float64

Sources

- Article 9: Automated Well-Log Pattern Alignment and Depth- Matching Techniques: An Empirical Review and Recommendations. Ezenkwu, Guntoro, Starkey, Vaziri, Addario (2023). DOI: 10.30632/PJV64N1-2023a9. Petrophysics Vol. 64 No. 1 (Feb 2023).

washburn_diameter

petrolib.capillary_pressure

```
washburn_diameter(pc, *, sigma, theta_deg=0.0, absolute=True)
```

Purpose

Pore-throat diameter $d = 4 \cdot \sigma \cdot \cos(\theta) / P_c$.

The diameter dialect of washburn_radius (src2022_10 uses it with $\theta=180$; src2018_02's $-4 \cdot \cos$ variant is this with the mercury angle's sign folded in — its facade passes `absolute=True`).

Input parameters

Parameter	Type	Default
pc	ArrayLike	required
sigma (kw-only)	ArrayLike	required
theta_deg (kw-only)	ArrayLike	0.0
absolute (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Petrophysics Vol. 63 No. 5 (Oct 2022) (issue-level reference).
- Petrophysics Vol. 59 No. 1 (Feb 2018) (issue-level reference).

washburn_radius

petrolib.capillary_pressure

```
washburn_radius(pc, *, sigma, theta_deg=0.0, absolute=True)
```

Purpose

Pore-throat radius from capillary pressure: $r = 2 \cdot \sigma \cdot \cos(\theta) / P_c$.

Inverse of young_laplace_pc (their round trip is a library property test).

Input parameters

Parameter	Type	Default
pc	ArrayLike	required
sigma (kw-only)	ArrayLike	required
theta_deg (kw-only)	ArrayLike	0.0
absolute (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).

water_filled_porosity

petrolib.em_dielectric

```
water_filled_porosity(eps_meas, *, eps_matrix, eps_w, alpha=0.5, clip=True)
```

Purpose

Water-filled porosity from a two-component (water/matrix) CRIM inversion.

$\phi_w = (\text{eps_meas}^a - \text{eps_m}^a) / (\text{eps_w}^a - \text{eps_m}^a)$; clip=True bounds the result to [0, 1].

Input parameters

Parameter	Type	Default
eps_meas	ArrayLike	required
eps_matrix (kw-only)	float	required
eps_w (kw-only)	float	required
alpha (kw-only)	float	0.5
clip (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 6: Application of an Oil-Movability Quicklook Technique Using Dielectric Measurements at Four Depths of Investigation. S.T. Grayson and J.L. Hemingway (2014). Petrophysics Vol. 55, No. 5 (October 2014), pp. 461-469. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 6: Improving Dielectric Interpretation by Calibrating Matrix Permittivity and Solving Dielectric Mixing Laws With a New Graphical Method. Wang, Wang, Toumelin, Brown, Crousse (2018). DOI: 10.30632/PJV59N2-2018a5. Petrophysics Vol. 59 No. 2 (Apr 2018).

water_permittivity

petrolib.em_dielectric

```
water_permittivity(rw, freq_hz, *, eps_real=55.0, eps_dl=15.0)
```

Purpose

Brine complex permittivity from rw (ohm.m) at GHz-band frequencies.

$\epsilon^* = \epsilon_{\text{real}} - j(\epsilon_{\text{dl}} + \sigma_w / (\omega \epsilon_0))$ with the brine conductivity $\sigma_w = 1/rw$. ϵ_{real} is the real permittivity of water and ϵ_{dl} an added dielectric-loss (double-layer) term.

Input parameters

Parameter	Type	Default
rw	ArrayLike	required
$freq_hz$	ArrayLike	required
ϵ_{real} (kw-only)	float	55.0
ϵ_{dl} (kw-only)	float	15.0

Output

numpy.ndarray of complex128

waxman_smits_conductivity

petrolib.saturation_resistivity

```
waxman_smits_conductivity(sw, *, cw, qv, b, phi, m_star=2.0, n_star=2.0)
```

Purpose

Waxman-Smits total conductivity.

$C_t = S_w^{n^*} * C_w * \phi^{m^*} + B * Q_v * \phi^{m^*} * S_w^{(n^*-1)}$ — the Archie term plus the cation-exchange term. Algebraically identical to the $(S_w^{n^*}/F^*) (C_w + B * Q_v / S_w)$ form some articles use. *b* is the equivalent cation conductance (S/m per meq/ml) and is an explicit argument — the articles use fixed constants, Waxman-Thomas temperature laws, and saturating correlations, so no default is safe. *m_star* and *n_star* are the shaly-rock calibrations (NOT Archie's *m*, *n*).

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
cw (kw-only)	ArrayLike	required
qv (kw-only)	ArrayLike	required
b (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required
m_star (kw-only)	float	2.0
n_star (kw-only)	float	2.0

Output

numpy.ndarray of float64

Sources

- Article 1: Review of Existing Shaly-Sand Models and Introduction of a New Method Based on Dry-Clay Parameters. Max Peeters and Antony Holmes (2014). Petrophysics Vol. 55, No. 6 (December 2014), pp. 543-553. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

waxman_smits_sw

petrolib.saturation_resistivity

```
waxman_smits_sw(ct, *, cw, qv, b, phi, m_star=2.0, n_star=2.0, lo=1e-06, hi=1.0,
n_iter=100, tol=1e-12, clip=None)
```

Purpose

Invert waxman_smits_conductivity for Sw by bisection.

ct is the measured total conductivity (S/m). The bracket [lo, hi] and iteration controls are explicit so facades can reproduce their article's solver settings exactly.

Input parameters

Parameter	Type	Default
ct	ArrayLike	required
cw (kw-only)	ArrayLike	required
qv (kw-only)	ArrayLike	required
b (kw-only)	ArrayLike	required
phi (kw-only)	ArrayLike	required
m_star (kw-only)	float	2.0
n_star (kw-only)	float	2.0
lo (kw-only)	float	1e-06
hi (kw-only)	float	1.0
n_iter (kw-only)	int	100
tol (kw-only)	float	1e-12
clip (kw-only)	tuple[float, float] None	None

Output

numpy.ndarray of float64

Sources

- Article 5: Graphical Solutions for Laminated and Dispersed Shaly Sands. Bootle (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 51-59. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Petrophysics Vol. 65 No. 5 (Oct 2024) (issue-level reference).

weight_to_volume_fractions

petrolib.porosity_lithology

```
weight_to_volume_fractions(w, rho)
```

Purpose

Convert weight fractions to volume fractions $v_i = (w_i/\rho_i)/\text{sum}(w/\rho)$.

Input parameters

Parameter	Type	Default
w	ArrayLike	required
rho	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 1: Thermal Maturity-Adjusted Log Interpretation (TMALI) in Organic Shales. Craddock, Miles, Lewis, Pomerantz (2019). DOI: 10.30632/PJV60N5-2019a1. Petrophysics Vol. 60 No. 5 (Oct 2019).

weights_to_yields

petrolib.nuclear

```
weights_to_yields(fy2w, s, w)
```

Purpose

Spectroscopy yield from a weight fraction $Y = W / (FY2W * S)$ (inverse).

Input parameters

Parameter	Type	Default
fy2w	ArrayLike	required
s	ArrayLike	required
w	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Formation Chlorine Measurement From Spectroscopy Enables Water Salinity Interpretation: Theory, Modeling, and Applications. Miles, Mosse, Grau (2020). DOI: 10.30632/PJV61N6-2020a2. Petrophysics Vol. 61 No. 6 (Dec 2020).

welge_shock

petrolib.relperm_wettability

```
welge_shock(sw, fw, swc)
```

Purpose

Welge tangent construction: (swf, fwf, sw_avg).

The Buckley-Leverett shock front is the tangent to $fw(S_w)$ drawn from $(S_{wc}, 0)$; found as the maximum secant slope $fw/(S_w - S_{wc})$ over the interior. $sw_avg = swf + (1 - fwf)/slope$ is the average saturation behind the front at breakthrough.

Input parameters

Parameter	Type	Default
sw	ArrayLike	required
fw	ArrayLike	required
swc	float	required

Output

tuple[float, float, float]

Sources

- Article 4: Low-Salinity Waterflooding: Facts, Inconsistencies and the Way Forward. Hamon (2016). Petrophysics Vol. 57, No. 1 (February 2016), pp. 41-50. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 4: Mud-Filtrate Invasion in Laminated and Spatially Heterogeneous Rocks: High-Resolution In-Situ Visualization and Analysis Using Time-Lapse X-Ray Microcomputed Tomography (Micro-CT). Schroeder, Torres-Verdin (2022). DOI: 10.30632/PJV63N5-2022a4. Petrophysics Vol. 63 No. 5 (Oct 2022).

WellboreData (class)

petrolib.data_qc_io.io

```
WellboreData(well_name, uwi)
```

Purpose

Hierarchical metadata + n-D channels with units and axes.

meta holds well_name, uwi, and the format version; channels maps name -> {"data", "units", "axes", "shape"}. Round-trip via to_json/from_json (or to_dict/from_dict); shape is recomputed on reload.

Input parameters

Parameter	Type	Default
well_name	str	required
uwi	str None	None

Output

WellboreData instance

Methods

add_channel(name, data, units, axes) — Store an n-D channel; axes must name every array dimension.

to_dict() — JSON-serializable dict; channel arrays become nested lists.

to_json(path=None) — Serialize to a JSON string (indent=2); optionally write path.

from_dict(d) — Rebuild from to_dict output (or parsed JSON).

from_json(s) — Rebuild from a JSON string produced by to_json.

Sources

- Bradley et al. (2023), Petrophysics 64(6): 823-836. A proposed universal format for storing/distributing wellbore data, capable of representing simple 1D logs and complex multi-dimensional measurements (e.g. ultradeep azimuthal resistivity).

wenzel_angle

petrolib.relperm_wettability

```
wenzel_angle(theta_young_deg, roughness)
```

Purpose

Wenzel apparent contact angle (degrees): $\cos(\text{theta_app}) = r \cdot \cos(\text{theta})$.

Roughness $r \geq 1$ amplifies the intrinsic wettability; the cosine is clipped to [-1, 1].

Input parameters

Parameter	Type	Default
theta_young_deg	ArrayLike	required
roughness	float	required

Output

numpy.ndarray of float64

Sources

- Article 5: Workflow for Upscaling Wettability From the Nanoscale to Core Scale. Rucker, Bartels, Bultreys, Boone, Singh, Garfi, Scanziani, Spurin, Yesufu-Rufai, Krevor, Blunt, Wilson, Mahani, Cnudde, Luckham, Georgiadis, Berg (2020). DOI: 10.30632/PJV61N2-2020a5. Petrophysics Vol. 61 No. 2 (Apr 2020).

wetness_ratio

petrolib.geochem_fluids.mudgas

```
wetness_ratio(c1, c2, c3, c4, c5, *, percent=True)
```

Purpose

Haworth wetness $Wh = (C2+C3+C4+C5)/(C1+...+C5)$ (percent by default).

Input parameters

Parameter	Type	Default
c1	ArrayLike	required
c2	ArrayLike	required
c3	ArrayLike	required
c4	ArrayLike	required
c5	ArrayLike	required
percent (kw-only)	bool	True

Output

numpy.ndarray of float64

wiener_bounds

petrolib.flow_transport

```
wiener_bounds(k)
```

Purpose

Wiener (harmonic-arithmetic) bounds on the effective permeability of a heterogeneous medium: (k_harmonic, k_arithmetic) — the series lower and parallel upper bounds.

Input parameters

Parameter	Type	Default
k	ArrayLike	required

Output

tuple[float, float]

Sources

- Article 3: Estimation of Permeability in the McMurray Formation Using High- Resolution Data Sources. Manchuk, Garner, Deutsch (2015). Petrophysics Vol. 56, No. 2 (April 2015), pp. 125-139. DOI: none assigned (this issue predates SPWLA DOI assignment).

wilson_k

petrolib.geochem_fluids.pvt

```
wilson_k(p_pa, t_k, pc_pa, tc_k, omega)
```

Purpose

Wilson (1969) K-value estimate $(P_c/P) \cdot \exp(5.373 \cdot (1 + \omega) \cdot (1 - T_c/T))$.

Input parameters

Parameter	Type	Default
p_pa	ArrayLike	required
t_k	ArrayLike	required
pc_pa	ArrayLike	required
tc_k	ArrayLike	required
omega	ArrayLike	required

Output

numpy.ndarray of float64

window_features

petrolib.data_qc_io.filt

```
window_features(x, window=11)
```

Purpose

Edge-padded sliding-window feature stack for depth-indexed ML.

A 1-D curve of length n becomes (n, window) — row i is the window centered on sample i , with `np.pad(mode='edge')` handling the ends so every depth keeps a full window. A 2-D (n, d) matrix stacks each column in turn to $(n, d*\text{window})$. `window` must be odd.

Input parameters

Parameter	Type	Default
<code>x</code>	ArrayLike	required
<code>window</code>	int	11

Output

`numpy.ndarray` of float64

Sources

- Article 10: Machine-Learning-Based Deconvolution Method Provides High- Resolution Fast Inversion of Induction Log Data. Hagiwara (2023). DOI: 10.30632/PJV64N2-2023a10. Petrophysics Vol. 64 No. 2 (Apr 2023) — SPWLA AI/ML Special Issue.

winland_permeability

petrolib.flow_transport

```
winland_permeability(r35, phi)
```

Purpose

Permeability (mD) from Winland r35 and porosity — the inverse of winland_r35: $\log_{10}(k) = (\log_{10}(r35) - 0.732 + 0.864 \cdot \log_{10}(\phi_{pct})) / 0.588$ (ϕ a fraction).

Input parameters

Parameter	Type	Default
r35	ArrayLike	required
phi	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).

winland_r35

petrolib.flow_transport

```
winland_r35(k_md, phi)
```

Purpose

Winland-Kolodzie r35 pore-throat radius (um) at 35% mercury saturation.
 $\log_{10}(r_{35}) = 0.732 + 0.588 \cdot \log_{10}(k) - 0.864 \cdot \log_{10}(\phi_{pct})$ — k in mD, phi a *fraction* (multiplied by 100 internally; the correlation is in percent).

Input parameters

Parameter	Type	Default
k_md	ArrayLike	required
phi	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 4: Petrophysical Characterization of the Pore Space in Permian Wolfcamp Rocks. Rafatian, Capsan (2015). Petrophysics Vol. 56, No. 1 (February 2015), pp. 45-57. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 2: Modeling Permeability in Different Carbonate Rock Types. Dernaika, Masalmeh, Mansour, Al Jallad, Koronfol (2023). DOI: 10.30632/PJV64N1-2023a2. Petrophysics Vol. 64 No. 1 (Feb 2023).
- Article 8: Presenting a Multifaceted Approach to Unconventional Rock Typing and Technical Validation - Case Study in the Permian Basin and Impacts on Reservoir Characterization Workflows. Perry, Hayes (2019). DOI: 10.30632/PJV60N5-2019a8. Petrophysics Vol. 60 No. 5 (Oct 2019).

within_drilling_window

petrolib.integrity_drilling

```
within_drilling_window(ecd, pore, frac)
```

Purpose

True when the ECD sits strictly inside the pore/frac window.

Input parameters

Parameter	Type	Default
ecd	float	required
pore	float	required
frac	float	required

Output

bool

Sources

- Article 5: Formation-Evaluation Challenges and Opportunities in Deepwater Roland Chemali, Wade Samec, Ron Balliet, Paul Cooper, David Torres, Chris Jones (2014). Petrophysics Vol. 55, No. 2 (April 2014), pp. 124-135. DOI: none assigned (this issue predates SPWLA DOI assignment).

wood_fluid_modulus

petrolib.acoustic_geomech

```
wood_fluid_modulus(saturations, moduli, *, axis=-1)
```

Purpose

Wood's law effective fluid modulus (Reuss average over fluid phases).

$1/K_{fl} = \sum(S_i/K_i)$ with saturations summing to one along axis.

Input parameters

Parameter	Type	Default
saturations	ArrayLike	required
moduli	ArrayLike	required
axis (kw-only)	int	-1

Output

numpy.ndarray of float64

Sources

- Article 4: Imaging Near-Wellbore Petrophysical Properties by Joint Inversion of Sonic, Resistivity, and Density Logging Data. Shetty, Liang, Simoes, Canesin, Boyd, Zeroug, Sinha, Habashy, Domingues, Amorim, Abbots (2017). Petrophysics Vol. 58, No. 5 (October 2017), pp. 501-516. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 7: Experimental Study on the Saturation Model of Volcanic Rock Based on Fluid Distribution. Pan, Zhou, Guo, Si, Lin (2021). DOI: 10.30632/PJV62N4-2021a6. Petrophysics Vol. 62 No. 4 (Aug 2021).

work_of_adhesion

petrolib.relperm_wettability

```
work_of_adhesion(sigma_wo, contact_angle_deg)
```

Purpose

Young-Dupre work of adhesion $W = \sigma_{wo} \cdot (1 + \cos(\theta))$.

Input parameters

Parameter	Type	Default
<code>sigma_wo</code>	float	required
<code>contact_angle_deg</code>	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: An Evaluation of Spontaneous Imbibition of Water into Oil-Wet Carbonate Reservoir Cores Using Nanofluid. Abbas Roustaei (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 31-37. DOI: none assigned (this issue predates SPWLA DOI assignment).

xcorr_shift

petrolib.depth_matching

```
xcorr_shift(ref, target, *, max_lag=50, edge='trim', return_curve=False)
```

Purpose

Integer-lag bulk shift of target onto ref by maximum correlation.

Scans lag in $[-\text{max_lag}, \text{max_lag}]$ maximising the Pearson correlation. `edge='trim'` correlates only the overlapping samples; `edge='wrap'` uses a circular `np.roll`. Returns the best lag, its correlation and (if `return_curve`) the full correlation-vs-lag curve.

Input parameters

Parameter	Type	Default
ref	ArrayLike	required
target	ArrayLike	required
max_lag (kw-only)	int	50
edge (kw-only)	str	'trim'
return_curve (kw-only)	bool	False

Output

ShiftResult

Sources

- Article 10: Machine-Learning-Based Automatic Well-Log Depth Matching. Zimmermann, Liang, Zeroug (2018). DOI: 10.30632/PJV59N6Y2018a9. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 1: A Machine-Learning-Based Approach to Assistive Well-Log Correlation. Brazell, Bayeh, Ashby, Burton (2019). DOI: 10.30632/PJV60N4-2019a1. Petrophysics Vol. 60 No. 4 (Aug 2019).
- Article 3: A Machine-Learning Framework for Automating Well-Log Depth Matching. Le, Liang, Zimmermann, Zeroug, Heliot (2019). DOI: 10.30632/PJV60N5-2019a3. Petrophysics Vol. 60 No. 5 (Oct 2019).
- Pan, W., Fu, L., Xu, C., Ashby, M., Lee, H., Lee, J., Meng, F., Chen, S., Ye, Y., Jiang, H., Kim, H., Kong, H., Baek, I., Baek, J., Sun, X., Sun, H., Li, S., Zhao, Z., Ke, Y., ... and Park, J. (2026). Automatic Well-Log Depth Shifting With Data-Driven Approaches: A Contest Summary. Petrophysics, 67(2), 437–462. DOI: 10.30632/PJV67N2-2026a12.

xcorr_shift_depth

petrolib.depth_matching

```
xcorr_shift_depth(depth_a, a, depth_b, b, *, max_shift=2.0, step=0.125, use_abs_corr=False)
```

Purpose

Physical-unit core-to-log homing by interpolated correlation.

Slides *a* (on *depth_a*) against *b* (on *depth_b*) over shifts in $[-\text{max_shift}, \text{max_shift}]$ in step increments, interpolating *b* onto the shifted *depth_a* and maximising the Pearson correlation (its absolute value if *use_abs_corr*). *b* may be a single curve or (*n*, *k*) multi-curve array. Returns (*best_shift*, *best_corr*).

Input parameters

Parameter	Type	Default
<i>depth_a</i>	ArrayLike	required
<i>a</i>	ArrayLike	required
<i>depth_b</i>	ArrayLike	required
<i>b</i>	ArrayLike	required
<i>max_shift</i> (kw-only)	float	2.0
<i>step</i> (kw-only)	float	0.125
<i>use_abs_corr</i> (kw-only)	bool	False

Output

tuple[float, float]

Sources

- Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning Implements the methodology from: Dong, M., Shang, J., Tian, L., Wu, M., and Nie, X., 2025, "Comparative Analysis of TOC Logging Evaluation Methods Using Machine Learning – A Case Study of the Ordos Basin-Yanchang Formation," *Petrophysics*, Vol. 66, No. 3, pp. 425–448.

yields_to_weights

petrolib.nuclear

```
yields_to_weights(fy2w, s, y)
```

Purpose

Elemental weight fraction from a spectroscopy yield $W = FY2W * S * Y$.

Input parameters

Parameter	Type	Default
fy2w	ArrayLike	required
s	ArrayLike	required
y	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 2: Formation Chlorine Measurement From Spectroscopy Enables Water Salinity Interpretation: Theory, Modeling, and Applications. Miles, Mosse, Grau (2020). DOI: 10.30632/PJV61N6-2020a2. Petrophysics Vol. 61 No. 6 (Dec 2020).
- Article 3: Self-Compensated Pulsed-Neutron Spectroscopy Measurements. Zhou, Rose, Miles, Gendur, Wang, Sullivan (2020). DOI: 10.30632/PJV61N6-2020a3. Petrophysics Vol. 61 No. 6 (Dec 2020).

young_contact_angle

petrolib.relperm_wettability

```
young_contact_angle(sigma_so, sigma_sw, sigma_wo)
```

Purpose

Young's-law contact angle (degrees) from the three interfacial tensions.

$\cos(\theta) = (\sigma_{so} - \sigma_{sw}) / \sigma_{wo}$; the cosine is clipped to $[-1, 1]$ before arccos.

Input parameters

Parameter	Type	Default
<code>sigma_so</code>	float	required
<code>sigma_sw</code>	float	required
<code>sigma_wo</code>	float	required

Output

numpy.ndarray of float64

Sources

- Article 3: An Evaluation of Spontaneous Imbibition of Water into Oil-Wet Carbonate Reservoir Cores Using Nanofluid. Abbas Roustaei (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 31-37. DOI: none assigned (this issue predates SPWLA DOI assignment).

young_laplace_pc

petrolib.capillary_pressure

```
young_laplace_pc(radius, *, sigma, theta_deg=0.0, absolute=True)
```

Purpose

Capillary entry pressure of a tube: $P_c = 2 \cdot \sigma \cdot \cos(\theta) / r$.

$\theta_{deg}=0$ gives the pure spherical-meniscus jump $2 \cdot \sigma / r$. $absolute=True$ (default) uses $|\cos|$ — the MICP convention; pass $absolute=False$ for signed physics.

Input parameters

Parameter	Type	Default
radius	ArrayLike	required
sigma (kw-only)	ArrayLike	required
theta_deg (kw-only)	ArrayLike	0.0
absolute (kw-only)	bool	True

Output

numpy.ndarray of float64

Sources

- Article 1 (Tutorial): Capillary Pressure Tutorial Part 1 - It's a Jungle in Here. Thomas (2018). DOI: 10.30632/PJV59V4-2018t1. Petrophysics Vol. 59 No. 4 (Aug 2018) - Special Issue on Flow Diagnostics.
- Article 2: Capillary Pressure and Resistivity Index Measurements in a Mixed- Wet Carbonate Reservoir Moustafa R. Dernaika, Mohamed S. Efnik, Safouh Koronfol, Svein M. Skjaeveland, Maisoon M. Al Mansoori, Hafez Hafez, Mohammed Z. Kalam (2014). Petrophysics Vol. 55, No. 1 (February 2014), pp. 24-30. DOI: none assigned (this issue predates SPWLA DOI assignment).
- Article 8: Spontaneous Gas-Water Imbibition in Mixed-Wet Pores. Wang, He, Xiao, Wang, Ma (2020). DOI: 10.30632/PJV61N2-2020a8. Petrophysics Vol. 61 No. 2 (Apr 2020).

youngs_from_kg

petrolib.acoustic_geomech

youngs_from_kg(k, g)

Purpose

Young's modulus from bulk and shear modulus $E = 9 \cdot K \cdot G / (3 \cdot K + G)$.

Input parameters

Parameter	Type	Default
k	ArrayLike	required
g	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: Integrating Measured Kerogen Properties With Log Analysis for Petrophysics and Geomechanics in Unconventional Resources. Craddock, Mosse, Prioul, Miles, Loan, Pirie, Rylander, Lewis, Pomerantz (2018). DOI: 10.30632/PJV59N5-2018a2.
- Article 7: Application of Digital Core Technology in Wellbore Stability Research. Zhou, Ye, Zhu, Cheng, Song, Wang, Cai (2022). DOI: 10.30632/PJV63N2-2022a7. Petrophysics Vol. 63 No. 2 (Apr 2022).

youngs_poisson_dynamic

petrolib.acoustic_geomech

```
youngs_poisson_dynamic(vp, vs, rho)
```

Purpose

Dynamic Young's modulus and Poisson's ratio from velocities -> (E, nu).

$$\nu = (V_p^2 - 2V_s^2) / (2(V_p^2 - V_s^2)); E = \rho V_s^2 (3V_p^2 - 4V_s^2) / (V_p^2 - V_s^2)$$
 (E in Pa with SI inputs).

Input parameters

Parameter	Type	Default
vp	ArrayLike	required
vs	ArrayLike	required
rho	ArrayLike	required

Output

tuple[ndarray[float64], ndarray[float64]]

Sources

- Article 3: Improvement of Petrophysical Properties of Tight Sandstone and Limestone Reservoirs Using Thermochemical Fluids. Mustafa, Mahmoud, Abdulraheem, Tariq, Al-Nakhli (2020). DOI: 10.30632/PJV61N4-2020a3. Petrophysics Vol. 61 No. 4 (Aug 2020).
- Article 6: Prediction of Sonic Wave Transit Times From Drilling Parameters While Horizontal Drilling in Carbonate Rocks Using Neural Networks. Gowida, Elkatatny (2020). DOI: 10.30632/PJV61N5-2020a6. Petrophysics Vol. 61 No. 5 (Oct 2020).
- Article 7: Effect of Fluids on the Elastic Properties of 3D-Printed Anisotropic Rock Models. Dande, Stewart, Dyaur (2021). DOI: 10.30632/PJV62N5-2021a7. Petrophysics Vol. 62 No. 5 (Oct 2021).

z_beggs_brill

petrolib.geochem_fluids.pvt

z_beggs_brill(ppr, tpr)

Purpose

Beggs-Brill (1973) gas compressibility factor from pseudo-reduced P/T.

Input parameters

Parameter	Type	Default
ppr	ArrayLike	required
tpr	ArrayLike	required

Output

numpy.ndarray of float64

Sources

- Article 3: The Compressibility Factor (Z) of Shale Gas at the Core Scale. Tran, Sakhaee-Pour (2019). DOI: 10.30632/PJV60N4-2019a3. Petrophysics Vol. 60 No. 4 (Aug 2019).

z_peng_robinson

petrolib.geochem_fluids.pvt

```
z_peng_robinson(p_pa, t_k, tc_k, pc_pa, omega, *, phase='vapor')
```

Purpose

Peng-Robinson z-factor (scalar) for the vapor or liquid root.

phase="vapor" returns the largest real root, "liquid" the smallest positive real root of the PR cubic.

Input parameters

Parameter	Type	Default
p_pa	float	required
t_k	float	required
tc_k	float	required
pc_pa	float	required
omega	float	required
phase (kw-only)	str	'vapor'

Output

float

zscore

petrolib.ml_stats

```
zscore(x, *, axis=None, eps=0.0)
```

Purpose

Standardize to zero mean and unit standard deviation.

`axis=None` scales the array as a whole (the flat-log idiom); `axis=0` scales feature columns independently. `eps` is the additive guard on the denominator (`std + eps`); default 0.0 yields NaN for a constant input. Population std (`ddof=0`), matching every article copy.

Input parameters

Parameter	Type	Default
<code>x</code>	ArrayLike	required
<code>axis</code> (kw-only)	int None	None
<code>eps</code> (kw-only)	float	0.0

Output

numpy.ndarray of float64

Sources

- Article 11: Data Preconditioning for Predictive and Interpretive Algorithms: Importance in Data-Driven Analytics and Methods for Application. Frost, Quinn (2018). DOI: 10.30632/PJV59N6Y2018a10. Petrophysics Vol. 59 No. 6 (Dec 2018) — Special Issue: Data-Driven Analytics in Logging and Petrophysics.
- Article 5: Synthetic Sonic Log Generation With Machine Learning - A Contest Summary From Five Methods. Yu, Xu, Misra, Li, Ashby, et al. (2021). DOI: 10.30632/PJV62N4-2021a4. Petrophysics Vol. 62 No. 4 (Aug 2021).
- Article 3: Automatic Detection of Anomalous Density Measurements due to Wellbore Cave-in. Sen, Ong, Kainkaryam, Sharma (2020). DOI: 10.30632/PJV61N5-2020a3. Petrophysics Vol. 61 No. 5 (Oct 2020).